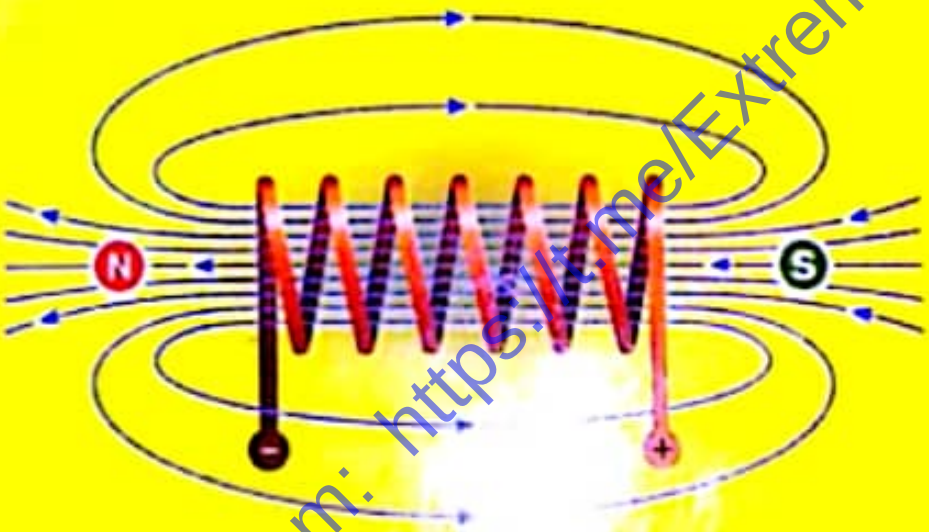
Extreme Series

PHYSICS

GRADE 11



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Based on Common Currently Used
National and International Curriculum

Extreme Series

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2016 E.C

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UNIT 1

1. Physics and Human Society

1.1 Importance of Physics to Society

1.2 Making of Physics Knowledge

1.3 The Mission of Physics

1.4 Current Status of Physics

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Physics and Human Society

Hello there ! how do you do ? welcome to this fascinating unit where you will know the importance of physics to society,learn about the mission of physics,and understand the current status of physics. For sure,you will love it. Enjoy !

1.1 Importance of Physics to Society

The struggle to live and the curiosity to know how things happen the way they are is human nature.

Did you know ?

The term physics is derived from the Greek word "physik"which means "knowledge of nature"Therefore,physics is the study of nature at particle and gigantic level and it is the basics of all sciences.

Physics : Is the branch of natural science that deals with matter,energy and their interaction.

physics is fundamental to other sciences since it contributes fundamental knowledge to understand nature and it brings about applied knowledge to develop technology so as to make life simple,safe and enjoyable.

(i) Contribution to Fundamental Knowledge

Fundamental knowledge is the knowledge gained while striving to know the natural world. At first,fundamental knowledge do not seem useful other than knowing how things work until it is proven to be applicable.

Now,let us see just few of physics fundamental knowledges that transformed the world.

☑ Newton's law of motion, F = ma was fundamental knowledge until it is proven applicable in designing vehicles,in launching satellites,and in moving bodies

☑ Einstein 's, E = mc² was fundamental knowledge before it is used for generating energy from nuclear reators and applied for production of atomic bombs and nuclear weapons.

☑ J.J Thompson's discovery of the electron was only fundamental knowledge until it is proven that electron is the basic for the working of telephone,computer,television and all the electronic devices.

Remember,fundamental knowledge is primarly focused in understanding the natural world for its own sake. As time goes, this knowledge brings practical out comes to the technological applications

(ii) Contribution to Applied Knowledge

Some of the contributions of physics to the applied knowledge are the following.

☑ Physics produce wel informed,rational, creative and innovative citizen.

☑ Physics helps to better understand other subjects. Such as chemistry, biology, engineering and medicine.

☑ Physics enhances the production of technology in the communication, transportation, military, health, economy and sectors.

☑ Physics improves quality of life by providing new devices for medical applications such as :

Computer Tomography Scanning, C T Scan

A device that uses a rotating x - ray source to examine disease or identify injured body part

Magnetic Resonance Imaging,MRI

A device that uses strong magnetic field and radio pulses to investigate damaged soft tissues and ligaments.

Ultrasonic Imaging

A device that uses sound wave to see inside patient's body for identification of cause of disease or damaged tissue.

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Laser Surgery

A device that uses gamma rays to kill cancer calls without surgery and speedy recovery time.

☑ Physics is a safe guard from hazards of outer space radiation and other yet to come natural hazards and catastrophic impacts.

Illustrative Examples

1. Is there any relation between fundamental and applied knowledges ?

   Explanation : Yes. for example,basic knowledge of optics,mirrors and lenses help the invension of microscopes and telescopes. These devices inturn help us to see things that are too small and too far to be seen by the naked eye

2. What is the benefit of physics to humanity as a whole ?

   Explanation : Physics helps to understand natural phenomena,to understand other science and to develop new technologies in all sectors that help us make life simple and intersting.

3. Why is physics considered as a basic science ?

   Explanation : Because,physics studies about matter,energy and the interaction between them, which are the fundamental constituents of nature.

4. What is the difference between fundamental knowledge and applied knowledge ?

   Explanation : Fundamental knowledge is the knowledge gained because of the human curiosity to know the natural world. While,applied knowledge is the use of fundamental knowledge to solve practical problems usually technological out put.

5. Mension some application of physics in medicine

   Explanation : - Computer Tomography scanning,CTS
   - Magnetic Resonance Imaging, MRI
   - Ultrasonic Imaging
   - Laser Surgery

1.2 Making of Physics knowledge

These days,the source of knowledge is plenty. In most cases however; culture, belief and science are sources of knowledge. Among these we have the following just to mension some.

Authority : Is the knowledge obtained from holy books or a significant figure

Example : Quran, Bible, respected elders

Intuition : Is the knowledge come from revelation

Example : I will win a lottery, it appears in my dream

Reason : Is the knowledge that relies on logic and mathematical proof

Example : 1+1+1=3

Sensory : Is the knowledge obtained from the feeling perceived by sense organs

Examples : A beautiful flower, a slow music, a deliconse food --- etc.

In general therefore, we can have experiential knowledge or non - experiential knowledge.

What is Experiential Knowledge?

Experiential or posteriori knowledge is the knowledge we acquire about the natural world through experience.

A Posteriori knowledge can be acquired through

Sensory Perception: It is the way of knowledge gathering via sense organs by seeing, hearing, tasting, touching and feeling.

Introspection: It is the way of acquiring knowledge via the sixth sense.

Momery: It is the way of capturing knowledge in the order of event or action occurance.

Testimony: It is the way of gathering knowledge from books, speachs.. etc.

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Non - Experiential Knowledge

Non - experiential or priori knowledge is the knowledge we acquire through rational insights of logic and mathematics.

Example: 1 + 3 = 4

Scientific Knowledge

Scientific knowledge is acquired by systematic observation and experimentation of natural event using logical reasonings and procedures.

Not all knowledge is valid and reliable unless critically examined and tested scientific knowledge has the following important features in this regard;

☑ Scientific knowledge is reliable and tentative

Having trust in scientific knowledge is fair and realizing that it can be falsified or disproved in light of new evidence is also a must.

☑ Scientific knowledge is supported by emperical evidence that are testable

Theoretical as well as experimental analysis give consistent results

☑ Scientific knowledge takes into account Rational argument, inferenee, skepticism, peer review and replicability.

What is more,the key element in scientific knowledge is the development of scientific attitude that requires the following characteristics to florish.

Open - Mindedness:- Respect and tolerate ideas and opinions of others and carefully examine ideas that seem against or odd with what is generally believed to be true

Skeptical: Doubt, question and reconsider conclusions in a wide range of perspectives.

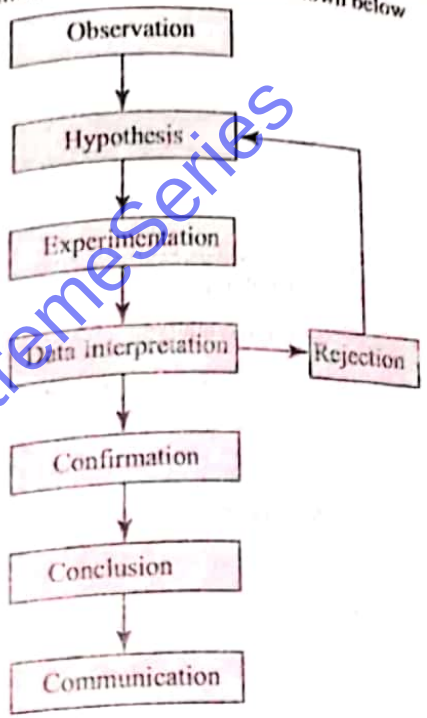
Objectivity: Make conclusions and decisions based on data facts with ww bias and subjective assumptions.

Cooperative:- Work and share ideas with others for commomon od

Perseverance:- Consistent effort to achieve desi

Science Process Skills

Science process skills are chief mechanisms of acquiring scientific knowledge and are cyclic procedures as shown below



☑ Science process skills are also called scientific method and are very crucial and common steps in most of the scientific investigations.

The primary goal of physics is the formation of theories,laws and principles making its intial step in hypothesis testing.

Illustrative Examples

6. Explain the difference between posteriori knowledge and priori knowledge.

Explanation: Posteriori knowledge is knowledge acquired through experience mostly via sense organs through seeing, touching, feeling, tasting. But, priori knowledge is knowledge acquired through reasoning

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via logic and mathematics.

Match the items in column "A" with those in column "B"

A	B
7. Experiential	A. Seeing things happening
8. Non - Experiential	B. Why things happen
9. Perception	C. Seeing, hearing feeling
10. Introspection	D. Testing variables
11. Momery	E. Scientific guess
12. Testimony	F. Series of logical steps
13. Observation	G. Posteriori knowledge
14. Question	H. Priori knowledge
15. Hypothesis	I. Sixth sense
16. Experiment	J. Capturing storing, retriving
17. Scientifi Method	K. Law of Nature

Answers

7. G	9. C	11. J	13. A	15. E	17. F
8. H	10. I	12. K	14. B	16. D	

Hypothesis : Is a tentative explanation about an event that can be accepted or rejected based on experimental result

Theory : Is an explanation that attempts to describe why a phenomenon occurs based on available data and observations

Example : Eistein theory of relativity

Law : Is a statement describing a phenomenon observed repeatedly and proven true.

Example : Newton's laws of motion

Principle : Is a basic truth,rule or law

Example : Pascal's principle

Wel - established laws and theories are :

- ☑ Consistent and compatible with existing facts
- ☑ Successfully tested against a range of conditions

Did you know ?

Laws,principles and theories in physics are made by individual human beings are influenced by : Existing social and cultural context. for example,in Gallilio period, thinking the earth revolves around the sun was irreligious. Expectation of the physicist. for example,initially Einstein assumed the universe is static,but later he found that the universe is expanding.

Illustrative Examples

18. What is scientific knowledge ?

Explanation : Is a knowledge obtaind through a regorious,critical and logical observation and experimentation about a natural phenomena

19. What are the desirable qualities of a scientific knowledge ?

Explanation : scientific knowledge is expected to be

- ➦ Reliable till it is desproved
- ➦ Supported by emperical evidence
- ➦ fascillitates peer review and replicability

20. What are the main implications of scientfic attitudes

Explanation : open - mindedness, skeptical,objectivity cooperative and perseverance.

21. What are the most common steps in a scientification method that we call science process skills ?

Explanation : observation → hypothesis → experiment → data analysis → data interpretation → rejection or confimation of hypothesis →conclusion → commonication

22. Explain the terms :

A. Hypothesis

B. Theory

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C. Law

Explanation:

A. Hypothesis : Is a tentative explanation or proposition that can be tested by observation or experimentation

B. Theory : Is a well - supported explanation of observation.

C. Law : Is a description of a natural phenomenon

23. What are the behaviors of a wel - established scientific law and theory ?

Explanation : A wel - established law or theory is consistent and compatible with the existing evidence and can be successfully tested against a range of conditions and assumptions.

Match the items in column A with those in column B

A	B
24. Hypothesis	A. Respect others ideas
25. Law	B . Doubt and question conclusions
26. Theory	C. Data fact only
27. Principle	D. Work and share ideas
28. Open - mindedness	E. Persist to achieve goal
29. Skeptical	F. Subjective assumptions
30. Objective	G. Tentative explanation
31. Cooperative	H. Proposition
32. Perseverance	I. Describtion of event
	J. Explanation of event
	k. Basic truth

Answers

12.H	15.k	18.C
13.I	16.A	19.D
14.J	17.B	20.E

1.3 The Mission of Physics

Physics studies about nature via a variety of specialization fields. The following are some of them.

Field of study	Studies About
Mechanics	Motion of things
Electricity	Electric Field
Magnetism	Magnetic Field
Electromagnetism	Electric and Magnetic fields
Optics	Light
Acoustics	Sound
Quantum Mechanics	Particles Smaller than atoms
Nuclear physics	Nuclei
Atomic physics	Atoms
Partic physics	Particles
Condensed Matter physics	Materials in solid form
Thermodynamics	Heat and Energy conversion
Cos mology	Universe
Astronomy	Sky objects : moo,star,planet

Moreover,the Career opportunities and the inter - disciplinary nature of physics in other fields that are available at the higher level are the following.

Field of Study	Studies About
Agrophysics	Agriculture
Astrophysics	Attraction,gravity, etc of planets
Bio physics	Neurology,electrical impulse, immone system
Econo physics	Economics
Environmental Physics	Environment
Geopphysics	Earth
Chemical Physics	Chemical reactions
Medical Physics	Medical Diognosis

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Socio physics	Social behavior
Physical Chemistry	Chemistry

Furthermore,there are also job opportunities for 12th grade complete physics students. The following are some of the fields of study.

Fields	Departments
Manufacturing	Metal works
	Automotive Technology
	Textile and Leathre Garment
	Wood Works
Construction	Electricity
	Plumbing
	Carpentry
	Finishing works
Information - Technology	Computer science
	Computer Maintenance
	Website Design
	Computer Graphics Design
Agriculture	Crop Production
	Animal production
	Natural Resource
	Agricultuer Technology
Health Science	Personal care
	child care
	patient care
	Nutrition

Illustrative Examples

33. What is optics ?

Explanation : Optics is the study of light wave

34. What is a coustics ?

Explanation : Acoustics is the study of sound wave

35. What is mechanics ?

Explanation : Mechanics is the study of motion and interaction of bodies through force or energy

36. What is thermodynamics ?

Explanation : Thermodynamics is the study of energy conversion between heat and other forms of energy.

37. What is a theoretical Physicistis ?

Explanation : A theoretical Physicist is one who uses intensive mathematical relations and modeling to explain a natural phenomena.

Example : Albert Einstein

38. What is an experimental physicist ?

Explanation : An experimental physicist is one who uses experimental results to explain a natural phenomena

Example: Isaac Newton

39. What are some of the branchs of physics.

Explanation : Mechanics,Electricity, Magnetism, wave optics, Acoustics, Thermodynamics, Quantum mechanics, Astronomy, Nuclear physics --- etc.

40. What are some of the interdisciplinaries with physics

Explanation : Agrophysics,Biophysics,medical physics,Geephysics,--- etc.

41. What are the possible job opportunity fields for a 12th grade complete natural science student ?

Explanation: Manufacturing, construction, Information technology,

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Agriculture and Health science

Match the items in Column A with those in Column B

A	B
42. Astronomy	A. Light
43. Cosmology	B Sound
44. Mechanics	C. Nucleus
45. Nuclear Physics	D. Motion
46. Optics	E. Universe
47. Acoustics	F. Space

Answers

42. F 43. E 44. D 45. C 46. A 47. B

1.4 Current Status of Physics

Physics studies the smallest particles such as electrons and protons as well as the most gigantic bodies such as planets, galaxies, the block hole. And, Some of the recent discoveries in physics are the following :

Exoplanets

Exoplanets also known as extrasolar planets are planets outside our solar system.

☑ As of october 2022, about 5197 - exoplanets are found in 3833 planetary systems.

Black hole

Black hole is a supermassive compact object at the center of the milky - way.

☑ The gravity of a black body is so strong that nothing can escape. not even light!

Quantum Entanglement

Quantum entanglement is the phenomena that if two particles are entangled at the start, they would be entangled forever.

☑ The possible future applications of quantum entanglement are quantum computing and quantum cryptography.

Quantum Computing

Quantum computing is a future application of quantum in place of transistors on and off states called bits.

Quantum Cryplography

Quantum Gyptography is a way of sending information in secret at a much higher level.

Particle Accelelerators

Particle accelelerater is a subatomic particle like photon,proton or electron at a very basic level.

Gravitational Wave

Gravitational wave is generated due to the collision of two massive black holes.

☑ Gravitational wave could be used as an additional instrument to radio telescope to detect evollution of the universe;

Global Warming

Global warming is an increase in earth's temperature

Illustrative Examples

48. What are the recent discoveries in physics ?

Explanation : Some of the recent discoveries in physics are : exoplanets, black hole, Quantum entanglement, particle accelerators, gravitational wave, global warming --- etc

49. What is an exoplanet ?

Explanation : Exoplanet is a planet that is located out of the solar

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system.

50. What is a black hole ?

Explanation : Black hole is a supermassive compact object at the center of the milky - way with a very strong gravitational force.

51. What is global warming ?

Explanation : Global warming is the increase in temperature of the earth.

52. What is Higgs boson ?

Explanation : Higgs boson is a particle at the very basic level

53. What is James Webb space Telescope ?

Explanation: It is an infrared telescope placed between the sun and the earth operating at - 223°c.

End of Unit Questions

1. Why do we study physics ?
2. What is a scientific knowledge ?
3. what is the scientific method ?
4. What is the difference between a scientific law and theory ?
5. What are the fields of physics
6. What are the interdisciplinaries to physics
7. What are the job opportunities in physics
8. What are the latest discoveries in physics
9. What is the use of gravitationel wave ?
10. What is the use of Quantum computing ?
11. What is the use of Quantum Cryptography ?

Short Answers

1. To get fundamental knowledge and apply it to solve practical problems
2. It is a knowledge gained through experimentation,observation and reasoning

3. Scientific method is a way of arriving to scientific knowledge using common and logical steps such as ; observation, hypothesis, experimentation, data analysis, result, conclusion, and evaluation.
4. Law summarizes a set of observations about natural phenomena while theory explains why natural phenomena occur
5. Mechanics, Electricity, Astronomy, magnetism, fluid, wave, oscillation, - etc
6. Astrophysics,medical physics,geophysics,--- etc
7. Engineering, technician, pilot, medical doctriate,--- etc
8. Black hole, quantum enactment, exoplanet --- etc
9. To detect the evolution of the universe
10. As a transistor to on and off state gates
11. To send information in secret.

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UNIT 2

2. Vectors

2.1 Vector Quantities

2.2 Graphical Method of Vector Addition

2.3 Resolution of Vectors

2.4 Vector Addition Algebrically

2.5 Vector Multiplication

Vectors

We are very excited to meet you in this wonderful unit where you will be able to define vector quantities, add vectors graphically, resolve vectors in to components, add vectors analytically, and multiply vectors. Have a nice time.

2.1 Vector Quantities

In our everyday life, we come across physical quantities that can be completly described by their magnitude alone and physical quantities that need both magnitude and direction for their complete description.

Scalar Quantity: Is a quantity that can fully be expressed using manitude or amount alone

For example, time mass, length, volume, density ..... etc. are scalar quantities.

Vector Quantity: Is a quantity that can fully be expressed using both magnitude and direction.

For example, displacement, acceleration, force momentom .... etc. are vector quantities.

In symbol, a vector quantity is represented using a letter with an on its head.

For example we can have the following vectors in symbols

Force, $\vec{F}$

Acceleration, $\vec{a}$

In graph, a vector is represented using an arrow so that the length of the arrow represents magnitude and the tip of the arrow represents direction.

For example, a force of 100N due North of east can be represented graphically as follows

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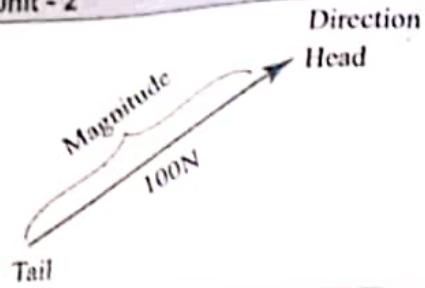


Illustrative Examples

1. Which one of the following is a vector quantity?
   A. Torque B. Pressure C. Speed D. Area
   Hint: Torque is turning effect of force clock wise or counter clockwise.
   Answer: A

2. Which of the following is a scalar quantity?
   A. Velocity C. Current
   B. Momentum D. Force
   Answer: C

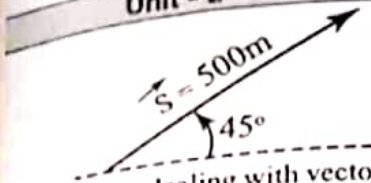
3. Which one of the following pairs represent vectors?
   A. Density and displacement C. Electric current and force
   B. Temperature and velocity D. Force and velocity
   Answer: D

4. Which one of the following pairs represent scalars?
   A. Electric field and mass C. Force and work
   B. Temperature and current D. Charge and force
   Answer: B

2.2 Graphical Method of Vector Addition

Vectors can be drawn graphically using an appropriate scale so as to specify their magnitude and direction.

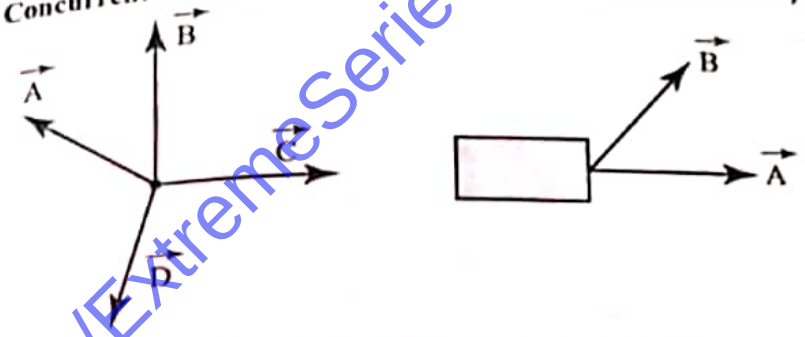
For example, if a car moved 500m north of east, then the displacement of the car can be drawn using a ruler and a protractor as follows:



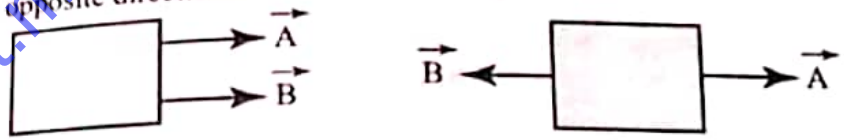
Scale
1cm = 100m
North of East = 45°

Now, before dealing with vector addition graphically, let us see concurrent, collinear, orthogonal and coplanar vectors.

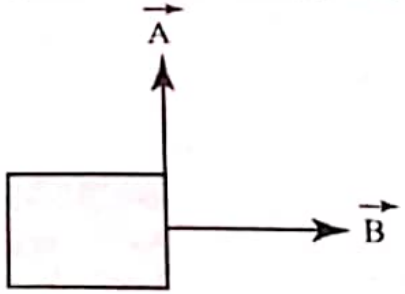
☑ **Concurrent vectors:** Are vectors that act on a body at the same point.



☑ **Collinear vectors:** Are vectors that act on a body in the same or opposite directions.



☑ **Orthogonal vectors:** Are vectors that act on a body perpendicularlly.



☑ **Coplanar vectors:** Are vectors that act on a body in the same plane.

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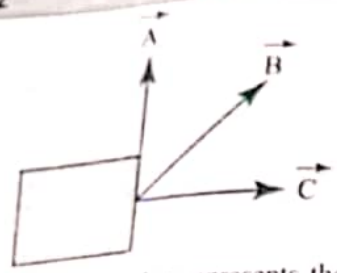
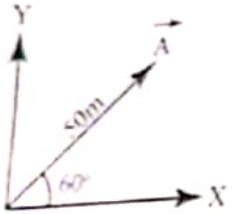


☑ **Position vector:** Is a vector that represents the location of body with respect to a reference point usually the origin. For example, the position of vector $\vec{A}$ can be reported in one of the following ways.

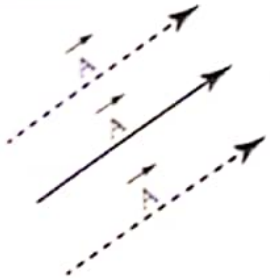


- $\vec{A}$ is 50m on a bearing of 30°
- $\vec{A}$ is 50m at 60° north of East
- $\vec{A}$ is 43.3m east and 25m North from the origin

Remember: we can move a vector from place to place keeping its magnitude and direction unchanged this process is called *translating a vector*.

Translating a vector: Is slidding or parallel shifting of a vector with out changing its magnitude and direction.

For example, vector $\vec{A}$ with solid line is the original vector, while the others are its translations.



Illustrative Examples

5. Collinear vectors act on a body at right angles.

A. True B. False

Hint: Collinear vectors act on a body in the same or opposite directions.

Answer: B

6. Concurrent vectors act on a body at different points

A. True B. False

Hint: concurrent vectors act on a body at the same point.

Answer: B

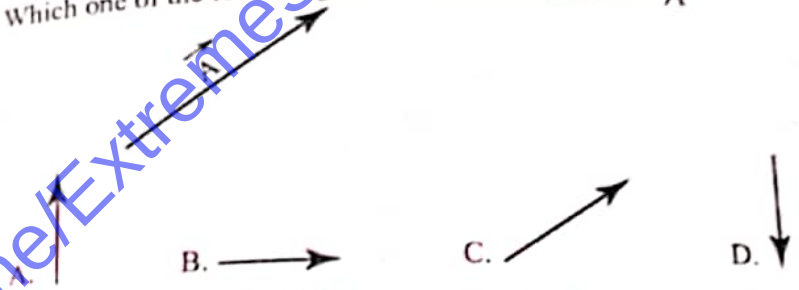
7. Coplanar vectors act on a body at different planes

A. True B. False

Hint: coplanar vectors act on a body at the same plant

Answer: B

8. Which one of the following is the translation of vector $\vec{A}$



Hint: Translation is shifting a vector keeping its magnitude and direction unchanged.

Answer: C

9. Position vector represents the position of a body from a specific direction alone.

A. True B. False

Hint: Position vector represents the position of a body from any direction.

Answer: B

Now, let us explain how we can add vectors graphically using the parallelogram, triangle and polygon methods.

Remember, resultant vector is that single vector it would have the same effect as all the original vectors taken together.

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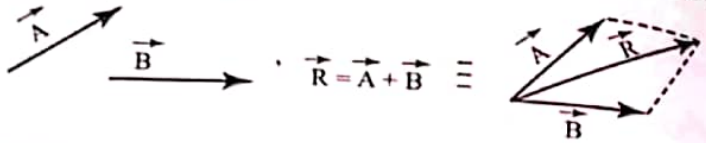
Parallelogram Method

Helps us to know the resultant of two vectors by joining them "tail to tail"

To get the resultant of two vectors using the parallelogram method, join the two vectors tail to tail to form a parallelogram. The diagonal of the parallelogram then represents the resultant vector.

For example, the resultant of vectors $\vec{A}$ and $\vec{B}$ can be obtained using the parallelogram method as follows

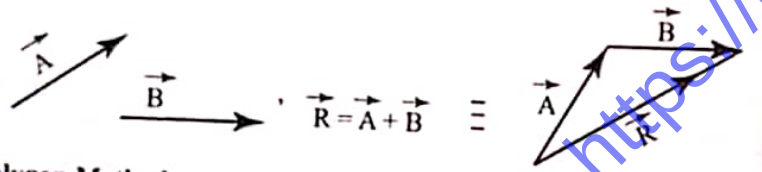


Triangle Method

Help us to get the resultant of two vectors by joining them "head to tail"

To get the resultant of two vectors using the triangle method, join them head to tail. The resultant vector is then drawn from the tail of the first vector to the head of the second vector.

For example, the resultant of vectors $\vec{A}$ and $\vec{B}$ can be obtained using the triangle method as followings.

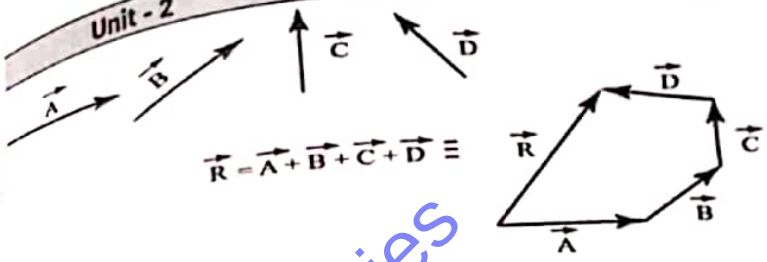


Polygon Method

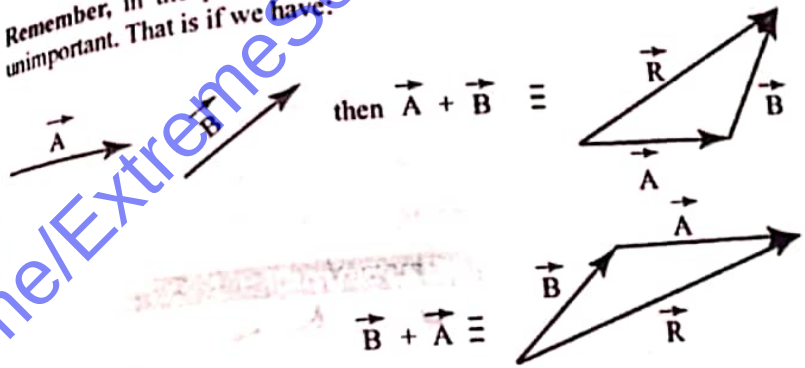
This method helps us to get the resultant of more that two vectors graphically by joining them "head to tail"

To get the resultant of more than two vectors suing the polygon method, join them head to tail. The resultant vector is then drawn from the tail of the first vector to the head of last vector.

For example, the resultant of vectors $\vec{A}$ and $\vec{B}$, $\vec{C}$ and $\vec{D}$ can be obtained using the polygon method as follows



Remember, in the process of vector addition graphically the order is unimportant. That is if we have:



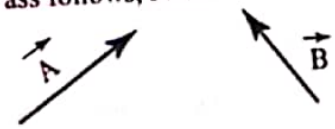
So, $\vec{A} + \vec{B} \equiv \vec{B} + \vec{A} = \vec{R}$ in both cases the resulant is the some

Illustrative Examples

10. Given two vectors $\vec{A}$ and $\vec{B}$ ass follows, Find:

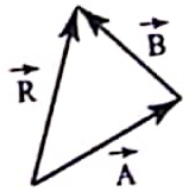
i. $\vec{A} + \vec{B}$

ii. $\vec{A} - \vec{B}$



Solution

i) $\vec{A} + \vec{B} \equiv$



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ii) $\vec{A} - \vec{B} = \vec{A} + (-\vec{B}) =$

☑ Subtraction of vectors is adding a vector to a negative vector.

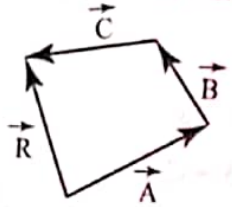
11. Given three vectors as shown below, Find

i. $\vec{A} + \vec{B} + \vec{C}$

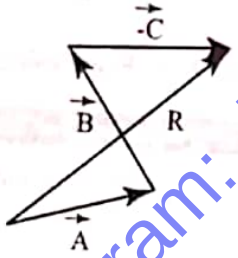
ii. $\vec{A} + \vec{B} - \vec{C}$

Solution

i. $\vec{A} + \vec{B} + \vec{C} =$

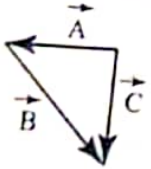


ii. $\vec{A} + \vec{B} - \vec{C} =$

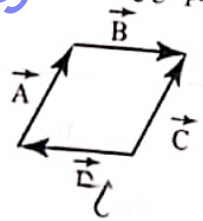


12. Identify the resultant from the following the following graphs

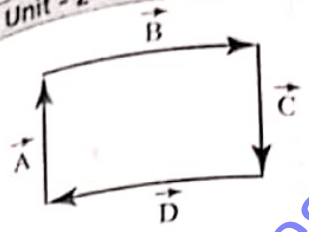
i)



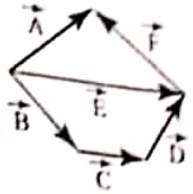
ii)



iii)



iv)



Hint: The resultant is drawn from the tail of the first vector to the last vector.

i. $\vec{C}$ ii. $\vec{C}$ iii. Zero iv. $\vec{A}$

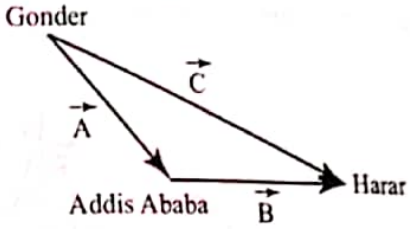
13. A Plane moves from Gonder to Addis Ababa then to Harar as shown. which vector represents the plane displacement.

A) $\vec{C} = \vec{A} + \vec{B}$

B) $\vec{C} = \vec{A} - \vec{B}$

C) $\vec{C} = \vec{B} - \vec{A}$

D) $\vec{A} = \vec{B} + \vec{C}$



Answer: A

2.3 Resolution of Vectors

A none parallel or a none perpendicular vector can be decomposed in to components using resolution of vector.

Resolution of vector: Is the process of splitting a vector in to components that are perpendicular to each other.

Vector resolution can be performed using the concepts of pythagoras theorm and the trigonometry relation. Pythagoras theorm applies for a right angled triangle and is stated as follows.

If two sides of triangle are perpendicular then, the sum of the squares of the two sides is equal to the third side. That is, if we have a right angle triangle with sides, a,b and c

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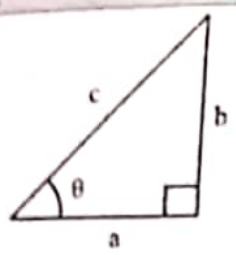
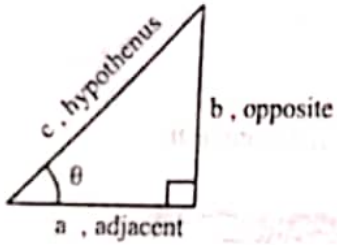


Then, according to pythagoras theorm, we have,

$$c^2 = a^2 + b^2$$

$$c = \sqrt{a^2 + b^2}$$

Trigonometry gives us the basic relations between angles and sides of a right angle triangle as follows.



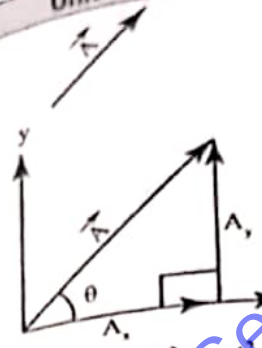
$$\text{Sin}\,\theta = \frac{\text{opposite}}{\text{Hypostens}} = \frac{b}{c} \Rightarrow b = c\sin\theta$$

$$\text{Cos}\,\theta = \frac{\text{Adjacent}}{\text{Hypostens}} = \frac{a}{c} \Rightarrow a = \cos\theta$$

$$tan\,\theta = \frac{\text{opposite}}{\text{Adjacent}} = \frac{b}{c} \Rightarrow \theta = \tan^{-1}\left(\frac{b}{c}\right)$$

Now, using the concepts of pythagorous theorm and the basic trigonometric relation, we can decompose say vector $\vec{A}$ in to horizontal or x component and vertical or y component as shown below.

Given vector



$$\cos\theta = \frac{A_x}{A} \Rightarrow A_x = A\cos\theta$$

$$\sin\theta = \frac{A_y}{A} \Rightarrow A_y = A\sin\theta$$

Moreover, $A^2 = A_x^2 + A_y^2$ magnitide of the vector

$$A = \sqrt{A_x^2 + A_y^2}$$

$$\tan\theta = \frac{A_y}{A_x}$$

$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$, the angle that the vector makes

Illustrative Examples

14. What are the components of a 400N force that acts on a body at an angle of 37° from the horizontal

Solution

$F_x = F\cos\theta = 400\cos 37^0 = 400(0.8) = 320N$

$F_y = F\sin\theta = 400\sin 37^0 = 400(0.6) = 240N$

15. A bird flys at 4m/s due East and then at 3 m/s due south. What is the resultant velocity of the bird?

Solution

$$V = \sqrt{V_x^2 + V_y^2}$$

$$V = \sqrt{4^2 + 3^2}$$

$V = 5\,m/s$ and direction of the bird is

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$$\tan\theta = \frac{V_y}{V_x} = \frac{3}{4} = 0.75$$

$$\theta = \tan^{-1}(0.75) = 37^0 \text{ south of east}$$

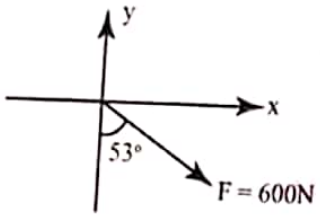
16. A force of 500N is acting on a body upward. What are the components of the force?

Solution

$F_x = F\cos\theta = 500\cos(90^0) = 500(0) = 0\ N$

$F_y = F\sin\theta = 500\sin(90^0) = 500(1) = 500N$

17. Find the components of the force vector shown below



Solution

$F_x = F\sin\theta = 600\sin(53^0) = 600(0.8) = 480N$

$F_y = F\cos\theta = 600\cos(53^0) = 600(0.6) = 360N$

Note, in this case, the x - component is to the right and the y - component is down hard.

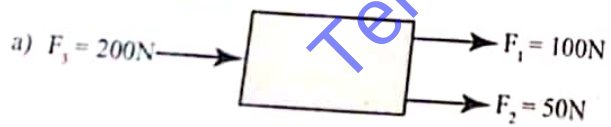
2.4 Vector Addition Algebrically

Two or more vectors can be added analytically as follows.

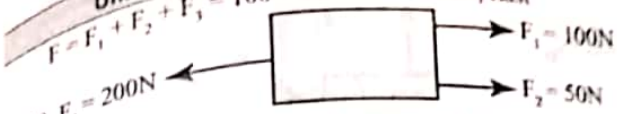
I. Adding parallel or anti - parallel vectors

The resultant is equal to the sum or the difference of the given vectors

For example, if we have

a) 

$F = F_1 + F_2 + F_3 = 100 + 50 + 200 = 350N$, East

b) 

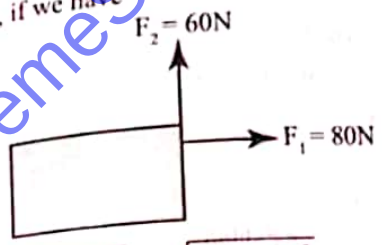
$F = F_1 + F_2 - F_3 = 100 + 50 - 200 = -50N$

$F = 50N$, west

II. Adding perpendicular vectors

The resultant is obtained using pythagoras' then

For example, if we have



$$F = \sqrt{F_1^2 + F_2^2} = \sqrt{60^2 + 80^2}$$

$F = 100N$, and its direction is obtained by

$$\tan\theta = \frac{F_2}{F_1} = \frac{60}{80} = \frac{3}{4}$$

$$\theta = \tan^{-1}(0.75)$$

$$\theta = 37^0$$

Adding any Vectors

To add vectors in any orientation, we need to first resolve them in to components, then add components acting in the same direction and finaly use pythagoms theorm to get the resultant.

For example, if we have.

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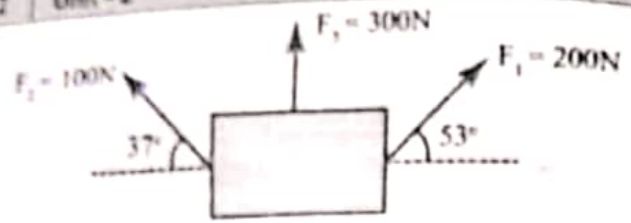


Then magnitude of the resultant is obtained by the equation

$$F = \sqrt{F_x^2 + F_y^2}$$

But, $F_x = F_{1x} + F_{2x} + F_{3x}$

$= F_1 \cos\theta_1 + F_2 \cos\theta_2 + F_{3x}$

$= 200(0.6) + (-100(0.8)) + 300\,(0)$

$= 120 - 80 + 0$

$F_x = 40N$

$F_y = F_{1y} + F_{2y} + F_{3y}$

$= F_1 \sin\theta_1 + F_2 \sin\theta_2 + F_3 \sin\theta_3$

$= 200(0.8) + 100(0.6) + 300\,(1)$

$= 160 + 60 + 300$

$F_y = 520N$

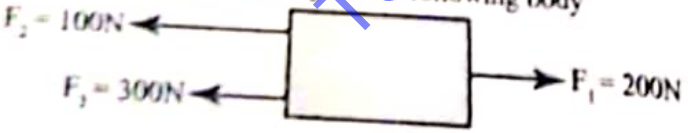
Therefore, $F = \sqrt{40^2 + 520^2} = 521.5N$

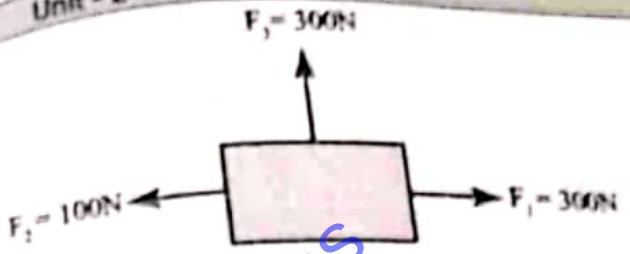
$$\tan\theta = \frac{520}{40} = 13,$$

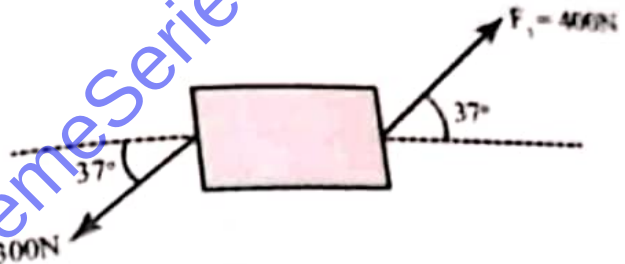
$$\theta = \tan^{-1}(13) = 85.6^\circ$$

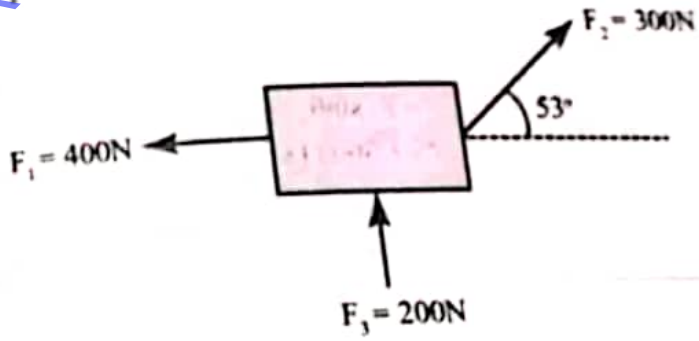
Illustrative Examples

18. Find the resultant force acting on the following body

i. 

ii. 

iii. 

iv. 

Solution

i) $F = F_1 - F_2 - F_3 = 200 - 100 = 300 = -200N$

$F = 200N$, Wast

ii) $F = \sqrt{F_x^2 + F_y^2}$

$F_x = F_1 - F_2 = 300 - 100 = 200\ N$

$F_y = F_3 = 300N$

$F = \sqrt{200^2 + 300^2} = 10\sqrt{13}\ N$

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iii) $F = \sqrt{F_x^2 + F_y^2}$

$F_x = F_{1x} + F_{2x} = F_1 \cos 37° - F_2 \cos 37°$

$F_x = 400(0.8) - 300(0.8) = 80N$

$F_y = F_{1y} + F_{2y} = F_1 \sin 37° + F_2 \sin 37°$

$F_y = 400(0.6) + 300(0.6) = 420N$

$F = \sqrt{F_x^2 + F_y^2} = \sqrt{80^2 + 420^2} = 427.5N$

iv) $F = \sqrt{F_x^2 + F_y^2}$

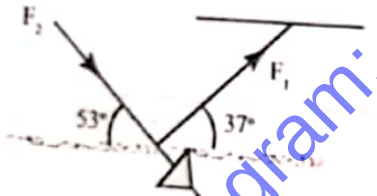
$F_x = F_{2x} - F_1 = 300(0.6) - 400 = 220N$

$F_y = F_{2y} + F_1 = 300(0.8) + 200 = 440N$

$F = \sqrt{220^2 + 440^2} = 492N$

$\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right) = \tan^{-1}\left(\frac{440}{220}\right) = 63.4°$

19. A farmer exerting a force of 100N on the handle at 53° is ploughing a field where the oxes are pulling yoke with a force of 1500N. Find the resultant force.



Solution

$F = \sqrt{F_x^2 + F_y^2}$

$F_x = F_{1x} + F_{2x} = F_1 \cos\theta_1 + F_2 \cos\theta_2$

$F_x = 1500(0.8) + 100(0.6) = 1260N$

$F_y = F_{1y} + F_{2y} = F_1 \sin\theta_1 + F_2 \sin\theta_2$

$F_y = 1500(0.6) + (-100)(0.8) = 820N$

$F = \sqrt{1260^2 + 820^2} = 1503N$

$\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right) = \tan^{-1}\left(\frac{820}{1260}\right) = 33°$

20. Which pair of forces can not give a resultant of 5N?

A. 1N and 4N
B. 1N and 6N
C. 3N and 4N
D. 1N and 7N

Solution

The magnitude of the resultant of any two vectors $\vec{A}$ and $\vec{B}$ lies in the range $|\vec{A} - \vec{B}| \le R \le |\vec{A} + \vec{B}|$

1N and 4N can give $|1 - 4| \le R \le |1 + 4| \Rightarrow 3 \le R \le 5$

1N and 6N can give $5 \le R \le 7$

3N and 4N can give $1 \le R \le 7$

1N and 7N can give $6 \le R \le 8$

Answer: D

21. A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of their sum can be?

A. 62
B. 3
C. 9
D. 47

Solution

$|A - B| \le R \le |A + B|$

$|20 - 25| \le R \le |20 + 25|$

$5 \le R \le 45$

Answer: C

22. What is the horizontal component displacement of a bus that moved 600m due east and then 800m 530 north of East.

A. 1000m
B. 1080m
C. 200m
D. 1240m

Solution

$S_x = S_1 + S_2 \cos\theta$

$= 600 + 800 \cos(53°)$

$= 600 + 800(0.6)$

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$= 600 + 480$

$S_x = 1080m$

Answer: B

Illustrative Example

Match the items in column "A" with those in column "B"

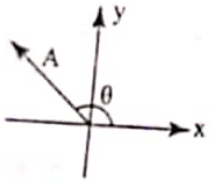
A	B
23. Equal vectors	A. Have same size and direction
24. Zero vector	B. Have same size but opposite direction
25. Negative vectors	C. Has zero size and undefined direction
26. Unit vector	D. Has magnitude one and specific direction
	E. Has magnitude zero and specific direction

Answer

23. A 24. C 25. B 26. D

27. A vector has an X component of -30units and a Y component of 40 units. Find the vector?

Solution



$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{(-30)^2 + (40)^2} = \sqrt{2500}$$

$A = 50$ unit

$$\tan\theta = \frac{A_y}{A_x} = \frac{-40}{30} = -1.33$$

$$\theta = \tan^{-1}(-1.33) = 127^\circ$$

28. **Given that :** $\vec{A} = 30$ units along North, and

$\vec{A} + \vec{B} = 10$ units along south

Find, $\vec{B}$

Solution

$\vec{A} + \vec{B} = -10 \Rightarrow \vec{B} = -10 - 30$

$\vec{B} = -40$ units

$\vec{B} = 40$ units south

2.5 Vector Multiplication

A vector can be multiplied by a scalar or another vector as shown below.

I. Multiplying a Vector by a Scalar

When a vector is multiplied by a scalar, a change in magnitude, direction or both of the vector may occur.

The production of a scalar k and a vector $\vec{A}$ is a new vector whose magnitude is KA and whose direction is the same as that of $\vec{A}$ if K is positive and opposite to $\vec{A}$ if K is negative.

For example, if we have a force vector $\vec{F} = 100$N, East then,

$3\vec{F} = 3(100) = 300$N, East

$-3\vec{F} = -3(100) = 300$N, West

$-\frac{1}{2}\vec{F} = -\frac{1}{2}(100) = 50$N, West

Illustrative Examples

29. Vector $\vec{P}$ has magnitude of 5m pointing south what are the magnitude and direction of ?

A. $3\vec{P}$ B. $-3\vec{P}$ C. $-\frac{1}{5}\vec{P}$

Solution

A. $-3\vec{P} = 3(5) = 15$m, South

B. $-3\vec{P} = -3(5) = 15$m, North

C. $\frac{-1}{5}\vec{P} = \frac{-1}{5}(5) = 1$m, North

30. Two non - zero vectors $\vec{A}$ and $\vec{B}$ are related by $\vec{A} = C\vec{B}$ where C is

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a scalar. If the two vectors have opposite directions, then one of the following is true about C?

A. C is positive
B. C is negative
C. C = 1
D. C = 0

Hint: A vector change its direction when multiplied by a negative number.

Answer: B

II. Multiplying a vector by Another Vector

When a vector is multiplied by another vector, the result is a scalar or a vector quantity.

Dot Product

The don product of two vectors is also known as **scalar product** since the product of the two vectors give a scalar quantity.

The dot product of two vectors $\vec{A}$ and $\vec{B}$ is denoted by $\vec{A} \bullet \vec{B}$ and is read as $\vec{A}$ dot $\vec{B}$ is defined as follows:

Dot product of two vectors is equal to the product of the magnitude of the two vectors and the cosine of the angle between them when they are joined tail to tail

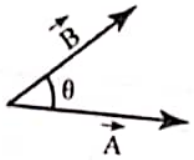
That is, if the have vectors $\vec{A}$ and $\vec{B}$ as shown

Then, their dot product is obtained using either of the following two cases

i) $\vec{A} \bullet \vec{B} = AB\,Cos\theta$

$$Cos\theta = \frac{\vec{A} \bullet \vec{B}}{AB}$$

ii) $\vec{A} \bullet \vec{B} = A_x B_x + A_y B_y$



Illustrative Examples

31. Find the dot product of the two vectors

$\vec{A} = (3,4)$ and $\vec{B} = (4,3)$

Solution

$\vec{A} \bullet \vec{B} = A_x B_x + A_y B_y$

$= (3)(4) + (2)(6)$

$\vec{A} \bullet \vec{B} = 24$

32. What is the work bone on a box when a horizontal force of 3.5N acts on it and displaced for 10.5m along the x - axis?

Solution

$W = F\cos\theta$

$= (3.5)(10.5)\cos(0^0)$

$W = 36.75J$

33. Which one of the following is the value of $\vec{A} \bullet \vec{B}$, where $\vec{A} = (3,-1)$ and $B = (5, 2)$

A. 13 B. 20 C. -13 D. -20

Solution

$\vec{A} \bullet \vec{B}, = A_x B_x + A_y B_y = (3)(5) + (-1)(2) = 13$

Answer: A

34. Given two vectors $\vec{A} = (2,-4)$ and $\vec{B} = (4,2)$. The angle between them is?

A. 0^0 B. 30^0 C. 45^0 D. 90^0

Solution

$$\vec{AB} = AB\,Cos\theta \Rightarrow Cos\theta = \frac{\vec{A} \bullet \vec{B}}{AB} = \frac{(2)(4) + (-4)(2)}{\sqrt{2^2 + -4^2} + \sqrt{2^2 + 4^2}}$$

$Cos\theta = 0^0 \Rightarrow \theta = \cos^{-1}(0) = 90^0$

Answer: D

given that: $\vec{A} = 4i - 3j$, $\vec{B} = 2i + j$ And, $\vec{A} + \vec{B} - \vec{C} = 0$, Then, find

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a) Vector $\vec{C}$

b) a unit vector along vector $\vec{A}$

Solution

a) $\vec{A} + \vec{B} - \vec{C} = 0$

$(A_x + B_x + C_x)i + (A_y + B_y - C_y)j = 0$

$(4 + 2 - C_x)i + (-3 + 1 - C_y)j = 0$

$6 - C_x = 0$

$C_x = 6$

$-2 - C_y = 0$

$C_y = -2$

Hence $\vec{C} = 6i - 2j$

b) $\hat{U}_A = \dfrac{\vec{A}}{|A|} = \dfrac{4i - 3j}{\sqrt{(4)^2 + (-3)^2}} = \dfrac{4i - 3j}{5} = \dfrac{1}{5}(4i - 3j)$

36. Find the angle between the vectors $\vec{A} = 3i - j$ and $\vec{B} = i - 2j$

Solution

$\vec{A} + \vec{B} = |A||B| \cos\theta$

$$\cos\theta = \frac{\vec{A} \bullet \vec{B}}{|A||B|} = \frac{(A_x B_x) + (A_y B_y)}{\left(\sqrt{A_x^2 + A_y^2}\right)\left(\sqrt{B_x^2 + B_y^2}\right)}$$

$$\cos\theta = \frac{(3)(1) + (-1)(-2)}{\left(\sqrt{(3)^2 + (-1)^2}\right)\left(\sqrt{(1)^2 + (-2)^2}\right)} = \frac{2}{\sqrt{50}} = \frac{1}{\sqrt{2}}$$

$$\cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \cos{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$\theta = 45^0$

37. Given that, $\vec{A} = (5i - 3j)$ and $\vec{B} = (2i + j)$.

Find the angle between $\vec{A} + \vec{B}$ and the x - axis

Solution

$\vec{A} + \vec{B} = (A_x + B_x)i + (A_y + B_y)j$

$\vec{A} + \vec{B} = (5 + 2)i + (-3 + 1)j$

$\vec{A} + \vec{B} = 7i - 2j$

And, the angle between $\vec{A} + \vec{B}$ and the x - axis is,

$(\vec{A} + \vec{B}) \cdot i = |A + B||i| \cos\theta$

$$\cos\theta = \frac{(A_x + B_x)(1) + (A_y + B_y)(0)}{\left(\sqrt{(7)^2 + (-2)^2}\right)(1)}$$

$$\cos\theta = \frac{7}{\sqrt{53}}$$

$$\theta = \cos^{-1} = \left(\frac{7}{\sqrt{53}}\right) = \cos^{-1}(0.962)$$

$\theta = 16^0$

38. Find vector of magnitude 10 units pointing along the vector $\vec{A} = 3i + 4j$

Solution

Let us find a unit vector along vector $\vec{A}$

$$\hat{U}_A = \frac{3i + 4j}{\sqrt{(3)^2 + (4)^2}} = \frac{3i + 4j}{5}$$

Then, the vector notation of the given vector of size 10 unit is obtained by

$$\vec{B} = |B| \hat{U}_A = 10\left(\frac{3i + 4j}{5}\right)$$

$B = 6i + 8j$

Unit Vector along the Direction of any Vector

The unit vector in the direction of a non - zero vector is defined as the vector divided by its magnitude. That is, if we have vector $\vec{A} = A_x i + A_y j$,

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then the unit vector the direction of $\vec{A}$ is obtained by:

$$\hat{U}_A = \frac{\vec{A}}{|A|}$$

Illustrative Examples

39. What is the unit vector in the direction of vector $\vec{A} = 2i - j$

Solution

$$\hat{U}_A = \frac{\vec{A}}{|A|} = \frac{2i - j}{\sqrt{(2)^2 + (-1)^2}} = \frac{2i - j}{\sqrt{5}} = \frac{1}{\sqrt{5}}(2i - j)$$

40. Give that $\vec{A} = i - j$ and $\vec{A} + \vec{B}$ is a unit vector along the positive x - axis. Find vector $\vec{B}$

Solution

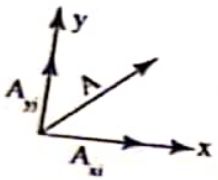
$$\vec{A} + \vec{B} = i$$
$$(A_x + B_x)i + (A_y + B_y)i = i$$
$$(1 + B_x)i + (-1 + B_y)i = i$$
$$\longrightarrow 1 + B_x = 1 \Rightarrow B_x = 0$$
$$\longrightarrow (-1 + B_y) = 0 \Rightarrow B_y = 1$$

Hence, $\vec{B} = 0i + 1(j) = j$

41. If $0.6j + mj$ is a unit vector, what is m ?

Solution

$$\sqrt{(0.6)^2 + (m)^2} = 1$$
$$m = \pm 0.8$$



$$\vec{A} = A_x + A_y$$

Moreover, the resoltant of the vectors $\vec{A}$ and $\vec{B}$ can be obtained by:

$\vec{A} = A_x + A_y$

$\vec{B} = B_x + B_y$, then

Their sum: $\vec{R} = \vec{A} + \vec{B} = (A_x + B_x)i + (A_y + B_y)j$

Their Difference: $\vec{R} = \vec{A} - \vec{B} = (A_x - B_x)i + (A_y - B_y)j$

Illustrative Examples

42. Given that: $\vec{A} = 2i + 5j$ and $\vec{B} = i - 3j$

Find: a) $\vec{A} + \vec{B}$

b) $\vec{A} - \vec{B}$

Solution

a) $\vec{A} + \vec{B} = (2i + j) + (i + 3j)$

$\vec{A} + \vec{B} = (2 + 1)i + (1 + (-3))j$

$\vec{A} + \vec{B} = 3i - 2j$

b) $\vec{A} - \vec{B} = (2 - 1)i - (i - 3j)$

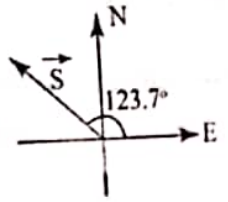
$\vec{A} - \vec{B} = (2 + 1)j + (1 - (-3))j$

$\vec{A} - \vec{B} = i + 4j$

43. If a car travelled 20km north and then 25km as 60^0 west of north, then what is the car's displacement

Solution

$$S = \sqrt{S_x^2 + S_y^2}$$
$$S_x = S_{1x} + S_{2x} = 0 + (-25)\sin 60^0$$
$$S_x = -21.65\text{km}$$
$$S_y = S_{1y} + S_{2y} = 20 + 25\cos 60^0$$



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$S_y = 32.5Km$

$S = \sqrt{(-21.65)^2 + (32.5)^2}$

$S = 39.05km$

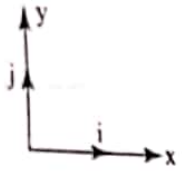
$\tan\theta = \frac{S_y}{S_x} = \frac{-32.5}{21.65} = (-1.5)$

$\theta = \tan^{-1}(-1.5) = 123.7^0$

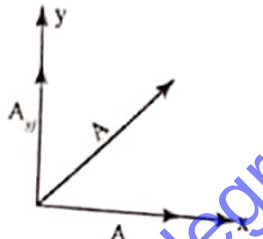
What are Unit Vectors?

Unit vector is a vector of magnitude one and pointing into a particular direction.

In the xy coordinate system, we define unit vector i that points in the positive x - axis and a unit vector j that point in the positive y - axis as shown below:



And the components of any vector $\vec{A}$ can be written using the unit vector notation as follows:



That is ;

$\vec{A} = A_x + A_y$

Moreover, the resultant of two vectors $\vec{A}$ and $\vec{B}$ can be obtained by :

$\vec{A} = A_x + A_y$

$\vec{B} = B_x + B_y$, then

Their sum; $\vec{R} = \vec{A} + \vec{B} = (A_x + B_x)i + (A_y + B_y)j$

Their Difference; $\vec{R} = \vec{A} - \vec{B} = (A_x - B_x)i + (A_y - B_y)j$

Illustrative Example

44. Given that : $\vec{A} = 2i + j$ and $\vec{B} = i - 3j$ Find

a) $\vec{A} + \vec{B}$

b) $\vec{A} - \vec{B}$

Solution

a) $\vec{A} + \vec{B} = (2i + j) + (i - 3j$

$\vec{A} + \vec{B} = (2 + 1)i + (1 + (-3)j$

$\vec{A} + \vec{B} = 3i - 2j$

b) $\vec{A} - \vec{B} = (2i + j) - (i - 3j$

$\vec{A} - \vec{B} = (2 - 1)i + (1 - (-3)j$

$\vec{A} - \vec{B} = i + 4j$

45. If $3\vec{A} - \vec{B} + \vec{C} = 0$, given that:

$\vec{A} = -i + 2j$ and $\vec{B} = 3i - 2j$.

Find $\vec{C}$

Solution

$3\vec{A} - \vec{B} + \vec{C} = 0$

$3(-i + 2j) - (3i - 2j) + C_x i + C_y j = 0$

$(-3 - 3 + C_x)i + (6 + 2 + C_y)i = 0$

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$-6 + C_x = 0$

$\longrightarrow C_x = 6$, and

$\longrightarrow 8 + C_y = 0$

$C_y = -8$

Hence, $\vec{C} = 6i - 8j$

46. Given that : $\vec{A} = 3i + 6j$ and $\vec{B} = -2i - 4j$.

Find a unit vector along:

$\vec{A} + \vec{B}$

Solution

$\vec{A} + \vec{B} = (A_x + B_x)i + (A_y + B_y)j$

$\vec{A} + \vec{B} = (3 + (-2))i + (6 + (-4))j$

$\vec{A} + \vec{B} = i + 2j$

Then, the unit vector along $\vec{A} + \vec{B}$ is

$$\hat{U}_{A+B} = \frac{\vec{A} + \vec{B}}{|\vec{A} + \vec{B}|}$$

$$= \frac{i + 2j}{\sqrt{(1)^2 + (2)^2}} = \frac{i + 2j}{\sqrt{5}}$$

$$= \frac{1}{\sqrt{5}}(i + 2j)$$

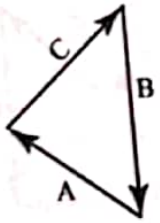
$\hat{U}_{A+B}$

End of unit Questions and problems

I. Give short Answer to the following Questions

1. Which one of the following is Not a scalar quantity?
   A. Electric current
   B. Temperature
   C. Work done
   D. Weight
2. Which one of the following is a scalar quantity?
   A. Lominceus intensity
   B. Electric field strength
   C. Acceleration
   D. Magnetic field stragth
3. Which one of the following pairs contain scalar quantises
   A. Work, force
   B. Work, Temperature
   C. Weight, density
   D. Weight, velocoty
4. Which one of the following pairs contain vector quantities
   A. Speed, velocity
   B. Velocity, Acceleration
   C. Force, Temperature
   D. Work, Energy
5. Which one of the following is not true about the three vectors shown below

   A) $-A = B + C$
   B) $C = -C - A$
   C) $C = -A - B$
   D) $A + B - C = O$



6. If three vectors, A,B and C form a triangle when joined head to tail, What will be their resultant
7. How will two vectors give a maximum results Explain
8. How will you add two vectors to get a least magnitude?
9. Can we add two vectors of different kind? If so, how?
10. Can the dot product of two vectors be a vector quantity?
11. What will be the scalar product of two collinear

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12. What will be the scalar product of two orthogonal vectors?

II. Solution the following Problems

13. If a vector has initial coordinate of (5,2) and a final coordinate of (-2,-3). What will be a the magnitude of the vector?

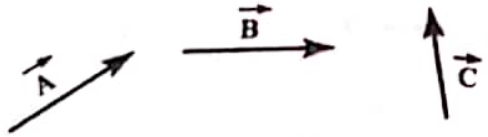
14. What are the components of a force of magnitude 50N acting on a body at an angle of 37°?

15. What is the dot product of $\vec{A} = (3,5)$ *and* $\vec{B} = (4,7)$

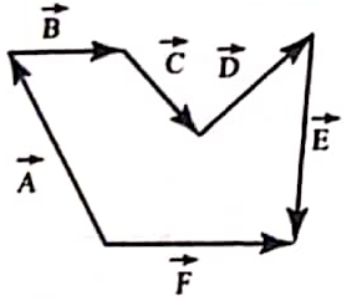
16. What should be the value of λ so that the vectors $\vec{A} = (2,5)$ *and* $\vec{B} = (10,\lambda)$ become perpendicular to each other.

17. Give the following three vectors $\vec{A}, \vec{B}$ *and* $\vec{C}$, what is the value of.

a) $\vec{A} + \vec{B} + \vec{C}$
b) $\vec{A} + -\vec{B} + \vec{C}$
c) $\vec{A} + \vec{B} + -\vec{C}$

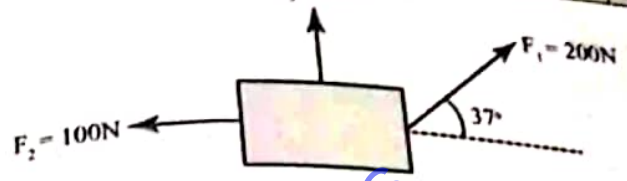


18. What is the resultant vector in the following graph



19. What is the magnitude of a vector that has a component of 8unit along the x - axis and 6 units along the y - axis

20. What is the resultant force acting on the block shown below



21. A horse displaces a cart 300m along the horizontal by applying a force of 800N at angle of 37° from the horizontal. What is the work done by the horse?

22. What should be the other component of a 500N force, if one of its component is 300N?

Short Answers and Explanations

1. D 2. A 3. B 4. B 5. D 6. Zero

7. When they act in the same directions

8. When they act in opposite directions

9. No. This is not possible

10. No. The dot product of two vectors is always a scalar

11. Collinear vectors act in the same or opposite directions. Hence, the angle between them is 0^0 or 180^0. There for:

$\vec{A} \cdot \vec{B} = AB \cos\theta$, $\cos(0^0) = \cos(180^0) = /1/$

Hence, $\vec{A} \cdot \vec{B} = /A/\ /B/$, maximum

12. Orthogonal vectors are vectors acting at right angle to each othere. Hence , the angle between them is 90^0, and $\cos(90^0) = 0$

$\vec{A} \cdot \vec{B} = AB\cos\theta$, $\cos(90^0) = 0$

$\vec{A} \cdot \vec{B} = O$, minimum

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Solution to End of unit problems

13. $A = \sqrt{(-2-5)^2 + (-3-2)^2} = \sqrt{74}$ units

14. $F_x = F\cos\theta = (50)\cos(37^0) = 40N$

$F_y = F\sin\theta = (50)\sin(37^0) = 30N$

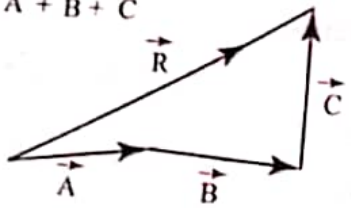
15. $\vec{A} \cdot \vec{B} = (3)(4) + (5)(7) = 47$

16. $\vec{A} \cdot \vec{B} = 0$

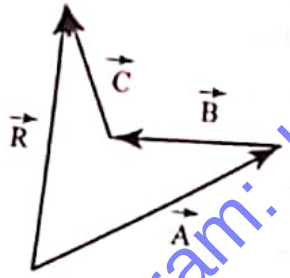
$(2)(10) + (5)(\lambda) = 0$

$\lambda = -4$

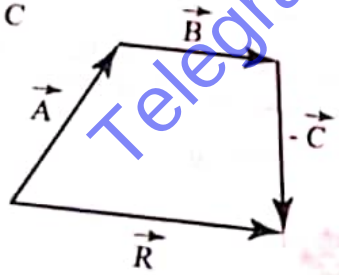
17. a) $\vec{A} + \vec{B} + \vec{C}$



b) $\vec{A} - \vec{B} + \vec{C}$



c) $\vec{A} + \vec{B} - \vec{C}$



18. $\vec{F}$

19. $A = \sqrt{A_x^2 + A_y^2} = \sqrt{6^2 + 8^2}$

$A = 10$ Units

20. $F = \sqrt{F_x^2 + F_y^2}$

$F_x = F_1\cos\theta - F_2 = 200\cos 37^0 - 100 = 60N$

$F_y = F_1\sin\theta + F_2 = 200\sin 37^0 + 50 = 170N$

$F = \sqrt{60^2 + 170^2} = 180.27N$

21. $W = Fs\cos\theta = (800)(300)\cos 37^0 = 192000J$

22. $A^2 = Ax^2 + Ay^2$

$(500)^2 = (300)^2 + Ay^2$

$Ay^2 = 1600$

$Ay = 400N$

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UNIT 3

3. Motion In One Dimension

3.1 Motion of a Body

3.2 Position and Displacement of a Body

3.3 Average Velocity and Average Speed

3.4 Average and Instantaneous Acceleration

3.5 Uniformly Accelerated Motion

3.6 Graphical Representation of Motion

3.7 Free Fall and Vertical Motion

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Motion In One Dimension

Dear learner, welcome to this incredible unit where you will be able to; define motion in one dimension, explain position, displacement and distance, distingush average speed from average velocity, explain the difference between average velocity and instantaneous velocity, differenciate average acceleration from instantaneous acceleration, describe uniformly accelerated motion, represent motion graphically. And, define free fall and solve plenty of problems. Enjoy !

3.1 Motion of a Body

Motion is a continues change in position of a body with respect to a certain frame of reference.

Motion of a body may be one dimensional like the motion of a car along a straight road, two dimensional like the motion of a body from the first floor to the third floor along the stair case, or three dimensional like the motion of a satellite around the earth.

One Dimensional motion : Is the motion of a body up ward or downward, to the right or to the left, or backward or forward along a straight line

For example, the following motions are one dimensional.

Motion of a car on a straight rode

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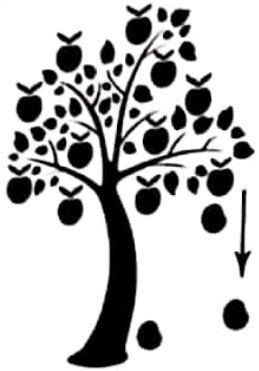
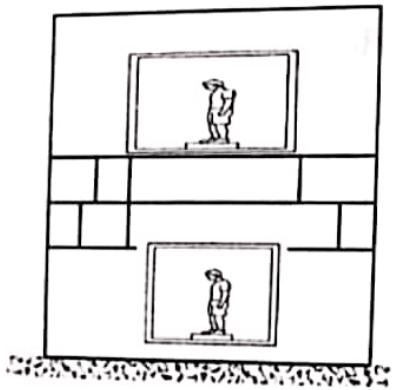
Free fall of a fruit from a tree



Motion of elevator in a building

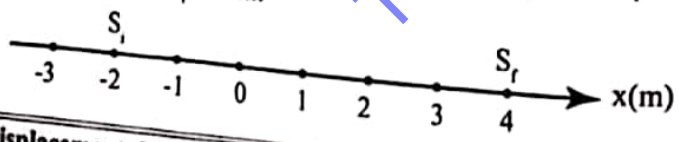


3.2 Position and Displacement of a body

You may ask, is there any difference between position and displacement of a body ? the answer is yes there is !

Position: Is the location of a body at a given instant of time.

For example, the initial position of a body shown below is $S_i = -2$ m and its final position is $S_f = 4$m.



Displacement: Is the change in position of a body as time passes

Therefore, the change in position or displacement of the body is obtained by :

$$\Delta S = S_f - S_i$$

$$\Delta S = 4 - (-2)$$

Displacement, $\Delta S = 6$m, to the right

Illustrative Examples

1. Where will be the final position of a body if it is displaced by 7m to the right from an initial position of - 3m ?

Solution

$$\Delta S = S_f - S_i$$

$$7 = S_f - (-3)$$

$$S_f = 4\text{m, to the right}$$

2. What is the displacement of a body if it is displaced from an initial position of $S_i = 3$m to a final position of $S_f = -2$m ?

Solution

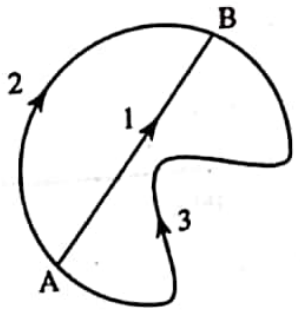
$$\Delta S = S_f - S_i$$

$$= -2 - (3)$$

$$= -5\text{m}$$

$\Delta S = 5$m, to the left

3. Which path represents displacement of a body that moved from point A to point B ?



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Explanation

Path - 1 represents the shortest path or displacement of the body, while path 2 and 3 including path - 1 represent the actual path length or distance moved by the body.

4. A particle starts from the origin and moved 2 cm to the right and then 9 cm to the left. What is its

A. Distance travelled

B. Displacement

Solution

A. $S = S_1 + S_2 = 2 + 9 = 11$ cm

B. $\Delta S = S_f - S_i = 9 - 2 = 7$ cm to the left

5. An athelet perform his training in a circular track of radius 100m. After 10 complete revolutions, what is the athelet's

A. Distance moved

B. Displacement

Solution

A. $S = 10\,(2\pi r) = 10\,(2\pi\,(100) = 2000\,\pi\,m = 6280m$

B. ΔS = zero

6. A girl starting from her home moved 400m due east and then 300m due north to a supermarket. What is the girl's

A. Distance covered

B. Displacement

Solution

A. $S = S_1 + S_2 = 400 + 300 = 700m$

B. $S = \sqrt{S_1^2 + S_2^2}$

$S = \sqrt{400^2 + 300^2}$

$S = 500m$

$\theta = \tan^{-1}(0.75) = 53^0$

3.3 Average Velocity and Average Speed

A body in motion changes its position with time. And the rate at which a body changes its position with time is called **velocity** of the body.

Average velocity : Is the rate at which a body changes its position per unit time.

$$\text{Average velocity} = \frac{\text{Displacement}}{\text{Time taken}}$$

$$\vec{V}av = \frac{\Delta \vec{S}}{\Delta t} \qquad \frac{S_f - S_i}{t_f - t_i}$$

Average velocity is a vector quantity and its SI unit is meter per second (m/s)

☑ Since velocity is a vector quantity, it can be positive, negative or zero. And, how fast a body moves is described using the concept of average speed

Average speed: Is total distance travelled divided by time taken

$$\text{Average speed} = \frac{\text{Total distance moved}}{\text{Time taken}}$$

$$Vav = \frac{S_T}{t_T} = \frac{S_1 + S_2}{t_1 + t_2}$$

Average speed is a scalar quantity and its SI unit is meter per second (m/s)

☑ Since speed is a scalar quantity, it is always positive

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Conceptual Examples

7. Is there any difference between speed and velocity ?

Explanation

Yes. Speed is distance moved per unit time and it is a scalar quantity. While, velocity is displacement per unit time and it is a vector quantity.

8. Under what condition will average velocity of a body be the same as its average speed ?

Explanation

When the body moves in a straight line its average velocity and its average speed become the same

9. Can a body have constant speed while having a varying velocity ?

Explanation

Yes. This happens when the body moves along a curved path at constant speed.

Illustrative Examples

10. A body changes its position from $S_i = -2m$ to $S_f = 10m$ in 4 seconds. What is the average velocity of the body ?

Solution

$$\bar{V}_{av} = \frac{\overrightarrow{\Delta S}}{\Delta t} = \frac{S_f - S_i}{\Delta t} = \frac{10 - (-2)}{4} = \frac{12}{4}$$

$$\bar{V}_{av} = 3m/s$$

11. A body moved 100m to the right and 40m to the left in 10 seconds. What is the body's

A. Average speed

B. Average velocity

Solution

A. $V_{av} = \frac{S_T}{t_T} = \frac{S_1 + S_2}{t} = \frac{100 + 40}{10} = 14m/s$

B. $\bar{V}_{av} = \frac{\Delta \vec{S}}{\Delta t} = \frac{S_2 - S_1}{\Delta t} = \frac{100 - 40}{10} = 6m/s$

12. A man runs around a circular track of radius 100 m and completes one revolution in 31.4 seconds.

A. Average speed

B. Average velocity

Solution

A. $V_{av} = \frac{S_T}{t_T} = \frac{2\pi r}{t} = \frac{2(3.14)(100)}{31.4} = 20m/s$

B. $\vec{V}_{av} = \frac{\Delta S}{\Delta t} = Zero$

13. A boy walks for 1 minute at a speed of 1m/s and then runs for 1 minute at 3m/s along a straight road. What is the average speed of the boy ?

Solution

$$V_{av} = \frac{S_T}{t_T} = \frac{S_1 + S_2}{t_1 + t_2}$$

$$V_{av} = \frac{(1)(60) + (3)(60)}{60 + 60}$$

$$V_{av} = 2m/s$$

14. A train travels from one station to another at a speed of 40 km/h and returns to the first station at a speed of 60 km/h. Calculate the train's

A. Average speed

B. Average velocity

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Solution

A. $V_{av} = \frac{S_T}{t_T} = \frac{S_1 + S_2}{t_1 + t_2} = \frac{S + S}{\frac{S_1}{V_1} + \frac{S_2}{V_2}} = \frac{S + S}{\frac{S}{40} + \frac{S}{60}}$

$V_{av} = 48$ km/h

B. $\vec{V}_{av} = \frac{\Delta \vec{S}}{\Delta t} =$ Zero

How fast a body is moving at an instant of time or at a specific position is described using the concept of instantaneous velocity

Instantaneous velocity : Is the velocity of a body at an instant of time or a particular position.

Instantaneous velocity = Limiting value of $\frac{\Delta \vec{S}}{\Delta t}$ as Δt approaches zero

$$\vec{V}_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \vec{S}}{\Delta t} = \frac{d\vec{S}}{dt}$$

Critical Thinking Question

15. Can the instantaneous velocity of a body be zero while its average velocity is none - zero?

Explanation

Yes. A body may stop moving at an instant of time on its journey history while moving from place to place.

16. Can the instantaneous velocity and the average velocity of a body ever be equal?

Explanation

Yes. This is so when the body moves at a constant velocity.

Illustrative Examples

17. The position of a body as a function of time is given by; $S(t) = 3t^2 - 2t$. Where displacement is in meters and time is in seconds. Find the body's

A. Average velocity in the time interval $1 \le t \le 3$

B. Instantaneous velocity at $t = 3$ second.

Solution

A. $\vec{V}_{av} = \frac{\Delta \vec{S}}{\Delta t} = \frac{S(3)-S(1)}{3-1} = \frac{[3(3)^2 - 2(3)] - [3(1)^2 - 2(1)]}{2} = 10$m/s

$\vec{V}_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \vec{S}}{\Delta t} \longrightarrow$ Method - I

B. $= \lim_{\Delta t \to 0} \frac{S(t+\Delta t) - S(t)}{\Delta t}$

$\Delta S = S(t+\Delta t) - S(t)$

$S(t+\Delta t) = 3(t+\Delta t)^2 - 2(t+\Delta t) = 3t^2 + 6\Delta t t + 3\Delta t^2 - 2t - 2\Delta t$

$S(t) = 3t^2 - 2t$

$\Delta S = 6\Delta t t + 3\Delta t^2 - 2\Delta t$

$\vec{V}_{inst.} = \lim_{\Delta t \to 0} \frac{\Delta \vec{S}}{\Delta t} = 6t - 2$

At $t = 3$ seconds

$\vec{V}_{inst} = 6(3) - 2$

$\vec{V}_{inst.} = 16$ m/s

$\vec{V}_{inst} = \frac{ds}{dt} \longrightarrow$ Method - II

$= \frac{d}{dt}(3t^2 - 2t)$

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$\vec{V}_{inst}$ $6t - 2$, at $t = 3$ second

$V_{inst} = 6(3) - 2 = 16 m/s$

Note; In elementary calculus of derivation, a function $y = x^n$ can be differienciated as a function of x as $\frac{d}{dx}(x^n) = nx^{n-1}$

18. A body moves along the x - axis in such a way that its distance as function of time is $S(t) = 12t - t^2$. What is its

A. Distance moved during the first 3 seconds

B. Instantaneous velocity at $t = 4$ second.

C. Time when direction changes.

Solution

A. $S(t) = 12t - t^2$, $t = 3$ second

$S = 12(3) - (3)^2 = 27 m$

B. $\vec{V}_{inst.} = \frac{ds}{dt} = \frac{d}{dt}(12t - t^2) = 12 - 2t$

At $t = 4$ second, $V_{inst.} = 12 - 2t = 12 - 2(4)$

$V_{inst} = 4 m/s$

C. When the body change its direction of motion, it momentary stops. Therefore; as direction changes, the instantaneous velocity of the body become zero.

$V_{inst} = 12 - 2t$

$0 = 12 - 2t$

$t = 6$ seconds

3.4 Average and Instantaneous Acceleration

When the velocity of a body changes in magnitude, in direction or in both the body is said to be **accelerating**

Average Acceleration : Is the rate at which the velocity of a body changes with time.

$$\text{Average acceleration} = \frac{\text{Change in velocity}}{\text{time taken}}$$

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v} - \vec{u}}{\Delta t}$$

Acceleration is a vector quantity and its SI unit is meter per second square (m/s^2)

Critical Thinking Question

19. What does it mean when we say a body is moving at a constant acceleration of $3 m/s^2$?

Explanation

It means, the velocity of the body is increasing at 3m/s every second.

That is , at $t = 1$ sec, $v = 3 m/s$

at $t = 2$ sec, $v = 6 m/s$

at $t = 3$ sec, $v = 9 m/s$ and so on

20. Can the acceleration of a body be zero while its velocity is none zero ?

Explanation

Yes. This happens when the body moves at constant velocity.

Illustrative Examples

21. What is the average acceleration of a body if its velocity changes from 10m/s to 30m/s in 5 seconds ?

Solution

$$a = \frac{\Delta v}{\Delta t} = \frac{v - u}{t} = \frac{30 - 10}{5}$$

$a = 4 m/s^2$

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22. What will be the final velocity of a body if it uniformly accelerates at $3m/s^2$ for 5 seconds starting from rest ?

Solution

$$a = \frac{v - u}{t} \Rightarrow v = u + at, \quad u = 0$$

$$v = 0 + (3)(5)$$

$$v = 15m/s$$

23. How long will a body take to reach a speed of 20m/s starting from rest if it constantly accelerats at $2m/s^2$?

Solution

$$a = \frac{v - u}{t} \Rightarrow t = \frac{v - u}{a}, \quad u = 0$$

$$t = \frac{20 - 0}{2}$$

$$t = 10 \text{ seconds}$$

The acceleration of a body at an instant of time is expressed using the concept of instantaneous acceleration

Instantaneous Acceleration : Is the acceleration of a body at a particular position or instant of time.

Instantaneous acceleration = limiting value of $\frac{\Delta \vec{v}}{\Delta t}$ as Δt approaches zero.

$$\vec{a}_{inst.} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

Illustrative Example

24. If the velocity of a body as a function of time is given by $v(t) = 4t - 2$. Where, velocity is in meter per second and time is in seconds. Find the body's

A. Average acceleration in the time interval $1 \le t \le 3$

B. Instantaneous acceleration at $t = 2$ second.

Solution

A. $\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \frac{v(3) - v(1)}{3-1}$

$$= \frac{[4(3) - 2] - [4(1) - 2]}{2} = 4m/s^2$$

B. $\vec{a}_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t}$, Method - I

$\Delta v = v(t+\Delta t) - v(t)$

$v(t+\Delta t) = 4(t+\Delta t) - 2 = 4t + 4\Delta t - 2$

$v(t) = 4t - 2$

$\Delta v = v(t+\Delta t) - v(t) = 4\Delta t$

$\vec{a}_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = 4$ At 2second, $\vec{a}_{inst} = 4m/s^2$

Note; the average and the instantaneous acceleration of the body are equal since the body is moving at constant accoleration.

$$a_{inst.} = \frac{dv}{dt}, \text{ Method - II}$$

$$\frac{d}{dt}(4t - 2) = 4m/s^2$$

25. If the displacement of a body is given by $x = 7t + 4t^2$. What is its acceleration at $t = 5$ second.

Solution

$$V = \frac{dx}{dt} = \frac{d}{dt}(7t + 4t^2)$$

$$V = 7 + 8t$$

$$v$$

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$$a = \frac{dv}{dt} = \frac{d}{dt}(7+8t) = 8$$

$$a = 8\,m/s^2$$

3.5 Uniformly Accelerated Motion

When a body moves at constant acceleration, we have simple but powerful equations that can help us to describe its motion.

The most common equations of motion of a body moving at constant acceleration are the following

☑ Average acceleration of the body

$$a = \frac{\Delta v}{\Delta t} = \frac{v-u}{t}$$

☑ Velocity of the body at any time

$$a = \frac{v-u}{t}$$

$$v = u + at$$

☑ Average velocity of the moving body

$$\vec{V}av = \frac{u+v}{2}$$

☑ Displacement of a body in terms of initial and final velocities at any time

$$\vec{S} = \vec{V}av\,t$$

$$S = \left(\frac{v+u}{2}\right) t$$

☑ Displacement of a body in terms of initial velocity, and acceleration at any time

$$S = \left(\frac{v+u}{2}\right) t, \qquad v = u + at$$

$$S = \frac{[(u+at) + u]\,t}{2}$$

$$S = ut + \frac{1}{2}at^2$$

☑ Displacement of a body in terms of its initial velocity, final velocity and acceleration

$$S = \left(\frac{v+u}{2}\right) t, \quad t = \frac{v-u}{a}$$

$$S = \left(\frac{v+u}{2}\right)\left(\frac{v-u}{a}\right)$$

$$v^2 = u^2 + 2as$$

Remember, in this unit, u stands for initial velocity and v stands for final velocity of a body

Illustrative Examples

26. A car moves uniformly from rest to a speed of 36 km/h in 2 seconds. What is

A. Acceleration of the car

B. Distance moved by the car

Solution

A. $a = \frac{v-u}{t} = \frac{10-0}{2}$, $36\,km/h = 10\,m/s$

B. $S = ut + \frac{1}{2}at^2$, $u = 0$

$S = 0 + \frac{1}{2}(5)(2)^2$

$S = 10m$

27. An airplane lands on a runway with a speed of 80m/s and stops after moving 1 km.

A. What time did it take to stop ?

B. What is the deceleration of the plane ?

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Solution

A. $S = \left(\frac{v+u}{2}\right) t$, 1km = 1000m

$1000 = \left(\frac{0+80}{2}\right) t$

$t = 25$ seconds

B. $a = \frac{v-u}{t} = \frac{0-80}{25}$

$a = -3.2 m/s^2$

28. A car moving at 30m/s can decelerate at $9m/s^2$ on a sunny day and at $4.5m/s^2$ on a rainy day. How far will the car move before stopping ?

A. In a sunny day

B. In a rainy day

Solution

A. $V^2 = u^2 + 2as$, $v = 0$

$S = \frac{v^2-u^2}{2a} = \frac{0-(30)^2}{2(-9)}$

$S = 50m$

B. $V^2 = u^2 + 2as$, $v = 0$

$S = \frac{v^2-u^2}{2a} = \frac{0-(30)^2}{2(-4.5)}$

$S = 100m$

29. An object starts from rest and moved with a constant acceleration of $4m/s^2$ along a straight line. Find

A. The average velocity of the body in 5 seconds

B. The distance moved in 5 second

C. The distance moved in the 5^{th} second

Solution

A. $V_{av} = \left(\frac{v+u}{2}\right)$, $u = 0$

$v = u + at = 0 + (4)(5)$

$v = 20m/s$

$V_{av} = \left(\frac{0+20}{2}\right) = 10m/s$

B. $S = ut + \frac{1}{2}at^2$

$S = 0 + \frac{1}{2}(4)(5)^2$

$S = 100m$

C. $S^{th}_n = u + \frac{a}{2}(2t-1) = 0 + \frac{4}{2}(2(5)-1) = 18m$

3.6 Graphical Representation of Motion

When the motion of a body is represented graphically the dependent variable such as displacement, velocity and acceleration of the body is plotted along the y - axis and the independent variable such as time is plotted along the x - axis.

Now, let us examine the displacement - time graph, the velocity - time graph and the acceleration - time graph of a body in a uniform motion and in a uniformly accelerated motion.

(i) Graphical Representation of Uniform Motion

When a body moves at a constant speed in a straight line it is said to be in a uniform motion and its acceleration is zero.

☑ Distance - time Graph

If the body starts motion from rest, its s - t graph is a straight line that passes through the origin.

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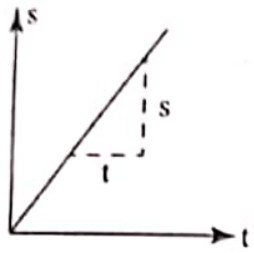
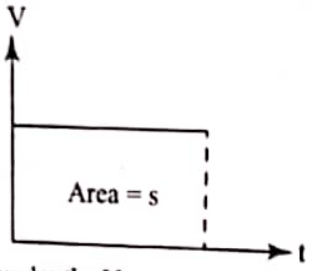


➥ The slope the S - t graph represent speed.

☑ **Speed - time Graph**

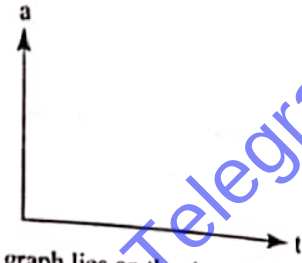
The speed of a body in a uniform motion is constant and the v - t graph is horizontal



➥ The area under the V - t graph represents the distance moved by the body.

☑ **Acceleration - time Graph**

For a body in a uniform motion, the acceleration is zero and the a - t graph lies on the time - axis.



➥ The a - t graph lies on the time - axis sine $\vec{a} = 0$

(ii) Graphical Representation of Uniformly Accelerated Motion

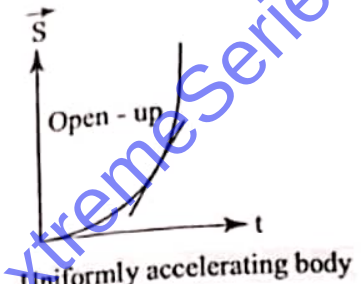
When a body moves in a uniformly accelerated motion the displacement - time graph, the velocity - time graph and the acceleration - time graph

looks the following

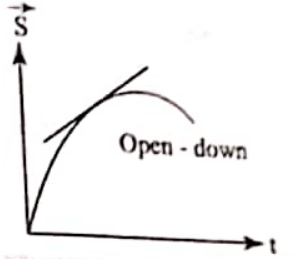
☑ **Displacement - time Graph**

When a body moves at a constant acceleration, its

$\vec{s}$ – t graph looks like the following.



Uniformly accelerating body



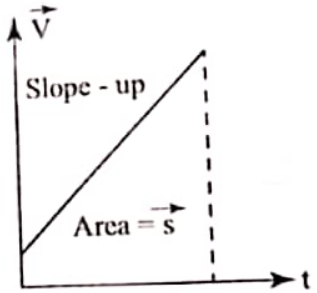
Uniformly decelerating body

The slope of the $\vec{s}$ - t graph represent velocity

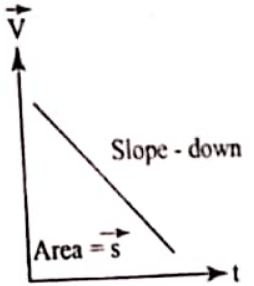
Velocity - time Graph

When a body moves at a constant acceleration, its

$\vec{V}$ – t graph looks like the following.



Uniformly accelerating body



Uniformly decelerating body

The area under the $\vec{V}$ - t graph represents displacement

☑ **Acceleration - time Graph**

When a body moves in a constant acceleration, its

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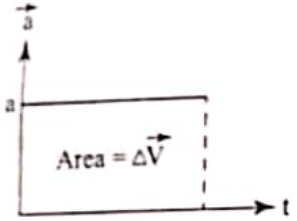
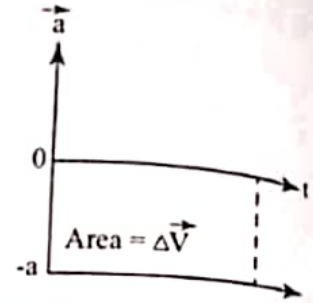
$\vec{a}$ -t graph is straight line parallel to the time - axis



Uniformly accelerating body



Uniformly decelerating body

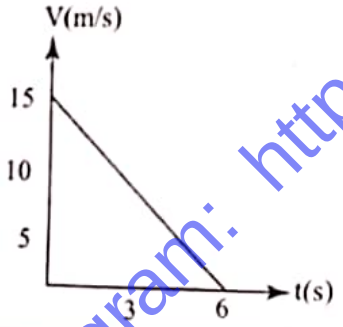
The area under the $\vec{a}$ - t graph represents change in velocity

Illustrative Examples

30. The velocity - time graph of a car that slows down at uniform rate is shown. Find

A) The acceleration of the car

B) The distance it travelled before it stops.



Solution

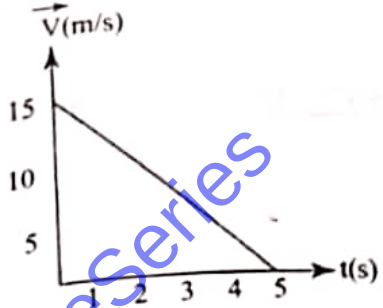
A. $\vec{a}$ = slope = $\frac{\Delta v}{\Delta t} = \frac{0-15}{6-0}$

$\vec{a} = -2.5 m/s^2$, deceleration

B. S = area = $\frac{1}{2} vt = \frac{1}{2}(15)(6)$

$S = 45m$

31. The velocity - time graph of a moving body is shown below. How far does the body move in the first two seconds?



Solution

$\vec{a}$ = slope = $\frac{\Delta v}{\Delta t} \quad \frac{0-15}{5-0} = -3 m/s^2$

at $t = 2$ second, $v = u + at = 15 - (3)(2) = 9 m/s^2$

$S = \frac{v^2 - u^2}{2a} = \frac{9^2 - 15^2}{-2(3)} = 26m$

3.7 Free Fall and Vertical Motion

In the absence of air resistance, all bodies fall towards the centre of the earth with constant acceleration regardless of their mass.

Near the surface of the earth, the acceleration due to gravity is $9.81 m/s^2$ which we take it as $10 m/s^2$ for calculation purpose.

Since free fall is one of the best examples of uniformly accelerated motion, the equations of motion can be used with some modifications. Therefore, we can replace (a) by (g) and (s) by (y) or (h) : moreover, we take g as positive for downward motion and as negative for up ward motion: For a free fall or a vertical motion, we have the following equations.

Dropped or Released	Downward Thrown	Up ward Thrown
$u = 0$	$u \neq 0$	$u \neq 0$

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$v = gt$	$v = u + gt$	$v = u - gt$
$y = \frac{1}{2} gt^2$	$y = ut + \frac{1}{2} gt^2$	$y = ut - \frac{1}{2} gt^2$

Illustrative Examples

32. A ball is dropped from the top of a building and reached the ground after 4 seconds. What is

A. The height of the building

B. The velocity with which it hits the ground

Solution

A. $y = \frac{1}{2} g t^2 = \frac{1}{2} (10) (4)^2$

$y = 80\,m$

B. $v = gt = (10)(4)$

$v = 40\,m/s$

33. A stone is thrown vertically downward at a speed of 10m/s from a cliff of 40m hight. What is

A. The speed of the stone at the ground level

B. The time it takes to hit the ground.

Solution

A. $V^2 = u^2 + 2gh = (10)^2 + 2(10)(40) = 900$

$V = 30\,m/s$

B. $y = ut + \frac{1}{2} gt^2$

$40 = 10t + 5t^2$

$t^2 + 2t - 8 = 0$

$t = \frac{-2 \pm \sqrt{22 - 4(1)(-8)}}{2(1)} = \frac{-2 \pm 6}{2}$

$t = 2$ seconds.

34. A ball is thrown up ward at a speed of 20 m/s what is

A. The maximum hight reached by the ball

B. The time it takes to reach the maximum hight

C. The speed of the ball at t = 3 seconds

D. The time it takes the ball to return to the ground

E. The speed with which the ball hits the ground

Solution

A. $V^2 = u^2 - 2hg$, $V = 0$ at h_{max}

$h = \frac{u^2}{2g} = \frac{(20)^2}{2(10)}$

$h = 20m$

B. $V = u - gt$, $v = 0$ at h_{max}

$t = \frac{u}{g} = \frac{20}{10}$

$t = 2$ seconds

C. $V = u - gt$, at $t = 3$ seconds

$V = 20 - (10)(3) = -10\,m/s$

$V = 10\,m/s$ downward

D. $y = ut - \frac{1}{2} gt^2$, $y = 0$ when it returns

$0 = ut - \frac{1}{2} gt^2 \Rightarrow ut = 5t^2$

$t = \frac{u}{5} = \frac{20}{5}$

$t = 4$ seconds

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E. $V^2 = u^2 - 2gh, \quad h = 0$

$V^2 = u^2 = (20)^2$

$V = 20$ m/s

Note, the speed the ball hits the ground is the same as the speed with which its is thrown upward. Okay !

35. A particle is dropped from a height of 4m. In what time will the particle travel the first meter distance ?

A. $\frac{1}{\sqrt{2}}$ Sec C. $\frac{1}{2}$ Sec

B. $\frac{1}{\sqrt{3}}$ Sec D. $\frac{1}{\sqrt{5}}$ Sec

Solution

$y = \frac{1}{2} gt^2 \Rightarrow t = \sqrt{\frac{2y}{5}}$

$t = \sqrt{\frac{2(1)}{10}} = \frac{1}{\sqrt{5}}$ sec

Answer : D

36. A ball released from a certain hight falls 5m in one second. In 4 seconds, it will fall through

A. 80m B. 40m C. 20m D. 10m

Solution

$y = \frac{1}{2} gt^2 = \frac{1}{2}(10)(4)^2$

$y = 80$m

Answer : A

37. The displacement of a freely falling body is proportional to

A. Time of fall C. Mass of the body

B. Square of time of fall D. Square of mass of the body.

Hint : $y = \frac{1}{2} gt^2 \Rightarrow y \sim t^2$

Answer : B

38. A body projected up - ward with a velocity u returns to the starting point in 4 seconds. The value of u is ?

A. 5m/s B. 4m/s C. 20m/s D. 10m/s

Solution

$t = \frac{2u}{g} \Rightarrow u = \frac{gt}{2} = \frac{(10)(4)}{2} = 20$m/s

Answer: C

39. An object moving with uniform acceleration has a velocity of 12m/s in the positive x - axis when it is 3m from the origin. If it is at after 2 seconds, what is its acceleration ?

A. -12m/s^2 B. -13m/s^2 C. -16m/s^2 D. 12m/s^2

Solution

$\Delta S = ut + \frac{1}{2} at^2, \quad \Delta S = S_f - S_i = -5 - 3 = -8$cm

$-8 = 12(2) + \frac{1}{2} a (2)^2$

$a = -16$m/s^2.

Answer : C

40. A driver moving at 20 m/s sees an accident up ahead and hits the breakes. If the car decelerates at a constant rate of 5m/s^2, how far will the car go before it comes to stop ?

A. 120m B. 100m C. 40m D. 20m

Solution

$V^2 = u^2 + 2as, \quad v = 0$

$0 = u^2 + 2as \quad S = \frac{-u^2}{2a} = -\frac{(20)^2}{2(-5)}$

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$S = 40m$

41. A body moving with constant acceleration covers the distance between two points that are 60m aparts in 5 sec. What is the acceleration of the body if its velocity at the second point is 15 m/s

A. $3m/s^2$ B. $2.5m/s^2$ C. $1.5m/s^2$ D. $1.2\ m/s^2$

Answer : C

Solution

$$S = \left(\frac{v+u}{2}\right) t \Rightarrow 60 = \left(\frac{15+4}{2}\right) 5$$

$u = 9m/s.$ $\quad a = \frac{v-u}{t}$

$$a = \frac{15-9}{5} = 1.2m/s^2$$

Answer : D

42. A car moves from point A to point B at a speed of 20m/s. It then returns at a speed of 30m/s to its initial position. What is the average speed of the car for the whole journey?

A. 25m/s B. 50m/s C. 24m/s D. 10m/s

Solution

$$V_{av} = \frac{2v_1 v_2}{v_1 + v_2} = \frac{2(20)(30)}{20+30} = \frac{1200}{50}$$

$V_{av} = 24m/s.$

Answer: C

43. If a ball is thrown upward with speed u, then the distance covered during the last 1 second of its ascent is ?

A. ut C. gt

B. $u + g/2$ D. $\frac{1}{2}gt$

Solution

$$y = ut - \frac{1}{2} g t^2 - [u(t-1) - \frac{1}{2} g (t-1)^2]$$

$$y = u + g/2$$

Answer : B

44. If a body moves with an initial velocity u and a constant acceleration a, then the distance it moved during the n^{th} second will be ?

A. $S = ut$ C. $S = a(2t-1)$

B. $S = ut + \frac{1}{2}(2t-1)$ D. $S = ut + \frac{1}{2} at^2$

Solution

$$S = ut + \frac{1}{2} at^2 - [u(t-1) + \frac{1}{2} a (t-1)^2]$$

$$S = u + \frac{a}{2} (2t-1)$$

Answer : B

45. A body moving with 10m/s has a constant acceleration of $2m/s^2$. In what time will the velocity of the body be doubled ?

A. 5 second B. 10 second C. 20 second D. 30 second

Solution

$V = u + at$

$20 = 10 + 2t$

$t = 5$ second.

Answer : A

46. The initial velocity of a particle is 10m/s and its acceleration is $2m/s^2$. The distance travelled by the particle in the 5^{th} second of its motion is ?

A. 20 m B. 10 m C. 5 m D. 1m

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Solution

$S_n = u + \frac{a}{2}(2t - 1)$

$= 10 + \frac{2}{2}(2(5) - 1)$

$= 1m$

Answer : D

47. A car at rest accelerates uniformly to reach a speed of 144 km / h. What time will it take to cover 400m ?

A. 5 second B. 10 second C. 15 second D. 20 second

Solution

$S = \left(\frac{v + u}{2}\right) t \Rightarrow 400 = \left(\frac{40+0}{2}\right) t$

$t = 20$ seconds

Answer : D

48. A ball is thrown vertically up ward with a speed of 18 km/h. What is the maximum height reached by the ball ?

A. 1.25m B. 2.50m C. 3.15m D. 5m

Solution

$V^2 = u^2 - 2gh$, $v = 0$

$h = \frac{u^2}{2g} = \frac{(5)^2}{2(10)}$

$h = 1.25$ m

Answer : A

49. A ball thrown up ward at a speed of 25m/s from a building takes 9 seconds to reach the ground. What is the hight of the building ?

A. 100m B. 180m C. 220m D. 320m

Solution

$y = -ut + \frac{1}{2} gt^2$ $y = -25(9) + \frac{1}{2}(10)(9)^2$

$y = 180m$

Answer : B

50. Two bodies begun a free fall from rest from the same hight 1 second apart. How long after the first body begun to fall will the bodies 10m apart ?

A. 3.5 second B. 2.5 second C. 1.5 second D. 0.5 second

Solution

$y_1 = \frac{1}{2} gt^2$ and $y_2 = \frac{1}{2} g (t-1)^2$

$\Delta y = y_1 - y_2 \Rightarrow 10$ m

$t = 1.5$ second.

Answer : C

51. An aircraft decelerates at $0.5m/s^2$. After 8 minutes, its velocity has dropped to 160m/s. What was the initial velocity of the craft ?

A. 100 m/s B. 200 m/s C. 300 m/s D. 400 m/s

Solution

$a = \frac{v - u}{t} \Rightarrow u = v - at$, $a = -0.5 m/s^2$,

$t = 8 \times 60$ sec

$u = (160) - (-0.5)(480)$

$u = 400 m/s$

Answer : D

52. When brakes are applied to a car moving at 30m/s, it uniformly slowed down and stops in 5 seconds. How far does it go before coming to rest ?

A. 75 m B. 50m C. 25m D. 5m

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Solution

$$S = \left(\frac{v+u}{2}\right) t = \left(\frac{0+30}{2}\right) 5$$

$$S = 75m$$

Answer : A

53. A car accelerates uniformly from 20m/s to 60m/s through a displacement of 2000m. What is the acceleration of the car ?

A. $0.5\ m/s^2$ B. $0.8m/s^2$ C. $5m/s^2$ D. $8m/s^2$

Solution

$$V^2 = u^2 + 2as$$

$$a = \frac{v^2-u^2}{25} = \frac{60^2-20^2}{2(2000)}$$

$$a = 0.8m/s^2$$

Answer : B

54. A bullet is shot from a rifle with a speed of 20m/s and start accelerating at a constant rate of $4m/s^2$. If The target is located at 200m, how long will it take the bullet to hit the target ?

A. - 6.18 second C. - 3.09 second

B. 6.18 second D. 3.09 second

Solution

$$S = ut + \frac{1}{2} at^2$$

$$200 = 20t + \frac{1}{2} (4) (t^2)$$

$$t^2 + 10t - 100 = 0$$

using the quadratic equation, $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$$t = \frac{-10 + \sqrt{10^2 - 4(-100)}}{2(1)}$$

$$t = -5 + 5\sqrt{5}$$

$$t = 6.18 \text{ second.}$$

Answer : B

55. The displacement - time history of a body is given bellow. What is the acceleration of the body ?

S (m)	0	2	8	18	32	50
t (sec)	0	1	2	3	4	5

A. $2m/s^2$ B. $3m/s^2$ C. $4m/s^2$ D. $-5m/s^2$

Solution

$$S = ut + \frac{1}{2} at^2, \quad u = 0$$

$$50 = \frac{1}{2} (a) (5)^2$$

$$a = 4m/s^2$$

Answer: C

56. The velocity - time history of a body is given below. What is the distance moved by the body in 4 seconds ?

Velocity (m /s)	0	3	6	9	12
Time (s)	0	1	2	3	4

A. 24m B. 12m C. 6m D. 2m

Solution

$$S = \frac{u+v}{2} \qquad S = \left(\frac{0+12}{2}\right) 4$$

$$S = 24m$$

Answer : A

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57. A stone is thrown vertically upward with an initial speed of 60m/s. The time it takes the stone to come back is ?

A. 6 second B. 12 second C. 24 second D. 18 second

Solution

$$y = ut - \frac{1}{2} g t^2, \quad y = 0$$

$$0 = 60(t) - \frac{1}{2}(10)t^2$$

$$t = \frac{120}{10} = 12 \text{ second}$$

Answer : B

58. A uniformly moving train covered a 24 km distance in 10 minutes. How far can it travel with in 1 hour and 30 minutes?

A. 144km B. 244km C. 345km D. 216km

Solution

The average speed of the train is,

$$V = \frac{S_1}{t_1}, \quad t_1 = 10 \text{ minute} = \frac{1}{6} \text{ hour}$$

$$V = 24/(1/6) = 144 \text{ km/hr}$$

$$S_2 = v t_2 = (144)(1.5) = 216 \text{km}$$

Answer : D

59. The displacement - time graph of a moving object is a straight line that makes an angle of 30° with the time - axis and passes through the origin. Then

A. The speed of the object is not constant

B. The slope of the line represents acceleration of the object

C. The slope of the line represents speed of the object

D. The area under the graph represents distance travelled.

Hint : s(cm) Slope = Speed t(s)

Answer : C

60. Which of the following is not correct about time ?

A. It can be measured using pendulum motion

B. It can be measured using atomic clock

C. Since time evolves from past to future, it is a vector

D. Time is a basic quantity

Hint : Time is a fundamental scalar quantity

Answer : C

61. A particle starts from rest and moves with a constant acceleration of 2m/s² for the first two seconds and then moves for the next 4 seconds with out accelerating. Finally it uniformly decelerates and comes to rest in 2 seconds. What Is the distance travelled by the particle ?

A. 16m B. 24m C. 32m D. 40m

Solution

$$s = s_1 + s_2 + s_3$$

$$s = \frac{1}{2} a t_1^2 + v t_2 + \left(\frac{v+u}{2}\right) t_3, \quad v = at = 4 \text{m/s}$$

$$s = \frac{1}{2}(2)(2)^2 + 4(4) + \left(\frac{4+0}{2}\right)2$$

$$s = 24 \text{m}$$

Answer : B

62. If a ball dropped from the top of a building reaches the ground after 3 seconds, what is the height of the building ?

A. 30m B. 45m C. 60m D. 90m

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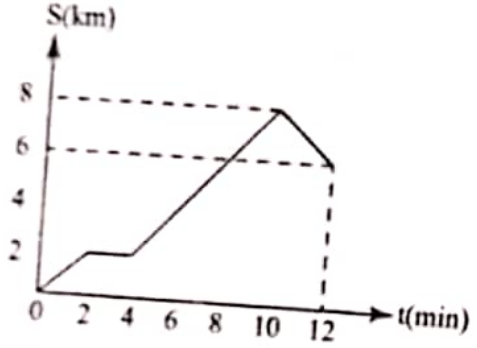
Solution

$$y = \frac{1}{2} g t^2 = \frac{1}{2} (10) (3)^2$$

$$y = 45m$$

63. A car starts from rest and moves along a straight line according to a displacement - time graph. What is the average velocity of the car during the 12 seconds ?

Answer : B



A. 6.3 km/min

B. 1.2 km/min

C. 0.5 km/min

D. 0.3 km/min

Solution

$$\vec{V_{av}} = \frac{\Delta \vec{S}}{\Delta t} = \frac{8-2}{12}$$

$$V_{av} = 0.5 km/min$$

64. A ball is thrown up ward into the air with a speed of 40m/s. What is the velocity and displacement of the ball after 2 seconds ?

Answer : C

A. 10m/s, 20m

B. 20m/s, 60m

C. 30m/s, 80m

D. 40m/s, 100m

Solution

$$v = u - gt = (40) - 10) (2)$$

$$v = 20 m/s$$

$$y = ut + \frac{1}{2} gt^2 = (40) (2) - \frac{1}{2} (10) (2)^2$$

$$y = 60m$$

Answer : B

65. Starting from rest, a car accelaretes to $2m/s^2$ How long will it take the car to speed up to $10m/s^2$

Solution

$$V = u + at, \quad u = 0$$

$$V = 0 + at$$

$$t = \frac{v}{a} = \frac{10}{2}$$

$$t = 5 \text{ seconds}$$

66. A car moving at 20 m/s slows down and stops in 10 seconds. What is

A. Deceleration of the car

B. Distance moved by the car

Solution

A. $a = \frac{v - u}{t} = \frac{0 - 20}{10}$

$$a = -2 m/s^2$$

B. $S = \left(\frac{v + u}{2}\right) t = \left(\frac{0 + 20}{2}\right) 10$

$$S = 100m$$

67. An automobile accelerates from rest to a speed of 72 km/h in 10 seconds. Find

A. The acceleration of car

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B. The distance the car travels

C. The deceleration of the car if it stop in 100m

Solution

A. $a = \frac{v - u}{t} = \frac{20 - 0}{10}$, $72km/h = 20m/s$

$a = 2 m/s^2$

B. $S = \frac{(u + v)}{2} t = \frac{(0 + 20)}{2} 10$

$S = 100m$

C. $V^2 = u^2 + 2as$, $V = 0$

$a = \frac{v^2 + u^2}{2S} t = \frac{0 + (20)^2}{2(100)}$

$a = -2 m/s^2$

68. A vihicle traveling at 20m/s stops in 200m. What is the deceleration of the car?

Solution

$V^2 = u^2 + 2as$, $V = 0$

$0 = u^2 + 2as$

$a = \frac{-u^2}{2S} = \frac{-(20)^2}{2(200)}$

$a = -1 m/s^2$

69. A ball is thrown upward with a velocity of 20m/s from the top of a 20m high building.

A. How high will the ball rise?

B. How long will it take the ball to hit the ground?

Solution

A. $y = \frac{(v + u)}{2} t$

$v = u + gt$, $V = 0$

$t = \frac{u}{g} = \frac{20}{10} = 2 sec$

$y = \left(\frac{0 + 20}{2}\right) 2$

$y = 20m$

B. $-y = ut - \frac{1}{2} gt^2$

$-20 = 20t - \frac{1}{2}(10) t^2$

$-20 = 20t - 5t^2$

$5t^2 - 20t + 20 = 0$

$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{20 \pm \sqrt{(-20)^2 - 4(5)(20)}}{2(5)}$

$t = 2 seconds$

70. A ball is thrown vertically upward with a velocity of 10m/s from a tower of 20m above the ground. Find

A. The time required the ball to hit the ground

B. The velocity with which the ball hits the ground

Solution

A. $-y = ut - \frac{1}{2} gt^2$

$-20 = 10t - 5t^2$

$5t^2 - 10t + 20 = 0$

$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{10 \pm \sqrt{(-10)^2 - 4(5)(20)}}{2(5)}$

$t = 1 seconds$

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B. $V^2 = u^2 + 2gh$

$V = \sqrt{u^2 + 2gh}$

$= \sqrt{(10)^2 + 2(10)(20)}$

$V = 22.4 m/s$

71. Given the displacement of a body as a function of time as $\vec{S} = (50t - 10t^2)$ m. Find

A. Velocity of the body at t = 5sec

B. Acceleration of the body at t = 2sec

Solution

A. $\vec{V}_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \vec{S}}{\Delta t} = \frac{d\vec{S}}{dt}$

$\vec{V}_{inst} = \frac{d}{dt}(50t - 10t^2) = 50 - 20t$

at t = 5sec

$\vec{V}_{inst} = 50 - 20(5) = -50 m/s = 50 m/s$, to the left

B. $\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{V}}{\Delta t} = \frac{d\vec{V}}{dt}$

$\vec{a} = \frac{d}{dt}(50 - 20t) = -20 m/s^2$

72. A car moving at 40m/s stops in 200m. What is the deceleration of the car?

Solution

$V^2 = u^2 - 2as$, $V = 0$

$a = \frac{-u^2}{2s} = \frac{-(40)^2}{2(200)} = -4 m/s^2$

73. Can a train moving at 10m/s stops in 400m if it slows down at a constant rate of 2m/s ?

Solution

$V^2 = u^2 - 2as$, $V = 0$

$S = \frac{u^2}{2a} = \frac{(40)^2}{2(2)}$

$S = 400 m$ Yes

74. What is the maximum speed with which a car can be driven safely along a curved road of radius 30m and super elevation of 30°?

Solution

$\tan\theta = \frac{V^2}{rg}$

$V = \sqrt{rg \tan\theta} = \sqrt{(30)(10)(0.6)}$

$V = 13 m/s$

75. A bicycle with wheel radius of 40cm is moving at a speed of 20km/h. Find

A. Its Period of revolution

B. Its angular velocity

Solution

A. $T = \frac{2\pi r}{v} = \frac{2(3.14)(0.4)}{5.6}$

$T = 0.5 sec$

B. $\omega = \frac{v}{r} = \frac{2.6}{0.4}$

$\omega = 14 rad/s$

76. What is the centripetal acceleration of a body moving at a speed of 10m/s around a circle of radius 100m?

Solution

$a_c = \frac{v^2}{r} = \frac{(10)^2}{100}$

$a_c = 1 m/s^2$

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77. If the coefficient of static friction for tires on a road is 0.25, at what maximum speed can a car turn around a level curve of 40m radius without slipping?

Solution

$f_s = f_c$

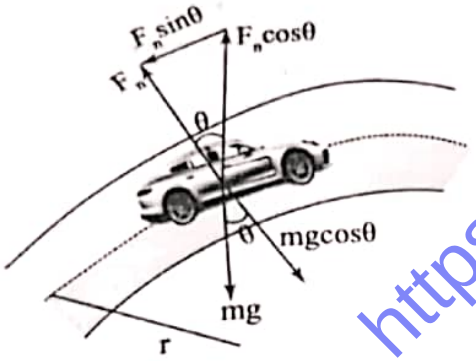
$$\mu mg = \frac{mv^2}{r}$$

$$V = \sqrt{\mu g r} = \sqrt{(0.25)(10)(40)}$$

$$V = 10m/s$$

78. A highway curve is banked at 37°. If in the absence of friction a car can move at a speed of 13.5 m/s without skidding, what is the radius of the curve?

Solution



$$\Sigma F_y = 0$$

$$F_N \sin\theta = mg$$

$$\sum F_x = 0$$

$$F_N \sin\theta = \frac{mv^2}{r}$$

Taking ratio

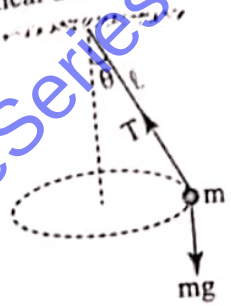
$$\frac{F_N \sin\theta}{F_N \cos\theta} = \frac{\frac{mv^2}{r}}{mg}$$

$$\frac{\sin\theta}{\cos\theta} = \frac{v^2}{rg} \Rightarrow \tan\theta = \frac{v^2}{rg}$$

$$r = \frac{v^2}{g\tan\theta} = \frac{(13.5)^2}{(10)\tan(37^\circ)} = \frac{(13.5)^2}{(10)(0.75)}$$

$$r = 24.3m$$

79. A conical pendulum with an 80kg bob on a 10m wire makes an angle of 5° from the vertical determine.



A. The horizontal and vertical components of the wire tension forces

B. The radial acceleration of the bob

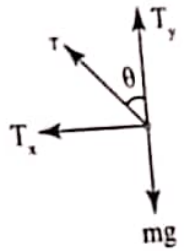
Solution

A. $T_y = mg$

$T\cos\theta = mg$

$$T = \frac{mg}{\cos\theta} = \frac{(80)(10)}{\cos(5^\circ)} = \frac{800}{0.996}$$

$T = 803N$



Horizontal component, $T_x = T\sin\theta = (803)\sin(5^\circ)$

$T_x = 70N$

Vertical component, $T_y = T\cos\theta = (803)\cos(5^\circ)$

$T_y = 800N$

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B. $F_c = a_c m$

$$a_c = \frac{F_c}{m} = \frac{T_x}{m} = \frac{70}{80}$$

$$a_c = 0.875 m/s^2$$

End of unit Questions and problems

(I) Give Short Answer to the Following Questions

1. What is the difference between distance and displacement
2. What is free fall
3. The slope of displacement - time graph represents ___
4. The area under acceleration - time graph gives___
5. The slope of velocity - time graph represents ___
6. The area under velocity - time graph represents ____
7. What is the difference between average speed and average velocity?
8. Can the instantaneous velocity of a body greater than its average velocity ?
9. If the average velocity is zero. Can the instantaneous velocity be different from zero ?
10. Is there a possibility for the velocity and acceleration to have opposite signs ?
11. Suppose you throw one ball upward and another down ward. Which ball will land with greater speed ?
12. A ball thrown upward from the ground reaches its maximum height and return to ground. In which case will it take longer time. Ascending time or descending time

(II) Solve the Following Problems

13. A car traveling at 5m/s starts to speed up. After 3 seconds, its velocity has become 11m/s. What is the acceleration of the car?

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14. The velocity of a particle is given by $v(t) = 3t^2 - 4t$. What Is the instantaneous acceleration of the particle at t = 1sec.
15. A city bus starts from rest and accelerates at $2m/s^2$ till it reaches a speed of 20m/s. It then travels for 20 second at constant speed until the brakes are applied and stop the bus in an additional 5 second.
    A. How long is the bus in motion ?
    B. What is the average velocity of the bus ?
16. A stone dropped from rest down a deep well and 3 second later, a splash is heard as it hits the water below
    A. How fast will it be moving as it hits the water ?
    B. What will its average speed over the three seconds ?
    C. How deep is the well ?
17. A ball dropped from a bridge strikes the ground in 5 second. What is
    A. The speed with which it strike the ground
    B. The height of the bridge
18. A ball is thrown up with 10m/s one second later, a stone is throum up with 20m/s. Find
    A. The time the stone takes to overtake the ball
    B. The height where the stone and the ball meet up
19. A car accelerates from rest at a rate of $2.4 m/s^2$ for 12 seconds. How far does the car travel during this time
20. A motorcycle starting from rest speeds up to 144 km/h in 4 seconds. What is the average acceleration of the motor in m/s^2
21. A helmet is dropped from the top of a 20m tall building.
    A. How long does it take the helmet to reach the ground
    B. How fast should the helmet be thrown upward to reach the top of the building
22. A ball thrown upward at 8m/s takes 4 seconds to reach the top of a building .How high is the building ?

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23. If a kangaro jimps 2.5m vertically. Then

A. What is the initial speed of the kangaro ?

B. How long will it stay in air ?

I. Short Answers and Explanation

1. Distance is the actual path length while displacement is the shortest path or short cut of the body
2. Free fall is the motion of a body under the infulence of gravity alone.
3. Velocity
4. Change in velocity
5. Acceleration
6. Displacement
7. Average speed is distance travelled per time taken while, average velocity is displacement per time taken
8. Yes
9. Yes
10. Both will have same speed
11. Time of ascend is equal to the time of descend.

II. Solution to the End of Unit Problems

12. $a = \frac{v-u}{t} = \frac{11-5}{3} = 2\ m/s^2$

13. $a = \frac{dv}{dt} = \frac{d}{dt}(3t^2 - 4t) = 6t - 4$

$a = 6(1) - 4$, at $t = 1$ second, $a = 2m/s^2$

14.A. $t_T = t_1 + t_2 + t_3$

$t_1 = \frac{v-u}{a}$, $u = 0$

$t_1 = \frac{20-0}{2} = 10$ seconds $t_T = 10 + 20 + 5 = 35$ seconds

B. $\vec{V}_{av} = \frac{\Delta s}{\Delta t}$, $\Delta t = 35$ seconds

$\Delta S = S_1 + S_2 + S_3 = \left(\frac{20+0}{2}\right)10 + (20)(20) + \left(\frac{20+0}{2}\right)5$

$\Delta S = 550\,m$, $\vec{V}_{av} = \frac{550}{35} = 15{,}7\,m/s$

15.A. $V = gt = (10)(3) = 30\,m/s$

B. $V_{av} = \frac{u+v}{2} = \frac{0+30}{2} = 15\,m/s$

C. $y = \frac{1}{2}gt^2 = \frac{1}{2}(10)(3)^2 = 45\,m$

16.A. $V = gt = (10)(5) = 50\,m/s$

B. $y = \frac{1}{2}gt^2 = \frac{1}{2}(10)(5)^2 = 125\,m$

17.A. $y_{ball} = y_{stone}$

$u_b t_b - \frac{1}{2}gt_b^2 = u_s t_s - \frac{1}{2}gt_s^2$, $t_s = t_b - 1$

$10t_b - 5t_b^2 = 20(t_b - 1) - 5(t_b - 1)^2$

$10t_b - 5t_b^2 = 20t_b - 20 - 5t_b^2 + 10t_b - 5$

$-20t_b = -25$

$t_b = 1.25$ second $t_s = 0.25$ second

B. $y_s = u_s t_s - \frac{1}{2}gt_s^2 = 0.25(20) - 5(0.25)^2$

$y_s = y_b = 4.7\,m$

18. $S = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(2.4)(12)^2$

$S = 172.8m$

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19. $a = \frac{v - u}{t}$, $u = 0$, $v = 40\,m/s$

$a = \frac{40 - 0}{4}$, $a = 10\,m/s^2$

20.A. $y = \frac{1}{2} g t^2 \Rightarrow t = \sqrt{\frac{2y}{g}} = \sqrt{\frac{2(20)}{10}}$

$t = 2$ seconds

B. $v^2 = u^2 - 2gh$, $v = 0$

$u = \sqrt{2gh} = \sqrt{2(10)(20)}$

$u = 20\,m/s$

21. $v^2 = u^2 - 2gh \Rightarrow h = \frac{u^2}{2g} = \frac{(8)^2}{2(10)} = 3.2\,m$

22.A. $V^2 = u^2 - 2gh \Rightarrow u^2 = 2gh = 2(10)(2.5)$

$u = 5\sqrt{2}$ m/s

B. $t = \frac{2u}{g} = \frac{2(5\sqrt{2})}{10} = \sqrt{2}$ seconds.

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UNIT 4

4. Dynamics

4.1 Force and Its Effects
4.2 Newton's Laws of motion
4.3 Mass and Weight
4.4 Frictional Force
4.5 First Condition of Equillibrium
4.6 Linear Momentum and Impulse
4.7 Work, Energy and Power
4.8 What is Center of Mass?

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Dynamics

Welcome to this impressive unit where you will be able to define force, mension types of forces, describe the basic forces in nature, state Newton's laws of motion, distingnsh mass from weight, define the first condition of equilibrium, linear momentum and impulse, state the law of conservation of momentum, differenciate between work, energy and power and define center of mass. Enjoy !

4.1 Force and its Effects

Force is a push or a pull interaction between bodies When ever, force is exerted on a body, different effects can be observed on it.

Effects of Forces

Some of the Effects of Force are :

- ☑ Change state of motion of a body. That is, it may start or stop motion of a body.
- ☑ Change shape of a body. That is, it may deform a body
- ☑ Change size of a body. That is, it may compress or stretch a body
- ☑ Change direction of motion of a body, That is, it may turn a body from left to right, or from up ward to down ward.

When a force acts on a body, it may have physical contact with it or it may be at a distant so we have contact and non - contact forces.

Contact force : Is a force that involves physical contact with the body.

Fore example, kicking a ball, pushing a table, throwing a stone are contact forces.

Non - contact force : Is a force that acts at a distance due to its field with out physical contact with the body.

For example, gravity, electric force, magnetic force are non - contact forces

In nature, there are four basic forces. These are gravitational force, electromagnetic force, weak nuclear force and strong nuclear force.

Gravitational force : Is an attractive force between any two masses and it acts over long range .

Electro magnetic force : Is an attractive or repulsive force between any two charged particles and it acts over long range.

Weak Nuclear force : Is a force that allows protons to turn into neutrons during a decaying process

Strong Nuclear force : Is a force that keeps protons and neutrons bond together in the nucleus and it acts over very short range.

Moreover, a force may be conservative or dissipative.

Conservative force : Is a force that is not path dependent

For example, gravitational and spring as well as electric and magnetic forces are conservative.

Dissipative force : Is a force that is path dependant

For example, friction and air resistance

4.2 Newton's Laws of Motion

Sir Isaac Newton formulated the three laws of motion. These laws are; first law, second law and third law.

Newton's First law states that an object at rest remains at rest and an object in motion continues moving in a straight line unless external force acts up on it.

Newton's first law is also known as the **law of inertia**

- ☑ **Inertia**, is the property of an object to remain at rest or to move at constant velocity.

The inertia of a body depends on its mass. The greater the mass the body, the greater its inertia

There fore, it become difficult to change the state of motion of a bigger body than a smaller body.

For example, it is easier to move a plastic ball than to move a huge rock and it is easier to stop a rolling plastic ball than to stop a big rolling stone.

In case of Newton's first law, the net force acting on the body is zero.

$F_{net} = 0$, Newtons 1st Law

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Conceptual Examples

1. Why do passagers advised to tie seat belts ?

Explanation

To keep them safe in case the vehicle stops its motion suddenly. A seat belt will help the body of passanger to be in its stable condition inspite of the change in state of motion of the vehicle.

2. According to Newton's first law, no force acts on a body. True or False

Explanation

False .Force may act but the net force is zero.

When a net force acts on a body, the body accelerates and this is explained in newton's 2nd law

Newton's second law states that The acceleration of a body is directly proportional to the net force acting on it and inversely proportional to its mass

In symbols, it can be written as;

$$a \sim \frac{F}{m} \quad \text{or} \quad F = ma$$

Where : F = Net force acting on the body

m = Mass of the body

a = Acceleration of the body

☑ Force is a vector quantity and its SI unit is Newton (N)

$$1N = 1kg / m/s^2$$

Illustrative Example

3. A force of 80 N acts of a body of mass 40 kg.

A) What is the acceleration of the body

B) What acceleration will it this force give to a 160kg mass

Solution

A) $a = \frac{F}{m} = \frac{80}{40}$

$a = 2m/s^2$

B) $a = \frac{F}{m} = \frac{80}{160}$

$a = 0.5m/s^2$

4. The acceleration that a force produces on a 20 kg is larger by $2m/s^2$ than the acceleration it produces on a 25 kg mass. What is magnitude of the force ?

Solution

$F = m_1 a_1 = m_2 a_2$

$a_2 = a_1 - 2$

$m_1 a_1 = m_2 (a_1 - 2)$

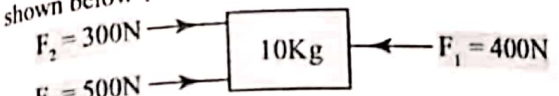
$20a_1 = 25 (a_1 - 2)$

$a_1 = 10m/s^2$

$F = m_1 a_1 = (10)(10)$

$F = 200N$

5. What acceleration will the system of forces produce on the 10kg box shown below ?



Solution

$a = \frac{F}{m} = \frac{(F_2 + F_3) - F_1}{m}$

$a = \frac{(300 + 500) - 400}{10}$

$a = 40m/s^2$

6. Two forces, F_1 and F_2 each of 15 N and 20N respectively are acting on a 50kg body Perpendiclarly. What acceleration will they produce on the body ?

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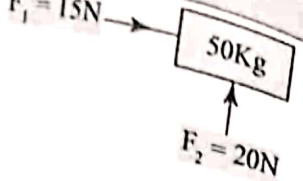


Solution

$$a = \frac{F}{m} = \frac{\sqrt{F_1^2 + F_2^2}}{m}$$

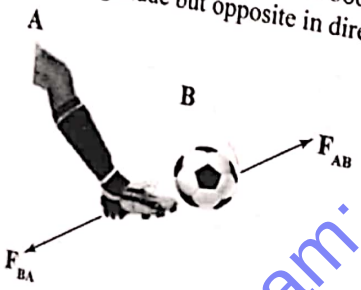
$$a = \frac{\sqrt{15^2 + 20^2}}{50} = \frac{25}{50}$$

$$a = 0.5 m/s^2$$

When you kick a ball, the ball kicks you back. With the same force you exerted on it . This is explained by Newton's 3rd law

Newton's Third law States states that "for every action, there is an equal and opposite reaction".

That is, if body A exerts a force on body B, then body B exerts a force on body A that is equal in magnitude but opposite in direction.



$$F_{AB} = - F_{BA}$$

Where : F_{AB} = The force body A exerts on body B

F_{BA} = The force body B exerts on body A

Conceptual Examples

7. According to Newton,s 3rd law forces exist in pairs

A) True B) False

Explanation

True in nature, forces exist in pairs, no single force exists.

8. If a Bajaji collides with a Truck, which one exerts greater force on the other ?

Explanation

According to Newton's 3rd law, they experience same forces.

9. Action and reaction forces act on one body. True or false ?

Explanation

False. They act on two different bodies.

4.3 Mass and Weight

In the every day language, mass and weight are terms used inter changebly. However, in physics they are different.

Mass (m) : Is the measure of amount of matter contained by a body.

It is a scalar quantity, measured using balance, constant every where and its SI unit is kilogram (kg).

Weight (w): Is the gravitational pull of the earth on a body.

It is a vector quantity, always acting to wards the center of the earth, measured using Newton meter, varies from place to place, and its SI unit is Newton (N)

Mass and weight of a body are related by :

Weight = Mass × Acceleration due to gravity

$$W = mg$$

Where : g = Acceleration due to gravity.

Illustrative Example

10. What is the weight of a 10 kg body on the surface of

A) Earth where $g = 10 m/s^2$

B) Moon where $g = 1.6 m/s^2$

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Solution

A) $W = mg = (10)(10)$

$W = 100N$

B) $W = mg = (10)(1.6)$

$W = 16\ N$

☑ Weight of a body varies from place to place

Apparent Weightlessness

True or real weightlessness of a body occurs when the body is out of the gravitational field, which is usually an ideal case. But apparent weightlessness occurs when the only force acting on the body is gravity.

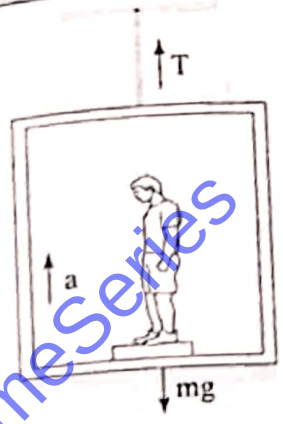
Now, let us examine the change in weight of a person inside an elevator when the elevator moves at constant velocity and when it accelerates.

☑ Elevator moves upward or downward at constant speed.

$T - mg = ma = 0$

$\boxed{T = mg}$, No Change in weight of the person

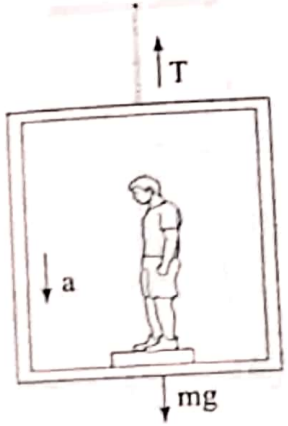
☑ Elevator moves upward with constant acceleration



$T - mg = ma$

$\boxed{T = mg + ma}$, Weight apppers as if it increased

☑ Elevator moves down ward with constant acceleration



$mg - T = ma$

$\boxed{T = mg - ma}$, Weight appears as if it decreased.

☑ Elevator moves down ward at an acceleration equal to the acceleration due to gravity.

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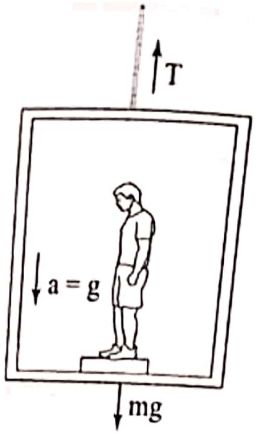
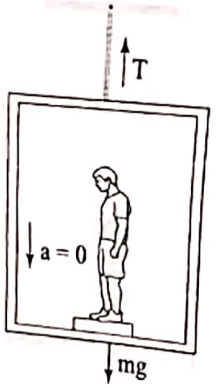


$mg - T = ma = mg$

$\boxed{T = 0}$, weght appers as if it is zero.

☑ Elevator at rest, V = 0 and a = 0



$T - mg = ma = 0$

$\boxed{T = mg}$, weight does not change

Illustrative Example

11. A 50 kg boy stands in an elevator. What will be this apparent weight when the elevator

A) Moves at a constant velocity of 5 m /s ²

B) Accelerates up ward at 2m/s ²

C) Accelerates downward at 2m/s²

D) Accelerates downward at10 m/s ²

Solution

A) $T = mg = (50)(10) = 500$ N

B) $T = mg + ma = (50)(10) + (50)(2) = 600$N

C) T = 400 N .

D) Zero

Illustrative Examples

12. A certain force gives mass m_1 an acceleration of 12 m /s ² and mass m_2 an acceleration of 3.3 m /s² . What acceleration will this force give to the combined mass ?

A) 1.5 m/ ² B) 2.5m/s² C) 3.5 m / s² D) 4.5 m /s ²

Solution

$$F = ma \Rightarrow a = \frac{F}{m}$$

$$m_1 a_1 = m_2 a_2 \Rightarrow m_2 = \frac{a_1}{a_2} \quad m_1 = \frac{12}{3.3} m_1$$

$$a = \frac{12m_1}{m_1 + m_2} = 2.5\,m/s^2.$$

Answer : B

13. A force gives an acceleration of 3m/s² to a 2kg mass. What acceleration will this force give to a 1kg mass ?

A) 6 m/s² B) 4 m/s² C) 2 m/s² D) 1 m/s²

Solution

$$m_1 a_1 = m_2 a_2 \Rightarrow a_2 = \frac{m_1 a_1}{m_2}$$

$$a_2 = \frac{(3)(2)}{1} = 6\,m/s^2$$

Answer : A

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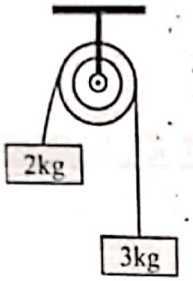
14. Two masses 2kg and 3kg are attached to the end of a light string passing over a pulley. What are the tension and acceleration in terms of g ?

A) $\frac{12}{5}$ g, g/5
C) $\frac{21}{8}$ g, g/8
B) $\frac{7}{8}$ g, g/5
D) $\frac{7}{8}$ g, g/5



Solution

For the 2 kg mass

T - mg = ma

T - 2g = 2a -------- (1)

Fore the 3 kg mass

3g - T = 3a -------- (2)

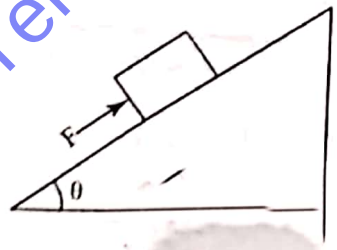
Solving equation (1) and equation(2),

$T = \frac{12}{5}$ g, a = g/5

Answer : A

15. A 100 kg box is pushed at constant speed up a smooth 30° incline plane. Force needed to move it up is ?

A) 577.35N
B) 1000. 35 N
C) 1732. 35 N
D) 1154. 70 N



Solution

F cos θ = mg sin θ

F = mg (tan θ) = (100) (10) tan 30°

F = 577 . 35 N

Answer : A

16. Which pair of forces can not give a result at of 5 N ?

A) 1N and 4 N
C) 3 N and 4 N
B) 1N and 6 N
D) 1 N and 7 N

Solution

$|A - B| \leq R \leq |A + B|$

$|1 - 7| \leq R \leq |1 + 7|$

$6 \leq R \leq 8.$

Answer : D

17. Which one of the following is true about action and reaction forces

A) They act on the same body
B) They are equal in magnitude but different line of action
C) They act on two different bodies
D) They are not equal in magnitude but have same line of action

Hint Action and reaction forces have same magnitude and line of action but opposite direction

Answer : C

18. A force of 6 N acts on a mass of 1 kg which acquires a velocity of 3m/s. The time for which the force acts is ?

A) 26 second
C) 0.5 second
B) 6 second
D) 2second

Solution

$F = ma = \frac{m\Delta v}{\Delta t}$

$\Delta t = \frac{m\Delta v}{F} = \frac{(1)(3)}{6}$

$\Delta t = 0.5$ ...

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Answer : C

19. When horse starts running all of a sudden, the rider on the horse falls back because ?

A) He is a fraid
B) He is pulled back ward
C) Of inertia of rest
D) Of inertia of motion

Hint : Newton's first law

Answer : C

20. A force of 5 N is applied to one end of a spring balance and a force of 5 N is applied simultaneously at the other end in the opposite direction. The reading of the balance will be

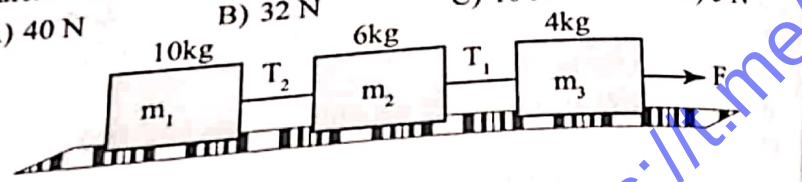
A) 2.5 N B) 5N C) 10 N D) Zero

Hint : Equal and opposite vectors cancel out

Answer : D

21. A 40 N force pulls a system of three masses on a horizontal frictionless surface. Tension T_1 is ?

A) 40 N B) 32 N C) 10 N D) 5 N



Solution

$F - T_1 = m_3 a$

$T_1 - T_2 = m_2 a$

$T_2 = m_1 a$

$$a = \frac{F}{m_1 + m_2 + m_3} = \frac{40}{4+6+10}$$

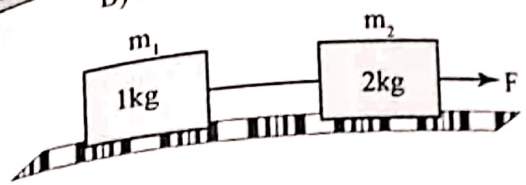
$= 2\ m/s^2$

$40 - T_1 = 4(2) \Rightarrow T_1 = 32\ N$

Answer : B

22. Two blocks of masses 1kg and 2 kg rest on a smooth surface. When the 2kg block is pulled by a certain force, the string tension become 1.5 N. What is the value of the applied force ?

A) 1.5 N B) 2.5 N C) 3.5 N D) 4.5 N



Solution

$F - T = m_2 a$

$T = m_1 a$

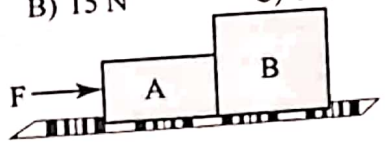
$$\Rightarrow F = \left(\frac{m_2}{m_1}\right) T + T$$

$$F = \left(\frac{2}{1}\right)(1.5) + 1.5N = 4.5N.$$

Answer : D

23. On smooth surface, two blocks A and are accelerated by applying a force of 15 N on A. If mass of B is twice that of A, the force on B is ?

A) 30 N B) 15 N C) 10 N D) 5 N



Solution

$F = ma = (m_A + m_B)\ a\ ,\ m_B = 2m_A$

$$a = \frac{F}{3m_A} \text{ and, the force on B is}$$

$$F_B = m_B a = (2m_A)\left(\frac{15}{3m_A}\right) = 10N$$

Answer : C

24. A girl in a lift will weigh more when the light

A) Is falling freely
B) Is moving down with acceleration, $a < g$
C) Is moving down with acceleration, $a > g$
D) Is about to move up

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Hint : up ward acceleration of light increase weight

Answer : D

25. A bag weighs 49 N when the elevator is at rest what will be the weight of the bag if the elevator accelerate down ward at 5 m/s² ?

A) 49 N B) 24 N C) 74 N D) 15 N

Solution

Apparent weight = mg - ma

$$m = \frac{\text{weight}}{g} = \frac{49}{10} = 4.9 \text{ kg}$$

Apparent weight = 49 -(4.9) (5) = 24 N

Answer : B

26. A man weighs 80 kg. He stands in a lift that accelerates up ward at 5m / s². What would be the reading on the scale ?

A) Zero B) 400 N C) 800 N D) 1200 N

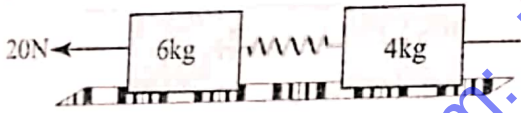
Solution

Apparent weight = mg + ma = (80) (10) + (80) (5) = 1200 N

Answer : D

27. A newton meter is attached to two blocks of masses 6 kg and 4 kg. Forces of 20 N and 10 N are applied on the blocks. What is meter reading ?

A) 10 N B) 20 N C) 6 N D) 14 N



Solution

20 - T = 6 a

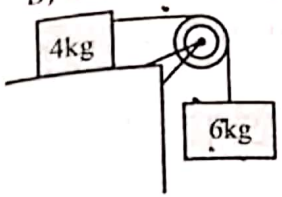
10 - T = 4 a

a = 1m/s² and T = 14 N

Answer : D

28. A block of mass 4 kg is placed on a frictionless table and is connected to a 6 kg block by a string that passes through a frictionless pulley. What is the tension of the string ?

A) 12 N B) 24 N C) 36 N D) 48 N



Solution

6 (10) - T = 6 a

T = 4 a

a = 6m/s²

T = 24 N

Answer: B

4.4 Frictional force

The force that acts in opposite direction of motion and resists the sliding or an attempt of sliding of a surface is called **friction**

Friction : Is a force that opposes the motion of one surface over the other surface.

The amount of opposition to the motion of a body depends on the roughness of the surfaces in contact and the normal force.

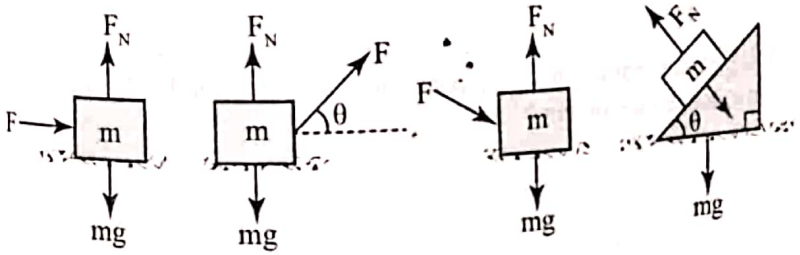
Friction = Coefficient of × Normal friction force

$$F = \mu F_N$$

Where : μ (mu) is the coefficient of friction and its value is : $0 < N \leq 1$

Normal force (F_N) : Is a supporting or pressing force which is perpendicular to the surfaces in contact.

The normal force of a body on a level surface is equal to the weight of the body Now let us find the normal force of some mass surface orentation



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$\Sigma F_y = 0$ $\Sigma F_y = 0$ $\Sigma F_Y = 0$ $\Sigma F_Y = 0$

$F_N = mg$ $F_N = mg - F\sin\theta$ $F_N = mg + F\sin\theta$ $F_N = mg\cos\theta$

And, there are two types of frictions. Static and kinetic

Static Friction (f_s) : Is the friction force between two bodies that are in contact and trying to move past each other but not yet moving

Mathematically, it is obtained by

$$f_s = \mu_s F_N$$

Where : μ_s = Coefficient of static friction.

Kinetic friction (f_k): Is the friction force between two surfaces when there is relative motion in them

Mathematically, it is obtained by :

$$f_k = \mu_k F_N$$

Where : μ_k = coefficient of kinetic friction.

Kinetic friction may occur when a surface rolls or slides over another surface.

Remember, static friction is usually greater than kinetic friction and the maximum static friction is called **limiting friction** and it arises when the body just begin to move.

$$f_s >= f_k \quad \Rightarrow \quad \mu_s > \mu_k$$

Illustrative Examples

29. A 4 kg block is placed on a table of friction coefficient 0.8 .What is the friction force when the body is forced to move ?

Solution

$f_s = \mu_s F_N = (0.8)(mg)$

$= (0.8)(4)(10)$

$f_s = 32\ N$

30. A 3kg object moving at 15m/s comes to rest in 5 seconds. What friction force stopped the object ?

Solution

$f = ma$

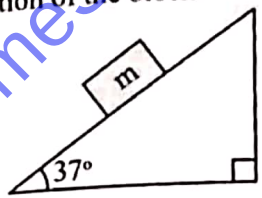
$a = \frac{v - u}{t} = \frac{0.15}{5}$ $a = -3m/s^2$

$f = (3)(3) = 9N$

31. A 4 kg block is moving down an inclined plane which inclination of 37°. Find

A) The frictional force if $\mu_k = 0.5$

B) The acceleration of the block



Solution

A) $f = \mu_k F_N = \mu mg \cos\theta$

$f = (0.5)(4)(10)(0.8)$

$f = 16\ N$

B) $mg\sin\theta - f = ma$

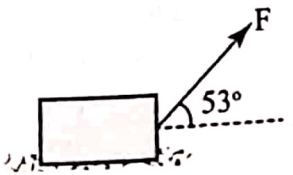
$a = \frac{mg\sin\theta - \mu mg\theta}{m} = \left(\frac{(40)(0.6) - 16}{4}\right)$

$a = 2m/s^2$

Illustrative Example

32. A force of magnitude 20 N is applied to a block of mass 2 kg . The coefficient of kinetic friction between the block and the surface is 0.4. What is acceleration of the block ?

A) $10m/s^2$ B) $5.2m/s^2$ C) $4m/s^2$ D) $2.8m/s^2$



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Solution

$$F\cos\theta - \mu_k (mg - F\sin\theta) = ma$$

$$a = \frac{F\cos\theta - \mu_k (mg - F\sin\theta)}{m}$$

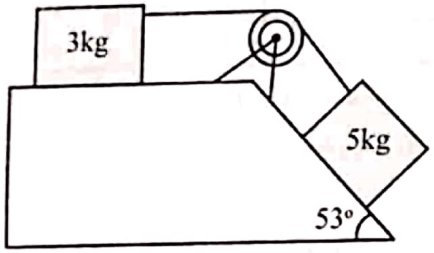
$$a = \frac{(20)(0.6) - 0.4[2(10) - 20(0.8)]}{2}$$

$$a = 5.2\,m/s^2$$

Answer : B

33. A block of mass 3 kg is connected to a block of mass 5 kg and are placed on an inclined plane of coefficient of friction 0.3 . What is the tension of the string ?

A) 29.4 N B) 17 .25 N C) 12.25 N D) 8.60 N



Solution

$T - f_1 = m_1 a$

$T - (0.3)(3)(10) = 3a$ .....(1)

$m_2 g\sin\theta - f_2 - T = m_2 a$

$(5)(10)(0.8) - (0.3)(5)(10)(0.6) - T = 5a$ .....(2)

Solving equation 1 and 2, T = 17.21 N

Answer : B

34. A body of mass 15 kg is pushed on a horizontal surface by applying a 100N horizontal force. Assuming the coefficient of friction to be 0.3, What is the acceleration of the body ?

A) 3.67m/s² B) 2.67m/s² C) 1.67m/s² D) 0.67m/s²

Solution

$F - f = ma$

$$a = \frac{F - f}{m} = \frac{F - \mu mg}{m}$$

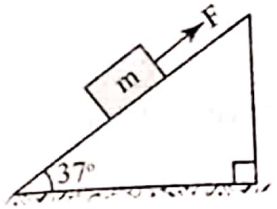
$$a = \frac{100 - (0.3)(15)(10)}{15}$$

$$a = 3.67\ m/s^2$$

Answer : A

35. A block having a weight of 100N is pulled up along an inclined plane of coefficient of friction 0.3 at constant speed. What is the value of the force F ?

A) 54 N B) 64 N C) 74 N D) 84 N



Solution

$F - mg\sin\theta - f = 0$

$F = mg\sin\theta + \mu mg\cos\theta$

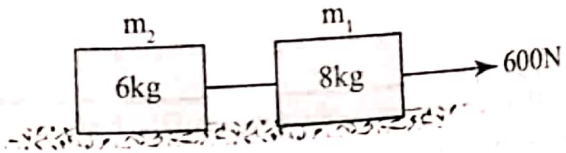
$F = (100)(0.6) + (0.3)(100)(0.8)$

$F = 84\ N$

Answer : D

36. Two blocks of masses 6 kg and 8 kg are connected by a light string. A 600 N force is applied on the 8 kg block as shown below. If there is a constant friction force of 40 N on the surfaces. What is the tension of the cable connecting the blocks ?

A) 62 N B) 162N C) 262 N D) 362 N



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Solution

$F - T - f = m_1 a$

$600 - T - 40 = 8\,a$ ---------- (1)

$T - f = m_2 a$

$T - 40 = 6\,a$ -------- (2)

Solving equation 1 and 2, $a = 37\ m/s^2$ and $T = 262\ N$

Answer : C

4.5 First condition of Equillibrium

When a net force acts on a body, the body accelerates. However, If the net force acting on a body is zero , the body became in equillibrium.

A body is said to be in the first condition of equillibrium if the sum of forces acting on it is zero.

A body is said translational or in first condition of equillibrium when :

$$\sum F = 0$$

$$\sum F_x = \sum F_y = \sum F_z = 0$$

Remember, A body at rest and a body moving at constant velocity are in equillibrium since in both cases there is no net force acting on the body.

Conceptual Examples

37. A body is said to be in equillibrium only when there is no force acting on it. True or False ?

Explanation

False. A body is in equillibrium when no net force and no net torque acts on it.

38. The normal force is always equal to the weight of the body. True or False ?

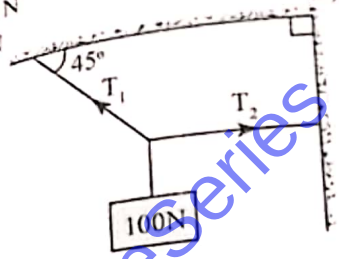
Explanation

False. Normal force is equal to the weight of the body only when the body lies on a level surface.

Illustrative Examples

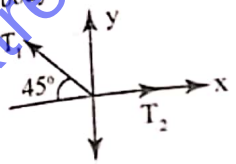
39. For the situation shown, is the tension T_2 -

A) 141.4 N
B) 100 N
C) 241.4 N
D) 41.4 N



Solution

Free body diagram



$\sum F_x = 0,\ T_2 - T_1 \cos 45^\circ = 0$

$T_2 = 0.707\ T_1$

$\sum F_y = 0,\ T_1 \sin\theta - W = 0$

$0.707\ T_1 = W = 100N$

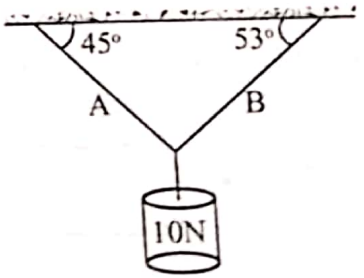
$T_1 = 141.4\ N$

$T_2 = 0.707\ (141.4) = 100\ N$

Answer : B

40. The tension in cord A is 30 N. The tension in cord B is ?

A) 20 N B) 30 N C) 40 N D) 50 N



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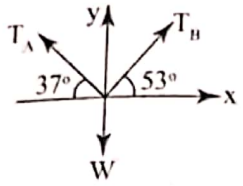
Solution

Free body diagram



$\sum F_x = 0$

$T_A \cos 37^0 = T_B \cos 53^0$

$(30)(0.8) = T_B (0.6)$

$T_B = 40$ N

Answer : C

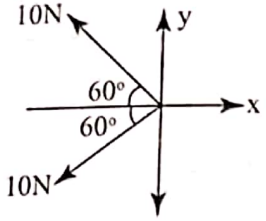
41. Particle is set to move with constant velocity under the action of three forces. Two forces with magnitude 10 N each are shown. What most be the third force ?

A) 10 N right

B) 5 N up ward

C) 10 N, up ward

D) 5 N, right



Solution

$\sum F_x = 0, 2(10)\cos 60^0 = F_x$

$F_x = 10$ N

$\sum F_y = 0, 10 \sin 60^0 - 10 \sin 60^0 = F_y$

$F_y = 0$

Answer : A

42. A body experiencing two forces of equal magnitude remains in equilibrium. The angle between them is ?

A) 0^0 B) 45^0 C) 90^0 D) 180^0

Hint : $\sum F = 0 \Rightarrow F_1 + F_2 = 0$

$F_1 = -F_2$

Answer : D

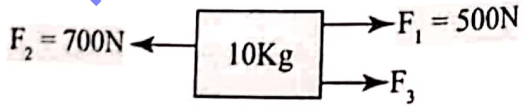
43. What should be the value of F_3 so that the block remains stationery on the smooth surface

A) 200 N

B) 300 N

C) 400 N

D) 500 N



Solution

$F_3 + F_1 = F_2$

$F_3 = F_2 - F_1 = 700 - 500$

$F_3 = 200$ N

Answer : A

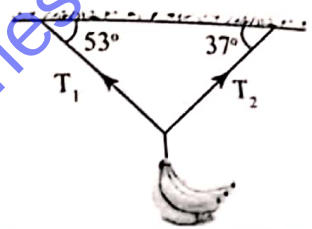
44. A bunch of banana of mass 25kg is suspended with two ropes. If $T_1 =$ 200 N, then T_2 is ?

A) 150 N

B) 250 N

C) 350 N

D) 450 N



Solution

$F_x = 0, 0.6\, T_1 = 0.8\, T_2$

$T_2 = 0.75\, T_1 = (0.75)(200) = 150$ N

Answer : A

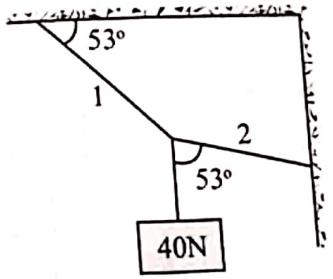
45. If the weight of the block suspended is 40 N. The tension in cable - 1 is

A) 14.3 N

B) 14.3 N

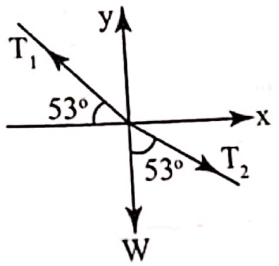
C) 114.3 N

D) 411 .3 N



Solution

Free body diagram



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$\sum F_x = 0,\ T_1 \cos 53^\circ = T_2 \sin 53^\circ$

$0.6\ T_1 = 0.8\ T_2$

$T_2 = 0.75\ T_1$ ------- (1)

$\sum F_y = 0,\ W + T_2 \cos 53^\circ$

$0.8\ T_1 = 40 + 0.6\ T_2$

$0.8\ T_1 = 40 + 0.6\ (0.75)\ T_1$

$0.35\ T_1 = 40$

$T_1 = 114.3$ N

Answer : C

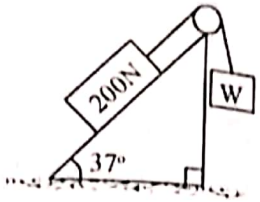
46. Assuming no friction , how much should the block hanging weigh if the 200 N block is to remain at rest ?

A) 120N

B) 200 N

C) 160 N

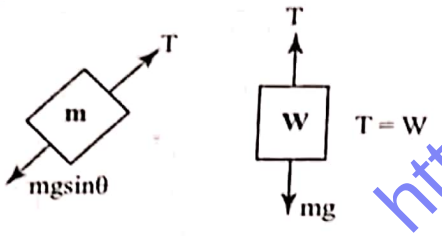
D) 220 N



Solution

Free body diagram

mg sin θ T = W



$T - mg \sin\theta \Rightarrow T = (200)(0.6) = 120$ N

Answer : A

47. A 75N force is require to start a 25kg block to move, and a 60N force is needed to keep the block moving on horizontal road. Find

A) Static coefficient of friction

B) Kinetic coefficient of friction

Solution

A) $F_s = \mu_s F_N$, $F_N = mg = (25)(10) = 250$

$\mu_s = \frac{f_s}{F_\mu} = \frac{75}{250} = 0.3$

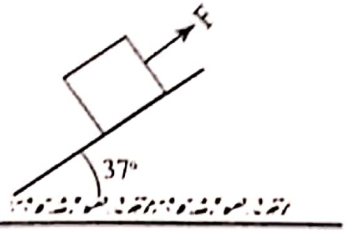
B) $f_k = \mu_k F_\mu$

$\mu_k = \frac{f_k}{F_\mu} = \frac{60}{250} = 0.24$

48. What force is required to move the 10kg block along the inclined plane of coefficient of kinetic friction 0.25 at

A) Constant speed

B) Constant acceleration of 0.5 m/s²



Solution

A) $F - f - mg \sin\theta = 0$

$F = \mu mg \cos\theta + mg \sin\theta$

$F = (0.25)(10)(10)(0.8) + (10)(10)(0.8)$

$F = 100$N

B) $F - f - mg \sin\theta = mg$

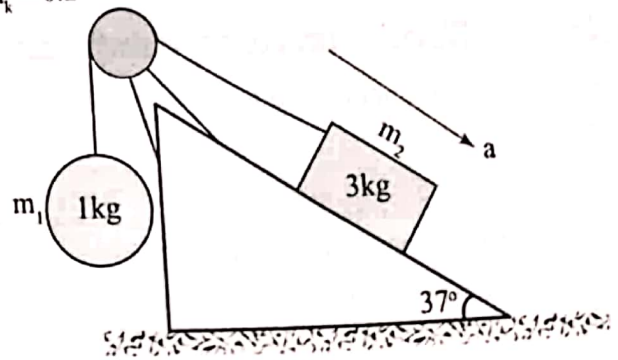
$F = \mu\ mg \cos\theta + mg \sin\theta + ma$

$F = 105$N

49. What is the acceleration of the system of mass show below if the inclined plane is

A) Frictionless

B) With $\mu_k = 0.2$



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Solution

A) $m_2 g \sin\theta - T = m_2 a$

$T - m_1 g = m_1 a$

$$a = \frac{m_2 g \sin\theta - m_1 g}{m_1 + m_2} = \frac{(3)(10)(0.6) - (1)(10)}{1+3}$$

$a = 2m/s^2$

B) $m_2 g\sin\theta - \mu m_2 g\cos\theta - T = m_2 a$

$T - m_1 = m_1 a$

$$a = \frac{m_2 g \sin\theta - \mu m_1 g \cos\theta - m_1 g}{m_1 + m_2}$$

$$a = \frac{(3)(10)(0.6) - (0.2)(3)(10)(0.8) - (1)(10)}{1+3}$$

$a = 0.875 m/s^2$

50. What is the value of T_1 and T_2 if the system is in equilibrium?

Solution

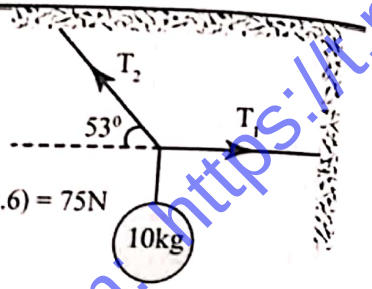
$\Sigma F_y = 0$

$T_2 \sin 53^0 = mg$

$0.8\, T_2 = (10)(10)$

$T_2 = 125N$

$\Sigma F_x = 0,\ T_1 = T_2 \cos 53^0 = (123)(0.6) = 75N$



4.6 Linear Momentum and Impulse

Momentum is a quantity of motion and it is a measure of how hard it is to stop a moving body.

Momentum : Is the product of mass and velocity of a body

Mathematically, liner momentum is obtained by :

Momentum = Mass × Velocity

$$\vec{P} = m\vec{v}$$

Momentum is a vector quantity and its SI unit is kilogram meter per second (kg m /s)

And, from Newton's second law, the net force acting on a body is obtained by

$$F = ma = m\frac{\Delta v}{\Delta t}$$

$$F = m\left(\frac{v-u}{\Delta t}\right) = \frac{mv - mu}{\Delta t}$$

$$F = \frac{\Delta P}{\Delta t}$$

$F\Delta t = \Delta P$ → Impulse

Remember, the change in momentum of a body is called *impulse* and is given as the product of average force and the time of contact.

Critical Thinking Question

51. A glass dropped on to a concrete floor is more likely to break than when it is dropped in to a pond of water. Why ?

Explanation

The shorter the contact time, the greater the average force acting on the body.

Illustrative Examples

52. What is the linear momentum of a 500 g ball moving at speed of 40 m /s ?

Solution

$P = m\,v = (0.5)\ (40)$

$P = 10$ kg m / s

53. If a body of mass 3 kg changes its speed from 2m /s to 12m / s in 5 seconds. Then, What is

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A) Impulse imparted on the body

B) Average force acting on the boy

Solution

A) Impulse $= \Delta P = m\,(v-u)$

$= 3\,(12-2)$

Impuse $= 30$ kgm/s

B) Impulse $= F\Delta t = \Delta P$

$F\Delta t = mv - mu$

$$F = \frac{mv - mu}{\Delta t} = \frac{3(12) - (3)(2)}{5}$$

$F = 6N$

54. A soccer player exerts a force of 1200 N that acts for 0.02 second on a ball of 400 g mass. What is the speed of the ball as it leaves his foot ?

Solution

Impulse $= F\Delta t = m\Delta v$

$$\Delta v = \frac{F\Delta t}{m} = \frac{(1200)(0.02)}{0.4}$$

$\Delta v = 60$ m/s

$v - u = 60,\ u = 0$

$v = 60$ m/s

Conservation of Linear Momentum

For an isolated system where there is no external force, the total liner momentum is constant.

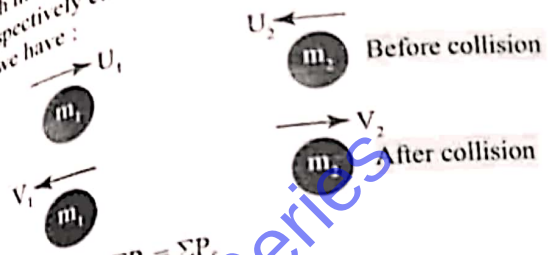
Law of conservation of linear momentum : states that for an isolated system, the initial momentum is equal to the final momentum

That is, if there is no external force, then

Momentum Initial = Momentum Final

$$\boxed{P_i = P_f}$$

Which means, if two bodies of mass m_1 and m_2 initially moving at u_1 and u_2 respectively collide, and move with velocities v_1 and v_2 after collision then we have :



$\Sigma P_i = \Sigma P_f$

$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$

Illustrative Examples

55. A 15 kg fish swimming at 1.1 m / s in lake Tana suddenly gets a 4.5 kg fish initially at rest. What is the speed of the larger fish after he eat the small fish ?

A) 1.1 m/s B) 2.2 m/s C) 0.85 m/s D) 1.85 m/s

Solution

$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$

$(15)(1.1) + 0 = (15 + 4.5)\,v$

$V = 0.85$ m/s

Answer : C

56. The momentum of the system is conserved

A) Always

B) Never

C) Only in absence of external force

D) Only when external force acts

Hint : Momentum of a system conserved when there is no external force acting on it.

Answer : C

57. In a rocket , fuel is ejected with a velocity of 6 km/s. If the fuel burns at a rate of 1kg/s, then the force exerted by the exhaust gas is ?

A) 6000 N B) 600 N C) 60 N D) 6 N

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Solution

$$F = \frac{v\Delta m}{\Delta t} = (6000)(1)$$

$F = 6000\,N$

Answer : A

58. If the liner momentum of a 2 kg cat is equal to the linear momentum of a 20 kg dog running at 10 m/s, What is the speed of cat ?

A) 10 m / s B) 100 m / s C) 5 m / s D) 25 m / s

Solution

$m_c v_c = m_d v_d$

$$v_c = \frac{m_d v_d}{m_c} = \frac{(20)(10)}{2} = 100\,m/s$$

Answer : B

59. A 25 kg object moving at 20 m / s to the right collides with a 10 kg object moving at 15 m /s to the left. If the final velocity of the 25 kg object is 15 m /s to the right, what will be the velocity of the 10 kg object after collision ?

A) 25 m / s to the left C) 2.5 m / s to the left

B) 2.5 m / s to the right D) 25 m / s to the right

Solution

$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$

$(25)(20) - (10)(15) = (25)(15) + 10 v_2$

$v_2 = 2.5$ m / s to the right.

Answer: B

60. What will happen to the linear momentum of a body if its mass is doubled and its velocity is halved ?

A) Halved C) Quadriplled

B) Doubled D) Remains the same

Hint : $P_1 = m_1 v_1$

$P_2 = m_2 v_2$, $m_2 = 2m_1$ and $v_2 = 1/2\, v_1$

$P_2 = (2m_1)(1/2\, v_1)$

$P_2 = m_1 v_1$

Answer : D

61. A ball of mass 0.2kg with a velocity of 5 m/s strikes a wall perpendicularly and bounces off straight back with a velocity of 3 m/s. The change in momentum of the ball is ?

A) 1 kg m / s C) 1.6 kg m / s

B) 0.6 kg m / s D) 2.6 kg m / s

Solution

$P_i = mv_i = (0.2)(5) = 1$ kg m / s

$P_f = mv_f = (0.2)(-3) = -0.6$ kg m / s

$\Delta P = P_f - P_i = -0.6 - 1 = -1.6$ kg m /s

Answer : C

62. A car of mass 500 kg is moving at 24 m / s. If a tiger of mass 100 kg drops on the roof of the car from an over hanging branch, What will be the speed of the car after carrying the tiger ?

A) 20 m / s B) 24 m/ s C) 30 m / s D) 35 m / s

Solution

$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$

$(500)(24) + 0 = (100 + 500)\, v$

$V = 20$ m / s

Answer : A

63. Two cars having masses of 150 kg and 70 kg respectively are initially at rest and are held together by a compressed massless spring. When released, the 150 kg car moves to the left with a velocity of 8 m /s. The velocity of the 70 kg will be ?

A) 17.14 m / s to the left C) 8.7 m/ s to the left

B) 17.14 m / s to be the right D) 8.7 m/s to the right

Solution

$P_i = P_f$

$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$

$0 + 0 = (150)(8) + 70(v_2)$

$v_2 = -17.14$ m / s = 17.14 m /s to the right

Answer : B

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What is Collision?

The situation in which two bodies strike one another and in which the net external force is zero is called collision.

Collision: Is an event where two or more bodies make physical contact for very short time and experience impulsive force.

Based on what happens to the kinetic energy of the system of masses after the collision, we have elastic collision and in **elastic** collision

Elastic collision: Is a collision in which both momentum and kinetic energy of the system are conserved.

That is for an elastic collision

$KE_i = KE_f$, and

$P_i = P_f$

For example, collision between atomic and subatomic particles and collision between billiard balls are elastic collisions.

Illustrative Examples

64. A ball of mass 0.8kg moving at 5m/s collides with another ball of mass 1.2kg moving in opposite direction with 4m/s. After collision, the first ball moves with 4m/s. What is the speed of the second ball after collision?

Solution

$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$

$(0.8)(5) - (1.2)(4) = (0.8)4 + (1.2)V_2$

$-0.8 = 3.2 + 1.2V_2$

$V_2 = -3.3$ m/s

$V_2 = 3.3$ m/s, Opposite to initial direction

65. A pool ball of mass 100g moving at (5, - 4) m/s collides with a stationary ball of the same mass. After the collision, one of the balls move at (2, -3) m/s. What is velocity of the other ball?

Solution

$$m_1\begin{bmatrix}u_{1x}\\u_{1y}\end{bmatrix} + m_2\begin{bmatrix}u_{2x}\\u_{2y}\end{bmatrix} = m_1\begin{bmatrix}v_{1x}\\v_{1y}\end{bmatrix} + m_2\begin{bmatrix}v_{2x}\\v_{1y}\end{bmatrix} + m_2\begin{bmatrix}v_{2x}\\v_{2y}\end{bmatrix}$$

$m_1 = m_2$

$$\begin{bmatrix}5\\-4\end{bmatrix} + \begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}2\\-3\end{bmatrix} + \begin{bmatrix}v_{2x}\\v_{2y}\end{bmatrix}$$

$$\begin{bmatrix}5\\-4\end{bmatrix} = \begin{bmatrix}2\\-3\end{bmatrix} + \begin{bmatrix}v_{2x}\\v_{2y}\end{bmatrix}$$

$$\begin{bmatrix}v_{2x}\\v_{2y}\end{bmatrix} = \begin{bmatrix}2\\-3\end{bmatrix} - \begin{bmatrix}5\\-4\end{bmatrix}$$

$$\begin{bmatrix}v_{2x}\\v_{2y}\end{bmatrix} = \begin{bmatrix}-3\\1\end{bmatrix}$$

$\overrightarrow{V_2} = [V_{2x}, V_{2y}] = [-3, 1]$ m/s

Inelastic Collision: Is a collision in which the momentum of the system is conserved but its kinetic energy is not conserved.

That is, for an inelasic collion

$KE_i \neq KE_f$ but,

$P_i = P_f$

For example, collision of a ball with mud and explosion are inelastic collisions.

Illustrative Examples

66. A truck of mass 20,000kg moving at 20m/s collides with a bajaji of mass 200kg moving at 10m/s in the same direction as the truck. If they stick together, what is their common velocity after the collision?

Solution

$m_1 u_1 + m_2 u_2 = (m_1 + m_2)v$

$(20{,}000)(20) + (200)(10) = (20{,}000 + 200)V$

$V = 19.9$m/s

67. A 3.5g bullet is fired in to a sand bag of 100kg initially at rest on a

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smooth floor. If the bag with the bullet inside it move at 0.5 m/s. What was the initial speed of the bullet?

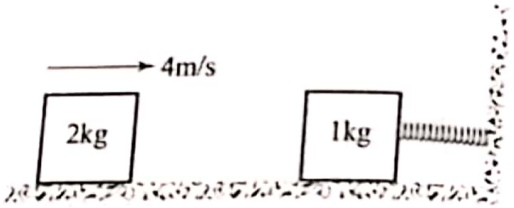
Solution

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

$$(3.5)\,10^{-3}\,u_1 + 0 = (100 + 0.0035)0.5$$

$$u_1 = 14{,}286 \text{m/s}$$

68. A 1kg block connected to a spring of force constant 400N/m rests on a frictionless surface. If a 2kg block moving at 4m/s collides with the 1kg block and stickes to it. What is the maximum compression of the spring?



Solution

$$m_1\,u_1 + m_2\,u_2 = (m_1 + m_2)v$$

$$(2)(4) + 0 = (1+2)v$$

$$V = 2.67 \text{ m/s}$$

$$\text{And, } \frac{1}{2}mv^2 = \frac{1}{2}Kx^2$$

$$x = \sqrt{\frac{mv^2}{K}} = \sqrt{\frac{(1+2)(2.67)^2}{400}}$$

$$x = 0.23\text{m}$$

69. *Collision occurs between a 1kg body moving with (-2i - 3j) m/s and 2kg body moving (3i - j)m/s. After the collision, the bodies stock together What is their common velocity?*

Solution

$$m_1\begin{bmatrix} u_{1x} \\ u_{1y} \end{bmatrix} + m_2\begin{bmatrix} u_{2x} \\ u_{2y} \end{bmatrix} = (m_1 + m_2)\begin{bmatrix} v_x \\ v_y \end{bmatrix}$$

$$1\begin{bmatrix} 2 \\ -3 \end{bmatrix} + 2\begin{bmatrix} 3 \\ -1 \end{bmatrix} = 3\begin{bmatrix} v_{1x} \\ v_{1y} \end{bmatrix}$$

$$\vec{V} = (V_x, V_y) = (2.3i - 1.7j)\text{m/s}$$

70. A bullet of mass 2 gram leaves the muzzle of a pistol at 600 m /s. The mass of the pistol is 400 gram. What is the recoil velocity of the pistol ?

A) 3m / s B) 5 m / s C) 0.5 m / s D) 0.3 m / s

Solution

$$m_1\,u_1 + m_2\,u_2 = m_1\,v_1 + m_2\,v_2$$

$$0 + 0 = (2)(600) + (400)\,v_2$$

$$v_2 = 3 \text{ m/s}$$

Answer : A

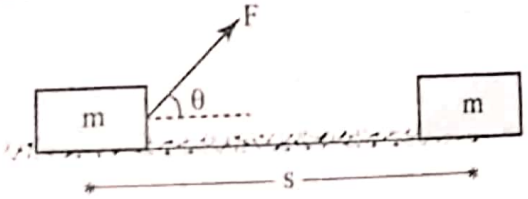
4.7 Work, Energy and Power

The terms work, energy and power are technical and different in physics even though they are used inter changeably in every day language.

Work Done by Force

In physics, work is said to be done when force acts on a body and the body is displaced in the force direction.

In general, if a block is displaced an amount s by a force F, then the work done on the block can be obtained by :



$$W = Fs\cos\theta$$

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Where : θ = Is the angle between the force and the displacement of the body.

Work is a scalar quantity and its SI unit is **joule (J)**

$1J = 1Nm$

Conceptual Example

71. A man carrying a 100 kg, box moves for 600m horizontally. How much is the work done by the man ?

Explanation

$W = F\,s\cos\theta, \quad \theta = 90^0$

$W = Fs\cos 90^0$

$W = 0$

Illustrative Examples

72. What is the work done by a 600 N force that makes 37^0 from the horizontal and displaced a 30 kg block by 60 m horizontally ?

Solution

$W = F\,s\cos\theta = (600)(60)\cos(37^0)$

$W = 2880\ J$

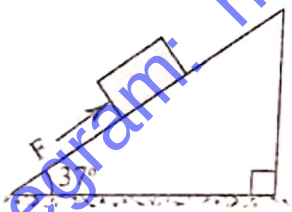
73. A force of 200 N pushes a block of 2 kg for 3 m along an inclined plane of 37^0 inclination and 0.5 coefficient of friction. Find the work done by :

A) Applied force

B) Gravity

C) Friction

D) Net force



Solution

A) $W_F = F_s\cos\theta \ , \ \theta = 0^0$

$W = (200)(3)$

$W = 600\ J$

B) $W_g = mgh = mgs\sin\theta$

$W = (2)(10)(3)(0.6)$

$W = 36\ J$

C) $W_f = f\,s\cos\theta \ , \ \theta = 180^0$

$W_f = \mu F_N S = \mu\, mg\cos\theta\, s$

$W_f = (0.5)(2)(10)(0.8)(3)$

$W_f = 24\ J$

D) $W_{net} = W_F - W_F - W_g = 600 - 36 - 24$

$W_{net} = 540\ J$

What is Energy ?

Energy enables humans, machines and vehicles to perform different activities and takes.

Energy: Is the ability or capacity to do work

Energy is a scalar quantity and its SI unit is joule (J).

Energy can exist in various forms but most of it is mechanical.

Mechanical Energy: Is the sum of kinetic energy and potential energy.

Kinetic Energy

Is the energy that a moving body possessed.

Kinetic Energy: Is the energy of motion

The kinetic energy of a body of mass m moving at a speed v is obtained by:

$$\text{Kinetic Energy} = \frac{1}{2}(\text{mass})(\text{speed})^2$$

$$\boxed{K_E = \frac{1}{2}mv^2}$$

Conceptual Example

74. Can kinetic energy be negative? Explain

Explanation: Never, in $KE = \frac{1}{2}mv^2$ all variables are positive

75. What will happen to the kinetic energy of a body if

A) Its speed is halved while its mass is constant

B) Its speed is doubled while its mass is constant

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Explanation

A) Quartered

B) Qualdpled

Illustrative Examples

76. What is the kinetic energy of a 1500kg mass moving at a speed of 10m/s

Solution

$$K_E = \frac{1}{2}mv^2 = \frac{1}{2}(1500)(10)^2$$

$$K_E = 75{,}000J = 75KJ$$

77. What is the speed of a 100kg body if its kinetic energy is 5000J?

Solution

$$K_E = \frac{1}{2}mv^2$$

$$V = \sqrt{\frac{2K_E}{m}} = \sqrt{\frac{(2)(5000)}{100}}$$

$$V = 10m/s$$

78. A 0.6kg particle has a speed of 2m/s at point A and a kinetic energy of 7.5J at point B What is the particle's

A) Kinetic energy at point A

B) Speed at point B

Solution

A) $K_E = \frac{1}{2}mv_A^2 = \frac{1}{2}(0.6)(2)^2$

$$K_{EA} = 1.2J$$

B) $K_{EB} = \frac{1}{2}mv_B^2$

$$V_B = \sqrt{\frac{2K_E}{m}} = \sqrt{\frac{2(7.5)}{0.6}}$$

$$V_B = \sqrt{25}$$

$$V_B = 5m/s$$

Potential Energy

is the energy stored by a body when it is at a particular position or condition

Potential Energy: Is the energy possessed by a body because of its position or configuration

Therefore, potential energy can be gravitational or elastic

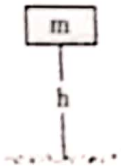
Gravitational Potential Energy: Is the potential energy of a body because of its position from a given surface

The gravitational potential energy of a body of mass m placed at a height h above the given reference surface is obtained by:

Gravitational potential energy = (mass) (acceleration due to gravity) (height)

$$GP_E = mgh$$

Where; $g = 10m/s^2$



Critical Thinking Question

79. For a body moving upward, the work done by gravity is positive. True or False?

Explanation

False, Gravity always acts down ward to wards the center of the earth. Therefore, for a body moving down ward, work done by gravity is positive.

Illustrative Example

80. What is the gravitational potential energy of a body of mass 100kg at a height of 10m from the ground?

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Solution

$GP_E = mgh = (100)(10)(10)$

$GP_E = 10{,}000J = 10KJ$

81. At what height from the ground will a 200kg mass have a gravitational potential energy of 4000J

Solution

$$GP_E = mgh$$

$$h = \frac{GP_E}{mg} = \frac{4000}{(200)(10)}$$

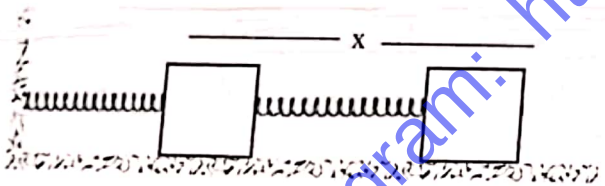
$$h = 2m$$

Elastic potential Energy: Is the potential energy stored by a body such as spring when it is stretched or compressed.

The elastic potential energy of a spring of force constant K stretched or compressed by an amount x from its rest position is obtained by:

Elastic potential energy $= \frac{1}{2}$ (Force constant) (deformation)^2

$$EP_E = \frac{1}{2}Kx^2$$



Where: x is the extension or compression of the material and we call it **deformation**

Illustrative Examples

82. What is the potential energy of a spring of force constant 500N/m when it is stretched 1cm?

Solution

$$EP_E = \frac{1}{2}Kx^2 = \frac{1}{2}(500)(0.01)^2$$

$$EP_E = 0.025J$$

83. What is the deformation of a spring of force constant 400N/m if it stored a potential energy of 8J?

Solution

$$EP_E = \frac{1}{2}Kx^2$$

$$x = \sqrt{\frac{2EP_E}{K}} = \sqrt{\frac{2(8)}{400}}$$

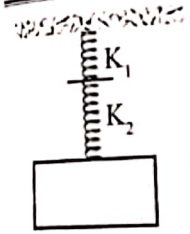
$$x = 0.2m$$

84. What potential energy is stored by a system of spring of force constants 300N/m and 600N/m when they ane stretchted by 10mm while they are connected in

A) Series

B) Paraiiei

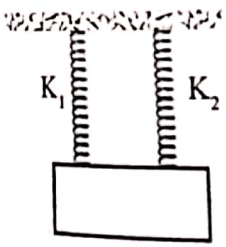
Solution



A) $EP_E = \frac{1}{2}Kx^2 = \frac{1}{2}(200)(0.01)^2$

$= 0.01J$

$$K = \frac{K_1 K_2}{K_1 + K_2} = \frac{(300)(600)}{300 + 600} = 200N/m$$



B) $EP_E = \frac{1}{2}Kx^2$, $K = K_1 + K_2 = 900N/m$

$$EP_E = \frac{1}{2}(900)(0.01)^2$$

$= 0.045J$

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Conservation of Mechanical Energy

According to the conservation of energy,in the absence of external force such as air resistance and friction, energy is neither created or destroyed but transferred from one form to the other.

Law of conservation of Mechanical Energy states that; "In the absence of dissipative force, the total mechanical energy of a body or a system of bodies remains constant.

That is; Mechanical Energy Initial = Mechanical Energy Final

$$ME_i = ME_f$$

$$KE_i + PE_i = KE_f + PE_f$$

Therefore, the change in mechanical energy is zero. $\Delta ME = 0$

Illustrative Example

85. A 500g ball is dropped from a 20m tall building what is the speed of the ball when it is

A) 10m from the ground

B) At the ground level

Solution

A) $KE_i + PE_i = KE_f + PE_f$

$$0 + mgh_i = \frac{1}{2}mv_f^2 + mgh_f$$

$$0 + (0.5)(10)(20) = \frac{1}{2}(0.5)v_f^2 + (0.5)(10)(10)$$

$$100 = \frac{1}{4}v_f^2 + 50$$

$$v_f^2 = 200$$

$$v_f = \sqrt{200} = 10\sqrt{2}\ m/s$$

B) $K_{Ei} + P_{Ei} = K_{Ef} + P_{Ef}$

$$0 + mgh_i = \frac{1}{2}mv_f^2 + 0$$

$$v_f^2 = 2gh = (2)(10)(20) = 400$$

$$V_f = 20m/s$$

86. A 3 kg block moving at a speed of 10m/s over a horizontal smooth floor hits a spring of force constant 300N/m. What will be maximum compression of the spring?

Solution

$$K_{Ei} + P_{Ei} = K_{Ef} + P_{Ef}$$

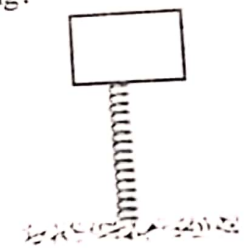
$$\frac{1}{2}mv_i^2 + 0 = 0 + \frac{1}{2}kx^2$$

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$$

$$x = v\sqrt{\frac{m}{k}} = (10)\sqrt{\frac{3}{300}}$$

$$x = 1m$$

87. A 5kg block resting on spring compressed the spring by 5cm. What is the force constant of the spring?



Solution

$$F = Kx = mg$$

$$K = \frac{mg}{x} = \frac{(5)(10)}{0.05} = 100N/m$$

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The Work Energy Theorem

The maximum work done by a force acting on a body is obtained by ;

$W = F\,s$, $F = ma$

$W = mas$

And from the relation, $v^2 = u^2 + 2\,as$

$$as = \frac{v^2 - u^2}{2}$$

$$W = m\left(\frac{v^2 - u^2}{2}\right)$$

$$W = \frac{1}{2}\,m\,V^2 - \frac{1}{2}\,mu^2$$

Where : $1/2\ m\,v^2$ = Final kinetic energy, and

$1/2\ mu^2$ = Initial kinetic energy

$W = k_{Ef} - k_{Ei}$

$\boxed{W = \Delta k_E}$

Therefore, according to the work - energy theorem, the work done by a net force is equal to the change in kinetic energy of the body.

Conceptual Examples

88. Work done by gravity on a body moving up ward is positive. True Or False ?

Explanation

False. $W = F\,s\cos\theta$, $\theta = 180^0$ and $\cos 180^0 = -1$. Therefore, the work done is negative.

89. Work done by a force is always positive, True or false ?

Explanation

False. Work done can be positive, negative or zero depending on the force - displacement orientation.

Mechanical Power

Sometime, work is done quickly and other times, it is done slowly. How fast a work is done is determined using the concept of power.

Power : Is the rate at which work is done or energy is transferred.

In symbols ; $\text{Power} = \frac{\text{Work done}}{\text{Time taken}} = \frac{\text{Energy Transfered}}{\text{Time taken}}$

$\boxed{P = \frac{W}{t}}$

Power is a scalar quantity and its SI unit is **watt (W)**

$1\,W = 1\,J/s$

Moreover, from the defination of work done, power can also be calculated by

$$\text{Power} = \frac{\text{Work}}{\text{time}} = \frac{(\text{Force})\ (\text{displacement})}{\text{time}}$$

$\text{Power} = \text{Force} \times \text{Velocity}$

$\boxed{P = FV}$

Critical Thinking Question

90. A strong person is always powerful. True or false ?

Explanation

False. A person is powerful only when he does a given work in a shorter time compared to others who did the same job.

91. Power is the ability to do work. True or False ?

Explanation

False. Power is the rate at which work is done. And, the ability to do work is called energy.

92. Power is always positive. True or false ?

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Explanation

False. Power can be positive if energy is added per unit time to the body and negative if energy is taken away from a body per unit time.

Illustrative Examples

93. A lifter picked up a 400 kg block to a height of 1.5 min 5 seconds. What is the power dissipated by him ?

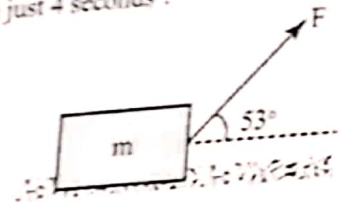
Solution

$$P = \frac{W}{t} = \frac{mgh}{t}$$

$$P = \frac{(400)(10)(1.5)}{5}$$

$$P = 120\,W$$

94. What is power developed by a force of 100N while displacing a box for 10m in just 4 seconds ?



Solution

$$P = \frac{W}{t} = \frac{Fs\cos\theta}{t}$$

$$P = \frac{(100)(10)\cos(53^\circ)}{4}$$

$$P = 150\ W$$

95. An engine develops 10 k w power. How much time will it take to lift a mass of 200 kg to a height of 40m ?

A) 4 second C) 5 second

B) 8 second D) 10 second

Solution

$$P \quad \frac{mgh}{t} \Rightarrow t = \frac{mgh}{P}$$

$$t = \frac{(200)(10)(40)}{(10)(1000)} = 8\,\text{second}$$

Answer : C

96. A 4 kg mass is moving on a smooth table with a speed of 20 m/s, and strikes a spring of force constant 100 N/m. What is the maximum compression of the spring ?

A) 4m B) 4cm C) 4mm D) 0.4m

Solution

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$$

$$X = \sqrt{\frac{mu^2}{k}} = \sqrt{\frac{4(20)^2}{100}}$$

$$X = 4m$$

Answer : A

97. A body of mass 2 kg is thrown up ward with a kinetic energy of 490 J. At What height will its kinetic energy become halved ?

A) 50 m B) 25 m C) 12.25 m D) 10.25m

Solution

$$490 = 245 + mgh$$

$$mgh = 245$$

$$h = \frac{245}{(mg)} = \frac{245}{2(10)} = 12.25m$$

Answer : C

98. A car of mass 1500 kg is travelling at a uniform velocity of 30m/s due east - The driver applies the brakes to slow down the car to 10m/s. What is the work done in slowing down the car ?

A) 15kJ B) 600 k J C) 300 k J D) 750k J

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Solution

$W = \Delta k_E = 1/2m\,[v^2 - u^2]$

$W = 1/2\,(1500)\,[30^2 - 10^2] = 600{,}000$ J

$W = 600$ k J

Answer : B

99. A bullet of mass 50 gram travelling at 80 m/s hits a sand bag and moves a distance of 10 cm before it stops. What force is exerted on the bullet by the sand bag ?

A) 600 N B) 800 N C) 1600 N D) 1800 N

Solution

$$W = \Delta kE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$-fs = -\frac{1}{2}mu^2$$

$$f = \frac{mu^2}{25} = \frac{(0.05)\,(80)^2}{2(0.1)}$$

$f = 1600$N

Answer : C

100. A 1.75m tall worker lifts a 50kg cement bag by himself and goes 1km carrying the bag along the horizontal. What is the work done by the man ?

A) 434.5 J B) 437.25 J C) 487.5 J D) 495.25 J

Solution

$W = Fs\cos\theta = (50)\,(10)\,(1.75) = 437.25$ J

Answer : B

101. The work done by a force is defined by $W = \vec{F} \cdot \vec{S}$. If $\vec{F}$ and $\vec{S}$ are not zero but work done is zero. We calculate that.

A) $\vec{F} > \vec{S}$

B) $\vec{F}$ and $\vec{S}$ are in the same direction

C) $\vec{F}$ and $\vec{S}$ in opposite direction

D) $\vec{F}$ and $\vec{S}$ are at right angle

Hint : $\vec{F} \cdot \vec{S} = Fs\cos\theta = 0$ if $\theta = 90^0$

Answer : D

102. A force $\vec{F} = (2, 2)$ N displaces a particle through $\vec{S} = (2, 0, 2)$ m in 16 second. The power developed is ?

A) 0.25 W B) 25 W C) 225 W D) 450 W

Solution

$$P = \frac{W}{t} = \frac{\vec{F}.\vec{S}}{t} = \frac{(2,2)\,(2,0,2}{16}$$

$P = 0.25$W

Answer : A

103. At a certain instant , a particle experiences of force $\vec{F}$ (4, -2, 9) N while having a velocity of $\vec{V}$ (- 2, 0, 4) m/s. What is the instantaneous rate at which the force does work on the particle ?

A) 28 W B) 54 W C) 14 W D) 36 W

Solution

$P = \vec{F} \cdot \vec{V} = (4,-2,9)\,(-2,0,4)$

$P = (4)\,(-2) + (-2)\,(0) + (9)\,(4)$

$P = 28$W

Answer : A

104. A force of $\vec{F} = (3, -5)$ N acts on a body and displaced it with $\vec{S} = (6, -1)$ m. The work done is ?

A) 23 J B) 18 J C) 5 J D) 13 J

Solution

$W = \vec{F} \cdot \vec{S} = (3,-5)\,(6,-1)$

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$W = (3)(6) + (-5)(-1)$

$W = 23\ J$

Answer : A

105. A 30 gram bullet initially travelling at 50m/s penetrates 10cm into a wooden block. The average force it exerts is ?

A) 4000N B) 2500N C) 375 N D) 4500 N

Solution

$$W = \Delta k_E \Rightarrow fs = \frac{1}{2} m (v^2 - u^2)$$

$$f = \frac{mv^2}{25} = \frac{(0.03)(2500)}{2(0.1)}$$

$f = 375\,N$

Answer : C

106. A body at rest may have

A) Energy C) Momentum

B) Speed D) Velocity

Hint : potential energy is stored energy.

Answer : A

107. If momentum of a body is numerically equal to its kinetic energy, then speed of the body is ?

A) 1m/s B) 2m/s C) 4m/s D) 8m/s

Solution

$P = k_E$

$m v = 1/2\ mv^2$

$V = 2m/s^2$

Answer : B

108. A boy pushes a cart with a force of $(150i - 40j)N$ and displace the cart to $(14i + 11j)m$ How much work does the boy do on the cart?

Solution

$$W = \vec{F}.\vec{S} = F_x S_x + F_y S_y$$

$W = (150)(14) + (-40)(11)$

$W = 1660J$

109. A boy mass 50kg and a girl of mass 40kg each take 20 seconds to run to the 3rd floor which is 10m from the 1st floor who is powerful?

Solution

$$P_{by} = \frac{mgh}{t} = \frac{(50)(10)(10)}{20} = 250W$$

$$P_{girl} = \frac{mgh}{t} = \frac{(40)(10)(10)}{20} = 200W$$

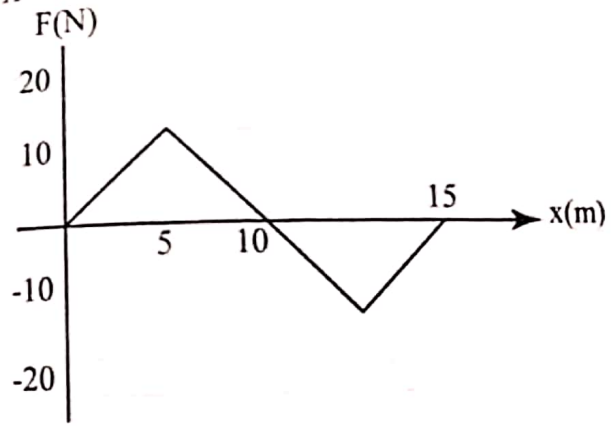
The boy!

110. The force acting on a body varies with position as shown. What is the work done by the force

A) From X = 0 to X = 10m

B) From X = 10m to X = 15m

C) From X = 0 to X = 15m



Solution

A) $W = \frac{1}{2} Fs = \frac{1}{2}(20)(10) = 100J$

B) $W = \frac{1}{2} Fs = \frac{1}{2}(-10)(5) = -25J$

C) $W = 100 - 25 = 75J$

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111. A ball of mass 1kg is released from a builds of 18m high. At what height above the ground will the kinetic energy of the ball became twice its potential energy?

Solution

$$KE_i + PE_i = KE_f + PE_f$$

$$0 + mgh_i = \frac{1}{2}mv_f^2 + mgh_f$$

$$mgh_i = 2mg_f + mgh_f$$

$$h_i = 3h_f$$

$$h_f = \frac{h_i}{3} = \frac{18}{3}$$

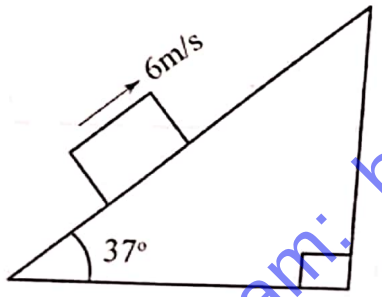
$$h = 6m$$

112. A 5kg block is set to move up an inclined plane of inclination 37° with an initial speed of 6m/s

How far up along the plane will the block move before coming to rest

A) when there is no friction

B) If $\mu_k = 0.3$



Solution

A) $\frac{1}{2}mv^2 = mgh = mg(S\sin\theta)$

$$S = \frac{V^2}{2g\sin\theta} = \frac{(6)^2}{2(10)(0.6)}$$

$$S = 3m$$

B) $\frac{1}{2}mv^2 = mgh + mg(\sin\theta)$

$$\frac{1}{2}v^2 = gs\sin\theta + \mu gs\cos\theta$$

$$S = \frac{V^2}{2g(\sin\theta + \mu\cos\theta)} = \frac{(6)^2}{2(10)[0.6 + (0.3(0.8)]} = 2.1m$$

113. What average power must be supplied to push a 10kg block from rest to 26m/s in 5 sec when the coefficient of friction is 0.2?

Solution

$$F - f = ma, a = \frac{v - u}{t} = \frac{20 - 0}{5} = 4m/s^2$$

$$F = ma + f$$

$$F = (10)(4) + \mu mg = 40 + (0.2)(10)(10)$$

$$F = 60N$$

$$P = Fv\cos\theta\ ,\ \theta = 0^\circ$$

$$P = Fv = (60)(20)$$

$$P = 1200W$$

114. If a 1200kg car accelerate from rest to 20m/s in 5seconds, what is the power developed by its motor?

Solution

$$F = ma, a = \frac{v - u}{t} = \frac{20 - 0}{5} = 4m/s^2$$

$$F = (1200)\ (4) = 4800N$$

$$P = (4800)(20) = 96{,}000W$$

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4.8 What is Center of Mass ?

The average location for the total mass of a system where the system can be balanced is called **center of mass** of the system.

Center of mass : Is a point at which all of the mass of the body or the system is considered to be concentrated.

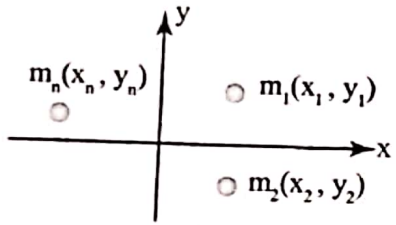
For a system of n particles in the x y plane with masses m_1, m_2 ---- m_n and coordinates $(x_1, y_1), (x_2, y_2)$ ----- (x_n, y_n) from any arbitrary origin, the x and y coordinate of the center of mass is obtained by ;

X - coordinate of the center of mass

$$X_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_n x_n}{m_1 + m_2 + \text{----} + m_n}$$

Y - coordinate of the center of mass

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + \text{--} + m_n y_n}{m_1 + m_2 + \text{---} + m_n}$$



The position of the center of mass of the system is then given by; cm = (x_{cm}, y_{cm})

Critical Thinking Question

115. A project is fired in to the air and suddenly explodes in to fragments. What can be said about the motion of the center of mass of the fragments ?

Explanation

The center of mass of the system will follow parabolic path as the unexploded projectile and the fragments will have the same center of mass as their mother

Illustrative Examples

116. Three masses m_1 = 1kg, m_2 = 500g and m_3 = 2 kg are placed at the x y plane coordinates of (-1, 2), (2,0) and (3, - 2) units respectively . What is the center of mass of the system ?

Solution

cm = (x_{cm}, y_{cm})

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

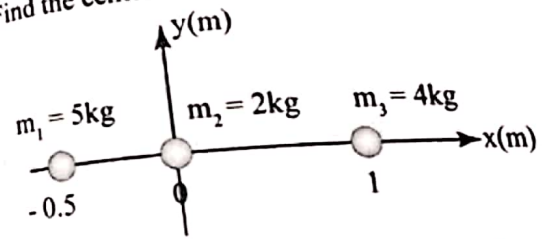
$$x_{cm} = \frac{(1)(-1) + (0.5)(2) + (2)(3)}{1 + 0.5 + 2} = 1.71$$

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{m_1 + m_2 + m_3}$$

$$y_{cm} = \frac{(1)(2) + (0.5)(0) + (2)(-2)}{1 + 0.5 + 2} = -0.57$$

cm = (1.71, - 0.57) units

117. Find the center of mass of the system shown below



Solution

cm = (x_{cm}, y_{cm})

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

$$x_{cm} = \frac{(5)(-0.5) + (2)(0) + (4)(1)}{5 + 2 + 4} = 0.14$$

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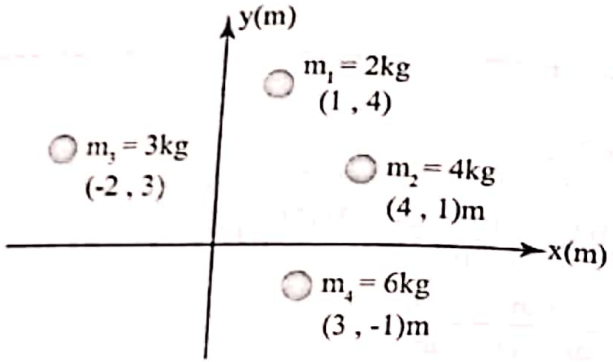
$$y_{cm} = \frac{(5)(0) + (2)(0) + (4)(0)}{5+2+4} = 0$$

$$cm = (0.14, 0)\,m$$

118. Find the position of the center of mass of the system of masses shown below



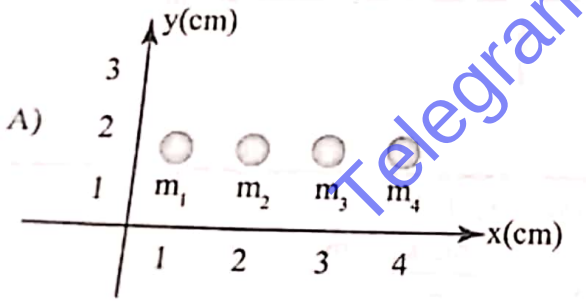
Solution

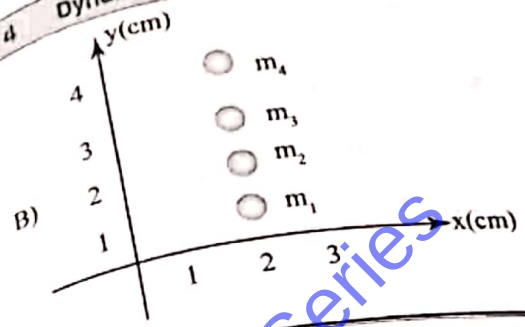
$$x_{cm} = \frac{(2)(1) + (4)(4) + (3)(-2) + (6)(3)}{2+4+3+6} = 2m$$

$$y_{cm} = \frac{(2)(4) + (4)(1) + (3)(3) + (6)(-1)}{2+4+3+6} = 1m$$

$$c.m = (x_{cm}, y_{cm}) = (2,1)\,m$$

119. What is the center of mass of the system of masses of each 2 kg





Solution

A) $x_{cm} = \dfrac{m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4}{m_1 + m_2 + m_3 + m_4}$

$x_{cm} = \dfrac{(2)(1) + (2)(2) + (2)(3) + (2)(4)}{2+2+2+2}$

$x_{cm} = 2.5$ cm

$y_{cm} = \dfrac{(2)(2) + (2)(2) + (2)(2) + (2)(2)}{8}$

$y_{cm} = 2$ cm

$cm = (x_{cm}, y_{cm}) = (2.5, 2)$ cm

B) $x_{cm} = \dfrac{m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4}{m_1 + m_2 + m_3 + m_4}$

$x_{cm} = \dfrac{(2)(2) + (2)(2) + (2)(2) + (2)(2)}{2+2+2+2}$

$x_{cm} = 2$ cm

$y_{cm} = \dfrac{m_1 y_1 + m_2 y_2 + m_3 y_3 + m_4 y_4}{m_1 + m_2 + m_3 + m_4}$

$y_{cm} = \dfrac{(2)(1) + (2)(2) + (2)(3) + (2)(4)}{2+2+2+2}$

$y_{cm} = 2.5$ cm

$cm = (X_{cm}, Y_{cm}) = (2, 2.5)cm$

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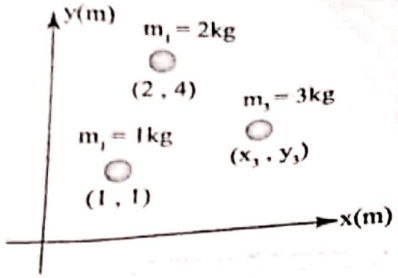
120. For what coordinates of (x3, y3) will the system of mass shown below will have a center of mass (2, 3) cm ?



Solution

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

$$2 = \frac{(1)(1) + (2)(2) + 3(x_3)}{1 + 2 + 3}$$

$$12 = 1 + 4 + 3x_3$$

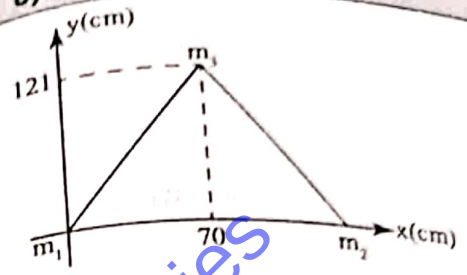
$$x_3 = 2.3\text{cm}$$

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{6}$$

$$3 = \frac{(1)(1) + (2)(4) + 3(y_3)}{6}$$

$$y_3 = 3\text{cm}$$

121. The particles of masses m_1 = 1.2 kg, m_2 = 2.5kg and m_3 = 3.4 kg are located at the corners of an equilateral triangle of side 140 cm. Where is the center of mass of the system ?



Solution

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

$$x_{cm} = \frac{(1.2)(0) + (2.5)(140) + (3.4)(70)}{1.2 + 2.5 + 3.4}$$

$$x_{cm} = 83\text{ cm}$$

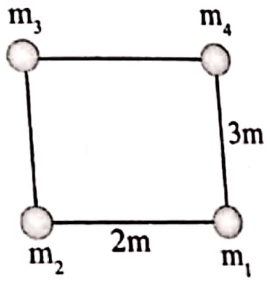
$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{m_1 + m_2 + m_3}$$

$$y_{cm} = \frac{(1.2)(0) + (2.5)(0) + (3.4)(121)}{1.2 + 2.5 + 3.4}$$

$$y_{cm} = 58\text{ cm}$$

$$cm = (x_{cm}, y_{cm}) = (83,58)\text{ cm}$$

122. Locate the center of mass of particles m_1 = 2 kg, m_2 = 4 kg, m_3 = 5 kg and m_4 = 3 kg each on the corners of a 2m × 3m rectangle and m1 is at the origin.



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Solution

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4}{m_1 + m_2 + m_3 + m_4}$$

$$x_{cm} = \frac{(2)(0) + (4)(-2) + (5)(-2) + (3)(0)}{2 + 4 + 5 + 3}$$

$$x_{cm} = -1.28 \text{ m}$$

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3 + m_4 y_4}{m_1 + m_2 + m_3 + m_4}$$

$$y_{cm} = \frac{(2)(0) + (4)(0) + (5)(3) + (3)(3)}{14} = 1.17 \text{ m}$$

cm = (-1.28, 1.71) m

End of unit questions and problems

I. Give Short answers to the following Questions.

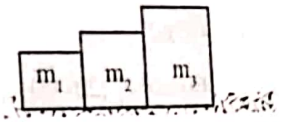
1. The speed of a falling body is proportional to its mass. True or False ?
2. Arrange the following bodies in the order of increasing inertia. Bicycle, Train Automobile, Truck
3. List down the effects of force
4. Why do dust particles removed from a carpet by shaking ?
5. Can the coefficient of friction be zero ?
6. State the law of conservation of liner momentum
7. Why a seat belt is tied in fast moving vehicle ?
8. Why is it recommended to follow a zig - zag path than a straight up while climbing a mountain ?
9. State the first condition of equillibrium
10. What is impulse ? Explain

II. Solve the following problems

11. A man pulls a box (20, - 1)m with a force of (35, 25) N. How much work does the man do on the box ?
12. A force of 20 N is applied on a body at an angle of 60^0 to the direction

of the displacement. If the body displaced by 8m, how much work is done by the force ?

13. Starting from rest, a 5kg block slides 2.5m down a rough 37° inclined plane. The coefficient of kinetic friction between the block and the plan is 0.4 Determine :
    A) Work done by gravity
    B) Work done by friction
14. How much work is required to increase the speed of a 4 kg object from 20m/s to 40m/s
15. A weight lifter lifts 200kg through 1.8m in 2 second. What is the power developed ?
16. A ball of mass 4 kg falls on to a floor with a velocity of (4, - 9) m/s and bounces off with (4, 9) m/s in 0.5 second. What is the
    A) Impulse of the ball
    B) Force exerted by the floor on the ball
17. A ball of mass 0.8 kg moving at 5m/s collides with another ball of mass 1.2kg moving at 4m/s in opposite direction. After collision, the first ball moves with 4m/s. What is the speed of the second ball after collision.
18. A force accelerates a body of mass m at 8m/s . What acceleration will this force give to a body of mass 4m ?
19. What will be the apparent weight of a 60 kg man standing in an elevator that accelerates down ward at $4 m/s^2$?
20. What force is needed to change the speed of a 10 kg body from 10m/s to 30m/s in 0.5 seconds ?
21. A force of 18 N pushes the mass system of m_1 = 11g, m_2 = 2 kg and m_3 = 3 kg along a frictionless surface. What is
    A) The acceleration of the system
    B) The contact force



22. What is the net work done on the box when a 120 N force displaced the box 10m against a 5N friction .

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I. Short Answer and Explanations

1. False . A body in free fall has a speed indepent of its mass Near the surface of the earth, all bodies accelerate at a rate of $9.81 m/s^2$ regadless of their mass
2. The bigger the mass, the larger the inertia of the body. Hence the following bodies are in the increasing order of their inertia.

   Bicycle, Automobile, Truck, train
3. Force starts motion, stops motion, change direction, change size, change shape of a body
4. Because a body at rest, according to newton's first law, remains at rest.
5. No, Friction can not be avoided but minimized Hence if the coefficient of friction is zero, it means friction is zero. Which Is an ideal case.
6. In the absence of eternal force, the initial momentum of a system is equal to its final momentum.
7. It helps passangers to remain intact with their seat in case the vihecle suddenly stops, as a body in motion wants to keep going at constant speed
8. When we go in a zig - zag fashion, we carry only part of our weight But when we go straight up, we carry the whole of our weight .
9. The sum of all forces acting on a body is zero
10. Impulse is the change in momentum of a body

II. Solution to the end of unit problems

11. $W = Fs\cos\theta = \vec{F}, \vec{S} = (20,-1).(35,25)$

    $W = (35)(20) + (-1)(25) \qquad W = 675\,J$

12. $W = Fs\cos\theta = (20).(8)(\cos 60^0 \qquad W = 80\,J$

13. A. $W_g = mgh = mgs\sin\theta = (5)(10)(2.5)\sin 37^0$

    $W_g = 75\,J$

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20. $F \Delta t = mv - mu$

$$F = \frac{mv - mu}{t} = \frac{(10)(30) - (10)(10)}{0.5} \qquad F = 1600\ N$$

21.A. $F = ma \Rightarrow a = \frac{F}{m} = \frac{18}{1+2+3}$

$a = 3 m/s^2$

B. $F - F^1 = m_1 a$

$F^1 = F - m_1 a = 18 - (1)(3)$

$F^1 = 15N$

22. $F - f = ma = F_{net}$

$120 - 5 = F_{net}$

$F_{net} = 115\ N$

$W = F s = (115)(10) \qquad W = 1150\ J$

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UNIT 5

5. Heat Conduction and Calorimetry

5.1 What is Heat?
5.2 Mechanisms of Heat Transfer
5.3 What is Heat Capacity?
5.4 Thermal Expansion
5.5 What is Phase or State of Matter?
5.6 What is Calorimetry?

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Heat Conduction and Calorimetry

Hello there! Welcome to this fashinating unit where you will be able to differentiate between heat and temperature, convert one temperature scale in to the other, describe heat transfer machanisms, explain phase or state of matter, explain calorimetry and apply the law of conservation of energy to solve practical problems, Enjoy!

5.1 What is Heat?

The terms heat and temperature are used interchangeably in the every day language . How ever, in physics, they are queit different.

Heat: Is a form of energy that fiows from a region of high temperature to a region of low temperature

Heat is a scaler quantity and its SI unit is joule *(J)*

Remember, heat exchange between bodies take place only when there is a temperature difference between them.

In the day to day langnage, temperature refers to the degree of hotness or coldness of a body. But, this description is not precise as it depends on individuals sensitivity. So, in physics it is defined as:

Temperature: Is the measure of average kinetic energy of particles of a body.

Therefore, a body having particles with greater kinetic energy is said to be hotter and at a higher temperature region.

- Temperature is a scalar quantity and its SI unit is the kelvin *(K)*. Other units of temperature include degree celsius (°C) and degree fahrenheit (°F)
- To convert one temperature scale to the other, we make use of the lower fixed point also called the ice, the melting or freezing point and the upper fixed point or boiling point of water which has the following values in the three scales.

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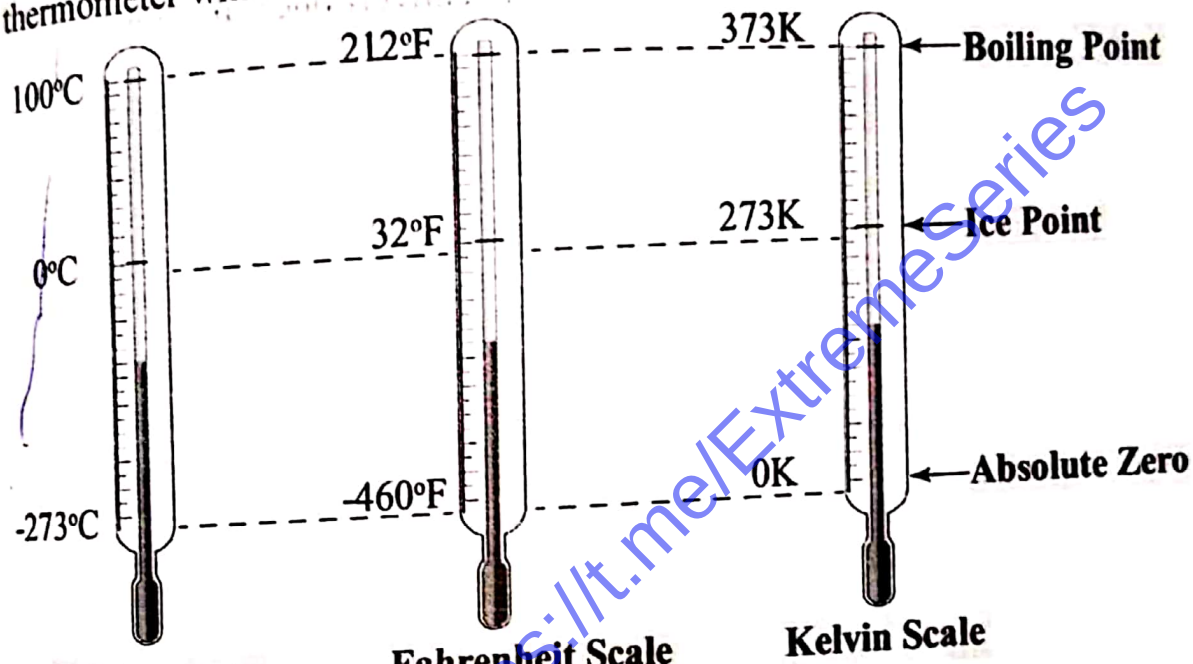
Fixed Point	Temperature Scale		
	Celsius (^{0}c)	Fahreheit (^{0}F)	Keluin (K)
Ice point	0	32	273
Boiling point	100	212	373

Temperature is measured using an instrument called **thermometer**.

Next, we have ice point and the boiling point of water using a mercury thermometer which has marks or labels in the three temperature scales.



➥ Absoilute zero (0k), is the temperature at which random motion of particles of a substance stops.

$$\text{Absolute zero} = 0k = -273^0c = -460^0F$$

Now, let us convert temperature scale X in to other temperature scale Y and vice - versa.

$$T_X \rightleftharpoons T_Y$$

To do so, use the following general relation

$$\frac{T_x - T_x\left(\begin{matrix}\text{ice}\\\text{point}\end{matrix}\right)}{T_x\left(\begin{matrix}\text{boiling}\\\text{point}\end{matrix}\right) - T_x\left(\begin{matrix}\text{ice}\\\text{point}\end{matrix}\right)} = \frac{T_Y - T_Y\left(\begin{matrix}\text{ice}\\\text{point}\end{matrix}\right)}{T_Y\left(\begin{matrix}\text{boiing}\\\text{point}\end{matrix}\right) - T_Y\left(\begin{matrix}\text{ice}\\\text{poin}\end{matrix}\right)}$$

For example, to convert celsius scale in to fahrenheit scale and vice - versa,

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we have:

$$T_C \rightleftharpoons T_F$$

$$\frac{T_C - 0}{100 - 0} = \frac{T_F - 32}{212 - 32}$$

Illustrative Examples

1. Explain the difference between heat and temperature

   Explanation:- Heat is a form of energy that flows from a hot body to a cold body while, temperature is the measure of average kinetic energy of particles of a body.

2. Covert 10^0c in to

   A) Kelvin

   B) Fahrenheit

Solution

A. $\frac{Tc - 0}{100 - 0} = \frac{T_K - 273}{373 - 273}$

$\frac{10 - 0}{100} = \frac{T_K - 273}{100}$

$T_K = 283K$

B. $\frac{Tc - 0}{100 - 0} = \frac{T_F - 32}{212 - 32}$

$\frac{10 - 0}{100} = \frac{T_F - 32}{180}$

$T_F = 50^0F$

3. Gold melts at 1336k. What is this value in

   A) Celsius

   B) Fahrenheit

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Solution

A. $$\frac{Tc - 0}{100 - 0} = \frac{T_K - 273}{373 - 273} = \frac{1336 - 273}{100}$$

$$Tc = 1063^0c$$

B. $$\frac{T_F - 32}{180} = \frac{1336 - 273}{100}$$

$$T_F = 1945.4^0F$$

4. At what temperature will the Celsius and Fahrenheit scales read the same?

Solution

$$\frac{T_c - 0}{100 - 0} = \frac{T_F - 32}{212 - 32}$$

Let $T_C = T_F = T$

$$\frac{T}{100} = \frac{T - 32}{180}$$

$$18T = 10T - 320$$

$$8T = -320$$

$$T = -40^0$$

Which means, at -40^0, the Celsius the Fahrenheit scales read the same.

5. If the temperature of a body changes from 40^0c to 60^0c, What is the change in temperature in the

A) Kelvin scale

B) Fahrenheit scale

Solution

A) First change the given temperature in to kelvin

$$\frac{40 - 0}{100 - 0} = \frac{T_{K1} - 327}{373 - 273}$$

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$$T_{K1} = 313K$$

$$\frac{60 - 0}{100 - 0} = \frac{T_{K2} - 273}{373 - 273}$$

$$T_{K2} = 333K$$

$$\Delta T_K = T_{K2} - T_{K1} = 333 - 313$$

$$\Delta T_K = 20K$$

B) First change the given temperature Fahrenheit

$$\frac{40 - 0}{100 - 0} = \frac{T_{F1} - 32}{212 - 32}$$

$$T_{F1} = 104^0F$$

$$\frac{60 - 0}{100 - 0} = \frac{T_{F2} - 32}{212 - 32}$$

$$T_{F2} = 140^0F$$

$$\Delta T_F = T_{F2} - T_{F1} = 140 - 104$$

$$\Delta T_F = 36^0F$$

5.2 Mechanisms of Heat Transfer

Heat flows from a region of high temperature to a region of low temperature in either of one of the following ways or mechanisms. Conduction, convition or radiation.

Conduction: Is the transfer of heat from a hotter body to a colder body due to vibration and collision of molecules of the interacting bodies

Conduction takes place in solids and no actual flow or movement of molecules during heat exchange.

For example, the heat we feel from a tea cup reached our hand via conduction.

➥ The rate at which heat flows from a hoter body to a colder body is obtained by

$$\frac{Q}{t} = KA \frac{\Delta T}{\ell}$$

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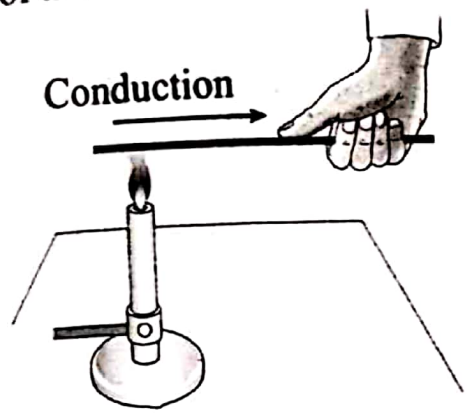
Where: K = Conduction constant

A = cross sectional area

ℓ = Length of the rod



Illustrative Example

6. What is the rate of heat flow through a glass of window with cross - section area $1m^2$, thickness 4mm and thermal conductivity of 0.84 W/mk where the inside temperature is 30^0c and that of the out side is - 20^0c?

Solution

$$\frac{Q}{t} = KA\frac{\Delta T}{\ell} = \frac{(0.84)(1)(30-(20))}{0.004}$$

$$\frac{Q}{t} = 10500W$$

$$\frac{Q}{t} = 10.5KW$$

Convection: Is the transfer of heat from a hotter body due to the actual movement or flow of molecules of the interacting bodies

Convection takes places in liquids and gases with actual movement of particles carrying energy.

For example, when a water in a vessel is heated we see the motion of water molecules with hight kinetic energy.

➥ The rate of heat flow by convection is obtained by

$$\boxed{\frac{Q}{t} = hA\ \Delta T}$$

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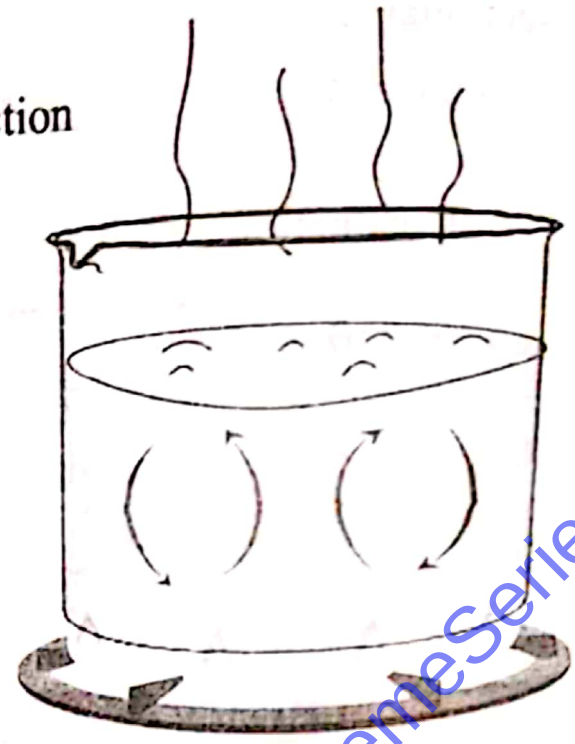
Where: h = convection constant

Convection



Illustrative Example

7. A heater with area of $25cm^2$, convection constant of 105 w/m^2K and temperature of 180^0c is immersed in to a cold water of temperature 20^0c. What is the rate of heat flow in the system?

Solution

$$\frac{Q}{t} = hA\Delta T = (105)(25 \times 10^{-4})(180 - 20)$$

$$\frac{Q}{t} = 42\ J/s = 42\ \text{Watt}$$

Radiation: Is the transfer of heat from a hotter body to a colder body via an electromagnetic wave

Radiation takes place in vacuum with out the need of medium to transport thermal energy

For example, the energy from the sun reaches the earth through radiation.

➥ The rate at which heat flows during radiation is,

$$\boxed{\frac{Q}{t} = \delta e A T^4}$$

Where: $\delta = 5.6 \times 10^{-8}\ N/m^2\ K^4$ is a constant

e = emissivity of the material

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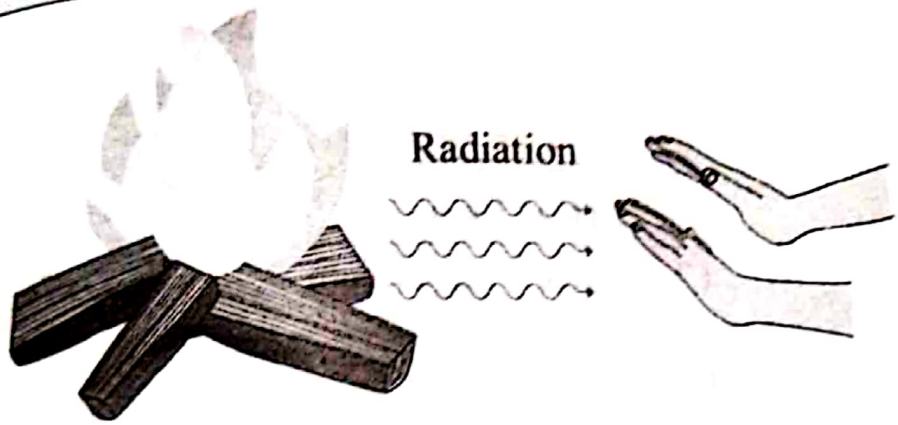

Illustrative Example

8. By what factor will the rate of radiation change when the absolute temperature of a source is doubled?

Solution

$P_1 = \delta\ eA\ T_1^4$, and

$P_2 = \delta\ eAT_2^4$, But, $T_2 = 2T_1$

$P_2 = \delta\ eA\ (2T_1)^4 = 16\ (\delta\ eA\ T_1^4)$

$P_2 = 16\ T_1$

9. A body has a heat transfer coefficient of $24\ \frac{J}{mk}$. If the temperature of the body is 40^0c and its area is $1.2m^2$. What will be its rate of heat lose if the surrounding has a temperature of 35^0c?

Solution

$\frac{Q}{t} = KA\ (T_1 - T_2) = (25)\ (1.2)\ (40 - 35)$

$\frac{Q}{t} = 144W$

5.3 What is Heat Capacity?

Different substance require different amount of thermal energy to change their temperature by one unit. And, this property depends on the heat capacity of the substance.

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Heat Capacity: Is the amount of heat energy required to change temperature of a body by 1^0c or 1K

In symbols:

$$\text{Heat capacity} = \frac{\text{Heat Flow}}{\text{Change in Temperature}}$$

$$\boxed{C = \frac{Q}{\Delta T}}$$

The unit of heat capacity is J/k or J/^{0}c

Specific Heat Capacity: Is the amount of heat energy required to change the temperature of 1kg of body by 1K or 1^0c

In symbols:

$$\text{Specific Heat capacity} = \frac{\text{Heat Flow}}{(\text{Mass})(\text{Change in Temperature})}$$

$$\boxed{C = \frac{Q}{m\Delta T}}$$

The unit of specific heat capacity is J/kg.k or J/kg^{0}c

Therefore, from the above relation, the amount of heat flow in to or out of a body is obtained by;

$$\boxed{Q = mC\Delta T}$$

The specific heat capacity of some substances is given in the following table

Substance	Specific heat capacity (J/kg.^{0}c)
Aluminuim	900
Brass	380
Iron	470
Copper	420
Lead	128
Glass	670
Meracury	140

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Ice	2090
Water	4200
Steam	2010
Gold	130

- Water has the largest specific heat capacity and this property makes it to be used as a cooling agent in many machineries.

Illustrative Explanation

10. How many joules of energy is required to raise the temperature of 100g of gold from 20^0c to 100^0c?

Solution

$Q = mc\Delta T = (0.1)\,(130)\,(100 - 20)$

$Q = 1040J$

11. A 0.5kg block of iron at a temperature of 27^0c is supplied 1600J of heat. What will be its final temperature?

Solution

$$Q = mc\Delta T \Rightarrow \Delta T = \frac{Q}{mc} = \frac{1600}{(0.5)(470)}$$

$\Delta T = 6.8$

$\Delta T = T_f - T_i \Rightarrow T_F = \Delta T + T_i$

$T_f = 27 + 6.8 = 33.8^0$c

12. When 8000J of heat is supplied to 0.5Kg body, its temperature increases from 15^0c to 55^0c. What is the specific heat capacity the body?

Solution

$Q = mc\Delta T$

$$c = \frac{Q}{m\Delta T} = \frac{8000}{(0.5)(55 - 15)}$$

$c = 400J/kg.^0c$

13. How many grams of iron at 20^0c must be heated to 100^0c so that it releases 1800 calories of heat?

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Solution

$$Q = mc\Delta T$$

$$m = \frac{Q}{c\Delta T}$$

But, 1cal = 4.2J

Hence, 1800cal = 7560J

$$\Rightarrow m = \frac{7560}{(470)(100 - 20)}$$

$$m = 0.2kg = 200g$$

14. A 70kg woman eat a chocolate having 1000cal. To what height should she run up stair to burn the chocolate in her body?

Solution

Heat energy = potential energy

$$mgh = Q$$

$$h = \frac{Q}{mg} = \frac{(1000)(4.2)}{(70)(10)}$$

$$h = 6m$$

Note: 1cal = 4.2J

5.4 What is Thermal Expansion ?

When a substance is heated, its molecules gain more kinetic energy and move further apart, as a result the body increases in size - Expand.

Thermal Expansion: Is the increase in size of a body as it is heated.

A good demonstration of expansion of materials is the ball and ring experiment, where the metal ball easily passes through the ring before heated but after heating, the ball expands and it is no longer able to pass through the ring.

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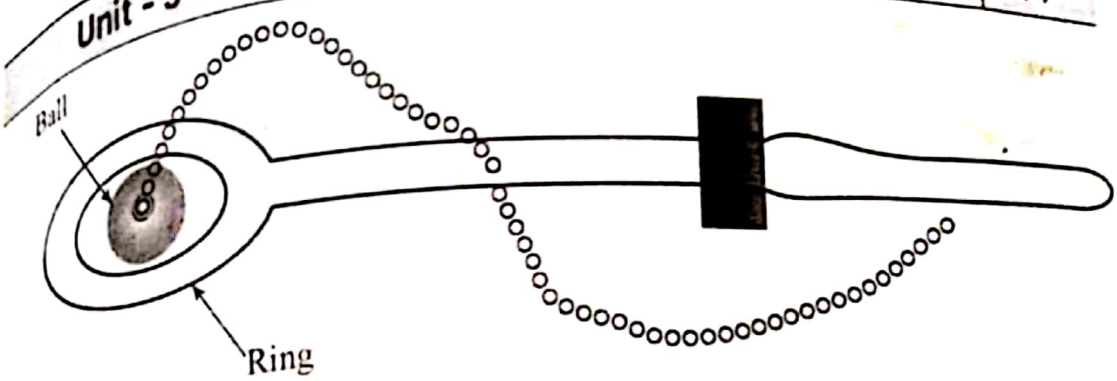


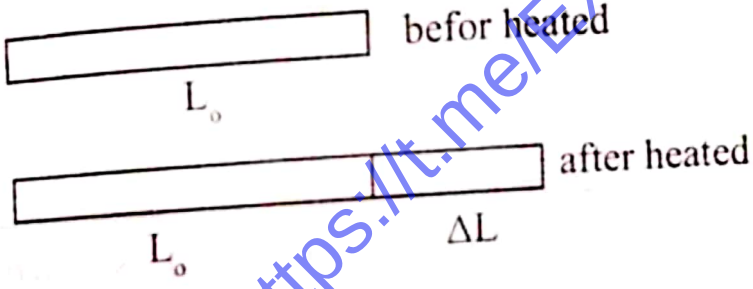
When a solid is heated, it may show linear expansion, area expansion or volume expansion depending on the dimension of the body exposed to heat.

Linear Expansion

When a metal rod or a wire is heated, it under goes one dimensional expansion

Liner Expansion: Is the increase in length of a body due to heating .

The increase in length of a metal rod of initial length L_o after heating is obtained by;



$$\Delta L = L_o \alpha \Delta T$$

Where ΔL = change in length

α = coefficient of linear expansion

ΔT = Change in temperature

Illustrative Examples

15. A steel measuring tape is 8m long at 20^0c. What will be its length at a temperature of 45^0c ($\acute{\alpha}$ steel = $1.2 \times 10^{-5}/^0$c)

Solution

$\Delta L = L_o \acute{\alpha} \Delta T$

$= (8)(1.2 \times 10^{-5})(45 - 20)$

$\Delta L = 0.0024$m

=

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16. A silver rod is 100cm long at 0°c and 100.19cm long at 100°c. What is the coefficient of linear expansion of silver?

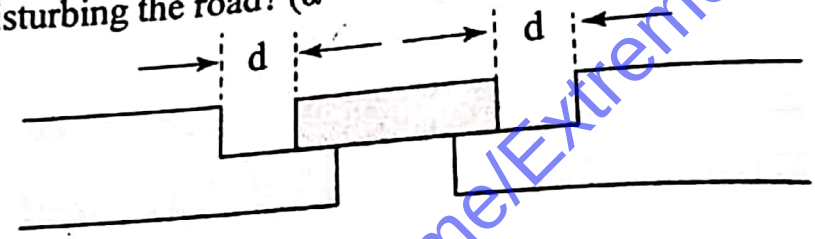
Solution

$$\Delta L = L_o \alpha \Delta T$$

$$\alpha = \frac{\Delta L}{L_o \Delta T} = \frac{L - L_o}{L_o \Delta T} = \frac{0.0019}{(1)(100)}$$

$$\alpha = 1.9 \times 10^{-5}/^0c$$

17. A long steel bridge at 5°c is constructed with some gap at its ends. If the range of temperature in the surrounding is from 5°c to 30°c, What must be the gap d at one end for the bridge to expand and contruct without disturbing the road? ($\acute{\alpha} = 1.2 \times 10\text{-}5/^0c$)



Solution

$$\Delta L = L_o \alpha \Delta T \quad (10)\,(1.2 \times 10 - 5)\,(30 - 5) = 0.003$$

$$\Delta L = 0.3cm$$

$$d = \frac{\Delta L}{2} = 0.15cm$$

Area Expansion

When a metal plate is heated, it under goes two dimensional expansion.

Area Expansion: Is the increase in area of a metal due to heating.

The increase in area of a metal plate of initial area A_o after heating is obtained by,

Before heated **After heated**

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$$\Delta A = A_o \beta \Delta T$$

Where : ΔA = change in Area

$\beta = 2\alpha$, coefficient of area expansion

ΔT = Change in termperature

Illustrative Examples

18. A 6m long square sheet of iron is heated from 20°c to 80°c. What is the change in area of the sheet ($\acute{\alpha}$ iron = 1.1×10^{-5}/°c)

Solution

$\Delta A = A_o B \Delta T$, $B = 2\acute{\alpha}$, $A_o = 6 \times 6 = 36m^2$

$\Delta A = (36)\ (2.2 \times 10^{-5})\ (80 - 60)$

$\Delta A = 0.05m^2$

19. A sheet of copper has an area of $500cm^2$ at 0°c, What will be the area of the sheet at 80°c ($\acute{\alpha} = 1.7 \times 10^{-5}$/0c)

Solution

$\Delta A = A_o \beta \Delta T$,

$\Delta A = (500)\ (2 \times 1.7 \times 10^{-5})\ (80 - 0)$

$\Delta A = 1.3cm^2$

And, the area of the plate at 80°c is;

$A = Ao + \Delta A = 500 + 1.3$

$A = 501.3cm^2$

Volume Expansion

When a metal block is heated, it under goes three dimensional expansion

Volume Expansion: Is the increase in volume of a body due to heating.

The increase in volume of a metal block of initial volume Vo, after heating is obtained by:

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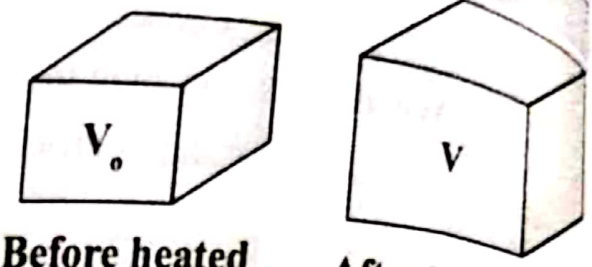
$\Delta v = V_o \delta \Delta T$



Before heated After heated

Where: Δv = Change in volume

$\delta = 3\acute{a}$, coefficient of volume expansion

ΔT = change in temperature

Illustrative Examples

20. A cubical block has a side length of 20cm at 20^0c What will be the change in volume of block if it is heated to 100^0c? ($\acute{a} = 1.9 \times 10^{-5}/^0c$)

Solution

$\Delta V = V_o \delta \Delta T$, $\delta = 3\acute{a}$, $V_o = \ell^3 = (20)^3 = 8000cm^3$

$\Delta V = (8000)(5.7 \times 10^{-5})(80)$

$\Delta V = 36.48cm^3$

21. A glass beaker with a volume of $20cm^3$ is filled with mercury at 20^0c. How much mercury will over flow when the temperature increases to 100^0c? $\delta_{mercury} = 1.8 \times 10^{-5}/^0c$ $\acute{a}_{glass} = 3 \times 10^{-6}/^0c$

Solution

ΔV = over flow = ΔV mercury - ΔV glass

ΔV mercury = $V_o \ell \Delta T = (20)(1.8 \times 10^{-5})(100 - 20) = 0.0288$

ΔV glass = $V_o \ell \Delta T = (20)(9 \times 10^{-6})(100 - 20) = 0.0144$

Over flow = $0.0288 - 0.0144 = 0.0144cm^3$

Coefficient of Expansion of Some Materials is Given Below

Material	Coefficient ($1/^0c$)
Aluminium	2.4×10^{-5}
Brass	1.9×10^{-5}
Copper	1.7×10^{-5}
Glass	0.9×10^{-5}
Lead	2.9×10^{-5}

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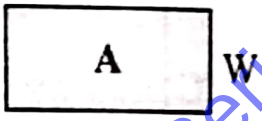
Iron	1.1×10^{-5}
Concrete	1.2×10^{-5}
Alcohol	1.1×10^{-4}
Mercury	1.8×10^{-4}
Air	3.7×10^{-4}

22. How is the coefficient of area expansion twice that of the linear expansion?

Final area of a metal plate after headed can be oblained by



$A = LW$

$A = (L_o + L_o \acute{\alpha}\Delta T)(W_o + W_o \acute{\alpha}\Delta T)$

$A = L_o W_o + L_o W_o \acute{\alpha}\Delta T + W_o L_o \acute{\alpha}\Delta T + L_o W_o \acute{\alpha}^2 \Delta T^2$

Since $\acute{\alpha}$ is small, $\acute{\alpha}^2 = 0$

$A = L_o W_o + 2LoWo\acute{\alpha}\Delta T$, $L_o W_o = A_o$

$A = A_0 [1 + 2\acute{\alpha}\Delta T] = A_o [1 + \beta\Delta T]$

$\Rightarrow \beta = 2\acute{\alpha}$

5.5 What is Phase or State of Matter?

Most substance exist in solid, liquid and gass states, water for example exists as ice in solid state, as water in liquid state, and as vapour in a gas state.

Phase: Is the state of matter of a body as a solid, liquid or gas form at a given temperature and pressure

At a given phase or state of matter, a body has unique physical properties such as density, refractive index and specific heat capacity. Therefore, the liquid water has different density and specific heat capacity from that of the ice.

As a body at a given state absorbs or releases heat, it changes its phase

➥ The transition from one phase to another is called phase change or phase transition, and the heat energy responsible for a change in phase is called **latent heat** or hidden heat.

The latent heat, that is required to change a substance from one state to another at constant temperature is obtained by

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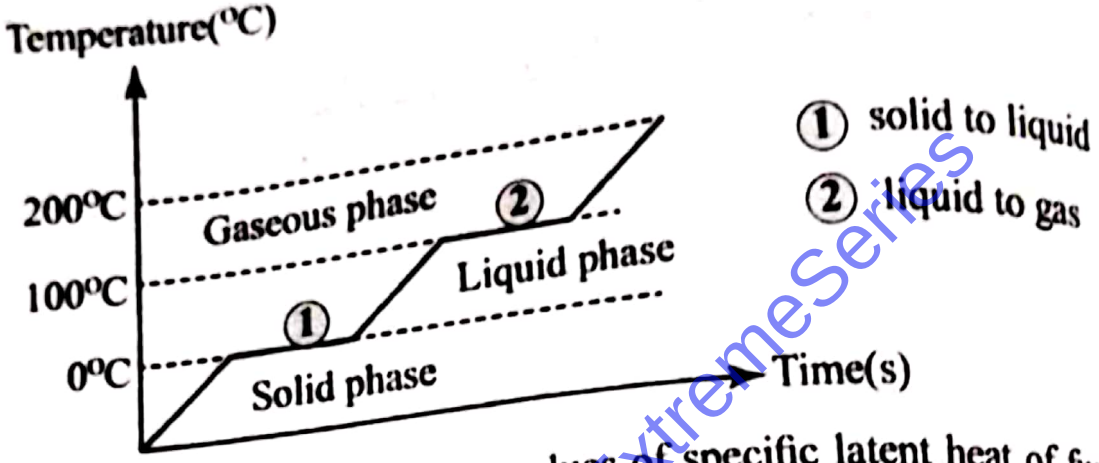
$Q = mL$

Where: m = mass of the body
L = specific latent heat

The heating curve is one way of illustrating the change in phase of a substance as temperature increases with time as shown below:



Different substance have different values of specific latent heat of fusion and specific latent heat of vaponisation as shown in the following table

Substance	Specific latent heat of fusion (J/kg)	Specific latent heat of vaporisation (J/kg)
Oxygen	1.38×10^4	2.13×10^5
Alcohol	1.04×10^5	8.54×10^5
Water	3.33×10^5	2.26×10^6
Lead	2.45×10^4	8.71×10^5
Copper	2.09×10^5	4.73×10^6
Aluminium	3.90×10^5	10.90×10^6
Iron	2.45×10^5	6.08×10^6
Mercury	1.18×10^4	2.7×10^4
Silver	8.83×10^4	2.34×10^6
Gold	6.37×10^4	1.65×10^6

There are two main types of changes of state; liquid to gas and solid to liquid. So we have latent heat of vaporisation and latent heat of fusion.

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Latent Heat of Fusion: Is the amount of heat energy required to change a solid in to a liquid at constant temperature.

Latent Heat of Vaporisation: Is the amount of heat energy required to change a liquid in to a gas at constant temperature.

Moreover, the specific latent heats of a substance are defined as follows:

Specific Latent Heat (L): Is the quantity of heat energy required to change 1kg of a substance from one state to another at constant temperature

Specific Latent Heat of Fusion (L_f): Is the quantity of heat energy required to change 1kg of a substance from a solid to a liquid at constant temperature

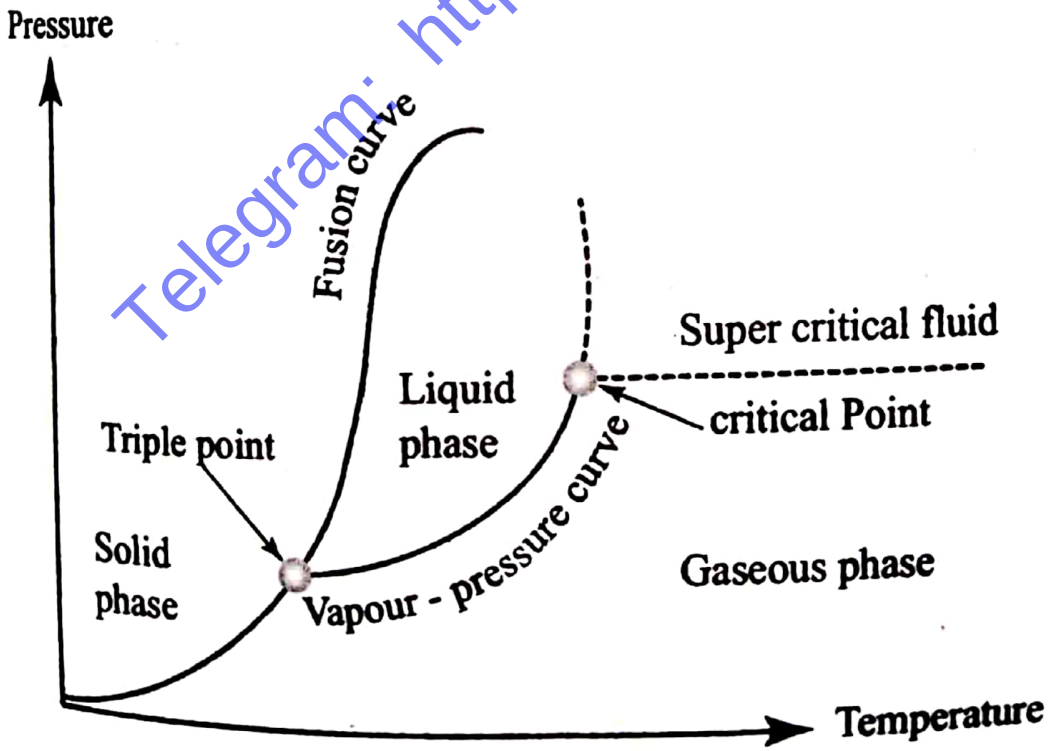
Specific Latent Ueat of Vaporisation (L_v): Is the quantity of heat energy required to change 1kg of a substance from liquid to gas at constant temperature.

There we have the following hidden heats during a phase change

Solid to liquid, $Q_f = mL_f$ --------- Latent Heat of Fusion

Liquid to Gas, $Q_v = mL_v$ -------- Latent Heat of Vaporisation

Furthemore, using the pressure versus temperature axis, phase of a substance can be visualized as Follows;



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Solution

$Q = mLf + mc\Delta T$

$Q = (2 \times 3.34 \times 10^5) + (2 \times 4200 \times 50)$

$Q = 1.09 \times 10^6 J$

26. Calculate the heat energy required to increase the temperature of 400g of water from 30^0c to 130^0c?

Solution

$Q = Q_1 + Q_v + Q_2$

$Q = mc_w\Delta T_1 + mL_v + mc_s\Delta T_2$

$Q = (0.4)(4200)(70) + (0.4)(2.5 \times 10^6) + (0.4)(2080)(30)$

$Q = 117600 + 1000000 + 24960$

$Q = 1.1 \times 10^6 J = 1.1MJ$

27. How many joules of heat is needed to change 1gram of ice from -25^0c to 125^0c?

Solution

$Q = Q_1 + Q_f + Q_2 + Qv + Q3$

$Q = mc_i\Delta T_1 + mL_f + mc_w\Delta T_2 + mLv + mC_s\Delta T_3$

$Q = (10^{-3})(2090)(25) + (10^{-3})(3.34 \times 10^5) + (10^{-3})(4200)(100) +$

$(10^{-3})(2010)(25)$

$Q = 856.5J$

28. How much heat does a refrigerator have to remove from 1.5kg water at 20^0c to make ice at -120^0c

Solution

$Q = Q_1 + Q_v + Q_2$

$Q = mc_w\Delta T_1 + mL_f + mc_i\Delta T_2$

$Q = (1.5)(4200)(20) + (1.5)(3.34 \times 10^5) + (1.5)(2090)(12)$

$Q = 126000 + 501000 + 37800$

$Q = 664800J$

29. 9750J of heat is needed to mclt 25 gram of piece of material. What is the material

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Solution

$$Q = mL_f \Rightarrow L_f = \frac{Q}{m} = \frac{9750}{0.025} = 390{,}000$$

$L_f = 390{,}000 J/kg$, *it is Aluminium*

5.6 What is Calorimetry?

To determiner the specific heat capacity of a substance, we use the concep of calorimetry.

Calorimetry: Is a method of determining the specific heat capacity of a substance using the concept of heat exchange.

Simple steps that we need to follow so as to determine the specific hea capacity of a body;

First, Heat the body of mass M_x you want to know its specific heat capacity to temperature, T_x

Second, Immerse the body in water of mass M_w and temperature of T_w

Third, determiner the final equilibrium temperature of the mass - water system, T_f

Finally, use the conservation of energy, and equate the heat lost by the hotter body, to the heat gained by the colder body.

$$Q_{gained} = Q_{lost}$$

Assuming, $T_x > T_W$

$$Q_{x(lost)} = Q_{W(gaind)}$$

$$M_x C_x(T_x - T_f) = M_w C_w (T_f - T_w)$$

From this, C_x is determined,

Illustrative Examples

30. A 50gram metal at 95°c is dropped in to 250gram of water at 17°c and warms it to 19.4°c. What is the specific heat of the metal?

Solution

Q_{lost} by metal = Q_{gaind} by water

$$m_m C_m \Delta Tm = m_w C_w \Delta T_w$$

$$(0.05)\, Cm\, (35 - 19.4) = (0.25)\, (4200)\, (19.4 - 17)$$

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$C_{metal} = 663 J/kg.K$

31. What mass of steam at 100^0c must be added to 1kg of ice at 0^0c to yield water at 20^0c?

Solution

$Q_{Lost} = Q_{gained}$

$m_s L_v + m_s C_w \Delta T_s = m_2 L_f + m_i c_w \Delta T_w$

$m_s (2.5 \times 10^6) + m_s (4200)(100 - 20) = 1(3.34 \times 10^5) + 1(4200)(20 - 0)$

$$m_{steam} = \frac{418000}{2836000} = 0.147Kg$$

Mass of steam needed = 147gram

32. If the amount of heat that raises the temperature of 1kg of a certain metal by 10^0c raises the temperature of 1kg of water by 2^0c. What is the specific heat capacity of the metal?

Solution

$Q_{Lost} = Q_{gained}$

$mC_{metal}\Delta T_{metal} = mC_{water}\Delta T_{water}$

$$m_{steam} = \left(\frac{\Delta T_{water}}{\Delta T_{metal}}\right)\left(C_{water}\right) = \left(\frac{2}{10}\right)4200$$

$C_{metal} = 840 J/kg.k$

Illustrative Examples

33. All of the following are correct EXCEPT

A) Water molecules are fart at 4^0c than are at 0^0c

B) Water molecules are far apart at 20^0s than they are at 4^0c

C) Water molecules expand when cooled from 4^0c 2^0c

D) Water molecules contract when cooled from 0^0c to 4^0c

Hint: As the strange behaviour of water molecules, they are denser at 40c, to 0^0c and they contract as heated from 0^0c to 4^0c

Answer: A

34. Latent heat of vaporization is the amount of heat required to change

A) Temperature of 1Kg body by 1K

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B) Temperature of a substance as a whole by 1k

C) Phase of a substance from solid to liquid at constant temperature

D) Phase of a substance from liquid to gas at constant temperature

Hint : Latent heat changes phase at constant temperature

Answer: D

35. A 0.3kg hot alloy of specific heat capacity 200j/kg0c cooled by 50°c when added in to water. What amount of heat is lost by in the process?

A) 1500J B) 1000J C) 5000J D) 3000J

Hint: $Q = mc\Delta T = (0.3)(200)(50) = 3000J$

Answer: D

36. A certain mass of ice at 00c is mixed with 0.1kg of water at 60c. How much ice will melt if the final temperature of the system is 00c?

A) 2×10^{-3}kg

B) 7.5×10^{-3}kg

C) 5×10^{-3}kg

D) 1.7×10^{-3}kg

Hint: $Q_{ice} = Q_{water}$

$$M_i L_f = m_w c\Delta T$$

$$M_i = \frac{m_w c\Delta T}{L_f} = \frac{(0.1)(4200)(6-0)}{3.34 \times 10^5}$$

$$M_{ice} = 7.5 \times 10^{-3}\text{kg}$$

Answer: B

37. A metal plate has an area of $0.09m^2$ at 30^0c. What will be the change in area of the plate at 600c ($\acute{\alpha} = 1.1 \times 10^{-5}/0c)^2$

A) $1.00012m^2$

B) $1.00056m^2$

C) $0.000594m^2$

D) $0.0000297m^2$

Hint: $\Delta A = A_o B\Delta T = A_o (2\acute{\alpha})\Delta T$

$\Delta A = (0.09)(2 \times 1.1 \times 10^{-5})(60 - 30)$

$\Delta A = 0.0000594m^2$

Answer: C

38. How much heat is required to change an ice of mass 720 gram at -10^0c to water of temperature 15^0c?

A) 6.12×10^4J

B) $6{,}42 \times 10^4$J

C) 3.0×10^5J

D) 2.7×10^5J

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Solution

$Q = Q_1 + Q_f + Q_2$

$Q = mci\Delta T_1 + mL_f + mc_w\Delta T_2$

$Q = (0.72)\ (2090)\ (0 - (-10)) + (0.72)\ (3.34 \times 10^5) + (0.72)\ (4200)\ (15 - 0)$

$Q = 3 \times 10^5 J$

Answer: C

39. A steel bean is 25m long on winter at 00c. How much does the length change in summer at 300c?($\acute{\alpha} = 1.2 \times 10\text{-}5/0c$)

A) 6mm B) 9mm C) 25mm D) 5mm

Solution

$\Delta L = L_o \acute{\alpha} \Delta T = (25)\ (1.2 \times 10\text{-}5)\ (30 - 0)$

$\Delta L = 9 \times 10^{-3} m = 9mm$

Answer: B

40. Water at 50^0c can be obtained by mixing

A) 1kg of ice at 0^0c and 1kg of water at 100^0c

B) 1kg of water at 0^0c and 1kg of water 100^0c

C) 1kg of ice at 0^0c and 1kg of steam at 100^0c

D) 1kg of water at 0^0c and 1kg of steam at 100^0c

Hint: $Q_{lost} = Q_{gained}$

$mc_w\Delta T_1 = mc_w\Delta T_2$

$(4200)\ (100 - T) = 4200\ (T - 0)$

$T = 50^0c$

Answer: B

41. How much water at 200c should be mixed to 5kg of water at 90^0c to get a mixture of 60^0c?

A) 3.75kg B) 2.25kg C) 5.25kg D) 6.75kg

Hint: $Q_{gained} = Q_{lost}$

$m_{cold}\ C\Delta T_1 = m_{hot}\ C\Delta T_2$

$m_{cold}\ (60 - 20) = 5\ (90 - 60)$

$m_{cold} = 3.75kg$

Answer: A

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42. Temperature of a body is the measure of average
A) Work
B) Power
C) K/netic energy
D) Pressure
Answer: C

43. When two bodies are in thermal eqaillibrium, they will have equal
A) Specific heat
B) Heat capacity
C) Conductivity
D) Temperature
Answer: D

44. Specific heat capacity
A) Determines direction of heat flow
B) Is different for different phases of a body
C) Is measured in joule per kilogram
D) Is the energy required to raise temperature of a body by 10c

Hint: $C = \frac{Q}{m\Delta T} = \frac{1J}{kg.K}$

Answer: B

45. When you are taking a bath, how many kilograms of hot water at 48^0c must you mix with cold water at 12^0c So that temperature of the bath is 36^0c and the total mass of water is 90kg
A) 40kg B) 50kg C) 20kg D) 60kg

Hint: $Q_{gained} = Q_{lost}$

$mcC\Delta T_1 = mhC\Delta T_2, m_c + m_n = 90$

$(90 - m_n)(36 - 12) = m_n (48 - 36)$

$m_n = 60kg$

Answer: D

46. Which one of the following remains constant during a change of state of matter?
A) Temperature
B) Pressure
C) Volume
D) Density

Answer: A

47. A copper cylinder of mass 0.5kg heated by supplying it with heat energy of 11700J. What is the change in temperature of the cylinder? ($c = 390J/kg^0c$)
A) 30^0c B) 60^0c C) 80^0c D) 120^0c

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Hint: $Q = Mc\Delta T$

$$\Delta T = \frac{Q}{mc} = \frac{11700}{(0.5)(390)}\ 60^0c$$

Answer: B

48. A metal rod is lm at 20^0c and 1.005m at 40^0c. What is the length of the rod at 60^0c?

A) 1.01m B) 1.05m C) 1.02m D) 1.10m

Hint: $\Delta\ell_1 = \ell o\alpha\Delta T_1 \Rightarrow lo\alpha = \frac{\Delta\ell_1}{\Delta T_1} = \frac{0.005}{40 - 20} = 2.5 \times 10^{-4}\ m/^0\ c$

$$\Delta\ell_2 = \ell o\alpha\Delta T_2 = (2.5 \times 10^{-4})\ (60 - 20) = 0.01m$$

$$\ell = \ell o + \Delta\ell_2 = 1.10m$$

Answer: A

49. The heat required to raise the temperature of an object from 250c to 500c is 125KJ. How much heat is needed to raise the temperature of the same object from 400c to 600c?

A) 100KJ B) 50KJ C) 150KJ D) 200KJ

Hint: $Q_1 = mc\Delta T_1 \Rightarrow mc = \frac{Q}{\Delta T} = \frac{125}{(50 - 25)}\ 5KJ/^0c$

$$Q_2 = mc\Delta T_2 = (5)(60 - 40)$$

$$Q_2 = 100KJ$$

Answer: A

50. An electrical heating system with an average power rating of 4KW is used to heat 400kg of water from 20^0c to 80^0c. How long will the process take?

A) 4hours B) 10hours C) 7hours D) 5hours

Hint: Power = Heat

$$Pt = mc\Delta T$$

$$t = \frac{mc\Delta T}{P} = \frac{(400)(4200)(80 - 20)}{4000} = 25{,}200\,\text{second}$$

$$t = 7\text{hours}$$

Answer: C

51. Which one of the following is a factor on which specific heat capacity

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of a conductor depends?

A) Heat
B) Substance
C) Mass
D) Temperature

Answer: B

52. The device the converts heat in to useful work is?

A) Transformer
B) Transistor
C) Heat Engine
D) Motor

Hint: Heat engine converts heat energy in to useful energy The other devices are electronic elements of different functions.

Answer: C

53. What is internal energy of a body?

A) Potential energy only
B) Kinetic energy only
C) All energy content
D) Chemical energy only

Hint:- Internal energy all energy content of a body.

Answer: C

54. If the end a rod is heated, then the rate of flow of heat does not depend on

A) Heat - sectional area of rod
B) Mass of rod
C) Temperature gradient
D) Longth of the rod

Hint:- $\frac{Q}{t} = KA\frac{\Delta T}{\ell}$

Answer: B

55. The coefficient of thermal conductivity depends on

A) Temperature gredient
B) Area of plate
C) Thickness of plate
D) Material of plate

Hint:- In, $\frac{Q}{t} = KA\frac{\Delta T}{t}$, K depends on the material type the conductor made from

Answer: D

56. Heat travel through vacuum by:

A) Conduction
B) Convection
C) Radiation
D) Can not travel

Hint:- Heat travels in vacuum by electromagnetic wave

Answer: C

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57. The radiation emitted by a body is proportional to
A) Temperature
B) Square of temperature
C) Fourth power of temperature
D) Square root of temperature

Hint:- $\frac{Q}{t} = \delta e A T^4 \Rightarrow P \sim T^4$

Answer: C

58. Which of the following does not occur through convection
A) Boiling of water
B) Land breeze and sea breeze
C) Circulation of air around fornance
D) Heating of glass bulb through filament
Hint: bulb heats up by radiation

Answer: D

59. Boiling point of water
A) Increases as attitude increases
B) Increases as pressure decreases
C) Decreases as pressure decreases
D) Is independent of pressure
Hint:- Boiling point decreases as altitude in creases

Answer: C

60. During melting process, the heat given to a body is utilized in
A) Increasing density
B) Increasing pressure
C) Increasing temperature
D) Increasing molecular distance

Answer: D

61. The rate of radiation of a black body at 0^0c is E. The rate of radiation of this body at 273^0c will be?
A) 16E
B) 8E
C) 4E
D) E

Hint: $\frac{Q}{t} \sim T4$, $\frac{Q}{t} = P =$ Power

$\frac{P_1}{P_2} = \frac{T_1^4}{T_2^4}$, $T_1 = 273K$ and $T_2 = 546K$

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$P_2 = 16P_1 = 16E$ **Answer: A**

62. The most suitable material for cooking is the one having
A) High specific heat and low conductivity
B) Low specific heat and low conductivity
C) Low specific heat and hight conductivity
D) High specific heat and low conductivity

Answer: C

63. How many joules of energy are required to raise the temperature of 100g of gold from 20^0c to 100^0c? (C_{gold} = 129 J/kg.k)

Solution

$Q = mc\Delta T$

$Q = (0.1)(129)(100 - 20)$

$Q = 1032J$

64. A0.8 kg aluminium pan on a stove is used to raise the temperature of 0.2 liter of water from 25^0 to 85^0c. If

$$\left(C_{water} = 4200\ {}^{J}\!/\!_{kg.k},\ C_{Almuinum} = 900\ {}^{J}\!/\!_{kg.k},\right)$$

Find

A) Heat required to bring this change in temperature

B) What percentage of the heat is used to raise temperature of the pan

Solution

A) $Q = mc\Delta T$

$Q = (0.2)(4200)(85 - 15)$

$Q = 58,800J$

B) $Q_{Al} = mc\Delta T = (0.8)(900)(70)$

$Q_{Al} = 50,400J$

$$\%Q = \left(\frac{50,400}{58,800}\right)100 = 86\%$$

65. What is the temperature raise of a 50kg steel bar falling from

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a 50m hight if 50% of its kinetic energy is changed in to heat?
$\left(C_{steel} = 450\ {}^{J}/_{kg.k}\right)$

Solution

$$Q = \frac{1}{2}mgh$$

$$mc\Delta T = \frac{1}{2}mgh$$

$$\Delta T = {}^{gh}/_{2c} = {}^{(10)(50)}/_{2(450)} = 0.6^0c$$

66. Part of steel railroad track has a length of 50m at 0^0c. What will be its length at 50^0c

Solution

$$L = Lo\,(1 + \acute{\alpha}\Delta T)$$
$$= 50\,[1 + (11 \times 10^{-6})\,(50)]$$
$$= 50.03m$$

67. To what temperature should a steel ring of diameter 9.9cm be heated so that a piston of diameter 10cm can pass through it?

Solution

$$A = A_o\,(1 + 2\acute{\alpha}\,\Delta T)$$
$$(10)^2 = (9.9)^2\,[1 + 2(11 \times 10^{-6})\Delta T]$$
$$1.02 = [1 + 22 \times 10^{-6}\,\Delta T]$$
$$22 \times 10^{-6}\,\Delta T = 0.02$$
$$\Delta T = 910^0c$$

68. A 300 litire steel barrel is filled Alcohol at 10^0c. How much alcohol will over flow at 100^0c?

Solution

$$V_{over\ flow} = V_{alcohol} - V_{barrel}$$
$$= V_0\,(1 + 3\acute{\alpha}_{alc}\,\Delta T) - V_o\,(1 + 3\acute{\alpha}_{steel}\Delta T)$$

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$= (\delta_{alcohol} - \delta_{steel})\ \Delta T V_o$

$= (112 \times 10^{-5} - 3.3 \times 10^{-5})\ (90)(300)$

$= 29.4$ Litire of alcohol

69. At 0°c the length of a steel wire is 100.05cm and that of a copper wire is 100 cm. At what temperature will they have the same length?

Solution

$L_{cu} = L_{steel}$

$Lo_{cu}\ [1 + \alpha_{cu}\ \Delta T] = Lo_{steel}\ [1 + \acute{\alpha}_{steel}\ \Delta T]$

$100(1 + 17 \times 10^{-6}\Delta T) = 100.05\ [1 + 11 \times 10^{-6}\Delta T]$

$599.45 \times 10^{-6}\ \Delta T = 0.05$

$\Delta T = 83.41^{0}c$

$T = 83.41^{0}c$

70. What is the change in area of a $0.05m^2$ plate when its heated from 0°c to 90°c ($\acute{\alpha} = 9 \times 10^{-6}/k$)

Solution

$\Delta A = 2\acute{\alpha}\ A_o\ \Delta T$

$= 2(9 \times 10^{-6})\ (0.05)\ (90)$

$= 0.000081m^2$

71. What is the change in volume of a 2L water when heated from 10°c to 100°c ($\acute{\alpha}_{water} = 2.1 \times 10^{-4}/k$)

Solution

$\Delta V = V_o\ 3\acute{\alpha}\ \Delta T$

$= 2\ (3 \times 2.1)10^{-4}\ (100 - 10)$

$= 0.1134L$

72. When a 1kg metal rod at 400°c is immersed in water of 2kg that is at 15°c, the equilibrium temperature is found to be 20.8°c. What is the specific heat capacity of the metal? ($C_{water} = 4200$ J/kg.k)

Solution

$Q_{lost} = Q_{gained}$

$m_{metal}\ C_{metal}\ (400 - 20.8) = m_{water}\ C_{water}\ (20.8 - 15)$

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(1) C_{metal} (379.2) = (4200) (2) (5.8)

C_{metal} = 128.5 J/kg.k

73. A caloimeter of 250g containing 200g of water is heated on 1000W heater for 5 minutes. If the system temperature is changed by 60°c, what is the specific heat capacity of the calorimeter?

Solution

$Q = Q_{calonineter} + Q_{water}$

$P_t = m_{cal}\, C_{cal}\, \Delta T + m_{water}\, C_{water}\, \Delta T$

$(1000)\,(5 \times 60) = 0.25\, C_{cal}\,(60) + 0.2(4200)\,(60)$

$300{,}000 = 15\, C_{cal} + 50{,}400$

$C_{calorimeter}$ = 16,640 J/kg.k

74. How much ice will melt if a copper block of mass 20kg at 500°c is placed on a large ice bloc?

Solution

$$m_{copper}\, C_{copper}\, \Delta T = m_{ice}\, L_f$$

$$m_{ice} = \frac{m_{cu} C_{cu}\, \Delta T}{L_f}$$

$$m_{ice} = \frac{(50)(380)(500)}{3.33 \times 10^5}$$

m_{ice} = 28.53kg

End of Unit Questions and Problems

I. Give Short Answer to the Following Questions

1. What is heat
2. Define temperature
3. How does heat transfer from one body to the other?
4. What are the mechanisms of heat transfer
5. What is absolute zero?
6. Define phase or state of matter

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7. What is latent heat?
8. What is calorimetry?
9. Define heat capacity
10. List the three temperature scales
11. Which is triple point?
12. which one is largest? 1^0c, 1K or 1^0F?
13. Which one is denser? Water at 2^0c or at 8^0c
14. Explain why materials expand on heating
15. Can a phase change with out a change temperature?
16. What is specific heat capacity?
17. Define specific latent heat of fusion?
18. What is specific latent heat of vaporization?
19. What is a critical point in a phase transition?

II. Match the Items in colum A with those in column B

A	B
20. Heat	A) One dimensional expansion
21. Temperature	B) Two dimensional expansion
22. Absolute zero	C) Three dimensional expansion
23. Latent capacity	D) Heat flow via particle collision
24. Heat capacity	E) Heat flow via particle motion
25. Calorimetry	F) Heat flow via electromagnetic wave
26. Triple point	G) Form of energy flow
27. Conduction	H) Average particles kinetic energy
28. Convection	I) Determine specific heat
29. Convection	J) Three phases coexis
30. Radiation	K) Liquid and gas merge together
31. Linear expansion	L) No particles movement
32. Area expansion	M) Changes body temperature by 1K
33. Volume expansion	N) Change phase of substance

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O) Change mass of substance

III. Solve the Following Problems

34. What is the value of 293k in fahreneit scale?
35. A metal rod has a length of 2m at 20^0c What be its length at 50^0c? (1.2 $\times 10^{-5}/^0$c)
36. Thermometer A and B have the same 0^0 ice point. If their upper point are 130^0 and 260^0 respectively. Then if A reads 50^0, then B will read?
37. If the volume expansion coefficient of a block is $3.3 \times 10^{-5}/^0$c, then what will be its coeffient of linear expansion?
38. What is the specific heat of a substance which requires 100J of heat to raise the temperature of a 2kg sample from 22^0c to 24^0c?
39. A bullet of mass 4kg (c = 128J/kg^{0}c) moving with 200m/s enters in to a bag of s and stops. If the kinetic energy of the bullet is charge to heat, what is its change in temperature ?
40. A metal bar of mass 100g is warmed from 20^0c to 80^0c. How much heat is absorbed by it (C = 450J/kg.K)
41. A metal block increases its temperature from 15^0c to 60^0c when supplied with 13500J of heat. What is its heat capacity?
42. What is the rate of energy flow in aluminium slab of area $40cm^2$, and width 30cm if the temperature is 5^0 and 200^0c at ends (K = 0.5 w/km)
43. How much energy is required to change 40g ice from ice from 100c to steam of 110^0c?

I. Short Answer and Explanations

1. Heat is a form of energy that flows from hotter body to colder body
2. Temperature is the measure of average kinetic energy of particles of a body
3. It flow from a region of high temperature to a region of low temperature
4. Conduction, convection and radiation
5. It is a temperature below which particles stop moving
6. Phase is the physical property of a substance at a particular pressure and temperature

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7. It is hidden heat that changes at constant temperature
8. Calorimety is a method of specific heat determination
9. It heat needed to raise body temperature by 1^0c
10. Kelvin, celsius and Fahrenheit
11. A point where the solid, liquid and gas coexist
12. 1^0c is largest
13. Water at 8^0c
14. Because, molecular bond break and particles get free
15. Yes, During phase change, temperature remains constant
16. Heat needed to change 1kg mass by 10c
17. Heat needed to melt 1kg mass at constant temperature
18. Heat needed to vapour 1Kg mass at constant temperature
19. At which the liquid and gas phases merge.

II. Answer to the Matching items

20. G
21. H
22. L
23. N
24. M
25. I
26. J
27. K
28. D
29. E
30. F
31. A
32. B
33. C

III Solution to the end of unit problems

34. $$\frac{T_K - 273}{373 - 273} = \frac{T_F - 32}{212 - 31}$$

$$\frac{293 - 273}{100} = \frac{T_F - 32}{180}$$

$$TF = 36^0F$$

35. $\Delta\ell = \ell_o \acute{\alpha} \Delta T = 2(1.2 \times 10 - 5)\,(50 - 20)$

$\Delta\ell = 7.2 \times 10\text{-}4m$

$\ell = \ell_o + \Delta\ell = 2 + 0.00072 = 2{,}000\ 72m$

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36. $\frac{T_A - 0}{130 - 0} = \frac{T_B - 0}{260 - 0}$

$\frac{50}{130} = \frac{T_B}{180}$

$T_B = 100^0$

37. $\delta = 3\alpha \Rightarrow \alpha = \frac{\delta}{3} = \frac{3.3 \times 10^{-5}}{3}$

$\alpha = 1.1 \times 10^{-5} /^0 c$

38. $Q = mc\Delta T \Rightarrow c = \frac{Q}{m\Delta T}$

$c = \frac{100}{2(24 - 22)} = \frac{100}{4}$

$c = 25 \text{ J/kg.}^0c$

39. $KE = Q$

$\frac{1}{2}mv^2 = mc\Delta T$

$\Delta T = \frac{v^2}{2c} = \frac{(200)^2}{2(128)}$

$\Delta T = 156.25^0c$

40. $Q = mc\Delta T = (0.1)\,(450)\,(80 - 20)$

$Q = 2700J$

41. $c = \frac{Q}{\Delta T} = \frac{13500}{(60 - 15)} = \frac{13500}{45} = 300J/K$

42. $\frac{Q}{\Delta T} = KA = \frac{\Delta T}{x} = \frac{(0.5)(40 \times 10\text{-}4)(200 - 5)}{30 \times 10^{-2}}$

$\frac{Q}{t} = 0.13J/S$

43. $Q = Q_f + Q_l + Q_v + Q_2$

$Q = (0.04)\,(3.34 \times 10^5) + (0.04)\,(4200)\,(100 - 10) + (0.04)\,(2.5 \times 10^6)$

$+ (0.04)\,(2010)(110 - 10) = 110848J$

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UNIT 6

6. Electrostatics and Electric Current

6.1 What is Electric Charge?

6.2 Coulomb's Law

6.3 What is Electric Field?

6.4 What is Electric Flux?

6.5 What is Electric Energy?

6.6 What is Electric Potential?

6.7 What is Electric Circuit?

6.8 Electric Measuring Instruments

6.9 Electric Measuring Instruments

6.10 Kirchhoff's Rules

6.11 Capacitors

6.12 Parallel Plate Capacitor

6.13 Colour code Resistors

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Electrostatics and Electric Current

Dear learner, we are very excited to meet you in this wonderful unit where you will be able to; define electric charge, state the fundamental law of electrostatics, state coulomb's law solve problems on static charge, explain electric field, plot electric field lines, define electric potential, define capacitance, describe electric current, state ohm's law, describe the capacitor and resistor network, determine the resistance of a carbon resistor using the concept of color code, and on and on. Have a nice time.

6.1 What is Electric Charge?

Under normal condition, a body is neutral as it has equal number of protons electrons. The electrical ballance is disturbed when the body gained or lost electrons.

A body is said to changed when it lost or gained electrons.

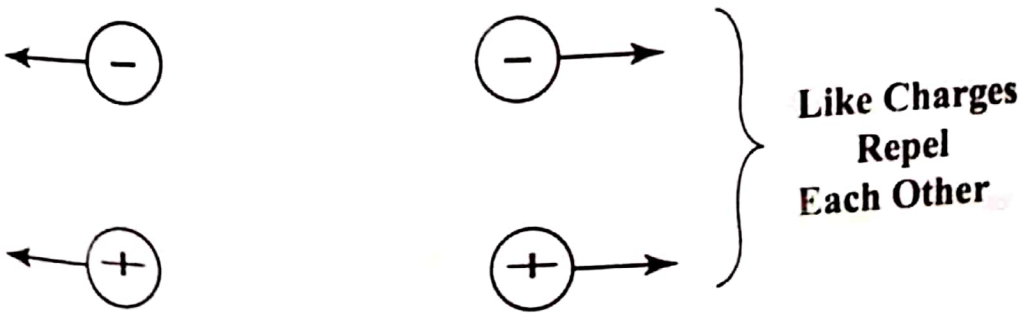
A body is said to be negatively charge when it galned electrons and it said to be positively changed when it lost electrons

As a result, a negatively charged body has surplus of electrons while a positively charged body has deficiency of electrons.

Electric charges interact with each other, And, their interaction is such that two positive charges and two negative charges repel one another, while, a negative and a positive charge attract one another. This fact is stated by the fundamental law of electrostatics

Fundamental law of Electrostatics states that, "Like charges repel each other and unlike charges attract each other

This



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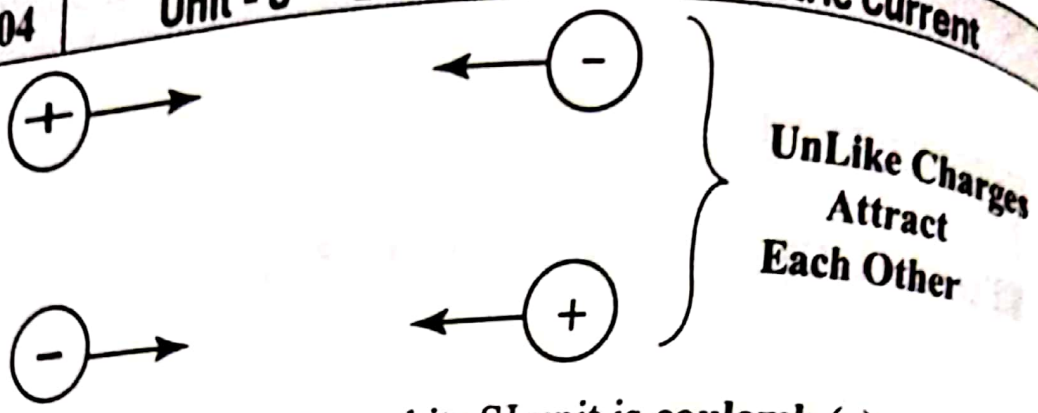

Charge is a scalar quantity and its SI unit is **coulomb (c)**

Critical Thinking Question

1. Can an object gain or lose proton under normal condition? Explain

   Explanation:- No. Protons are inside the nucleus bonded with the neutron by the strong nuclear force and can not be readily displaced with out applying energy.

2. What will happen to the universe if things were naturally like charged or oppositely charged? Explain

   Explanation: If things were made to carry like charges in nature then they will push each other and the universe would be an ever expanding gaseous cloud. But, if they were made to carry opposite charges, then they will attract each other and the universe would be pulled in to a tight ball.

Properties of Electric Charges

Some of the well known properties of electric charge are the following:

- There are two types of charges

  Positive and negative

- Charge is conserved

  Charge is neither created nor destroyed. Rather, it is transferred from one body to the other.

- The smallest charge known so far is the charge carried by an electron or a proton. With magnitude, $e = 1.6 \times 10^{-19}C$
- Charge is quantized

  Charge always occurs as some integral multiple of the charge of an electron

  $$\boxed{Q = ne}$$

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Where: Q = Charge carried by a body

n = Number of electron gained or lost by the body

$e = 1.6 \times 10^{-19}$c, basic charge

Critical Thinking Questions

3. Explain what electro statics deals with

Explanation: Electrostatics or static electricity deals with charges that are at rest

4. When do we say a body is charged?

Explanation: We say a body is charged when it lost or gained electrons

5. What does the fundamental law of electrostatics says?

Explanation: The fundamental law of electrostatics states that, like charges repel each other and unlike charges attract each other .

6. What do we mean by charge is quantized?

Explanation: We mean, charge of a body is the integral multiple of the charge of an electron. So we can have ± 1e, ±2e, ±3e and so on, But we can not have a charge of $\pm 1.5e$, $\pm\frac{3}{4}e$, $\pm 3\frac{1}{2}e$ etc.OK!

7. We said that, a body is said to be charged when it lost or gained electrons. So, can we say that electrons are created to or destroyed from the body?

Explanation: No! when a body is negatively charged, electrons are transferred from some other body to it and when a body is positively charged, electrons are transferred from it to some other body. In the given system there fore, the number of electrons gained by one body is equal to the number of electrons lost by the other body. The system. then remains neutral.

8. If a body lost 100,000 electrons, What is the charge carried by the body?

Solution

$$Q = ne = (100{,}000)\,(1.6 \times 10^{-19}\ c)$$

$$Q = 1.6 \times 10^{-14} c$$

9. How many electrons are there in 1 colons of charge?

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Solution

$$Q = ne \Rightarrow n = \frac{Q}{e}$$

$$n = \frac{1}{1.6 \times 10^{-19}}$$

$$n = 6.25 \times 10^{18} \text{ electrons}$$

10. What is the net charge on a conductor having 20protons and 12 electrons?

Solution

$$Q = ne = (20 - 12)(1.6 \times 10^{-19})$$

$$Q = 1.28 \times 10^{-18}c$$

11. Two spheres carried charges of 5×10^{-11}c and -11×10^{-11}c. If they are made contact

A) What will be the final charge on each sphere?

B) How many excess electrons are there on each sphere?

Solution

$$\text{A. } Q = \frac{Q_1 + Q_2}{2} = \frac{5 \times 10^{-11} + (-11 \times 10^{-11})}{2} = -3 \times 10^{-11}c$$

$$\text{B. } Q = ne \Rightarrow n = \frac{Q}{e} = \frac{-3 \times 10^{-11}}{-1.6 \times 10^{-19}}$$

$$n = 1.875 \times 10^{8} \text{ electrons}$$

6.2 Coulomb's Law

Qualitatively, we said that like charges repel each other and unlike charges attract each other. Quantitatively, the magnitude of the electrostatic force between two point charges is obtained by coulomb's law

Coulomb's law states that "The magnitude of the electrostatic force between two point charges is directly proportional to the product of the magnitude of their in charges and inversely proportional to the square of the distance between them"

That is, if we have

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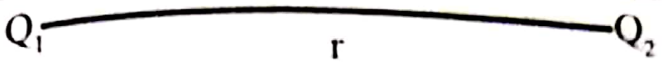

$$F \sim \frac{Q_1 Q_2}{r^2}$$

$$F = \frac{1}{4\pi\varepsilon_o} \frac{Q_1 Q_2}{r^2}$$

Where $\varepsilon_o = 8.85 \times 10^{-12}\ c^2/N.m^2$ is called the **permittivity of free space**

$$K = \frac{1}{4\pi\varepsilon_o} = \frac{1}{4(3.14)\ (8.85 \times 10\text{-}12)} = 9 \times 10^9\ \frac{Nm^2}{c^2}$$

Therefore, $\boxed{F = \frac{KQ_1Q_2}{r^2}}$

Where: $K = 9 \times 10^9\ \frac{Nm^2}{C^2}$ is called the electrostatic constant

Critical Thinking Questions

12. If a 0.001c and a 1000,000c bodies are brought closer, which body will exert greater force on the other? Explain

Explanation:- Using coulomb's law;

$$F_{12} = F_{21} = \frac{KQ_1Q_2}{r^2}$$

And, according to Newton's 3rd law;

$F_{12} = -F_{21}$, same!

13. Compare the electrostatic and gravitational forces between a proton and electron. Which one is Big?

Explanation

$$F_E = \frac{KQ_1Q_2}{r^2}\ ,\ F_G = \frac{Gm_1\ m_2}{r^2}$$

$$\frac{F_E}{F_G} = \frac{Ke^2}{Gm_e m_p} = \frac{(9 \times 10^9)\ (1.6 \times 10^{-19})^2}{(6.67)10^{-11}\ (9.1 \times 10^{-31})\ (1.67 \times 10^{-27})}$$

$F_E = 10^{39} F_G$. The Electrostatic Force is Bigger!

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14. Two charge 6nc and 3nc are placed 30mm apart. What is the magnitude of the force of interaction between the charge?

Solution

$$F = \frac{KQ_1Q_2}{r^2} = \frac{(9\times10^{9})(6\times10^{-9})(3\times10^{-9})}{(3\times10^{-2})^2}$$

$$F = 1.8\times10^{-4}N$$

15. The electrostatic force between two equal point charges separated by 3cm is 3.6N. What is the charge carried by each particle?

Solution

$$F = \frac{KQ_1Q_2}{r^2} = \frac{KQ^2}{r^2}$$

$$Q^2 = \frac{Fr^2}{K} = \frac{(3.6)(3\times10^{-2})^2}{9\times10^{9}}$$

$$Q^2 = 4\times10^{-14}$$

$$Q_1 = Q_2 = 2\times10^{-7}c$$

16. When charges $Q_1 = 6nc$ and $Q_2 = 15nc$ are separated by some distance, they experience a force of 900N. What is the separation between the charges?

Solution

$$F = \frac{KQ_1Q_2}{r^2} \Rightarrow r^2 = \frac{KQ_2Q_2}{F}$$

$$r = \sqrt{\frac{KQ_1Q_2}{F}} = \sqrt{\frac{(9\times10^{9})(6\times10^{-9})(15\times10^{-9})}{900}}$$

$$r = 3\times10^{-2}\,m = 3cm$$

17. Two small spheres carry charges of $6\times10^{-5}c$ and $-4\times10^{-5}c$. If they are brought in to contact and then placed 40mm, what will be the electrostatic force between them?

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Solution

$$Q = \frac{Q_1 + Q_2}{Q} = \frac{6 \times 10^{-5} + (-4 \times 10^{-5})}{2}$$

$$Q = 1 \times 10^{-5} c$$

$$F = \frac{KQ_1Q_2}{r^2} = \frac{KQ^2}{r^2} = \frac{(9 \times 10^9)(1 \times 10^{-5})^2}{(4 \times 10^{-2})^2}$$

$$F = 562.6N$$

18. The electrostatic force between two equal point charges when they are 50mm apart is 80N. What will be the force when they are 5mm apart?

Solution

$$F_1 = \frac{KQ_1Q_2}{r_1^2} \text{ --------- (1)}$$

$$F_2 = \frac{KQ_1Q_2}{r^2} \text{ --------- (2)}$$

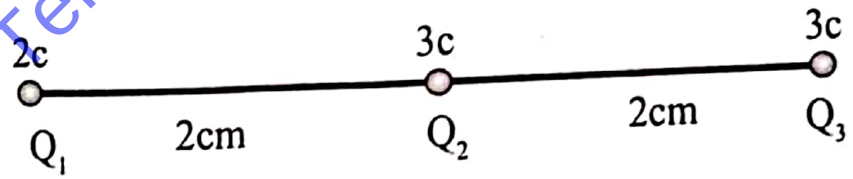
Taking *ratio*;

$$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2} \Rightarrow F_2 = F_1 \left(\frac{r_1}{r_2}\right)^2$$

$$F_2 = 80 \left(\frac{50}{5}\right)^2$$

$$F_2 = 8000N$$

19. What is the net electrostatic force on charge Q_2 by Q_1 and Q_3?



Solution

The force that Q_1 exerts on Q_2 $\xrightarrow{F_{12}}$

$$F_{12} = \frac{KQ_1Q_2}{r_1^2} = \frac{(9 \times 10^9)(2)(3)}{(2 \times 10^{-2})^2}$$

$$F_{12} = 13.5 \times 10^{13} N, \text{ to the right}$$

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The force that Q_3 exerts on Q_2 $\leftarrow F_{32}$ Q_2

$$F_{32} = \frac{KQ_1Q_2}{r_2^2} = \frac{(9\times10^9)(3)(3)}{(2\times10^{-2})^2}$$

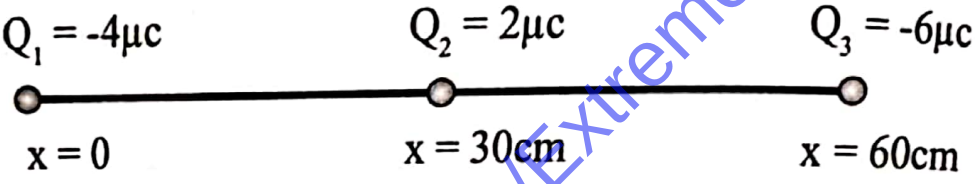
$F_{32} = 20.25 \times 10^{13}$N, to the left

Therefore, then net force acting on Q_2 is obtained by

$F = F_{32} - F_{12} = 20.25 \times 10^{13} - 13.5 \times 10^{13}$

$F = 6.75 \times 10^{13}$N, to the left.

20. What is the electrostatic force acting on Q_2?



Solution

The force that Q_1 exerts on Q_2, $\leftarrow F_{12}$ Q_2

$$F_{12} = \frac{KQ_1Q_2}{r_1^2} = \frac{(9\times10^9)(4\times10^{-6})(2\times10^{-6})}{(0.3)^2} = 0.8\text{N, to the left}$$

The force that Q_3 exerts on Q_2 $Q_2 \rightarrow F_{32}$

$$F_{23} = \frac{KQ_1Q_2}{r_{12}^2} = \frac{(9\times10^9)(2\times10^{-6})(6\times10^{-6})}{(0.3)^2} = 1.2\text{N, to the tight}$$

Therefore, the net force acting on Q_2 is obtained by;

$F = F_{32} - F_{12} = 1.2\text{N} - 0.8\text{N}$

$F = 0.4$N, to the Right!

21. Determine the net force on Q_2 given that $Q_1 = -3\mu c$, $Q_2 = 6\mu c$ and $Q_3 = 1.5\mu c$

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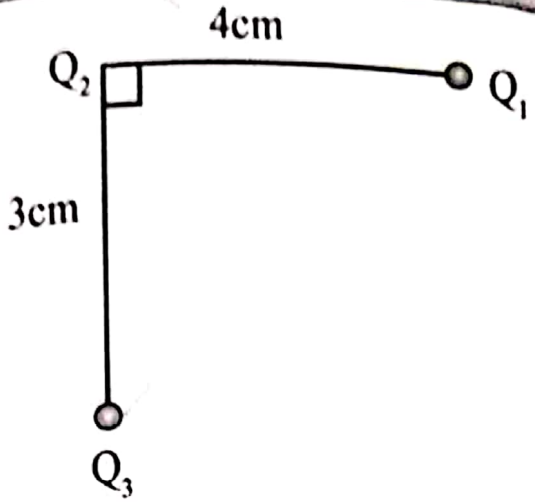

Solution

The force that Q_1 exerts on Q_2 is, F_{12}

$$F_{12} = \frac{KQ_1Q_2}{r_1^2} = \frac{(9\times10^9)(3\times10^{-6})(6\times10^{-6})}{(4\times10^{-2})^2} = 100N,$$

F_{12}

The force that Q_3 exerts on Q_2 is, F_{32}

F_{32}

$$F_{32} = \frac{KQ_3Q_2}{r_2^2} = \frac{(9\times10^9)(1.5\times10^{-6})(6\times10^{-6})}{(3\times10^{-2})^2} = 90N,$$

The net force acting on Q_2 is obtained by Pythagoras theoren as

$$F = \sqrt{F_{12}^2 + F_{32}^2} = \sqrt{100^2 + 90^2}$$

$F = 130N$

And, its direction is, $\theta = \tan^{-1}\left(\frac{90}{100}\right) = 42^0$ above $x-$ axis

22. Which of the following is in disagreement with coulomb's law?

A) The electrostatic force between two charges is proportional to the square of their separation

B) The electrostatic force between two charges is proportional to the product the charges

C) Bodies of the same charge exert repulsing force on each other

D) Bodies of opposite charge exert attractive force on each other

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Hint: $F = \frac{KQ_1Q_2}{r^2}$

Answer: A

23. Which of the following is a false statement?

A) Charge is quantized
B) Charge is conserved
C) Electrostatic force applies for moving charges
D) The smallest charge known to exist that of the electron

Hint: Electrostatic means charges at rest.

Answer: C

24. Two electrons are 10^{-8}m apart. What is the electrostatic force that exert on each other?

A) 2.3×10^{-12}N
B) 1.4×10^{-17}N
C) 1.8×10^{-37}N
D) 0.4×10^{-10}N

Solution

$$F = \frac{KQ_1Q_2}{r^2} = \frac{(9 \times 10^9)(1.6 \times 10^{-19})(1.6 \times 10^{-19})}{(10^{-8})^2}$$

$$F = 2.3 \times 10^{-12}\text{N}$$

Answer: A

25. Charge of q located between Q_1 and Q_2. If of is closer to Q_1 and the net force it is zero, What can you say about Q_1 and Q_2?

A) Q_1 and Q_2 are equal in magnitude and same in sign
B) Q_1 and Q_2 are equal in magnitude and opposite in sign
C) Q_1 and Q_2 are unequal in magnitude and opposite in sign
D) Q_1 and Q_2 are unequal in magnitude and sane in sign

Hint: Q_1 —— r_1 —— q —— r_2 —— Q_2 $\quad F_{net} = 0$

$$\frac{KQ_1q}{r_1^2} = \frac{KQ_2q}{r_2^2} \Rightarrow \frac{Q_1}{r_1^2} = \frac{Q_2}{r_2^2}$$

Since $r_1 < r_2$, $Q_1 < Q_2$

Answer: D

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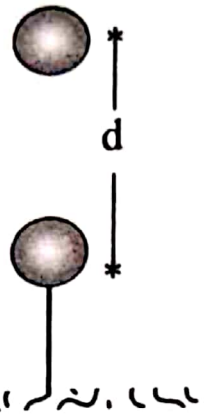
26. The charge on each of the pith balls shown is 20nc. What should be the distance between them for the electrostatic force to just balance the weight of the upper pith ball of mass 0.1 gram?



A) 1.8mm
B) 1.8mm
C) 6mm
D) 6cm

Solution

$$F = w \Rightarrow \frac{KQ_1Q_2}{d^2} = mg$$

$$\frac{KQ^2}{d^2} = mg$$

$$d = \sqrt{\frac{KQ^2}{mg}} = \sqrt{\frac{(9\times10^9)(20\times10^{-9})^2}{(0.1\times10^{-3})(10)}} = 0.06m$$

$$d = 6cm$$

Answer: D

27. Two positive point charges repel each other with a force of 0.16N when their separation is 1.5m. What force do they exert on each other when their separation is 1m?

A) 0.54N B) 1.64N C) 0.24N D) 0.36N

Solution

$$F_1 = \frac{KQ_1Q_2}{r_1^2} \text{ ---------(1)}$$

$$F_2 = \frac{KQ_1Q_2}{r_2^2} \text{ --------- (2)}$$

Taking ratio

$$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2} \Rightarrow F_2 = F_1\left(\frac{r_1}{r_2}\right)^2$$

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$$F_2 = (0.16)\left(\frac{1.5}{2}\right)^2 = 0.36N$$

Answer: D

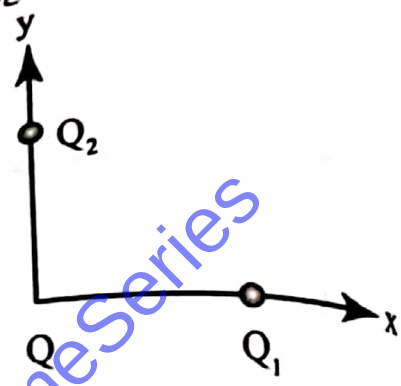
28. Charge $Q_1 = -4\mu c$ and $Q_2 = -3\mu c$ are located at (3,0) m on the x - axis and (0.3)m on the y - axis respectively. What are the x and y components of the net force on 1μc charge placed st the origin?

A) $F_x = -4 \times 10^{-3}N$ and $F_y = -3 \times 10^{-3}N$

B) $F_x = -3 \times 10^{-3}N$ and $F_y = -4 \times 10^{-3}N$

C) $F_x = 4 \times 10^{-3}N$ and $F_y = 3 \times 10^{-3}N$

D) $F_x = 3 \times 10^{-3}N$ and $F_y = 4 \times 10^{-3}N$



Solution

$$F_x = \frac{KQ_1Q}{r_1^2} = \frac{(9\times10^9)(4\times10^{-6})(1\times10^{-6})}{(3)2} = 4\times10^{-3}N$$

$$F_y = \frac{KQ_1Q_2}{r_2^2} = \frac{(9\times10^9)(3\times10^{-6})(1\times10^{-6})}{(3)^2} = 3\times10^{-3}N$$

Answer: C

29. Two negative point charge are 2m apart and repel each other with a force of 2N. When the distance between them is doubled the force between them is?

A) One fourth as great

B) One half as great

C) Twice as great

D) Four times as great

Solution

$$F_1 = \frac{KQ_1Q_2}{r_2^2} = \text{----------}(1)$$

$$F_2 = \frac{KQ_1Q_2}{r_2^2} = \text{----------}(2)$$

Taking ration;

$$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2} \Rightarrow F_2 = F_1\left(\frac{r_1}{r_2}\right)^2 = 2\left(\frac{2}{4}\right)^2 = \frac{1}{4}(2)$$

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$$F_2 = \frac{1}{2} F_1$$

Answer: B

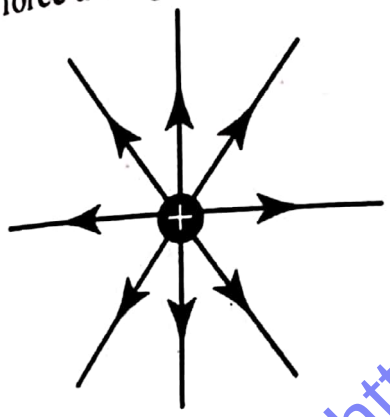
6.3 What is Electric Field?

As the name suggests, field is a region where force of a body acts at a distance

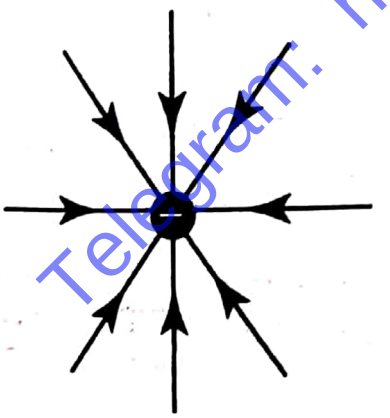
Electric Field: Is a region around a charged body where a test charge placed at that point experience a force

To visualize the nature of the electric field acting on a test charge, imaginary lines called **electric field line** or electric line of force are used.

- The direction of the electric field at a point is the same as the direction of the force acting on a test charge placed at that point.



The field lines are radially out ward from a Positive point charge



The field lines are radially in ward to a negative point charge

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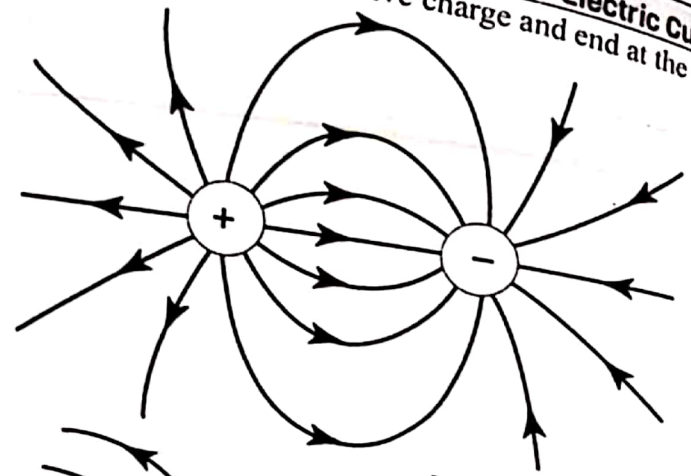
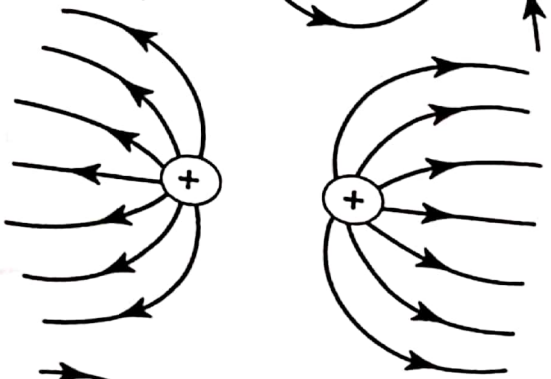
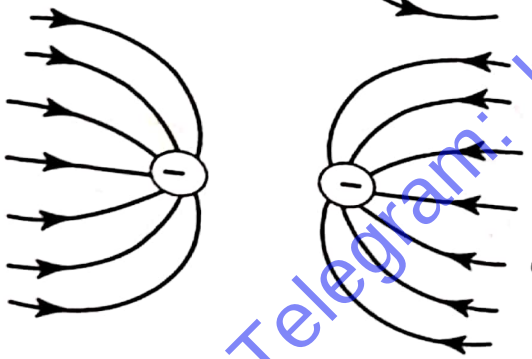
The field lines start at the positive charge and end at the negative charge

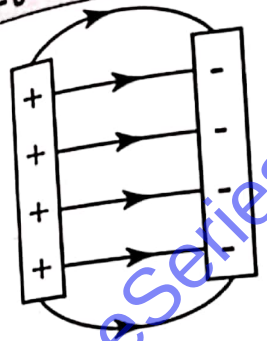




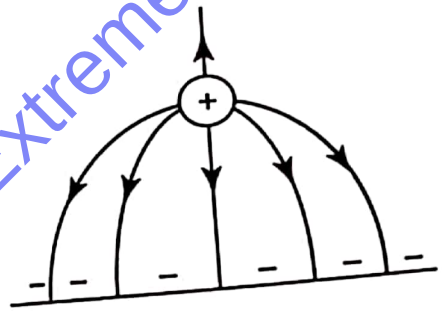
Like charges repel each other



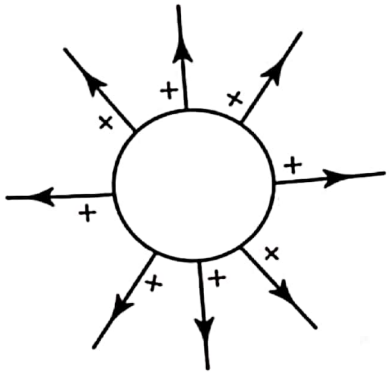
Like charge repel each other



Oppositely charged parallel plates field pattern



Electric field between a point charge and a charged plate



Electric field of a charged sphere

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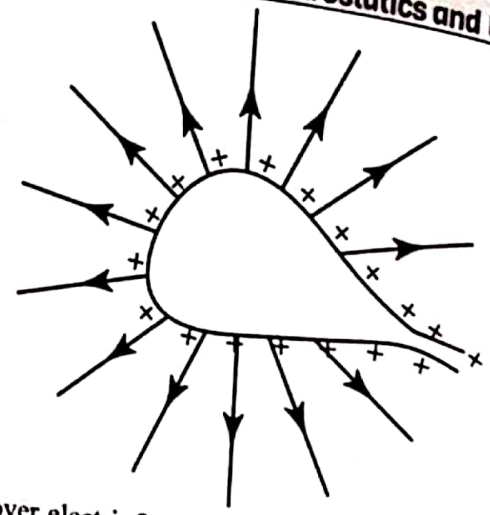
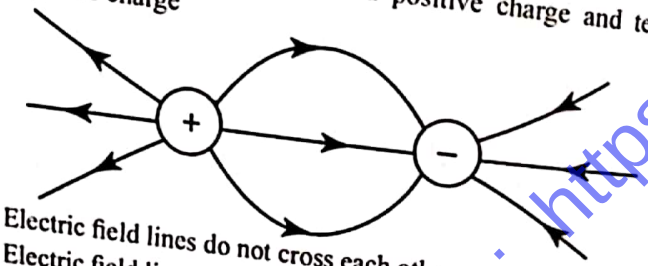


Electric field of an irregular charged conductor

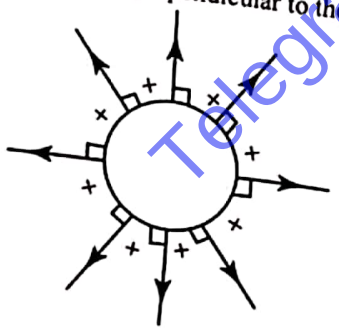
Moreover, electric field lines do have the following important characteristics.

Properties of Electric Field Lines

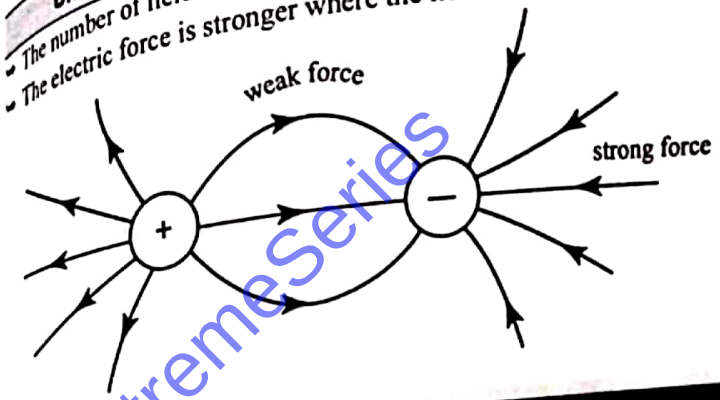
- Electric field lines start from a positive charge and terminate at a negative charge



- Electric field lines do not cross each other
- Electric field lines are symmetrical to each other
- Electric field line are perpendicular to the surface of the charged body



- The number of field lines is proportional to the magnitude the charge
- The electric force is stronger where the field lines are closer together



Critical Thinking Questions

30. Why do electric field lines points radially out ward from a positive point charge and radially in ward to a negative point charge? Explain

Explanation: Because, the direction of the field lines is the same as the direction of the electric force on a test charge. Therefore, when a test charge which is positive is brought near a positive point charge, it is repelled outward. And, when the test charge is brought closer a negative point charge, it is attracted inward. Okay!

Electric Field Strength

The strength or intensity of the electric field at a point tells us the magnitude of the electric effect at that point.

Electric Field Strength: Is the electrostatic force per unit positive charge acting on a test charge placed in the field

That is

$$\text{Electric Field} = \frac{\text{Electrostatic Force}}{\text{Charge}}$$

$$\vec{E} = \frac{\vec{F}}{Q^+}$$

Electric field strength is a vector quantity and its SI unit is newton per coulomb (N/c)

The electric field strength of a point charge Q at a distance r is obtained by

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$$E = \frac{F}{Q^+} = \frac{KQ\delta^+}{r^2/\delta^+}$$

$$\boxed{E = \frac{KQ}{r^2}}$$

The electric field strength due to a point charge is variable as it depends on the inverse square of the distance from the charge

Illustrative Examples

31. A charge of 2μc placed in an electric field experiences a force of 8×10^{-4}N. What is the field strength at that point.

Solution

$$E = \frac{F}{Q} = \frac{8 \times 10^{-4}}{2 \times 10^{-6}} = 400 N/c$$

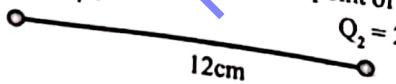
32. What is the field strength at a distance of 30cm from a point charge of 2μc?

Solution

$$E = \frac{KQ}{r^2} = \frac{(9 \times 10^9)(2 \times 10^{-6})}{(30 \times 10^{-2})^2}$$

$$E = 2 \times 10^5 N/c$$

33. What is the net electric field at the mid - point of the two point charges?



Solution

The field strength of Q_1 at r = 6cm is

$$E_1 = \frac{KQ_1}{r_1^2} = \frac{(9 \times 10^9)(2 \times 10^{-6})}{(6 \times 10^{-2})^2} = 0.5 \times 10^7 N/c, left$$

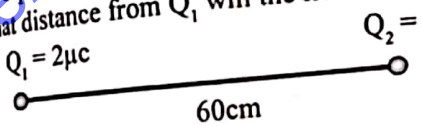
The field strength of Q_2 at r = 6cm is

$$E_2 = \frac{KQ_2}{r_2^2} = \frac{(9 \times 10^9)(2 \times 10^{-6})}{(6 \times 10^{-2})^2} = 0.5 \times 10^7 N/c, left$$

$$E = E_1 + E_2 = 0.5 \times 10^7 + 0.5 \times 10^7$$

$$E = 1 \times 10^7 N/c, \text{ to the left}$$

34. Two point charges $Q_1 = 2\mu c$ and $Q_2 = 8\mu c$ are placed 60cm apart. At what distance from Q_1 will the field be zero?



Solution

Let the field be zero at a distance x from Q_1 so we have:

$$E_1 = \frac{KQ_1}{x^2} = \frac{(9 \times 10^9)(2 \times 10^{-6})}{x^2}, \text{ to the right}$$

$$E_2 = \frac{KQ_2}{(0.6 - x)^2} = \frac{(9 \times 10^9)(8 \times 10^{-6})}{(0.6 - x)^2} \text{ to the left}$$

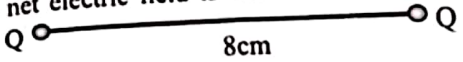
$$E = E_1 + E_2 = 0$$

$$\frac{(9 \times 10^9)(2 \times 10^{-6})}{x^2} + \frac{(9 \times 10^9)(8 \times 10^{-6})}{(0.6 - x)^2} = 0$$

$$3x^2 + 1.2x - 0.36 = 0$$

$$x = 0.2m = 20cm$$

35. Two point charges of equal magnitude are 8cm apart. At the mid - point of the line joining them, the net electric field is 45N/c. What is the magnitude of the charge



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Solution

The net electric field at the mid - point is

$E = E_1 + E_2$

$E = \frac{KQ}{r^2} + \frac{KQ}{r^2}$

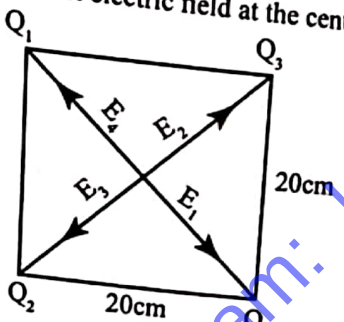
$E = \frac{2KQ}{r^2}$

$45 = \frac{2(9\times 10^9)Q}{(4\times 10^{-2})^2}$

$Q = 4\times 10^{-12}c$

$Q = 4Pc$

36. Four point charges of 4μc are placed at the four corners of a square of side 20cm. What is the net electric field at the centre of the square?



Solution

$E = E_1 + E_2 + E_3 + E_4 = 0$

37. An oil drop of 3.2×10^{-13}N weight is balanced by a field of 5×10^5 N/c. What is the charge of the drop?

Solution

$EQ = mg \Rightarrow Q = mg/E = 3.2\times 10^{-13} / 5\times 10^5 = 6.4\times 10^{-19}c$

6.4 What is Electric Flux?

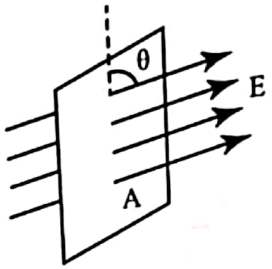
Electric flux is the measure of electric field lines flowing through a surface.

Electric Flux: Is the number of electric field lines penetrating a given area perpendicularly.

Electric Flux = (Electric field strength). (Surface area perpendicular to the field)

$\phi = \vec{E}\cdot\vec{A}$

$\boxed{\phi = EA\cos\theta}$



Where: Ø(Phi) is electric flux

E = the electric field strength

A = the surface area

θ = the angle between the normal to the surface and electric field lines.

Electric flux is a scalar quantity and its unit is Nm^2/c

Critical Thinking Questions

38. Can electric flux ever be negative? Defened you answer.

Explanation: Yes! When the charge stored on a surface is negative, the flux become negative implying that the electric field lines are entering in to the surface.

39. A surface of area $400cm^2$ is placed in a uniform electric field of 600N/c. Calculate the electric flux when the electric field is

A) Perpendicular to the surface

B) Parallel to the surface

C) 30^0 to the surface

Solution

A) $\phi = EA\cos,\ \theta = 0^0$

$\phi = (600)(4\times 10^{-2})\cos(0^0)$

$\phi = 24\ \frac{Nm^2}{c}$

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B) $\phi = EA\cos\theta, \theta = 90^0$

$\phi = (600)(4 \times 10^{-2})\cos(90^0)$

$\phi = 0 \frac{Nm^2}{c}$

C) $\phi = EA\cos\theta, \theta = 60^0$

$\phi = (600)(4 \times 10^{-2})\cos(60^0)$

$\phi = 12 \frac{Nm^2}{c}$

40. A 5cm × 8cm rectangular plate lies in the xy plane. What is the electric flux when the plate is subjected to an electric field of $\vec{E} = (100j + 10\hat{K})\frac{N}{C}$

Solution

$\phi = \vec{E}.\vec{A} = E_x A_x + E_y A_y + E_z A_z$

$\vec{A} = (5cm \times 8cm)\hat{K} = 40cm^2 \hat{K} = 0.004m^2 \hat{K}$

$\phi = (oi + 100j + 10\hat{K})(oi + oj + 0.004K)$

$\phi = (0)(0) + (100)(0) + (10)(0.004)$

$\phi = 0.04 \frac{Nm^2}{c}$

41. What is the maximum electric flux through a circle of radius R when subjected to a uniform electric field of strength E?

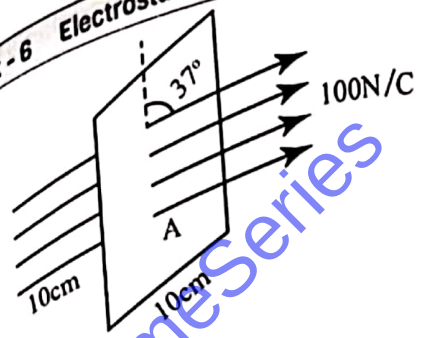
Solution

$\phi = EA\cos\theta$

$\phi_{max} = EA, A = \pi R^2$

$\phi_{max} = E\pi R^2$

42. What is the electric flux for the arrangement shown below



Solution

$\phi = EA\cos\theta, \theta = 37^0$

$\phi = (100)(10^{-2})\cos(37^0), A = (10cm)(10cm)$

$\phi = (1)(0.8)$

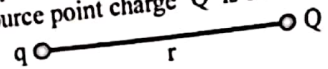
$\phi = 0.8 \frac{Nm^2}{c}$

6.5 Electric Potential Energy

The work done to move a test charge towards a positive point charge is stored as potential energy by the test charge.

Electric Potential Energy: Is the work done to bring a test charge from infinity to its current position.

The electric potential energy of a test charge 'q' as it brought to a distance 'r' from a source point charge 'Q' is obtained by:



$$U = \frac{KqQ}{r}$$

Electric potential energy is a scalar quantity and its SI unit is **joule (J)**

Critical Thinking Questions

43. What will happen to the electric potential energy of a test charge in bringing it towards

A) Positive point charge B) Negative point charge

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Explanation: A. $u = \frac{KqQ}{r}$, here both are positive

$u \longrightarrow$ increases

B. $u = \frac{kgQ}{r}$, here Q is negative

$u \longrightarrow$ decreases

Illustrative Examples

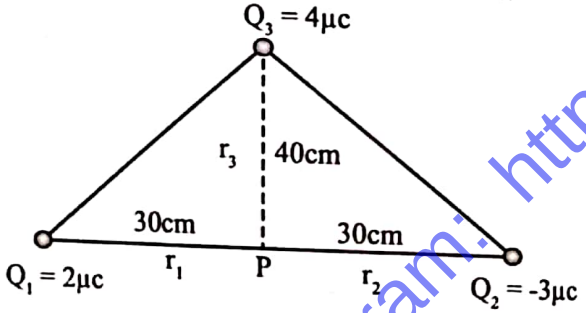
44. What is the electric potential energy of a 2nc charge as it is moved 20cm closer to 4μc point charge

Solution

$$u = \frac{kgQ}{r} = \frac{(9\times 10^{9})(2\times 10^{-9})(4\times 10^{-6})}{0.2}$$

$$u = 3.6\times 10^{-4}J$$

45. What is the electric potential energy of a 5μc charge as it moves from infinity to point P in the charge system shown blow



Solution

$$u = u_1 + u_2 + u_3$$

$$u = \frac{kQ_1q}{r_1} + \frac{KQ_2q}{r_2} + \frac{KQ_3q}{r_3}$$

$$u = Kq\left[\frac{Q_1}{r_1} + \frac{Q_2}{r_2} + \frac{Q_3}{r_3}\right]$$

$$u = 45\times 10^{3}\left[\frac{2\times 10^{-5}}{3} - \frac{3\times 10^{-5}}{3} + \frac{4\times 10^{-5}}{4}\right]$$

$$u = 45\times 10^{-2}\left[\frac{2}{3}\right]$$

$$u = 0.3J$$

46. What is the electric field strength at a distance of 20m from a point charge of 8×10^{-10}c?

A) 1.8×10^{-2}N/c
B) 0.72×10^{-3}N/c
C) 1.44×10^{-2}N/c
D) 3.5×10^{-2} N/c

Answer: A

Solution

$$E = \frac{KQ}{r^2} = \frac{(9\times 10^{9})(8\times 10^{-10})}{(20)^2} = 1.8\times 10^{-2}N/c$$

6.6 What is Electric Potential?

Electric potential or absolute potential of a point charge is defined as follows.

Electric Potential: Is the work done per unit positive charge needed to move a test charge from infinity to its present location.

That is;

$$\text{Electric potential} = \frac{\text{Work done}}{\text{Charge moved}}$$

$$\boxed{V = \frac{W}{Q^{+}}}$$

Electric potential is a scalar quantity and its SI unit is **volt (v)**

$$1v = 1J/c$$

The electric potential of a point charge Q at a distance r is obtained by

Q ——— r

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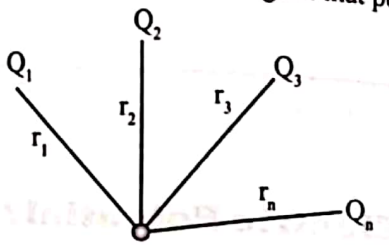
$$V = \frac{W}{Q^+} = \frac{Fr}{Q^+}$$

$$V = \left(\frac{KQq^+}{\frac{r^2}{q^+}}\right) r$$

$$\boxed{V = \frac{KQ}{r}}$$

The total electric potential of n point charges at a point is equal to the sum of the electric potentials of each charge at that point.



$$V = V_1 + V_2 + \text{----} + Vn$$

$$V = \frac{KQ_1}{r_1} + \frac{KQ_2}{r_2} + \text{-----} + \frac{KQ_n}{r_n}$$

$$\boxed{V = K\left[\frac{Q_1}{r_1} + \frac{Q_2}{r_2} + \text{----} + \frac{Q_n}{r_n}\right]}$$

Illustrative Examples

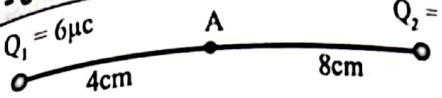
47. What is the electric potential of a 6μc point charge at distance of 30cm?

Solution

$$V = \frac{KQ}{r} = \frac{(9\times10^9)(6\times10^{-6})}{30\times10^{-2}}$$

$$V = 18000V$$

48. What is the total electric potential at point A?



Solution

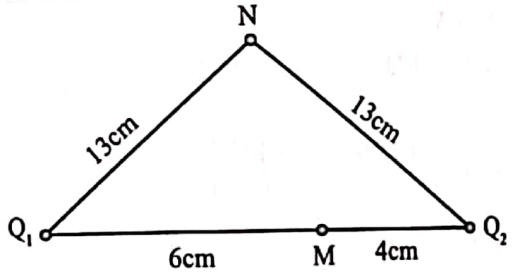
$$V_A = V_1 + V_2$$

$$V_A = \frac{KQ_1}{r_1} + \frac{KQ_2}{r_2} = K\left[\frac{Q_1}{r_1} + \frac{Q_2}{r_2}\right]$$

$$V_A = 9\times10^9\left[\frac{6\times10^{-6}}{4\times10^{-2}} - \frac{6\times10^{-6}}{8\times10^{-2}}\right]$$

$$V = 6.75\times10^5 V$$

49. Given the charge $Q_1 = 12nc$ and $Q_2 = -12nc$, what is the electric potential at point
A) N
B) M



Solution

A. $V_N = V_1 + V_2$

$$V_N = \frac{KQ_1}{r_1} + \frac{KQ_2}{r_2} = 9\times10^9\left[\frac{12\times10^{-9}}{13\times10^{-2}} + \frac{12\times10^{-9}}{13\times10^{-2}}\right]$$

$$V_N = 0$$

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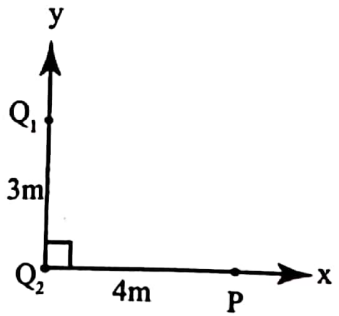
B. $V_M = V_1 + V_2$

$$V_M = \frac{KQ_1}{r_1} + \frac{KQ_2}{r_2} = 9 \times 10^9 \left[\frac{12 \times 10^{-9}}{6 \times 10^{-2}} - \frac{12 \times 10^{-9}}{4 \times 10^{-2}}\right]$$

$$V_M = -900V$$

50. What is the electric potential at point P due to $Q_1 = -6\mu c$ and $Q_2 = 2\mu c$ point charges shown below



Solution

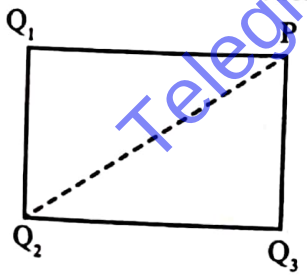
$$V_P = V_1 + V_2$$

$$= \frac{KQ_1}{r_1} + \frac{KQ_2}{r_2} \quad , \quad r_1 = \sqrt{3^2 + 4^2} = 5m$$

$$= 9 \times 10^9 \left[\frac{-6 \times 10^{-6}}{5} + \frac{2 \times 10^{-6}}{4}\right]$$

$$V_P = -6.3 \times 10^3 V$$

51. Three equal charges of each $2\mu c$ are placed at the three corners of a square of side 5cm. What is the electric potential at the fourth corner



Solution

$$V_P = V_1 + V_2 + V_3$$

$$= \frac{KQ_1}{r_1} + \frac{KQ_2}{r_2} + \frac{KQ_3}{r_3} \quad , \quad r_2 = \sqrt{5^2 + 5^2} = 5\sqrt{2}$$

$$V_P = (9 \times 10^9)(2 \times 10^{-6}) \left[\frac{1}{5 \times 10^{-2}} + \frac{1}{5\sqrt{2} \times 10^{-2}} + \frac{1}{5 \times 10^{-2}}\right]$$

$$V_P = 9.7 \times 10^5 V$$

52. What amount of work is done to bring a test charge towards a point charge of $2\mu c$ where the electric potential is 100v?

Solution

$$V = \frac{W}{Q} \Rightarrow W = vQ$$

$$W = (100)(2 \times 10^{-6})$$

$$W = 2 \times 10^{-4} J$$

53. The electric potential at a point in space is

A) The work done to move a point charge from point to point
B) Force per unit charge
C) Work done times charge moved
D) Work done to move a test charge from infinity to its current position

Hint: $V = \frac{W}{Q^+}$

Answer: D

54. What is the potential difference between parallel plates of uniform field strength 100N/c and plate separation 20cm?

A) 100V B) 20V C) 200V D) 2000V

Solution

$V = Ed = (100)(0.2) = 20v$

Answer: B

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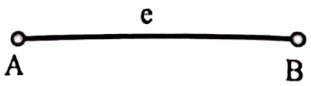
55. The electric potential at point A is 10v and that at point B is 30v. Which of the following is true about an electron placed mid - way between A and B?

A) I will stay at rest
B) It will move towards A
C) It will move toward B
D) It will move towards A and the towards B



Hint: The electron move from higher potential to lower potential

Answer: B

Electric Potential Difference

The difference in electric potential of a test charge at two different points in an electric field is called **electric potential difference**

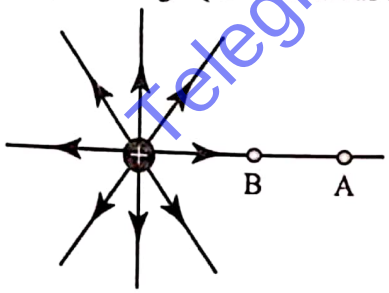
Electric potential Difference: Is the work done per unit positive charge required to move a test charge from a point of lower potential to a point of higher potential in an electric field.

That is, potential difference $= \frac{\text{Work done}}{\text{charge moved}}$

$$v = \frac{W}{Q}$$

Potential Difference in a Point charge Field

The potential difference of two points say A and B that are at a distance of r_A and r_B from a point charge Q is obtained as follows:

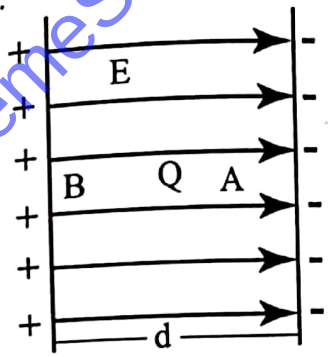


$$\Delta V = V_{AB} = V_B - V_A = \frac{KQ}{r_B} - \frac{KQ}{r_A}$$

$$\boxed{V_{AB} = KQ\left[\frac{1}{r_B} - \frac{1}{r_A}\right]} \longleftarrow \text{Potential difference in a point charge field}$$

Potential Difference in a Parallel Plate Field

The potential difference of two points say A and B that are at a distance of d in an oppositely charged. Parallel plates of uniform electric field E is obtained as follows:



Potential difference between two points is given by:

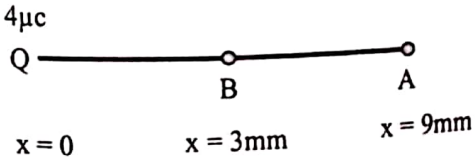
$$V = \Delta V = V_{AB} = \frac{W}{Q} = \frac{Fd}{Q}, \quad F = EQ$$

$$V = \frac{EQd}{Q}$$

$\boxed{V = Ed}$ ⟵ potential difference b/n parallel plates

Illustrative Examples

56. What is the electric potential difference between point A and B shown below?



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Solution

$$V_{AB} = KQ\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$

$$V_{AB} = (9 \times 10^9)(4 \times 10^{-6})\left[\frac{1}{3 \times 10^{-3}} - \frac{1}{9 \times 10^{-3}}\right]$$

$$V_{AB} = 8 \times 10^6 V$$

57. What is the potential difference between parallel plates of separation 5mm and field intensity 2×10^6N/c?

Solution

$V = Ed = (2 \times 10^6)(5 \times 10^{-3})$

$V = 10,000V$

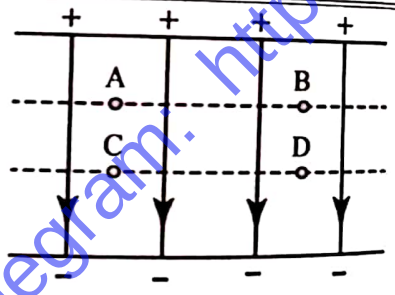
$V = 10KV$

Equipotential Lines or Surfaces

Equi means equal or same. Hence

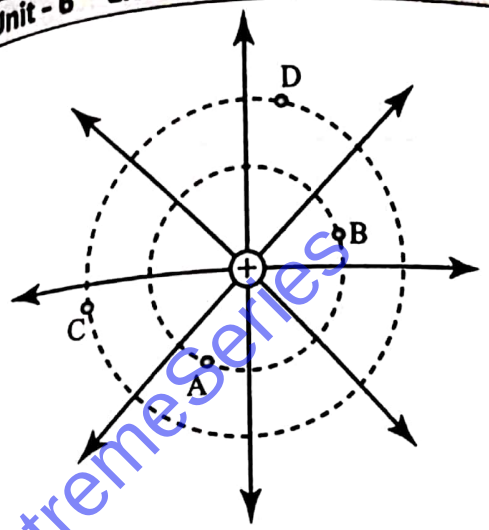
Equipotential line: Is a line joining points with in an electric field that have the same electric potentials

Uniform Field: In a parallel plate, the equipotential lines are parallel and equidistant.



For example, the line joining point A and point B is equipotential line. And the line joining points C and D is another equipotential line.

Variable Field: In a field of a point charge, the equipotential lines are concentric circles.



For example, the circle joining point A and B is equipotential line. And the circle joing points C and D is another equipotential line

Equipotential Surface: Is a two dimensional object like a plate or a three dimensional object like a sphere containing equipotential lines.

Characteristics of Equipotential Lines

- No work is done in moving a charge along an equi poetical line.
- At every point, electric field lines and equipotential lines are per perpendicular
- Equipotential line move closer together as the electric field strength increases.

Illustrative Examples

58. Positively charged balloon A repels ballon B and balloon B attracts ballon C. ballon B must be charged?

A) Positively
B) Negatively
C) Not charged
D) Can not be determined

Hint: opposite charges attract each other.

Answer: B

59. Two metal spheres of charges > c and -3c are in to contact and then separated. What is the suplus or deficiency of electrons on them after contact.

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A) 3.15×10^{19} electrons
B) 2.15×10^{19} electrons
C) 4.25×10^{29} electrons
D) 1.25×10^{29} electrons

Solution

$$Q = \frac{7-3}{2} = 2c$$

$$Q = ne$$

$$n = \frac{Q}{e} = \frac{2}{1.6 \times 10^{-19}}$$

$$n = 1.25 \times 10^{19}$$

Answer: D

60. Two point charges are separated by some distance. What will happen to the electrostatic force between them if their charges are doubled and their separation is halved?

A. $\frac{1}{4}F$ B. $\frac{1}{16}F$ C. 4F D. 16F

Hint:- $F_1 = \frac{KQ_1Q_2}{r_1^2}$ and $F_2 = \frac{KQ_1' Q_2'}{r_2^2} = \frac{K(2Q_1))(2Q_2)}{\left(\frac{1}{2}r_2\right)^2}$

$$F_2 = 16\frac{KQ_1Q_2}{r_1^2} = 16F_1$$

Answer: D

61. Volt per meter is equvalent to;

A) Joule per meter
B) Joule per coulomb
C) Newton per meter
D) Newton per coulomb

Hint: $E = \frac{V}{d} = \frac{F}{Q}$

Answer: D

62. Two capacitors have capacitances C_1 and C_2 where $C_1 < C_2$. When they are connected in series, their equivalent capacitance is?

A) Less than C_1
B) Greater than C_1

C) Greater than C_2
D) Between C_1 and C_2

Hint: When capacitors are connected in series their capacitance is smaller the capacitance of each capacitor.

Answer: A

63. How many electrons should two spheres of equal charges lose in order that, the force of repulsion became 10^{-3} when they are 3mm apart?

A) 6.25×10^6 B) 6.25×10^9 C) 6.25×10^{11} D) 6.25×10^{13}

Solution

$$F = \frac{KQ_1Q_2}{r^2} = \frac{KQ^2}{r^2}$$

$$10^{-3} = \frac{9 \times 10^9 Q^2}{(3 \times 10^{-3})^2}$$

$$Q = 10^{-9}c$$

$$Q = ne \Rightarrow n = \frac{Q}{e} = \frac{10^{-9}}{1.6 \times 10^{-19}}$$

$$n = 6.25 \times 10^9 \text{ electrons}$$

Answer: B

64. At what distance from a point charge of 1μc will the electric potential be 3×10^4V?

A) 1m B) 3m C) 0.3m D) 3.5m

Solution

$$V = \frac{Q}{r} \Rightarrow r = \frac{KQ}{V}$$

$$r = \frac{\left(9 \times 10^9\right)\left(1 \times 10^{-6}\right)}{3 \times 10^4} = 0.3m$$

Answer: C

65. At what separation will the force between a 2nc and 2μc charge be 1N

A) 6m B) 6cm C) 6mm D) 0.6m

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$$V = \frac{KQ_1Q_2}{r^2} \Rightarrow \frac{(9\times10^9)(12\times10^{-9})(2\times10^{-6})}{r^2} = 1$$

$$r = 6\times10^{-3}m = 6mm$$

66. A particle with charge of 6μc moves in a uniform electric field strength of 3×10^4μc. What electrostatic force is exerted on the particle?

A) 18N B) 1.8N C) 0.8N D) 180N

Answer: C

Solution

$$E = \frac{F}{Q} \Rightarrow F = EQ = (3\times10^4)(6\times10^{-6})$$

$$F = 0.18N$$

67. What is the capacitance of a capacitor that should be connected with 2 other capacitors having capacitance of 4μF and 8μF all in series to get an effective capacitance of $\frac{24}{11}\mu F$?

A) 12μF B) 10μF C) 8μF D) 4μF

Answer: C

Solution

$$\frac{1}{C_T} = \frac{1}{C} + \frac{1}{4} + \frac{1}{8}$$

$$\frac{1}{24/11} = \frac{1}{C} + \frac{3}{8}$$

$$\frac{1}{C} = \frac{3}{8} - \frac{11}{24} = \frac{2}{24}$$

$$C = 12\mu F$$

Answer: A

68. Two metal balls with charge 2Q and - Q are separated by some distance and exert a force F on each other. They are joined by a wire which is then removed. What will be the force between them after the process?

A. F B. $\frac{F}{2}$ C. $\frac{F}{4}$ D. $\frac{F}{8}$

Solution

$$F_1 = \frac{KQ_1Q_2}{T^2} \Rightarrow F = \frac{2KQ^2}{r^2}$$

$$F_2 = \frac{K\left(\frac{Q}{2}\right)\left(\frac{Q}{2}\right)}{r^2} = \frac{KQ^2}{4r^2}$$

$$F_2 = \frac{1}{8}F$$

Answer: D

69. A cube of side b has charge Q at each of its vertices. The field at the canter of the cube will be

A. $\frac{Q}{b^2}$ B. $\frac{Q}{2b^2}$ C. $32\frac{Q}{b^2}$ D. Zero

Hint: From symmetry, the fields cancel out at the centrer of the cube.

Answer: D

70. The ration of the electrostatic force between a 1μc and 100μc charge will be?

A) 1:100 B) 1:50 C) 1:25 D) 1:1

Hint: $E = \frac{v}{d} = \frac{KQ_1Q_2}{r^2}$

Answer: D

71. Two plates are 1cm apart. If the potential difference between them is 10v, What is the field strength?

A) 10N/c B) 500N/c C) 1000N/c D) 250N/c

Solution

$$E = \frac{v}{d} = \frac{10}{0.01} = 1000N/c$$

Answer: C

72. Charge Q is divided in to two parts which are then kept some distance

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apart. The force between them will be maximum if the two parts are:

A. $\frac{Q}{2}$ each C. $\frac{Q}{3}$ and $\frac{2}{3}Q$

B. $\frac{Q}{4}$ and $\frac{3}{4}Q$ D. Can not be determined

Hint:- $F \sim q\,(Q - q)$

The force is maximum when $q = Q - q$

$$2q = Q \Rightarrow q = \frac{Q}{2}$$

Answer: A

73. When 10^{19} electrons are removed from a metal plate, the charge on it became?

A) 1.6c B) -1.6c C) 16c D) -16c

Solution

$Q = ne = (10^{19})(1.6 \times 10^{-9})$

$Q = 1.6c$

Answer: A

74. Proton is about 1836 times massive than an electron. If the repulsive force between two protons is F, then the electrostatic force between two electrons will be?

A. F B. 1836F C. 1836 D. $\frac{F}{1836}$

Hint: Electrostatic force does not depend on mass

Answer: A

75. The electric field required to exert a force on a proton equal to its weight is ?

A) 1.67×10^{-27} N/c C) 1.6×10^{-9} v/m

B) 1.02×10^{-7} N/c D) 1.04×10^{-27} v/m

Solution

$EQ = mg$

$$E = \frac{mg}{Q} = \frac{(1.67 \times 10^{-27})(10)}{1.6 \times 10^{-19}} = 1.02 \times 10^{-7}\ v/m$$

Answer: B

76. Which of the following is equivalent to SI unit of electric potential?

A) Nmc^{-1} B) N^2mc^{-1} C) N^3mc^{-2} D) Nm

Solution

$$1V = 1\frac{J}{c} = \frac{1N}{C} = 1Nmc^{-1}$$

Answer: A

77. A -5c of charge is carried from infinity to a point in a field and perform 100J of work. The potential at the given point is?

A) 10V B) -10V C) 20V D) -20V

Solution

$$V = \frac{W}{Q} = \frac{-100}{5} = -20V$$

Answer: D

78. A nucleus has a charge of 50e. What is its electric potential at a distance of 10^{-12}m?

A) 14.4×10^4v C) 7.4×10^8v

B) 7.2×10^4v D) 14.4×10^8v

Solution

$$V = \frac{KQ}{r} = \frac{(9 \times 10^9)(50 \times 1.6 \times 10^{-19})}{10^{-12}}$$

$V = 7.2 \times 10^4 V$

Answer: B

79. Charges q of and -q are placed at the two vertices of an isosceles triangle. What is the electric potential at point A

A. $K\frac{2q}{\sqrt{a^2 + b^2}}$ B. Zero C. $\frac{Kq}{\sqrt{a^2 + b^2}}$ D. $\frac{-Kq}{\sqrt{a^2 + b^2}}$

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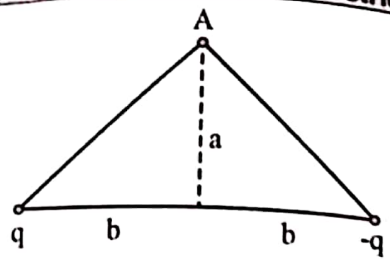


Solution

$$V = V_1 + V_2$$

$$V = \frac{Kq}{\sqrt{a^2 + b^2}} \quad \frac{-Kq}{\sqrt{a^2 + b^2}} = 0$$

Answer: B

80. Two charges of 1μc each are kept at a distance of 1m from each other. Their potential energy is?

A) 1J B) 1.6×10^{-19}J C) 9×10^{-3} J D) 9×10^{-9}J

Solution

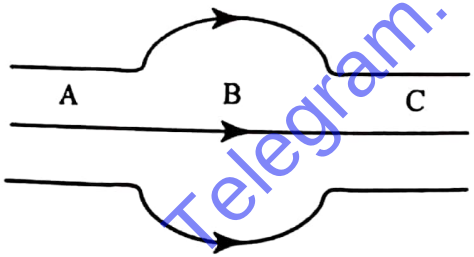
$$u = \frac{KQ_1Q_2}{r} = \frac{(9 \times 10^9)(10^{-6})(10^{-6})}{1} = 9 \times 10^{-3}\ J$$

Answer: C

81. The figure represents electric field lines. The figure suggests that:

A) $E_A > E_B > E_C$ C) $E_A = E_C > E_B$

B) $E_A = E_B = E_C$ D) $E_A = E_c < E_B$



Hint : The close the field lines, the strong the intensity

Answer: C

82. On bringing 0.4c of charge from infinity to point P, the work done is 20J. The potential at point P is?

A) 50V B) 8V C) 1000V D) 160V

Solution

$$V = \frac{W}{Q} = \frac{20}{0.4} = 50V$$

Answer: A

83. Absolute potential of a point charge is?

A) Inversely proportional to the square of the distance

B) Directly proportional to the square of the distance

C) Inversely proportional to the distance

D) Directly proportional to the distance

Hint: $V = \frac{KQ}{r} \Rightarrow V \sim \frac{1}{r}$

Answer: C

84. Work done in moving a positive charge on an equipotential surface is?

A) Positive B) Negative C) Zero D) Infinite

Hint: W = Qv, for equipotential surface V = 0

W = 0

Answer: C

6.7 What is Electric Current?

The flow of charge particles through a conductor just like water molecules flow through a pipe is called **electric current**

Electric current: Is the flow of charge through a conductor per unit time

If a charge of ΔQ flows across a given cross sectional area of a conductor in a time Δt, then the current flowing in the conductor is obtained by.

$$\boxed{I = \frac{\Delta Q}{\Delta t}}$$

Electric current is a scalar quantity and its SI unit is ampere (A)

$$IA = \frac{1c}{s}$$

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Critical Thinking Question

85. If electrons keep going through a wire, will the wire get thinner and thinner as electrons move from it? Explain

Explanation:- No! The wire thickness will not be affected because, the electrons only circulate through the wire and do not escape from it.

Current Density and Drift Velocity

Current Density

The quantity that is used to describe the magnitude of electric current through a given cross - sectional area is called **current density**

Current Density: Is the magnitude of electric current per cross - sectional area

That is

$$\text{Current density} = \frac{\text{Electric current}}{\text{Cross - sectional area}}$$

$$\boxed{J = \frac{I}{A}}$$

Current density is found to be proportional to the electric field that produces it and the conductivity of the conductor through which it flows

Current density = (conductivity) (Electric field)

$$\boxed{J = \delta E = \frac{E}{\delta}}$$

Where: p (rho) is the opposite of conductivity and is called resistivity of the material. Current density is a vector quantity and its SI unit is amper per square meter **(A/m²)**

86. How is the direction of current density determined?

Explanation: The direction of current density is in the direction of the electric field.

87. What is the current density if a current of 5mA flows through a wire of cross - sectional area 10mm²

Solution

$$J = \frac{I}{A} = \frac{5 \times 10^{-3}}{10 \times 10^{-6}}$$

$$J = 500 A/m^2$$

88. What is the current density in a wire of conductivity $2 \times 10^{-2}/\Omega$ When a 6×10^5N/c of electric field is supplied across its terminals?

Solution

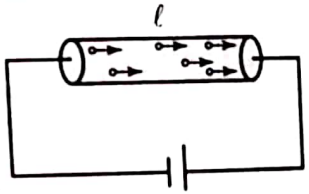
$$J = \delta E \ (2 \times 10^{-2}) \ (6 \times 10^5)$$

$$J = 1.2 \times 10^4 A/m^2$$

Drift Velocity

When an electric field is established inside a conductor by connecting its ends with a battery, the free electrons are forced to move with an average velocity called **drift velocity.**

Drift Velocity: Is the average velocity with which electrons move through a conductor when an electric field is applied across the conductor



To find the drift velocity (V_d) of electrons, suppose there are n free electrons per volume, and during time t, electrons flow across a conductor of length ℓ andcross - sectional area A

The total charge across the given volume of conductor is then obtained by:

Total charge = (Number of electrons) (Charge of electron)

$$\Delta Q = (nA V_d \ \Delta t)(q)$$

$$Vd = \frac{\Delta Q}{nA\Delta tq} = \frac{\Delta Q}{\Delta t}\left(\frac{1}{Aqn}\right)$$

$$\boxed{V_d = \frac{I}{Aqn} = \frac{J}{nq}} \longleftarrow \text{dreift velocity}$$

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Critical Thinking Question

89. The current density of wire A and B is the same. wire A has twice the number density of wire B. What will be the drift velocity of electrons in wire A as compared to that of wire B?

Explanation

$$Vd = \frac{J}{qn}$$

$$\text{Wire A, } V_A = \frac{J_A}{qn_A}$$

$$\text{Wire B, } V_B = \frac{J_B}{qn_B}$$

$$\text{And, } J_A = J_B = J,\ n_A = 2n_B$$

$$V_A = \frac{J}{q2n_B} = \frac{1}{2}\frac{J}{qn_B} = \frac{V_B}{2}$$

$$V_A = \frac{1}{2}V_B$$

So you see, drift velocity is inversely proportional to the number density (n)

90. Wire 4mm in diameter transfers a charge of 960c in 80min. If there are 5.8×10^{28} free electrons per cubic meter, find

A) The current

B) The drift velocity

Solution

$$\text{A) } I = \frac{Q}{t} = \frac{960}{(80)(60)}$$

$$I = 0.24A$$

$$\text{B) } V_d = \frac{I}{nqA} = \frac{0.2}{(5.8 \times 1028)(1.6 \times 10^{-19})(3.14(2 \times 10^{-3}))}$$

$$V_d = 1.7 \times 10^{-6} m/s$$

91. What is the drift velocity of electron in a wire of cross - sectional area $8mm^2$ carrying a current 0.5A when there are 5×10^{28} free electrons per cubic meter?

Solution

$$V_d = \frac{I}{nqA}$$

$$V_d = \frac{0.5}{(5 \times 10^{28})(1.6 \times 10^{-19})(8 \times 10^{-6})}$$

$$V_d = 7.8 \times 10^{-6} m/s$$

So you see, electrons move through a conductor slowly!

Illustrative Examples

92. How long will it take for a charge of 60c to pass through a conductor carrying a current of 3A?

Solution

$$Q = \frac{Q}{t} \Rightarrow t = \frac{Q}{I}$$

$$t = \frac{60}{3} = 20 \text{seconds}$$

93. How many electrons pass through a conductor in 5 seconds, if a current of 8A is observed?

Solution

$$I = It, \text{ and}$$

$$Q = ne$$

$$n = \frac{It}{e} = \frac{(8)(5)}{1.6 \times 10^{-19}}$$

$$n = 2.5 \times 10^{20} \text{ electrons}$$

Ohm's Law

For a metal wire at consant temperature, the current that flows through the wire is directly proportional to the voltage supplied across the ends of the

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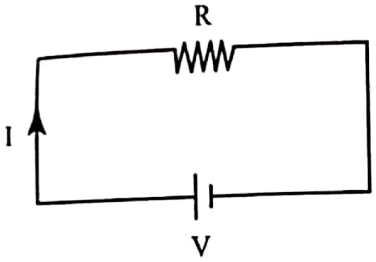
conductors. And, this fact is explained by ohm's law

Ohm's Law States that "The current that flows through a conductor at a given temperature is directly proportional to the voltage supplied across the ends of the conductor.

That is, for a given simple circuit of the form



$$\text{Current} = \frac{\text{Voltage}}{\text{Resistance}}$$

$$I = \frac{V}{R}$$

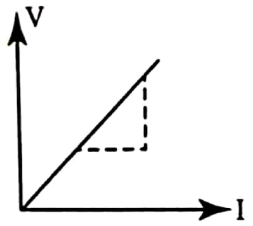
$$\boxed{V = IR} \leftarrow \text{Ohm's law}$$

Where R = is called resistance of the conductor which is the measure of opposition of a material for the flow of electric current through it.

Resistance is scalar quantity and its SI unit is ohm (Ω)

$$1\Omega = \frac{1V}{A}$$

Materials such as Aluminium and copper that obey ohm's law are called **ohmic**, and the current versus voltage graph is linear



$$\text{Slope} = \frac{\Delta y}{\Delta x}$$

$$R = \frac{V}{I}$$

Slope of the V - I curve represents resistance of the conductor

Illustrative Examples

94. What voltage is needed to pass 8A of current through a conductor of resistance 40Ω

Solution

$$V = IR = (80)(40)$$

$$V = 320V$$

95. How long will it take for 60c of charge to pass through a resistor of 12Ω resistance when a voltage of 3v is applied across its ends?

Solution

$$I = \frac{V}{R} = \frac{3}{12} = 0.25A$$

$$I = \frac{Q}{t} \Rightarrow t = \frac{Q}{I}$$

$$t = \frac{60}{0.25} = 240\text{second}$$

96. When a potential difference of 27V is applied on a resistor a charge of 5μc passes in half hour. What is the resistance of the resistor?

Solution

$$I = \frac{Q}{t} = \frac{5 \times 10^{-6}}{(30)(60)} = 0.27 \times 10^{-9}A$$

$$V = IR \Rightarrow R = \frac{V}{I} = \frac{27}{0.27} \times 10^{-9}$$

$$R = 10^{-11}\Omega$$

Critical Thinking Question

97. A bird can harmlessly stand on one wire of a high voltage line. But, it has better not reach over and grab a near by wire. why not?

Explanation: Electric shock requires a difference in potential between one part and the other part of a body. So, as long as the bird touch nothing else except a wire, it will receive no shock at all. But if it reach over and grab the other wire, a sudden explosion that leave the bird in to pieces will be heard

6.8 What is Electric Circuit?

A set of electrical components may be connected so that they provide a circuit which is a closed path for the flow of electric path.

Electric circuit: Is a complete path along which an electric current flows.

Some of the most common electrical elements found in a simple circuit;

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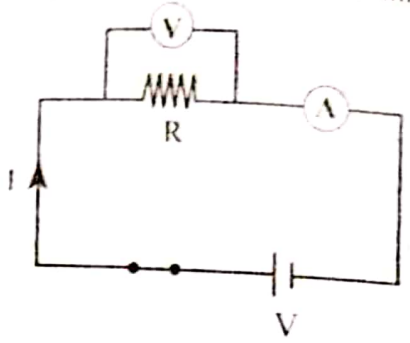
light bulb, battery, switch, voltmeter, and ammeter.



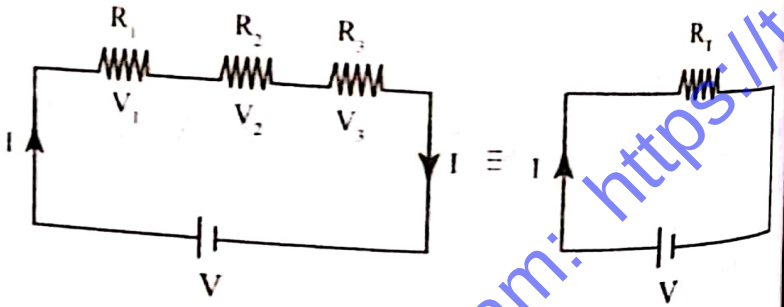
The switch helps us to open or make a circuit OFF or to close or make it ON.

Combination of Resistors

Depending on the desired amount of resistance in a circuit, resistor may be connected in series, in parallel or both.

Resistors in Series

When resistors are connected in series they are connected one after the other as shown below



In the series combination of resistors:

- The total current in the system is the same as the current that flows through each resistor.

$$I_1 = I_2 = I_n = I_T = I$$

- The total potential drop in the net work is equal to the sum of the potential drops a cross each resistor.

$$V_T = V_1 + V_2 + \text{------} + V_n$$

And, from ohm's law; $V = IR$. So we have

$$I_T R_T = I_1 R_1 + I_2 R_2 + \text{--------} + I_n R_n$$

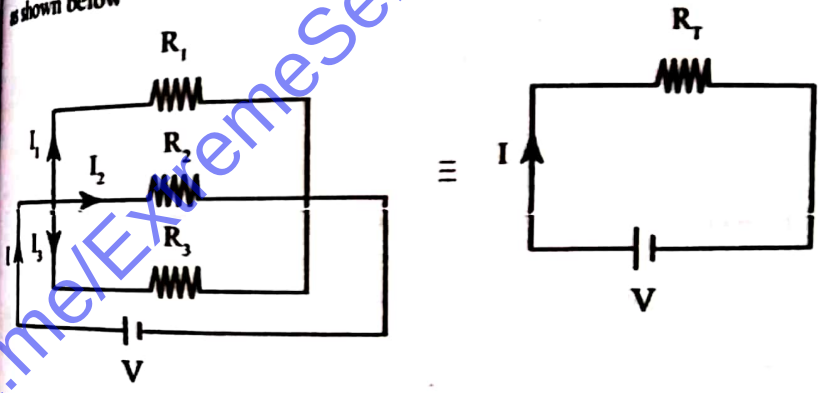
But the current is the same and can be cancelled out from both sides of the above equation

$$R_T = R_1 + R_2 + \text{--------} + R_n$$

This the equivalent resistance of n resistors connected in series

Resistors in parallel

When resistors are connected in parallel, they are connected face to face as shown below



In the parallel combination of resistors

- The total potential drop in the system is the same as the potential drop across each resistor.

$$V_1 = V_2 = V_n = V_T = V$$

- The total current flowing in the network is equal to the sum of the current that flows through each resistor.

$$I_T = I_1 + I_2 + \text{---} + I_n$$

And from ohm's law, $I = \frac{V}{R}$ so law have

$$\frac{V_T}{R_T} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \text{------} + \frac{V_n}{R_n}$$

But, the potential drop is the same and can be cancelled out from both sides of the above equation

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \text{------} + \frac{1}{R_n}$$

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This is equivalent resistance of n resistors connected in parallel.

Critical Thinking Question

98. What happens to the other lamps if one lamp burns out in a
A) Series circuit
B) Parallel circuit

Explanation:

A) If one of the lamps in a series circuits burns out, then the circuit will break and the light of all the other lamps will go out.

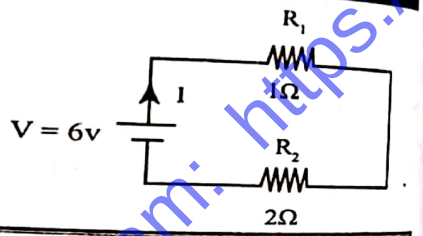
B) If one of the lamps in a parallel circuit burn out, then the other lamps will not be affected and hence they remain lit.

99. Is it the series or the parallel combination of resistors that gives a larger equivalent resistance? Explain

Explanation: The equivalent resistance of resistors in series is greater than that of resistors in parallel. The series combination gives a resistance that is the sum of the resistance of the individual resistors. But, parallel combination gives a resistance that is reciprocal sum of the individual resistances in the circuit and it is always less than the resistances of each resistor.

Illustrative Examples

100. For the circuit shown, find
A) The total resistance
B) The potential drop across each resistor



Solution

A) $R = R_1 + R_2 = 1 + 2$

$R = 3\Omega$

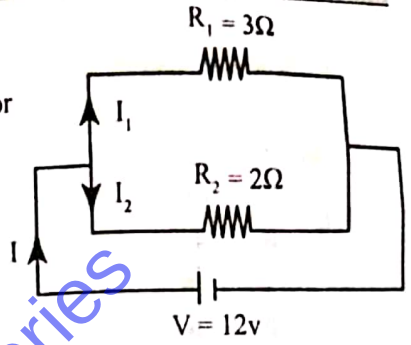
B) $I = \frac{V}{R} = \frac{6}{3}$

$I = 2A$

$V_1 = IR_1 = (2)(1) = 2V$

$V_2 = IR_2 = (2)(2) = 4V$

101. For the circuit shown, find
A) The total resistance
B) The current through each resistor



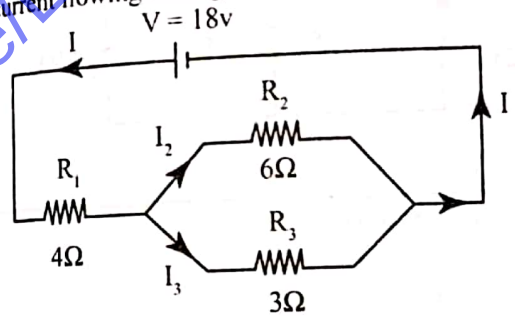
A) $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$

$R = 2\Omega$

B) $I_1 = \frac{V}{R_1} = \frac{12}{3} = 4A$

$I_2 = \frac{V}{R_2} = \frac{12}{6} = 2A$

102. From the following circuit, find
A) The total resistance
B) The potential drop across each resistor
C) The current flowing through each resistor



Solution

A) R_2 and R_3 are in parallel

$\frac{1}{R_{23}} = \frac{V}{R_2} + \frac{1}{R_3} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$

$R_{23} = 2\Omega$

$R = R_1 + R_{23} = 4 + 2$

$R = 6\Omega$

B) The total current, $I = \frac{V}{R} = \frac{18}{6} = 3A$

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p.d across R_1, $V_1 = I_1R_1 = I_1 R_1 = (3)(4) = 12v$

P.d across R_2 and R_3 are same

$$V_2 = V_3 = V - V_1 = 18 - 12 = 6V$$

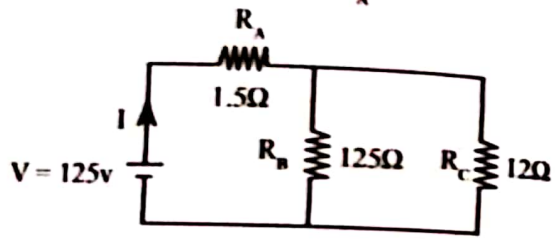
Therefore, $V_1 = 12V$, $V_2 = 6V$ and $V_3 = 6V$

C) $I_1 = \frac{V_1}{R_1} = \frac{12}{4} = 3A$

$I_2 = \frac{V_2}{R_2} = \frac{16}{6} = 1A$

$I_3 = \frac{V_3}{R_3} = \frac{6}{3} = 2A$

103. What is the potential drop across R_A?



Solution

R_B and R_C are in parallel,

$$\frac{1}{R_{BC}} = \frac{1}{R_B} + \frac{1}{R_C} = \frac{1}{125} + \frac{1}{12}$$

$R_{BC} = 11\Omega$

Then, R_A and R_{BC} became in series

$R = R_A + R_{BC} = 1.5 + 11$

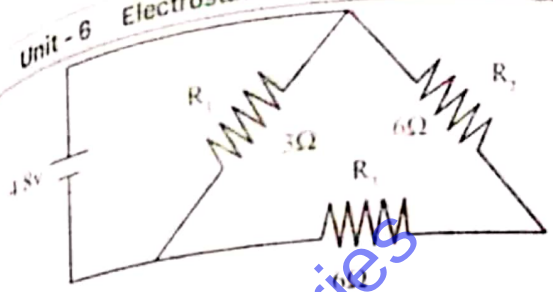
$R = 12.5\Omega$

The circuit current, $I = \frac{V}{R} = \frac{125}{12.5} = 10A$

And, $V_A = IR_A = (10)(1.5)$

$V_A = 15V$

104. What is the current in the circuit shown?



Solution

R_2 and R_3 are in series

$R_{23} = R_2 + R_3 = 6 + 6 = 12$

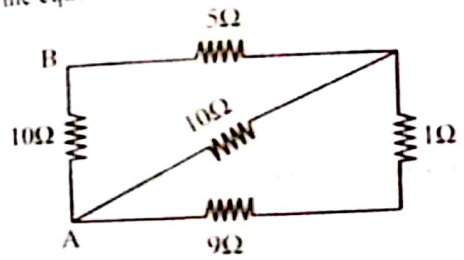
Then, R_1 and R_{23} became in parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_{23}} = \frac{1}{3} + \frac{1}{12} = \frac{5}{12}$$

From ohm's law, $I = \frac{V}{R} = \frac{4.8}{2.4}$

$I = 2A$

105. What is the equivalent resistance between points A and B?



Solution

The 1Ω and the 9Ω are in series and their result is 10Ω

The two 10Ω resistors are in parallel and their result is 5Ω

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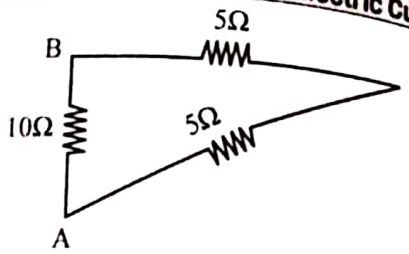
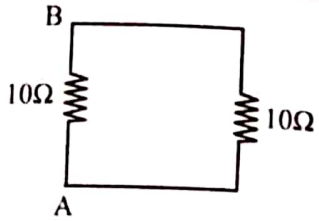
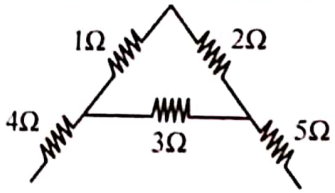


Their two 5Ω resistors are in series and their result is 10Ω



The last two 10Ω resistors are in parallel and their result is 5Ω

106. What is the equivalent resistance of the circuit shown



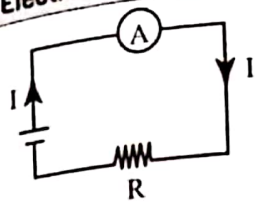
Answer: R = 10.5Ω

6.9 Electric Measuring Instruments

Current, voltage and resistance in a circuit are measured using Ammeter, voltmeter and ohmmeter respectively.

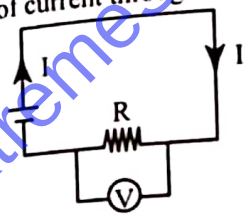
Ammeter —(A)—

A current measuring instrument is called an **Ammeter**. It is connected in series with the resistor and should have a very small internal resistance so that its presence should not alter the current in the circuit

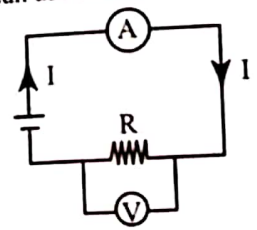


Voltmeter, —(V)—

A voltage measuring instrument is called **Voltmeter** It is connected in parallel with the resistor and should have a very large internal resistance so that the passage of current through it should not desturb its function.



Therefore, when we want to measure both the current and the voltage at the same time, we can use multimeter or ammeter and voltmeter as follows



Critical Thinking Questions

107. What should be the internal resitance of an ideal

A) Ammeter
B) Voltmeter

Explanation A) zero B) Infinite

108. Which one of the following is the correct connection of ammeter and voltmeter in a circuit with the resistor?

A) Ammeter and voltmeter in series
B) Ammeter in series, voltmeter in parallel
C) Both in series

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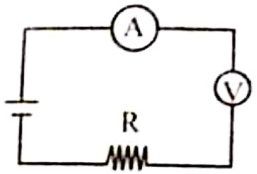
D) Both in parallel

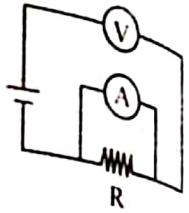
109. Choose the correct connection of Ammeter and voltmeter

Answer: B

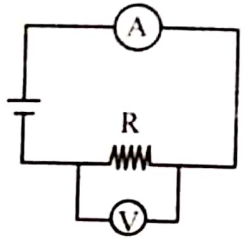
A)



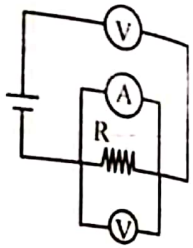
B)



C)



D)



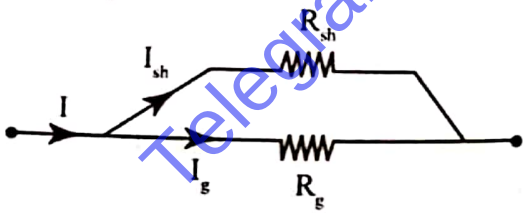
Answer: C

Conversion of Galvanometer in to an Ammeter and Voltmeter

Galvanometer is a very small usually 5mA electric current measuring instrument

Converting a Galvanometer in to an Ammeter

To modify a galvanometer in to an Ammeter, a resistor called *shunt* of very small resistance (R_{sh}) is connected in parallel with the galvanometer of internal resistance (R_g) as shown below:



Since voltage is constant in parallel combination of resistors, we have;

$$V_{sh} = V_g$$

$$I_{sh} R_{sh} = I_g R_g$$

$$R_{sh} = \frac{I_g R_g}{I_{sh}}, \quad I_{sh} = I - I_g$$

$$\boxed{R_{sh} = \frac{I_g R_g}{I - I_g}}$$

Where: R_{sh} = Shunt resistance

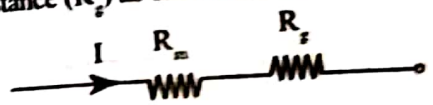
I_g = Fullscale deflection current

I = Current to be measured

R_g = Internal resistance of the galvanometer

Converting a Galvanometer in to a Voltmeter

To modify a galvanometer in to a voltmeter, a resistor called *multiplier* of very large resistance (R_m) is connected in series with the galvanometer of internal resistance (R_g) as shown below



Since current is constant in series combination of resistors, We have;

$$I_m = I_g$$

$$\frac{V_m}{R_m} = \frac{V_g}{R_g}$$

$$R_m = \frac{V_m R_g}{V_g}, \quad V_m = V - V_g$$

$$R_m = \frac{(V - V_g)}{V_g}, \quad V_g = I_g R_g$$

$$\boxed{R_m = \frac{V - I_g R_g}{I_g}}$$

Where: R_m = Multiplier resistance

I_g = Fullscale deflection current

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V = Voltage to be measured

R_g = Internal resistance of the galvanometer

Illustrative Examples

110. A galvanometer of full scale deflection 5mA is to be converted in to a 10A Ammeter. If the internal resistance of the galvanometer is 50Ω. What value of shunt must be connected?

Solution

$$R_{sh} = \frac{I_g R_g}{I - I_g} = \frac{(0.0005)(50)}{(10-0.0005)}$$

$$R_{sh} = 0.025\Omega$$

111. A galvanometer of full scale deflection 5mA is to be converted in to a 10V voltmeter. If the internal resistance of the galvanometer is 50Ω. What value of multiplier should be connected?

Solution

$$R_m = \frac{V - I_g R_g}{I_g} = \frac{10 - (0.0005)(50)}{0.005}$$

$$R_m = 1950\Omega$$

112. Which on of the following is true statement?

A) To convert galvanometer in to Ammeter, shunt resistor of large resistance is connected in series with it

B) To convert galvanometer in to Ammeter, shunt resistor of small resistance is connected in series with it

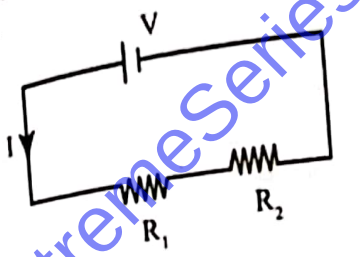
C) To convert galvanometer in to Ammeter, shunt resistor of large resistance is connected in parallel with it

D) To convert galvanometer in to Ammeter, shut resistor of small resistance is connected in parallel with it

Answer: D

Potential Divides

A potential divider consists of two or more resistors in series and a source of potential difference that is divided between the resistors in the circuit as shown below.



The total resistance of the circuit is;

$$R_T = R_1 + R_2$$

The current flowing through the circuit is;

$$I = \frac{V}{R_T} = \frac{V}{R_1 + R_2}$$

The potential drop across resistor R_2 is;

$$V_1 = IR_1 = \left(\frac{R_1}{R_1 + R_2}\right)V$$

The potential across resistor R_2 is;

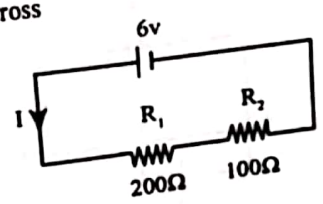
$$V_2 = IR_2 = \left(\frac{R_2}{R_1 + R_2}\right)V$$

Illustrative Example

113. What is the output voltage across

A) The 200Ω resistor

B) The 100Ω resistor



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Solution

A) $V_2 = \left(\frac{R_2}{R_1 + R_2}\right)V = \left(\frac{200}{200 + 100}\right)6 = 4V$

B) $V_1 = \left(\frac{R_1}{R_1 + R_2}\right)V = \left(\frac{100}{200 + 100}\right)6 = 2V$

What is Potentiometer ?

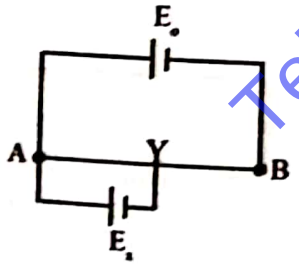
Potentiometer is a three terminal resistor with a sliding contact that forms an adjustable potential divider.

A Potentiometer can be used:

- To measure emf of a cell
- To measure internal resistance of a cell
- To compare emf of different cells
- As a variable resistor

Potentiometer to Measure emf of a Cell

Suppose we want to measure emf of a cell E_x given the emf of a driver cell E_o



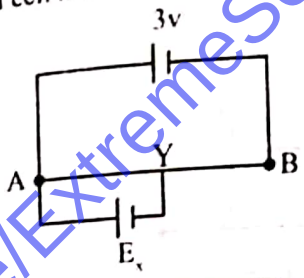
Let point Y is a point of zero voltage, so we can have

$$\frac{E_x}{AY} = \frac{E_o}{AB}$$

$$\boxed{E_x = E_o\left(\frac{AY}{AB}\right)} \leftarrow \text{emf of cell x}$$

Illustrative Example

114. Find the emp of cell x shown below given that $AY = \frac{1}{3}AB$



Solution

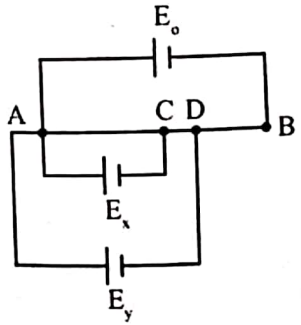
$$E_x = Eo\left(\frac{AY}{AB}\right)$$

$$E_x = 3\left(\frac{1}{3}\frac{AB}{AB}\right) = \frac{3}{3}$$

$$E_x = 1V$$

Potentiometer to Compare Emf of Two Cells

Suppose we have two cells of emfs E_x and E_y connected to potentiometer and give balance or zero voltage points at C and D



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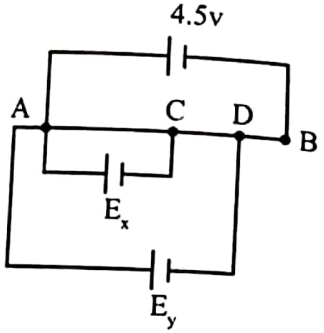
Then we have:

$$\frac{E_x}{E_y} = \frac{AC}{AD}$$

Hence, the ratio of the emf of two cells is equal to the ratio of the lengths of balance points

Illustrative Examples

115. Compare the values of the emfs of cell x and cell y given that $Ac = \frac{1}{3}AB$ and $AD = \frac{2}{3}AB$



Solution

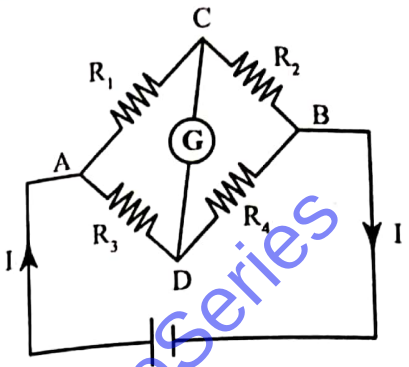
$$\frac{E_x}{E_y} = \frac{AC}{AD}$$

$$\frac{E_x}{E_y} = \frac{\frac{1}{3}AB}{\frac{2}{3}AB} = \frac{1}{2}$$

$$\frac{E_x}{E_y} = \frac{1}{2} = 1:2$$

The Wheatstone Bridge

A Wheatstone bridge is an arrangement of four resistors as shown below and is said to be balanced when there is no potential difference between points C and D and hence no current flows through the galvanometer

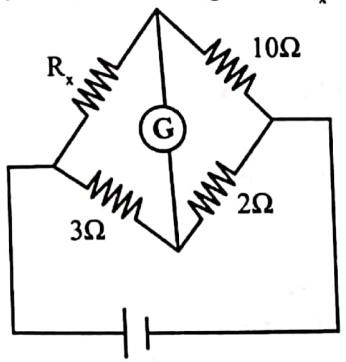


For a balanced Wheatstone, we have

$$\frac{R_1}{R_3} = \frac{R_2}{R_4}$$

Illustrative Examples

116. For the balanced Wheatstone bridge, find R_x



Solution

$$\frac{R_x}{3} = \frac{10}{2}$$

$$R_x = 15\Omega$$

117. The circuit shown represents a balanced Wheatstone bridge. Calculate the value of resistance R

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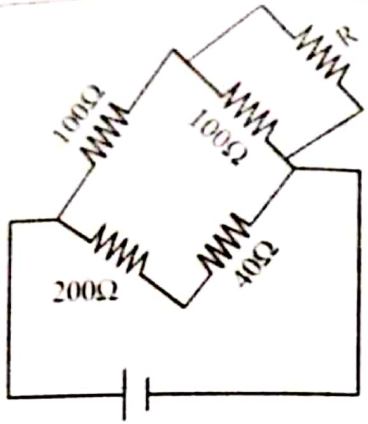


Solution

$$\frac{100}{200} = \frac{R'}{40}$$

$$R' = 20\Omega$$

$$\frac{1}{R'} = \frac{1}{R} + \frac{1}{100} \Rightarrow \frac{1}{20} = \frac{1}{100} + \frac{1}{R}$$

$$R = 25\Omega$$

6.10 Kirchhoff's Rules

To analyse the current, voltage and resistance of complex circuits, kirchhoff established two simple rules named as junction rule and loop rule.

Kirchhoff's Fist Rule

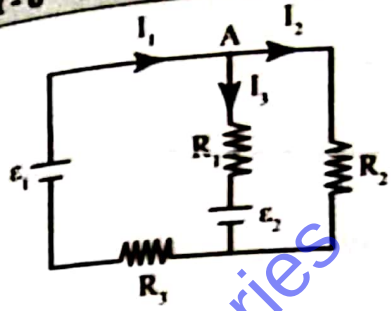
Kirchhoff's first rule also known as *junction or point rule* is based on the conservation of charge and is stated as follows:

> The total current entering a junction is equal to the total current leaving the junction

That is, the sum of currents flowing in to a point is equal to the sum of currents flowing out of that point.

$$\boxed{\Sigma I_{in} = \Sigma I_{out}}$$

As an example, consider the following simple circuit



Now, according to kirchhoff's junction rule, current I_1 enters point A while, current I_2 and I_3 leave the point. Therefore, We can have:

$$\Sigma I_{in} = \Sigma I_{out}$$

$$I_1 = I_2 + I_3$$

Kirchhoff's Second Rule

Kirchhoff's second rule also known as loop rule is based on the conservation energy and is stated as follows:

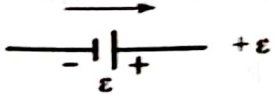
> The total potential rise in a closed loop of a circuit is equal to the total potential drop in that loop

That is, the sum of emfs in a loop is equal to the sum of potential drops in that loop.

$$\boxed{\Sigma\varepsilon = \Sigma IR}$$

While using kirchhoff's loop rule, we need to stick to the following sign conventions as we move clock wise or counter - clock wise around the loop.

- If we move from the negative terminal to the positive terminal of an electric cell, emf is taken as positive.



- If we move from the positive terminal to the negative terminal of an electric cell, emf is taken as negative

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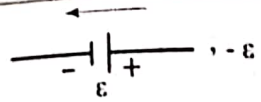
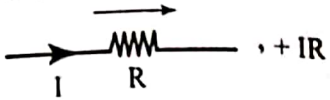
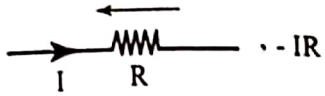


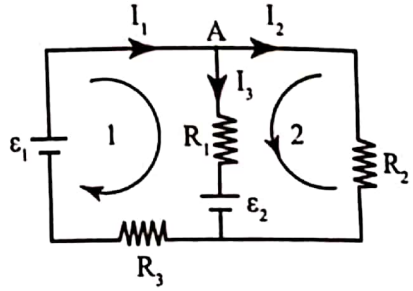
- If a resistor is traversed in the direction of the current flowing through it, potential drop is taken as positive



- If a resistor is traversed opposite to the direction of the current flowing through it, potential brop is negative



As an example, consider the following simple circuit



Now, applying kirchhoff's loop rule we can have the following

For loop - 1, which is traversed clock - wise

$$\Sigma\varepsilon = \Sigma IR$$

$$\varepsilon_1 - \varepsilon_2 = I_1 R_3 + I_3 R_1$$

For loop - 2, which is traversed counter - clockwise

$$\Sigma\varepsilon = \Sigma IR$$

$$-\varepsilon_2 = -I_2 R_2 + I_3 R_1$$

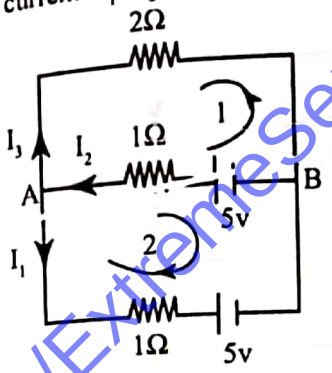
Critical Thinking Question

118. Explain how the current that enters a junction become the same as the current that leaves the junction

Explanation: Because, current is the rate of flow of charge per unit time across a conductor and as charge is neither created nor destroyed, the number of charged particles that flow in to a point will be equal to the charged particles that flow out of that point.

Illustrative Examples

119. Find the currents I_1, I_2 and I_3



Solution

Applying kirchhoff's rules

Junction rule, choose point A

$$\Sigma I_{in} = \Sigma I_{out}$$

$$I_2 = I_1 + I_3 \text{ ------ (1)}$$

Loop rule, loop - 1

$$\Sigma\varepsilon = \Sigma IR$$

$$-5 = -I_2 - 2I_3$$

$$I_2 + 2I_3 = 5 \text{ ---------- (2)}$$

Loop rule, loop - 2

$$\Sigma\varepsilon = \Sigma IR$$

$$-5 + 5 = -I_1 - I_2$$

$$I_1 = -I_2 \text{ ---------- (3)}$$

Substitute equation (3) in to equation (1)

$$I_2 = -I_2 + I_3$$

$$2I_2 = I_3 \text{ ---------- (4)}$$

Again, substance equation (4) in to equation (2)

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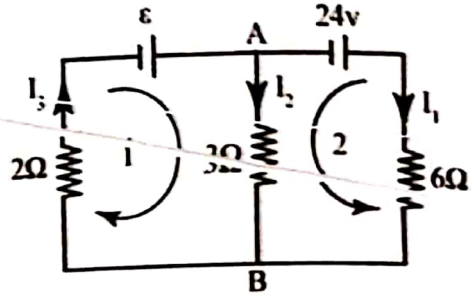
$I_2 + 2(I_2) = 5$

$I_2 = 1A$

$I_1 = -I_2 = -1A$

$I_3 = 2I_2 = 2A$

120. In the circuit shown, $I_1 = 3A$. What are the values of ε, I_2 and I_3?



Solution

Junction rule, point A

$\Sigma I_{in} = \Sigma I_{out}$

$I_3 = I_1 + I_2$ ——— (1)

Loop rule, loop - 2

$\Sigma\varepsilon = \Sigma IR$

$-24 = -6I_1 + 3I_2$

$-24 = -6(3) + 3I_2$

$3I_2 = -6$

$I_2 = -2A$

$I_3 = I_1 + I_2 = (3) + (-2)$

$I_3 = 1A$

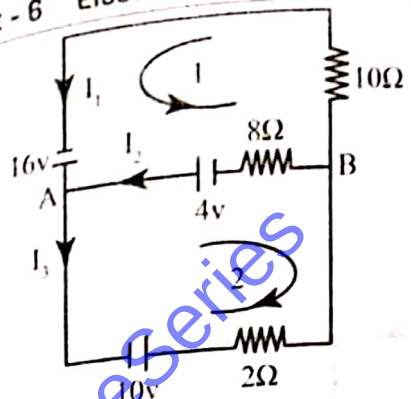
Loop rule, Loop - 1

$\Sigma\varepsilon = \Sigma IR$

$\varepsilon = 2 - 6 = -4$

$\varepsilon = -4v$

121. Find the currents I_1, I_2 and I_3



Solution

Junction rule, point

$\Sigma I_{in} = \Sigma I_{out}$

$I_1 + I_2 = I_3$ ---------- (1)

Loop rule, loop - 1

$\Sigma\varepsilon = \Sigma IR$

$16 - 4 = -10I_1 - 8I_2$

$16 - 4 = -10I_1 - 8I_2$

$12 = -10I_1 - 8I_2$

$5I_1 + 4I_2 = -6$ ---------- (2)

Loop rule, loop - 2

$\Sigma\varepsilon = \Sigma IR$

$-10 - 4 = -8I_2 - 2I_3$

$-14 = -8I_2 - 2I_3$

$4I_2 + I_3 = 7$ ---------- (3)

From equation (1) and equation (3)

$4I_2 + (I_1 + I_2) = 7$

$5I_2 + I_1 = 7$ ---------- (4)

Again, from equation (2) and equation (4) we have;

$5I1 + 4I2 = -6$

$-5(I_1 + 5I_2 = 7)$

$-21I_2 = -41$

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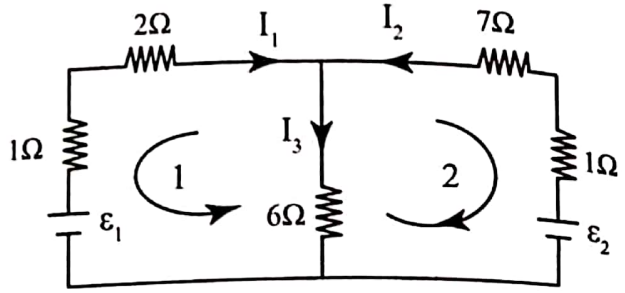
$I_2 = 2A$

$I_1 = -3A$

$I_3 = -1A$

122. For the circuit shown below, determine ε_1 and ε_2 if $I_1 = \frac{1}{3}A$ and $I_2 = \frac{1}{2}A$



Solution

Junction rule

$$\Sigma I_{in} = \Sigma I_{out}$$

$$I_1 + I_2 = I_3$$

$$I_3 = \frac{1}{3} + \frac{1}{2}$$

$$I_3 = \frac{5}{6}A$$

Loop rule, loop - 1

$$\Sigma\varepsilon = \Sigma IR$$

$$-\varepsilon_1 = -6I_3 - 2I_1 - I_1$$

$$-\varepsilon_1 = -6\left(\frac{5}{6}\right) - 2\left(\frac{1}{3}\right) - \frac{1}{3}$$

$$-\varepsilon = \frac{-18}{3}$$

$$\varepsilon_1 = 6V$$

Loop rule, loop - 2

$$\Sigma\varepsilon = \Sigma IR$$

$$-\varepsilon_2 = -6I_3 - 7I_1 - I_2$$

$$-\varepsilon_2 = -6\left(\frac{5}{6}\right) - 7\left(\frac{1}{2}\right) - \frac{1}{2}$$

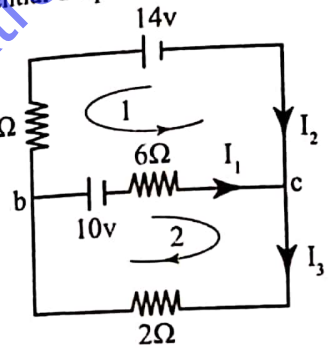
$$-\varepsilon_2 = \frac{-18}{2}$$

$$\varepsilon_2 = 9V$$

123. For the circuit shown, find

A) The currents I_1, I_2 and I_3

B) The potential drop between point b and c



Solution

A) Point rule

$$I_1 + I_2 = I_3 \text{ --------- (1)}$$

Loop - 1

$$10 + 14 = 6I_1 - 4I_2$$

$$3I_1 - 2I_2 = 12 \text{ -------- (2)}$$

Loop - 2

$$10 = 6I_1 + 2I_3$$

$$3I_1 + I_3 = 5 \text{ --------- (3)}$$

Substitute equation (1) in to equation (3)

$$3I_1 + (I_1 + I_2) = 5$$

$$4I_1 + I_2 = 5 \text{ ---------- (4)}$$

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Solve equation (2) and equation (4) at a time

$3I_1 - 2I_2 = 12$

$2(4I_1 + I_2 = 5)$

$11I_1 = 22$

$I_1 = 2A$

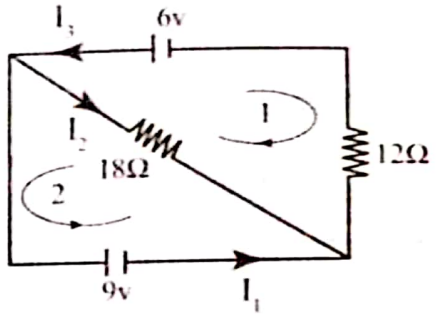
$I_2 = -3A$

$I_3 = -1A$

B) $V_c - V_b = 10 - 6I_1 = 10 - 6(2)$

$V_{cb} = -2V$

124. Find the currents in all the wires of the circuit shown below



Solution

Point rule

$I_3 = I_1 + I_2$ ---------- (1)

Loop - 1

$-6 = -18I_2 - 12I_3$

$3I_2 + 2I_3 = 1$ ---------- (2)

Loop - 2

$9 = -18I_2$

$I_2 = -0.5A$

Substitute equation (1) in to equation (2)

$3I_2 + 2(I_1 + I_2) = 1$

$5I_2 + 2I_1 = 1$

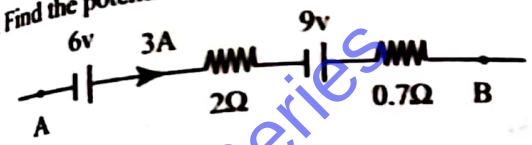
$5(-0.5) + 2I_1 = 1$

$I_1 = 1.75A$

$I_3 = I_1 + I_2 = (1.75) + (-0.5)$

$I_3 = 1.25A$

125. Find the potential difference between points A and B



Solution

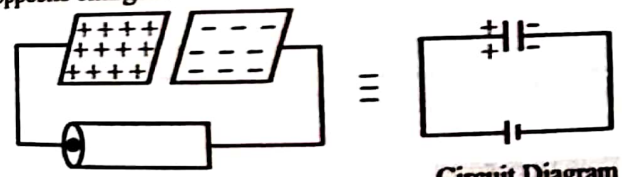
$V_B - V_A = 6 - (3)(2) + 9 - (3)(0.7)$

$V_{BA} = 5.1V$

6.11 Capacitors

Capacitors are devices usually formed by two parallel conductors and are used to store charge and electric energy.

Charging a capacitor is done by connecting the plates with a battery. The battery then pumps charge on to the plates. The plates finally carry equal and opposite charges.



Physical Circuit ≡ Circuit Diagram

The charge stored by the capacitor is obtained by:

$$Q = Cv$$

Where: Q = charge stored

V= potential difference between the plates

C = capacitance of the capacitor

Capacitance: Is the ability of a capacitor to store charge

Capacitance is a scalar quantity and its SI unit is the farad (F)

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$$1F = \frac{1c}{V}$$

Critical Thinking Question

126. What is the net charge stored by a capacitor?

Explanation: The plates of a capacitor carry equal and opposite charges. Therefore, the over all charge stored on the capacitor is zero. Okay!

Illustrative Examples

127. When a voltage of 4.5v is supplied to a capacitor, 5.4μc charge is stored on it. What is the capacitance of this capacitor?

Solution

$$Q = CV$$

$$C = \frac{Q}{V} = \frac{5.4}{4.5} \times 10^{-6}$$

$$C = 1.2\mu F$$

128. A charge of 15μc is stored on each plate of a capacitor of capacitance 5μF. What is the potential difference between the plates of the capacitor?

Solution

$$Q = CV$$

$$C = \frac{Q}{V} = \frac{15}{5}$$

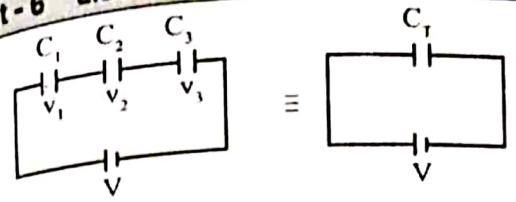
$$C = 3V$$

Combination of Capacitors

Based on the desired value of capacitance in a circuit, capacitors may be connected in series, in parallel or in both.

Capacitors in Series

When capacitors are connected in series, they are joined one after the other as shown below



In the series combination of two or more capacitors.

- The charge stored in the system as a whole is same as the charge stored by each individual capacitor

$$Q_1 = Q_2 = Q_n = Q_T = Q$$

- The total potential drop in the net work is equal to the sum of the potential groups across each capacitor

$$V_T = V_1 + V_2 + \text{------} + Vn$$

And, from the capacitor equation; $Q = CV$, $V = \frac{Q}{C}$

$$\frac{Q_T}{C_T} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \text{----------} \frac{Q_n}{C_n}$$

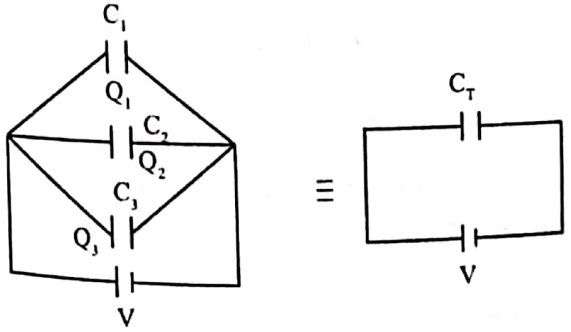
Since charge is constant, we can cancel it out from both side to get:

$$\boxed{\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \text{----------} + \frac{1}{C_n}}$$

This is the total or equivalent capacitance of capacitors connected on series.

Capacitors in Parallel

When capacitors are connected in parallel, they are joined face to face as shown below;



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In the parallel combination of two or more capacitors

- The potential drop across the net work is the same as the potential drop across each individual capacitor

$$V_1 = V_2 = V_n = V_T = V$$

- The total charge stored in the system is equal to the sun of the charges stored by each capacitor

$$Q_T = Q_1 + Q_2 + \text{———} + Q_n$$

And, from the capacitor equation, $Q = cv$

$$C_T V_T = C_1 V_1 + C_2 V_2 + \text{———} + C_n Vn$$

Since is voltage constant, we can cancel it out from both sides to get:

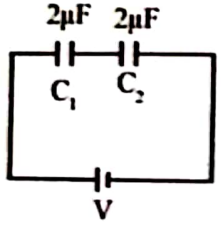
$$\boxed{C_T = C_1 + C_2 + \text{———} + C_n}$$

This is the total or equivalent capacitance of n capacitors connected in parallel

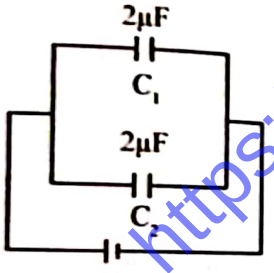
Critical Thinking Question

129. Which capacitor net work has greater capacitance?

A.



B.



Explanation: In case A, the capacitors are connected in series. The total capacitance is then:

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{2} + \frac{1}{2} = \frac{2}{2} = \frac{1}{1}$$

$$C_T = 1\mu F$$

In case B, the capacitors are connected in Parallel. The total capacitance is then;

$$C_T = C_1 + C_2 = 2 + 2$$

$$C_T = 4\mu F$$

Answer: B

- In general, capacitance is greater when capacitors are connected in parallel.

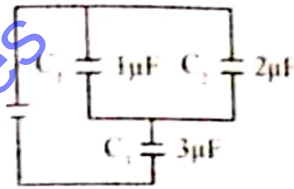
Illustrative Example

130. From the capacitor net work shown below, Find:

A) The capacitance

B) The potential across each capacitor

C) The charge stored on each capacitor



Solution

A) The 1μF and 2μF are in parallel

$C = 1 + 2 = 3\mu F$

C is then in series with the 3μF

$$\frac{1}{C_T} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

$$C_T = 1.5\mu F$$

B) The total charge is equal to the charge stored by the C_3 capacitor

$$Q = Cv = (1.5)(9) = 13.5\mu c$$

$$V_3 = \frac{Q_3}{C_3} = \frac{13.5}{3} = 4.5V$$

$$V_1 = V_2 = V - V_3 = 9 - 4.5 = 4.5V$$

That means, V1 = 4.5v, v2 = 4.5v and v3 = 4.5v

C) $Q_1 = C_1 V_1 = (1)(4.5) = 4.5\mu c$

$Q_2 = C_2 V_2 = (2)(4.5) = 9\mu c$

$Q_3 = C_3 V_3 = (3)(4.5) = 13.5\mu c$

131. For the capacitor system shown below; Find

A) The equivalent capacitance

B) The potential drop across each capacitor

C) The charge stored on each capacitor

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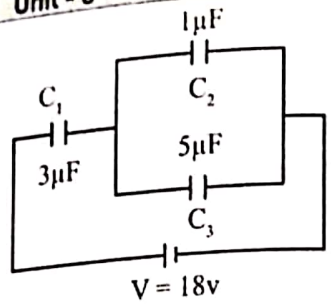


Solution

A) C_2 and C_3 are in parallel

$C_{23} = C_2 + C_3 = 1 + 5 = 6\mu F$

Then, C_1 and C_{23} become series

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_{23}}$$

$$\frac{1}{C_T} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

$C_T = 2\mu F$

B) The total charge is equal to the charge stored by C_1

$Q = CV = 2\,(18) = 36\mu F$

$$V_1 = \frac{Q}{C_1} = \frac{36}{3} = 12V$$

$V_2 = V_3 = V - V_1 = 18 - 12 = 6V$

That means, $V_1 = 12V$, $V_2 = 6V$ and $V_3 = 6V$

C) $Q_1 = C_1 V_1 = (3)(12) = 36\mu c$

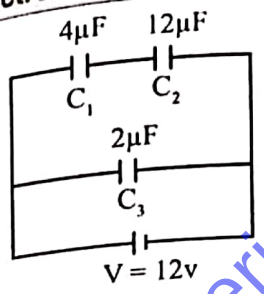
$Q_2 = C_2 V_2 = (1)\,(6) = 6\mu c$

$Q_3 = C_3 V_3 = (5)\,(6) = 30\mu c$

132. For the capacitors net work shown, find

A) The effective capacitance

B) The potential across each capacitor



Solution

A) C_1 and C_2 are in series

$$\frac{1}{C_{12}} = \frac{1}{C_1} + \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$$

$C_{12} = 3NF$

Then C_3 and C_{12} are in parallel

C_T = and $C_{12} + C_3 = 3 + 2 = 5\mu F$

B. $Q = CV = (5)\,(12) = 60\mu F$

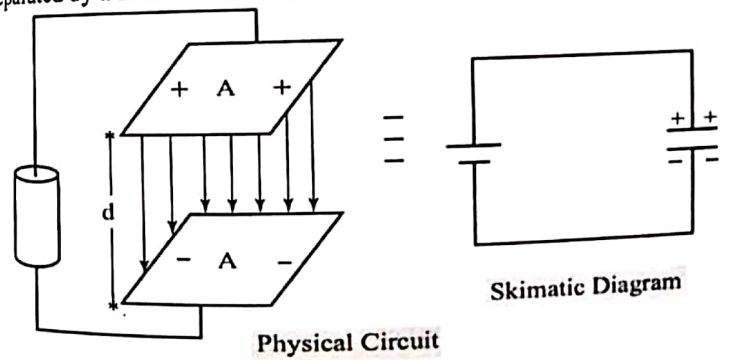
$$V_1 = \frac{Q_1}{C_1} = \frac{36}{4} = 9V$$

$$V_2 = \frac{Q_2}{C_2} = \frac{36}{12} = 3V$$

$V_3 = 12V$

6.12 Parallel Plate Capacitor

A parallel plate capacitor consists parallel plates of equal areas (A) separated by a small distance (d) in which uniform electric field develops



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Each plate of the parallel plate capacity carry the same charges of opposite signs and the charge density on the plates is obtained by

$$\text{Chargedensity} = \frac{\text{Charge stored}}{\text{plate area}}$$

$$\boxed{\delta = \frac{Q}{A}}$$

And, the electric field between the plates is given by

$$\varepsilon = \frac{\delta}{\varepsilon_o} = \frac{Q}{A\varepsilon_o}$$

The potential difference between a parallel plate is

$$V = \varepsilon d \ , \ \varepsilon = \frac{Q}{A\varepsilon_o}$$

$$V = \frac{Qd}{A\varepsilon_o}$$

And, from a capacitor equation we have:

$$Q = CV$$

$$C = \frac{Q}{V} \ , \ V = \frac{Qd}{A\varepsilon_o}$$

$$C = \frac{Q}{\frac{Qd}{A\varepsilon_o}}$$

$$\boxed{C = \frac{\varepsilon_o A}{d}} \longleftarrow \text{Air filled capacitor}$$

This is equation of parallel plate capacitor when the gap between the plates is vacuum or air filled, However, when there is insulating material called dielectrics between the plates, the equation of the capacitance will be modified as follows

$$\boxed{C = K\varepsilon_o \frac{A}{d}} \longleftarrow \text{Capacitor with permitivity}$$

Where: $\varepsilon_o = 8.85 \times 10^{-12} \ \frac{C^2}{Nm^2}$, is called permittivity of free space

K = dielectric constanstent

What is Dielectric?

A dielectric is insulating material such as mica, wax, glass, plastic, rubber .. etc inserted between parallel plates, so as to increase capacitance of the capacitor.

Dielectric Strength

Dielectrics block the flow of charges between plates until the electric field between the plates exceeds a certain value, above that value of electric field, they become conductors.

Dielectric strength: Is the maximum electric field that the dielectric material can with stand without undergoing breakdown and become conductor

The dielectric strength of some materials is given below

Material	Dielectric strength (V/m)
Air	3×10^6
Glass	3×10^7
Mica	2×10^8
Oiled paper	4×10^7
Rubber	2.1×10^7
Water	3×10^6

For example, if the electric field in air exceeds 3000, 000 v/m, dielectric breakdown will occur and a spark is observed

Conceptual Example

133. How can we increase the capacitance of parallel plate capacitor? explain

Explanation: The capacitance of a parallel - plate capacitor can be increased by the following intervensions

- Increasing plates area
- Decreasing plates separation
- Inserting dielectric materials

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Critical Thinking Question

134. If you were asked to design a capacitor that has small size and large capacitance, what will you do?

Explanation: One possible way is a parallel plate capacitor with small separation and dielectric of large dielectric constant.

135. When a dielectric is inserted in a parallel - plate, what is its effect on the

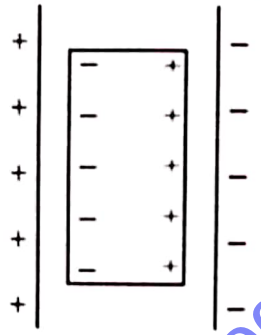
A) Capacitance of the capacitor
B) Potential difference between the plates
C) Electric field between the plates
D) Charge stored on the capacitor

Explanation

A) Increases
B) Decreases
C) Remains the same
D) Decreases

136. Explain why the electric field and the potential difference of parallel plate decreases when dielectric is inserted between the plates?

Explanation: Because, when a dielectric is inserted, there exists polarization that is separation of charges of the dielectric to opposite sides of the plates of due to electrostatic force as shown below



As a result, the net electric field decreases leading to a decrease in potential difference between the plates

Illustrative Examples

137. A parallel plate of plates area $100cm^2$ has plates separation of 2mm. What is its capacitance when it is

A) $C = \varepsilon_o \frac{A}{d} = (8.85)\frac{(10^{-12}) \times 10^{-12}}{2 \times 10^{-3}}$

$C = 4.425 \times 10^{-11}F$

B) $C = K\varepsilon_o \frac{A}{d} = \frac{4(8.85 \times 10^{-12})(10^{-12})}{3 \times 10^{-3}}$

$C = 1.77 \times 10^{-10}F$

138. What charge is stored on a capacitor of $5000cm^2$ plates area and 2mm separation with 5dielectric constant insulator between them as a 12v is supplied

Solution

$Q = CV$

$C = K\varepsilon o \frac{A}{d} = \frac{(5)(8.85 \times 10^{-12})(0.5)}{2 \times 10^{-3}}$

$C = 1.1 \times 10^{-10}F$

$Q = (1.1 \times 10^{-10})(12)$

$Q = 1.32 \times 10^{-9}c = 1.32nc$

139. What is the capacitance of an air filled parallel plate capacitor plates area $100cm^2$ and plates separation 0.1mm?

Solution

$C = K\varepsilon o \frac{A}{d} = \frac{(8.85 \times 10^{-12})(10^{-2})}{0.1 \times 10^{-3}}$

$C = 8.85 \times 10^{-10}F$

140. An air filled parallel plate capacitor has a capacitance of 1PF. When the plates separation is doubled and filled with wax, the capacitance increase to 2PF. What is the dielectric constant of wax?

A) 2
B) 4
C) 6
D) 8

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Solution

$$C = K\varepsilon_o \frac{A}{d} = \frac{KC_o}{2}$$

$$2 = \frac{K}{2}$$

$$K = 4$$

141. When a p.d of 150v is applied to the plates of a parallel plate capacitor, the plates carry a charge density of 30nc/cm² What is the spacing between the plates?

Answer: B

A) 2.24μm B) 3.52μm C) 12.24μm D) 4.42μm

Solution

$$V = Ed, \varepsilon = \frac{\delta}{\varepsilon_o}$$

$$V = \frac{\delta d}{\varepsilon_o} \Rightarrow d = \frac{V\varepsilon_o}{\delta}$$

$$d = \frac{(150)(8.85 \times 10^{-12})}{3 \times 10^{-4}} = 4.42 \times 10^{-6} m$$

$$d = 4.42Nm$$

Answer: D

142. A parallel plate capacity has a capacitance of 10μF. What will be its capacitance if the plates separation is double?

A) 1.2μF B) 5μF C) 10μF D) 20μF

Solution

$$C_1 = K\varepsilon_o \frac{A}{d} = 10NF$$

$$C_2 = \frac{K\varepsilon_o A}{2d} = \frac{10}{2} = 5NF$$

Answer: B

143. What dielectric constant causes a 12V p.d to 5v p.d?

A) 2 B) 4 C) 2.4 D) 4.2

Solution

$$V = \frac{Vo}{K} \Rightarrow K = \frac{Vo}{V} = 12/5 = 2.4$$

Answer: C

144. The electric potential along x - axis varies with x according to $v(x) = 4x + 5x^2$. What is the electric field at x = 2m?

A) 20v/m along the tue x - axis
B) 20v/m along the -ve x - axis
C) 24v/m along the - ve x - axis
D) 24v/m along the tue x - axis

Solution

$$E = \frac{-\Delta v}{\Delta x} = \frac{-dv}{dx} = 4 + 10x$$

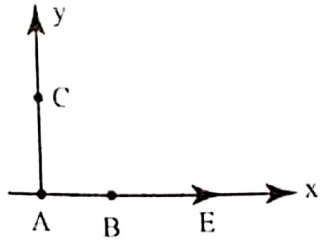
at x = 2, E = 4 + 10(2) = 24

E = 24 v/m

Answer: C

145. A uniform electric field pointing in x - direction exists let A be the origin, B is at x = 1cm and C is at y = 1cm Then the potential at points A,B and C satisfy?

A) $V_A < V_B$ B) $V_A > V_B$ C) $V_A < V_C$ D) $V_A > V_C$



Hint: Potential decreases in the field direction

Answer:

146. An electron is released in a region of field. it then experience?

A) Increasing potential C) Zero potential
B) Decreasing potential D) Constance potential

Hint: The electron moves in opposite direction to the applied electric

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field

147. If we treat the earth as a conducting sphere of radius 6400km, its capacitance will be the order of

Answer: A

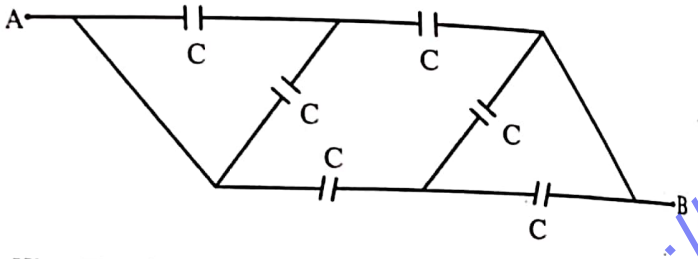
A) 1μF B) 1mF C) 1F D) 1000F

Hint: $C = \frac{Q}{V} = \frac{Q}{\frac{KQ}{r}} = \frac{r}{k} = \frac{6400,000}{9 \times 10^9} \approx 1mF$

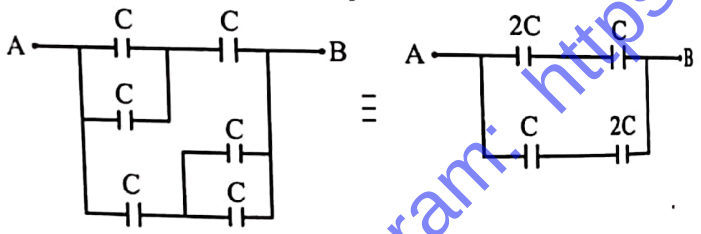
148. What is the equivalent capacitance between A and B?

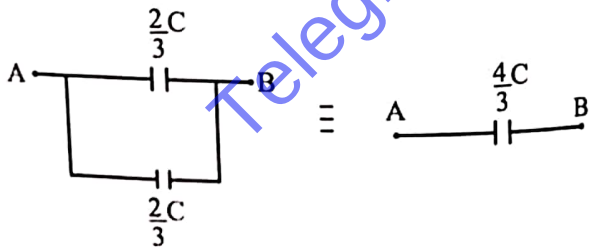
Answer: B

A. $\frac{C}{4}$ B. $\frac{4}{3}c$ C. $\frac{3C}{4}$ D. 3c



Hint: The given net work is equivalent to

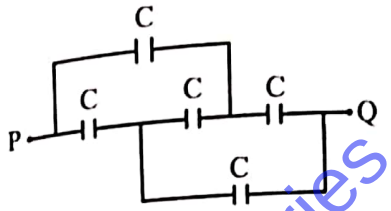




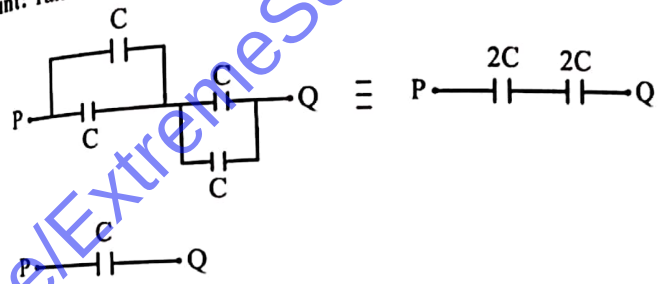
Answer: C

149. What is the equivalent capacitance between P and Q

A. $\frac{C}{5}$ B. $\frac{C}{2}$ C. $\frac{C}{3}$ D. C



Hint: Take out the common capacitor to get;



Answer: D

150. A capacitor of capacitance 1F will

A) Be fully charged in 1 second by a current of 1A

B) Store 1C of charge at a potential difference of 1V

C) Discharge in 1second when connected to 1Ω resistor

D) Gain 1J of energy when 1c of charge is stored on it

Hint: $C = \frac{Q}{V} \Rightarrow 1F = \frac{1C}{V}$

Answer: B

151. Two capacitors of equal capacitance are first connected in series and then in parallel. The ratio of their capacitance, series to parallel will be?

A) 2:1 B) 1:2 C) 4:1 D) 1:4

Hint: $\frac{1}{Cseries} = \frac{1}{C} + \frac{1}{C} = \frac{2}{C}$

$Cseries = \frac{C}{2}$

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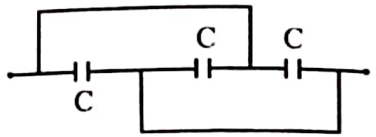
$C_{parallel} = C + C = 2C$

$$\frac{C_{series}}{C_{parallel}} = \frac{C}{2C} = \frac{1}{4} = 1:4$$

152. Three equal capacitors are connected as shown. What is the equivalent capacitance? **Answer: D**

A. C B. 3C C. $\frac{C}{3}$ D. $\frac{3}{2}C$



Hint: They are connected in parallel

153. Two capacitors of each 2NF are connected in parallel and their combination is connected to 12NF in series. What is the equivalent capacitance of the system? **Answer: B**

A) 16μF B) 13μF C) 4μF D) 3μF

Solution

$C = C_1 + C_2 = 2 + 2 = 4$

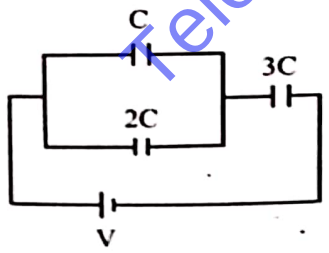
$$\frac{1}{C_T} = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$$

$C_T = 3NF$

Answer: A

154. What is the charge stored in capacitor C? **Answer: A**

A. $\frac{1}{2}CV$ B. CV C. 2CV D. $\frac{3}{2}CV$



Hint: $C_T = \frac{3}{2}C$

$Q_T = C_T V = \frac{3}{2}CV$

P.d across 3c, $\Rightarrow \frac{\frac{2}{3}CV}{3V} = \frac{V}{2}$

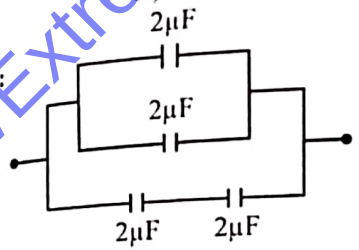
Charge on C, $\Rightarrow C\frac{V}{2} = \frac{1}{2}CV$

Answer: A

155. The minimum number of 2μF capacitors to get a capacitance of 5μF is?

A) 3 B) 4 C) 5 D) 6

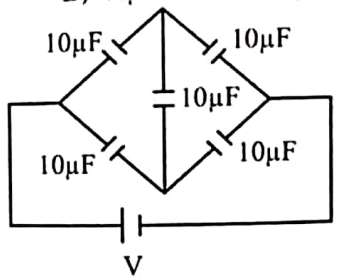
Hint:



Answer: B

156. What is the equivalent capacitance of the network shown below

A) 10μF B) 40μF C) 30μF D) 20μF



Hint: Take out the middle one

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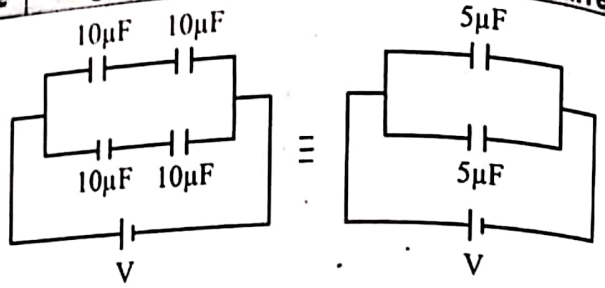


$C_T = 10\mu F$

Answer: A

157. Three capacitors of capacitance 6μF each are available The minimum and maximum capacitance obtained are?

A) 6μF and 19μF
B) 3μF and 12μF
C) 12μF, 26μF
D) 2μF, 18μF

Solution

$$\frac{1}{C\text{series}} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

$C\text{series} = 2\mu F$

$C_P = 6 + 6 + 6 = 18\mu F$

Answer: D

158. Eight identical capacitors when connected in series have equivalent capacitance of C. The equivalent capacitance when they are connected in parallel is?

A) 8C
B) 64C
C) C/8
D) C/64

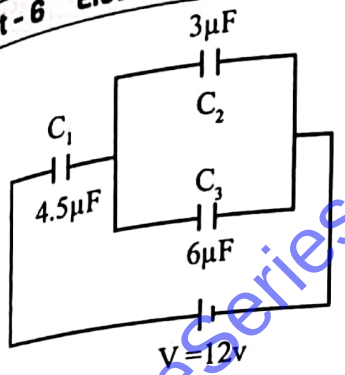
Solution

$C_T = C + C + C + C + C + C + C + C = 8C$

Answer: A

159. The P.d across C_1 is?

A) 3V
B) 6V
C) 4V
D) 8V



Solution

$CT = 3\mu F$

$$V_1 = \frac{Q}{C_1} = \frac{(3)(12)}{4.5}$$

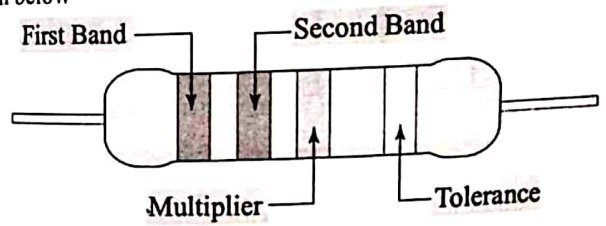
$V_1 = 8V$

Answer: D

6.13 Color Code Resistors

The resistance value of resistor is usually lablled by a color code. The different color band's have their own meaning.

For example, in a four band resistor, the first band represents the first digit, the second band represents the second digit, the third band represent s multiplier to the base 10 and the fourth band represents the tolerance as shown below



In reading color code, in a four band resistor for example, the first three strips have equal gaps and the fourth strip that represent the tolerance is located a bit further away.

The following chart gives the value of the bands in a five band resistor

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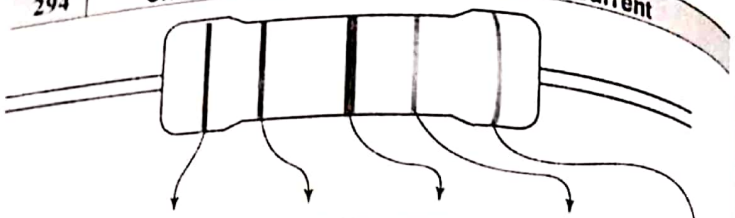
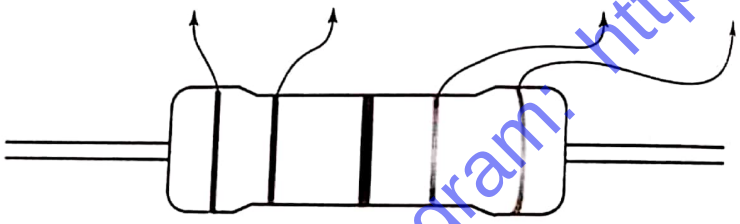


Color	1st Bane	2nd Band	3rd Band	Multiplier	Tolerance
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	±1% (F)
Red	2	2	2	100Ω	±2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	±0.5% (D)
Blue	6	6	6	1MΩ	±0.25%(C)
Violet	7	7	7	10MΩ	
Gray	8	8	8		± 0.05%
White	9	9	9		
Gold				0.1	±5% (J)
Silver				0.01	±10% (K)



Illustrative Examples

160. Which is the resistance of a carbon resistor with strip colors of Green, Yellow and Gold in a four band color resistor?

Solution

From the color chart, we have:

1st Band, Green = 5 Multiplier, Yellow = ×10KΩ

2nd Blue = 6 Tolerance, Gold = ± 5Ω

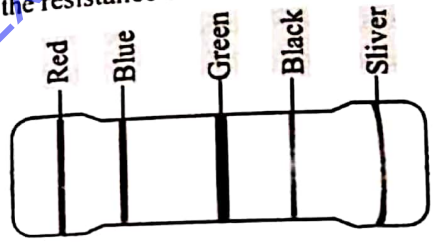
Therefore, the resistance of the resistor is

$R = 56 \times 10K\Omega \pm 5\%$

$R = 560,000\Omega \pm 5\%$

$R = 560,000\Omega \pm 28,000\Omega$

161. What is the resistance of the resistor shown below



Solution

From the color chart, We have

1ST Band, Red = 2 Multiplier, Black = 1Ω

2nd Band, Blue = 6 Tolerance, silver = ± 10%

3rd Band, Green = 5

Therefore, resistance of the resistors is;

$R = 265\Omega \pm 10\%$

$R = 2605\Omega \pm 26.5\Omega$

Illustrative Examples

162. A current in a torch bulb is 0.1A. How will it take for one coulomb

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of charge to pass through the bulb?

A) 5sec B) 10sec C) 0.1 sec D) 1.0 sec

Solution

$$I = \frac{Q}{t} = \frac{1}{0,1} = 10\text{sec}$$

Answer: B

163. A current of one Ampere corresponds to the flow of one coulomb of charge

A) In one second
B) In a uniform electric field
C) Across one ohm resistor
D) A cross one volt P.d

Hint: $I = \frac{Q}{t} \Rightarrow 1A = \frac{1C}{S}$

Answer: A

164. For an electric current in a wire

A) The current direction is opposite electron flow direction
B) Current is due to random motion of electrons in the wire
C) Current is due to perpendicular motion of electrons in the wire
D) Conventional current is the flow of electrons in the wire

Hint: Conventional current is opposite to electrons flow direction

Answer: A

165. A current of 16A flows through a resistor for 4minutes. What is the number of electrons flowing through the resistor?

A) 6.8×10^{22} B) 1.4×10^{22} C) 2.4×10^{22} D) 3.4×10^{22}

Solution

$$I = \frac{Q}{t} \Rightarrow Q = It = ne$$

$$n = \frac{It}{e} = \frac{(4)(60)(16)}{1.6 \times 10^{-19}}$$

$$n = 2.4 \times 10^{22} \text{ electrons}$$

Answer: C

166. Suppose a resistor is connected to a voltage supply. When will the current in the resistor decrease?

A) When length of the resistor is increased
B) When length of the resistor is decreased
C) When cross sectional area of the resistor increased
D) When the resistance is decreased

Hint: $R = \rho\frac{\ell}{A}$

Answer: A

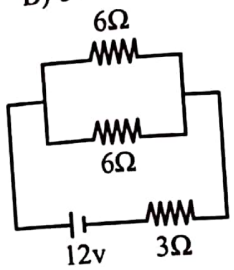
167. You are given three 1Ω resistors. All of the following equivalent resistors can be produced using all three except?

A. $\frac{1}{3}\Omega$ B. 1.5Ω C. 1Ω D. 3Ω

Answer: C

168. What is the potential drop across the 3Ω resistor

A) 6v B) 3v C) 12v D) 9v



Solution

The two 6Ω are in parallel and their result is 3Ωwhich is in series with resistor. R = 6Ω

$$I = \frac{v}{R} = \frac{12}{6} = 2A$$

$$V_3 = (3)(2) = 6v$$

Answer: A

169. What is the current through the 4Ω resistor

A) 0.125 A
B) 0.250A
C) 1.5A
D) 0.5A

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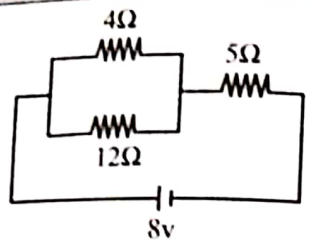


Solution

The 4Ω and 12Ω are in parallel and their resultant is 3Ω which is in series with the 5Ω.

$$R = 8\Omega\ ,\ I = \frac{V}{R} = \frac{12}{6} = 1.5A$$

$$I_4 = \frac{0.5}{4} = 0.125A$$

Answer: A

170. Ten identical resistance each having a resistance of 1Ω are joined parallel. What is their equivalent resistance?

A) 10Ω B) 1Ω C) 0.1Ω D) 0.01Ω

Solution

$$\frac{1}{R} = \frac{1}{1}+\frac{1}{1}+\frac{1}{1}+\frac{1}{1}+\frac{1}{1}+\frac{1}{1}+\frac{1}{1}+\frac{1}{1}+\frac{1}{1}+\frac{1}{1}$$

$$\frac{1}{R} = \frac{1}{10}$$

$$R = 0.1\Omega$$

Answer: C

171. The resistance of a conductor carrying a current of 2A when the potential difference across it is 20V

A) 0.1Ω B) 1Ω C) 10Ω D) 40Ω

Solution

$$V = IR \Rightarrow R = \frac{V}{I} = \frac{20}{2}$$

$$R = 10\Omega$$

Answer: C

172. Three resistors are given. How many combinations can be made from them?

A) 3 B) 4 C) 5 D) 6

Solution

All in series, all in parallel, two in series and parallel with third, two in parallel and series with the third

Answer: B

173. A total charge of 5cpasses through a conductor in 10ms. What is the current in the current?

A) 50A B) 125A C) 500A D) 250A

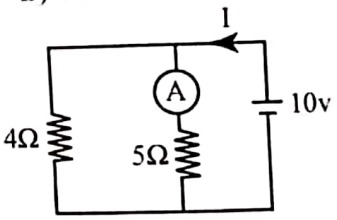
Solution

$$I = \frac{Q}{t} = \frac{5}{10\times10^{-3}}\ 500A$$

Answer: C

174. What will be the reading of the ammeter in the circuit shown below?

A) 40A B) 10A C) 5A D) 2A



Solution

$$I = \frac{V}{R} = \frac{10}{5} = 2A$$

Answer: D

175. Two resistors of 15Ω and 30Ω resistance are connected in parallel. What should be the value of R to be connected in series with the two so that net resistance will be 20Ω

A) 5Ω B) 10Ω C) 15Ω D) 120Ω

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Solution

$$\frac{1}{R} = \frac{1}{15} + \frac{1}{30} = \frac{3}{30}$$

$R = 10\Omega$

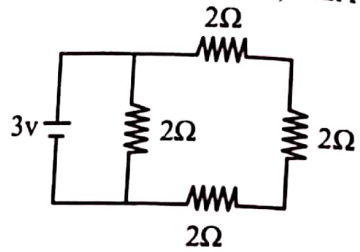
$R_T = 10 + R_3 = 20$

$R_3 = 10\Omega$

176. What is the current flowing through the following circuit?

Answer: B

A) 1A B) 0.5A C) 1.2A D) 2A



Solution

The three 2Ω are in series, with a result of 6Ω which is in parallel with the 2Ω

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{6} = \frac{4}{6}$$

$$R = \frac{3}{2}\Omega$$

From ohm's law, $I = V/R = 3/3/2 = 2A$

$I = 2A$

Answer: D

177. A wire of the resistance R is cut in to n equal parts These parts are then connected in parallel. The equivalent resistance of the combination is?

A) nR B) R/n C) n/R D) R/n^2

Solution

Resistance of each part is, $\frac{R}{n}$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \text{-----} + \frac{1}{R_n}$$

$$\frac{1}{R_T} = \frac{1}{R/n} + \frac{1}{R/n} + \text{-----} + \frac{1}{R_n}$$

$$\frac{1}{R_T} = \frac{1}{R/n} = \frac{n^2}{R}$$

$$R_T = \frac{R}{n^2}$$

Answer: D

178. A student has 10identical resistors each of resistance R. The minimum resistance made by him will be?

A) 10R B) R/10 C) R/100 D) R/5

Solution

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \text{-----} + \frac{1}{R_{10}}$$

$$\frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} + \text{-----} + \frac{1}{R}$$

$$\frac{1}{R_T} = \frac{10}{R} = \frac{n^2}{R}$$

$R_T = R/10$

Answer: B

179. What is the number of equal parts in to which a conductor having a resistance of 100Ω should be cut to obtain the resistance of 1Ω when the parts are connected in parallel?

A) 5 B) 10 C) 20 D) 2

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Solution

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \text{----} + \frac{1}{Rn}$$

Each piece has a resistance of R/n

$$\frac{1}{1} = \frac{1}{R/n} + \frac{1}{R/n} + \text{-----} + \frac{1}{R/n}$$

$$\frac{1}{1} = \frac{n^2}{R}$$

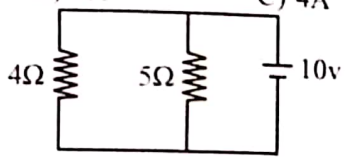
$$n = \sqrt{R} = \sqrt{100}$$

$$n = 10$$

180. What is the current flowing through the 4Ω resistor

Answer: B

A) 10A B) 5A C) 4A D) 2.5A



Solution

$$I = \frac{V}{R} = \frac{10}{4}$$

$$I = 2.5A$$

Answer: D

181. What is the equivalent resistance between x and Y

A) 12Ω B) 24Ω C) 8Ω D) 36Ω

Solution

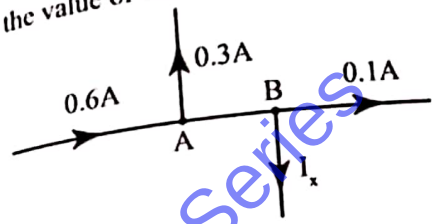
The first 12Ω resistors in series and their resultant is 24Ω which is in parallel with the third 12Ω

$$\frac{1}{R} = \frac{1}{12} + \frac{1}{24} = \frac{3}{424}$$

$$R = 8\Omega$$

Answer: C

182. What is the value of current Ix



Solution

Using kirchhoff's point rule at B

$$\Sigma I_{in} = \Sigma I_{out}$$

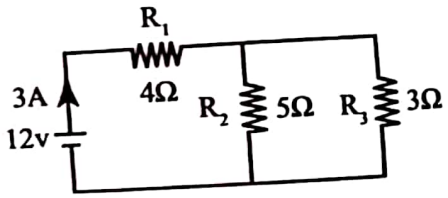
$$0.3 = I_x + 0.1$$

$$I_x = 0.2A$$

183. For the give circuit find

A) R_3

B) V_3



Solution

A) $R = \frac{V}{I} = \frac{12}{3} = 4\Omega$

$$R = R_1 + \frac{R_2 R_3}{R_1 + R_3}$$

$$4 = 2 + \frac{3R_2}{3+R_2}$$

$$\frac{3R_2}{3+R_2} = 2$$

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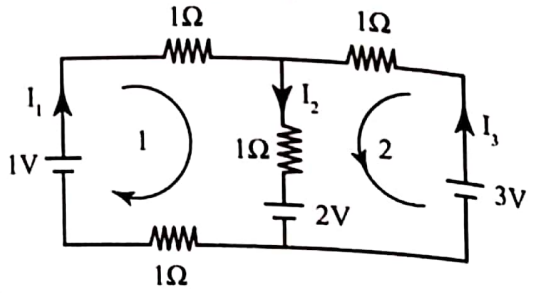
$3R_2 = 6 + 2R_2$

$R_2 = 6\Omega$

B) $V_3 = V - V1 = 12 - (3)(2)$

$V_3 = 6V$

184. Determine the values of I_1, I_2 and I_3



Solution

Point rule

$I_1 + I_3 = I_2$ -------- (1)

Loop - 1

$1 - 2 = I_1 + I_1 + 2$

$-1 = 2I_1 + I_2$ -------- (2)

Loop - 2

$3 - 2 = I_2 + I_3$

$1 = I_2 + I_3$ ---------- (3)

185. Solve the given equitations by substitution and elimination as follows

$-I_2 + I_3 = 1$

$I_2 + 2I_1 = -1$

$I3 - 2I_1 = 2$

And from equation (1), $I_3 = I_2 - I_1$

$(I_2 - I_1) - 2I_1 = 2$

$I_2 - 3I_1 = 2$ ---------- (4)

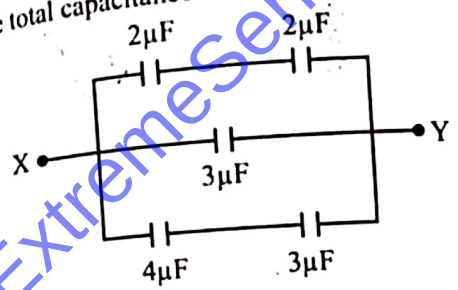
From equation (2) $2I1 + I2 = -1$

$I_2 - 3I_1 = 2$

$I_2 + 2I_1 = -1$

$I_1 = -0.6A$, $I_2 = 0.2A$ and $I_3 = 0.8$

186. Find the total capacitance of the network shown below



Solution

$\frac{1}{C_1} = \frac{1}{2} + \frac{1}{2} = \frac{2}{2}$

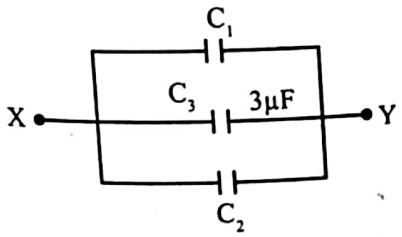
$C_1 = 1\mu F$

$\frac{1}{C_2} = \frac{1}{4} + \frac{1}{3} = \frac{7}{12}$

$C_2 = \frac{12\mu F}{7}$

$C = C_1 + C_2 + C_3 = 1 + \frac{12}{7} + 3$

$C = \frac{40}{7}\mu F$



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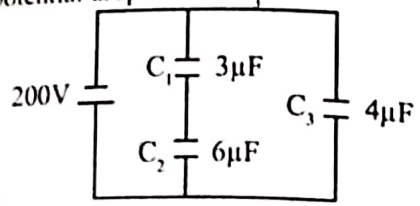
187. For the net work shown; find

A) The charge stored on C_3

B) The potential drop a cross C_1



Solution

A) $\frac{1}{C_{12}} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$

$C_{12} = 2\mu F$

$C = C_{12} + C_3 = 2 + 4 = 6\mu F$

$Q = Cv = (6)(200) = 1.2 \times 10^{-3} C$

$Q_3 = v_3 c_3 = (200)(4 \times 10^{-6}) = 0.8 \times 10^{-3} c$

B) $V_3 = V = 200V$

End of Unit Questions and Problems

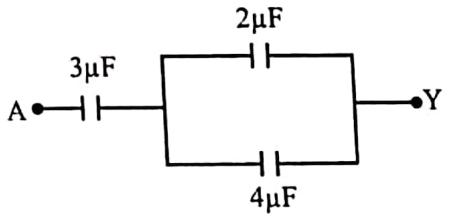
I. Give short answer to the Following Problems

1. What is electric charge ?
2. List properties of electric charge
3. State coulombls law in symbols
4. Define electric field
5. Mension characteristic of field lines
6. What is electric potential
7. Define capacitance
8. Why do plastics charge lose sooner on a damp day?
9. Why do plastics frequently absorb dust?
10. What is electric current?
11. State ohm's law

12. Is there any physical quantity whose unit is Ampere second
13. Is it possible to connect a group of resistors each of resistance R such that the equivalent resistance is less than R

II. Solve the Following Problems

14. How fer a part should two electrons be if the electric force each exert on the other equals weight of an electron?
15. How big a force will a charge of 2c experience if placed in an electric field of strength 18N/c?
16. What is the electric field strength 0.4m from a 7×10^{-5} charge?
17. Two charge 4μc and 8Nc are 2m apart
    A) What is the electric field mid - way between them?
    B) Where will the field be zero?
18. A body of mass 10^{-16}kg carries a charge of 10^{-6}c. What field ballances its weight?
19. If a 2μF capacitor is charged from a 6v battery, how much charge will it hold?
20. Given a 3μF and a 6μF capacitors. What will be their equivalent capacitance when they are connected in
    A) Series
    B) Parallel
21. What is the equivalent capacitance of the net work shown?



22. Charge flows along a wire at a rate of 3A. How many coulombs of charge will pass a point in the wire in 2minutes?
23. What are the largest and the smallest resistance obtained while connecting 1Ω, 2Ω and 3Ω resistors
24. What are the
    A) Current through the 6Ω resistor
    B) Potential drop a cross the 2Ω resistor

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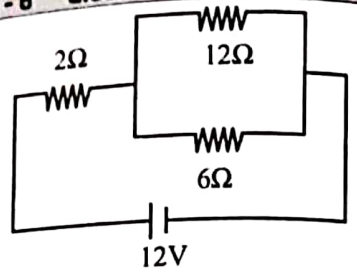
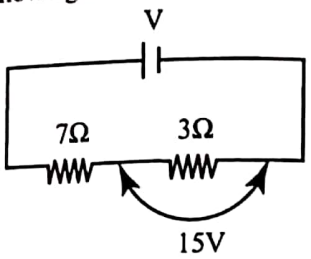


25. What current is flowing through the 7Ω resistor



I. Short Answer and Explanations

1. Electric charge a nature property of a body which is evident when a body gained or lost electrons
2. Electric is quantized, conserved,and the smallest charge known so far is the charge of an electron or a proton
3. Coulomb's law states that, the magnitude of the electrostatic force between two point charges is directly proportional to the product of the magnitude of the charges and inversely proportional to the square of the distance between them
4. Electric field is a region around a charged body where a test charge felt an electric force
5. Field lines never cross each other, they are symmetrical, start from the positively charged body and go on to the negatively charged body.
6. Electric potential is the work done per unit charge needed to bring a test charge from infinity to its current location
7. Capacitance is the ability of a capacitor to store charge
8. Because, moisture causes the charged particles to leak off in to the air
9. Because, plastics are insulators and once static charge store on them, it

stay there for some time.

10. Electric current is the flow of charge through a conductor per unit time.
11. Ohm's law states that, constant temperature the voltage supplied across its ends
12. Yes, charge
13. Yes, Parallel combination of resistors

II. Solution to the End of Unit Problems

14. $F_c = F_g$

$$\frac{ke^2}{r^2} = mg$$

$$r = \sqrt{\frac{ke^2}{mg}} = \sqrt{\frac{(9\times10^{9})(1.6\times10^{-19})^2}{(9.1\times10^{-31})(10)}} = 5m$$

15. $E = \frac{F}{Q}$

$F = EQ = (18)(2)$

$F = 36N$

16. $E = \frac{KQ}{r^2} = \frac{(9\times10^{9})(7\times10^{-5})}{(0.4)^2}$

$E = 3.9\times10^{-3}$ N/c

17. A. $E = E_1 - E_2$

$$E = \frac{KQ_1}{r^2} - \frac{KQ_2}{r^2} = 9\times10^{9}\left[\frac{8\times10^{-6}}{12} = \frac{4\times10^{-6}}{12}\right]$$

$E = 3.6\times10^{4}$ N/c

B. $E_1 = E_2 \Rightarrow 9\times10^{9}\left[\frac{8\times10^{-6}}{(2-x)} = \frac{4\times10^{-6}}{x2}\right]$

$x = 0.8$m from the 4μc charge

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18. $Fe = W$

$EQ = mg$

$$E = \frac{mg}{Q} = \frac{(10-6)(10)}{10^{-6}}$$

$E = 10N/C$

19. $Q = Cv = (2 \times 10^{-6})\ (6)$

$Q = 12\mu c$

20. A. $\frac{1}{C} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$

$C = 2\mu F$

B. $C = 3 + 6 = 9$

$C = 9\mu F$

21. The 2μF are in parallel and their resultant is 6μF. Which then became in series with the 3μF capacitance.

$$\frac{1}{C} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

$C = 2\mu F$

22. $Q = It = (3)\ (2)\ (60)$

$Q = 360c$

23. Largest, $1 + 2 + 3 = 6\Omega$

Smallest, $\frac{1}{1} + \frac{1}{2} + \frac{1}{3} = \frac{1}{R} \Rightarrow r = 0.6\Omega$

24.

A) The 12Ω and the 6Ω are in parallel and their resultant is 4Ω which is in series with the 2Ω and the given a total of 6Ω

$$I = \frac{V}{R} = \frac{12}{6} = 2A$$

B) $V_2 = (2)(2) = 4V$

25. $I = \frac{V}{R} = \frac{15}{3} = 5A$, which also pass through the 7Ω

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UNIT 7

7. Nuclear Physics

7.1 What is Nucleus?

7.2 What is Radioactivity?

7.3 Nuclear Reaction

7.4 Use of Nuclear Radiation?

7.5 Safety Rules Against Nuclear Radiation

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Nuclear physics

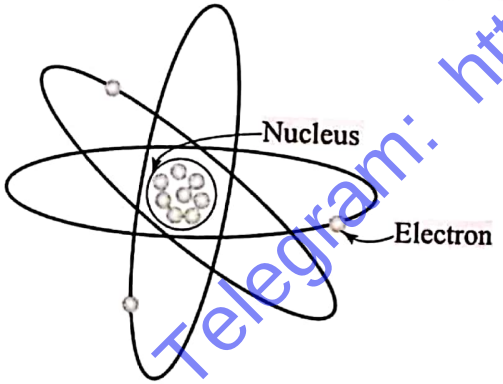
Dear learner, we are glad to meet you in this fascinating unit where you will be able to,describe the nucleus , explain radioactivity, determine types of radioactive decays, define nuclear fission and nuclear fusion, get to know application of the nuclear radiation and understand the safety rules aginst hazards of nuclear radiation. Have a Nice Time.

7.1 What is Nucleur?

Matter is made up of an element, an element is made up of atoms, the atom in turn is made up of tiny particles such as electron ,proton and the neutron.

By convention ,electrons carry negative charges, protons carry positive charge and neutrons carry no charge.

At the center of the atom, protons and neutrons form the nucleus. Therefore, an atom is made up of a positively charged nucleus and negatively charged electrons that revolve around the nucleus as shown below:



The two subatomic particles, protons and neutrons are collectively called **nucleons**.

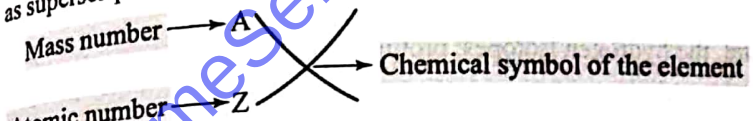
Critical Thinking Question

1. What is the difference between nucleus and nucleon?

Explanation :

The nucleus is the positively charged central part of an atom that is formed from nucleons, that is protons and neutrons which are held together by the strong nuclear force.

To describe atoms of elements,we combine chemical symbol of the elements with atomic number Z as a subscript and mass number A as superscript as shown below.



Z Atomic number = Number of Protons

A Mass number = Number of Protons + Number of Neutrons

For example ,the chemical symbol of Aluminium is Al ,its atomic number is 13 and its mass number is 27 and hence we can write it in the form of the above natation as: $^{27}_{13}Al$

The nucleus is approximately spherical and has an average radius obtained by:

$$R = R_o A^{\frac{1}{3}}$$

Where $R_o = 1.2\times10^{-15}$m is a constant

A = mass number

For example ,the radius of nitrogen is,

$$R=(1.2\times 10^{-15})\ (14)^{\frac{1}{3}} = 2.89\times 10^{-15}\ m$$

The number of protons in the nucleus determines what element the atom is .For example ,carbon atoms have a different number of protons from nitrogen atoms.

What is more ,the number of protons in a neutral atom is the same as the number electrons and the number of neutrons is equal to or the greater than the number of protons. Some elements have the same number of protons but different number of neutron. For example,carbon can have atom with 6

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neutrons ,atoms with 7 neutrons and atoms with 8 neutrons but all have 6 protons such atoms are called **isotopes**

Isotopes : Are atoms of the same element with the same atomic number but different mass number.

For example ,we have:

$^{12}_{6}C$, $^{13}_{6}C$, $^{14}_{6}C$

and, the hydrogen isotopes; protuim, deuterium and tritium respectively

$^{1}_{1}H$, $^{2}_{1}H$ and $^{3}_{1}H$

Atomic masses are usually expressed in atomic mass unit ,amu or u a system in which one atom of ^{12}C is exactly 12u where 1u = 1.660559×10^{-27}Kg

Masses of proton, neutron and electron in kilogram(kg) and the atomic mass unit (amu,u) is given below

Particle	Kg	amu,u
Proton	1.6726×10^{-27}	1.007276
Neutron	1.6750×10^{-27}	1.008665
Electron	9.109×10^{-31}	0.0005486

Did you know?

J.J Thomson discovered the electron ,discovered the proton and Chadruick discovered the neutron. What is more ,the size of the atom is estimated to be 1×10^{-10}m , the size of the nucleus is estimated to be 5×10^{-15}m and the proton is 1836 times massive than the electron.

Critical thinking Question

2. What keeps the nucleus stable?

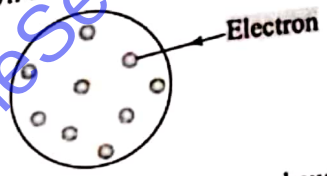
Explain

The electrostatic force between the protons is repulsive .In spite of this , protons and neutrons bond tightly together and form the nucleus due to the strong nuclear force. The driving force for the strong nuclear force is the chromodynamics of the different colors

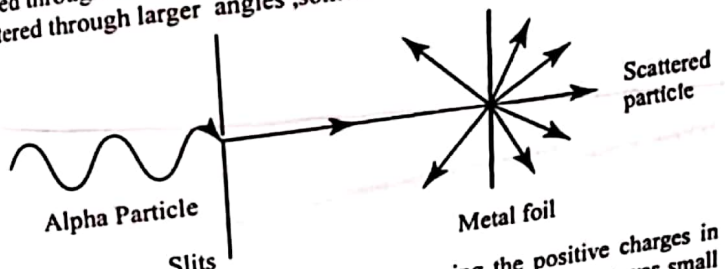
in the nucleons.

Rutherford Experiment

J.J Thomson suggested a model of the atom as a volume of positive charges with electrons embedded throughout the volume,much like the seeds in a watermelon as shown below.



Rutherford ,performed a critical experiment showing that Thomson's model couldn't be correct . In his experiment,a beam of positively charged alpha particles was projected against a thin metal foil. Most of the alpha particles passed through the foil as if it were empty space ,but a few particles were scattered through larger angles ,some even traveling backward.



Rutherford explained these results by assuming the positive charges in an atom was concentrated in a region called the **nucleus** that was small relative to the size of the atom. Any electrons belonging to the atom were visualized as orbiting the nucleus ,much as planets orbit the sun as shown below.

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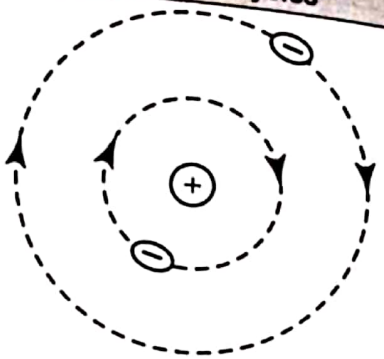


The Bohr model

Neil Bohr explained the electron shells in terms of the principal quantum number based on the planetary model.

According to him,electrons orbit the nucleus of an atom at different energy level And, when an electron in the higher energy level, E_2 drops to a lower energy level. E_1 it emits radiation obtained by

$$E_2 - E_1 = hf = \frac{hc}{\lambda}$$

Where :h 6.63×10^{-34} J.s is called **planck's constant**

f = frequency of radiation

λ = wavelength of radiation

c = 3×10^{8} m/s is speed of light in vacuum

And the wavelength of radiation is obtained by the emperical equation

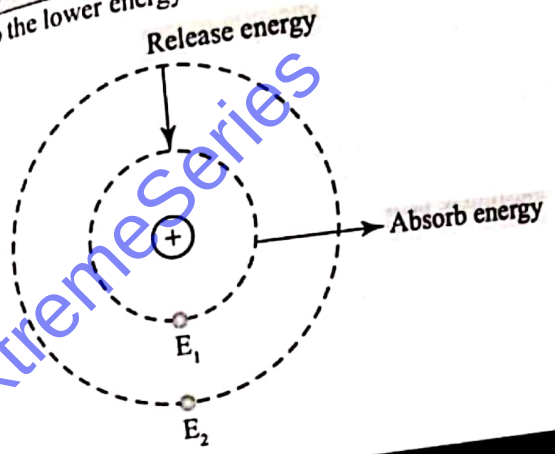
$$\frac{hc}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Where: $R_H = 1.097\times10^{7}$/m is called **Rydberg constant**

n_1 and n_2 are quantum numbers of states of the lower and higher energy levels.

✓ When an electron moves from the lower to the higher level, it absorb energy and when the electron drops from a higher

level to the lower energy level,it emits energy.



Illustrative Example

3. Calculate the wavelength, frequency and energy of a photon during an electron transition from n_3 to n_2

Solution

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{\lambda} = 1.097\times10^{7}\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$\lambda = 656.3nm$$

$$f = \frac{c}{\lambda} = \frac{3\times10^{8}}{6.563\times10^{-7}}$$

$$f = 4{,}568\times10^{14}\ HZ$$

$$E = hf = (6.63\times10^{-34})(4{,}568\times10^{14})$$

$$E = 3.027\times10^{-19}\ J$$

$$E = 1{,}892ev$$

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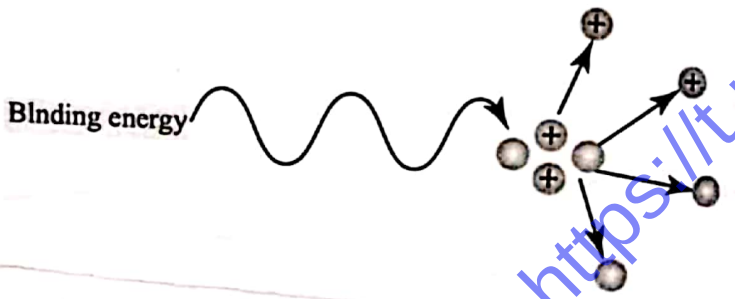
What is Binding Energy?

The stability of a nuclei is an indication of strong binding energy between nucleons. The mass of a nucleus is always less than the sum of the masses of its nucleons .And ,from the mass energy equivalence, the total energy of the nucleus is less than the combined energy of the separated nucleons. This difference in energy is called the **binging energy** of the nucleus and can be thought of as the energy that must be added to a nucleus to break it up into its constituent neutrons and proton

Binding Energy :Is the energy required to separate a nucleus into its constituent nucleons that is neutrons and protons

Binding energy



Mathematically ,binding energy is obtained by Eilstein's mass - energy equivallence equation as

Binding energy =(Mass defect) c^2

$$E_b = \Delta mc^2$$

Mass defect :Is the difference between the mass of a nucleus and the sum of masses of the neutron and the proton.

If a nucleus of mass m is composed of protons with mass m_p and neutrons with mass m_n,then the mass defect is obtained by:

$$\Delta m = Zm_p + (A - z)\, m_n - m$$

Where : z = number of protons

Therefore ,the binging energy is calculated using the relation :

$$E_b = \Delta mc^2$$

$$E_b = (Zm_p + (A - z)\, m_n - m)\, c^2$$

In nuclear physics, energy is often expressed interms of atomic energy unit, aeu that corresponds to the atomic mass unit, amu

$$1aeu = 1amuc^2 = (1.66\times10^{-27})(3\times10^8)^2$$

$$1aeu = 1.491\times10^{-10}J$$

Since $1ev = 1.6\times10^{-19}J$, then

$$1aeu = 931.1Mev$$

Where: aeu =atomic energy unit

Mev = Mega electron volt

Illustrative Example

4. Calculate the binding energy of deuterium atom that consistes of a proton, a neutron and nucleus plus electron mass of 2.014102u?

Solution:

$$E_b = \Delta mc^2 = ((m_p + m_n) - m)c^2$$

$$= ((1.007825u + 1.008665u) - 2.01402u)\, 9\times10^{16}$$

$$= (0.002388u)\, 931.1 = 2.224Mev$$

✓ On the average, the binding energy per nucleon is 8Mev

The plot of the binding energy per nucleon E_b/A as a function of mass number, A is given below

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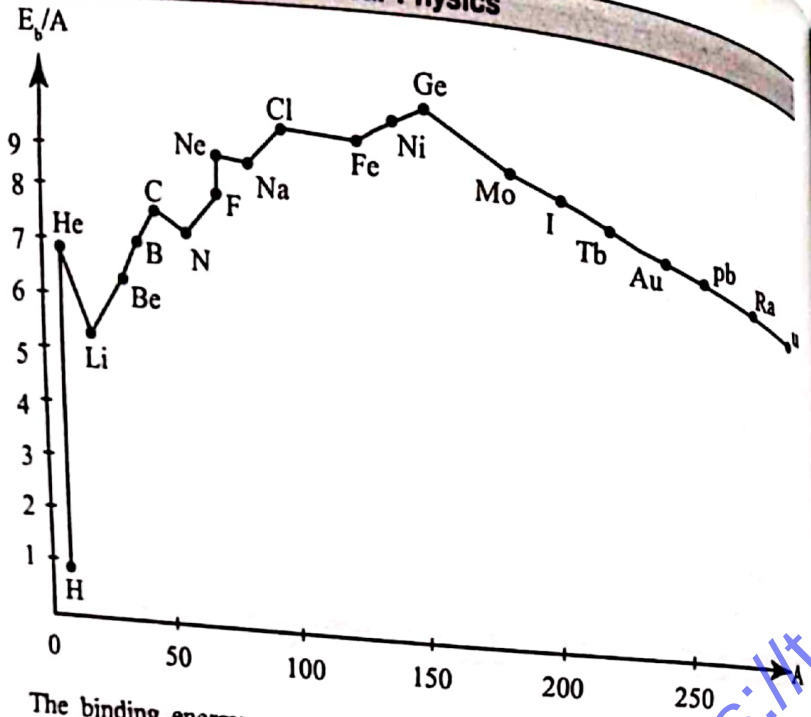

The binding energy per nucleon decreases as the number of nucleons increases. It is strongest in the range $28 < A < 138$ in these nuclei it is close to 8.7Mev and it is about 7.6Mev for the nuclei at the end the periodic table such as uranium.

Moreover ,for small mass numbers, the binding energy per nucleon is minimal for odd neucleon,

Example $^{6}_{3}Li$, $^{10}_{5}B$ and $^{14}_{7}Nc$

And,for even neucleon, the binding energy per nucleon is maxima

Example $^{4}_{2}He$, $^{12}_{6}C$ and $^{16}_{8}O$

Remember ,the binding energy per nucleon versus mass number curve tells us that ,for stability, nuclear energy can be released by fission of heavy nuclei and by the fusion of light nuclei.

Illustrative Example

5. What is an atom ?

Explanation:

It is the constituting block of a body having negatively charged electrons and positively charged nucleus

6. What are nucleons?

Explanation:

The neutrons and protons are collectively known as nucleons

7. What are isotopes ?

Explanation :

Isotopes are atoms of the same element having the same number of protons and different number of neutrons.

8. What is atomic mass unit ?

Explanation :

Is a mass unit where $1amu = 1.66053 \times 10^{-27}kg$

9. What is binding energy?

Explanation :

It is the energy required to break up a nucleus into its constituent parts that is protons and neutrons.

10. What is the mass defect?

Explanation :

It is the difference in mass of nucleus and the mass of the nucleons.

11. What does $E = mc^2$ tells us ?

Explanation

A mass- energy equivalence

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What is Radioactivity ?

Some elements such as uranium ,radium and isotopes of potassium and carbon to name but a few are unstable and disintegrated with time.

Radioactivity: Is the process of spontaneous emission of radiation by unstable nuclei

So as to get stability, radioactive elements emits radiations in the form of alpha,beta and gamma rays.

Alpha (α) Decay

When a nucleus emit an alpha particles (^4_2He), its mass number decrease by 4 and its atomic number decrease by 2.

$$^A_ZX \xrightarrow{\alpha - decay} {}^{A-4}_{Z-2}Y + {}^4_2He$$

Where : X = Parent Nucleus

Y = Daughter Nucleus

^4_2He = Helium is the end product

Example: Uranium in to Thorium

$$^{238}_{92}U \xrightarrow{\alpha - decay} {}^{234}_{90}Th + {}^4_2He$$

Beta (β) Decay

When a radioactive nucleus undergo beta decay ,the mass number remains the same but the atomic number is changed by 1

Beta minus(β^-) decay

In a beta minus decay ,the mass number remains constant and the atomic number increase by 1

$$^A_ZX \xrightarrow{B^- decay} {}^A_{Z+1}Y + e^-$$

Example Carbon into Nitrogen

$$^{14}_6C \xrightarrow{B^- decay} {}^{14}_7N + e^-$$

Beta plus (β^+) decay

In a beta plus decay, the mass number remains constant and the atomic number decrease by 1

$$^A_ZX \xrightarrow{B^+ decay} {}^A_{Z-1}Y + e^+$$

Example: Nitrogen into carbon

$$^{12}_7N \xrightarrow{B^+ - decay} {}^{12}_6C + e^+$$

Gamma (δ) decay

When a nucleons undergo gamma decay, neither its mass number nor its atomic number changes

$$^A_ZX \xrightarrow{\partial\ decay} {}^A_ZY + \partial$$

Example: Exited carbon nucleus into ground state

$$^{12}_6C \xrightarrow{\partial - decay} {}^{12}_6C + \partial$$

The penetration power and the particles emitted in the three decays is summarized below:

Name	Symbol	Emission	Penetration
Alpha	α	Helium	Sheet of paper
Beta minus	β^-	Electron	Sheet of Aluminium
Beta plus	β^+	Positron	Sheet of Aluminium
Gamma	δ	Photon	Sheet of lead

✓ Therefore, gamma is the most penetrating and alpha is the least penetrating ray.

Illustrative Example

12. What is the decay type for:

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A) $^{186}_{73}Ta \longrightarrow {}^{186}_{74}W$

B) $^{157}_{72}Hf \longrightarrow {}^{153}_{70}Yb$

C) $^{18}_{10}Ne \longrightarrow {}^{18}_{9}F$

D) $^{14}_{7}N \longrightarrow {}^{14}_{7}N$

Solution

A) Beta minus

B) Alpha

C) Beta plus

D) Gamma

Radioactive Decay

The rate at which a radioactive nuclei decays is found to be proportional to the inital nuclei ,N and is obtained by:

$$\frac{dN}{dt} = -\lambda N$$

$$\ln N - \ln N_o = \lambda t$$

$$\ln \frac{N}{N_o} = -\lambda t$$

$$\frac{N}{N_o} = e^{-\lambda t}$$

$$\boxed{N = N_o e^{-\lambda t}}$$

Where N = number of nuclei present at time t

N_o = initial number of nuclei at t = 0

ʎ = Constant and its unit is becquerel (Bq)

1Bq = 1decay/s

What is Half - life, $t\frac{1}{2}$?

Half - life is the time it takes for half of a given number of radioactive nuclei to decay, Therefore

Therefore, at $t\frac{1}{2}$, $N = \frac{N_o}{2}$

$$N = N_o e^{-\lambda t_{1/2}}$$

$$\frac{N_o}{2} = N_o e^{-\lambda t_{1/2}}$$

$$\frac{1}{2} = e^{-\lambda t_{1/2}}$$

$$-\lambda t\frac{1}{2} = \ln\frac{1}{2}$$

$$\lambda t\frac{1}{2} = \ln 2$$

$$\boxed{t\frac{1}{2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}}$$

Half - life of some radioactive elements is given below;

Uranium	4500 million years
Radium	1590 years
Protactium	3200 years
Radon	3825 years
Polonium isotopes	0.00015 years

Critical Thinking Question

13. If the half life of a radioactive nuclei is $t\frac{1}{2}$, then it will 100% decay in two half lifes .

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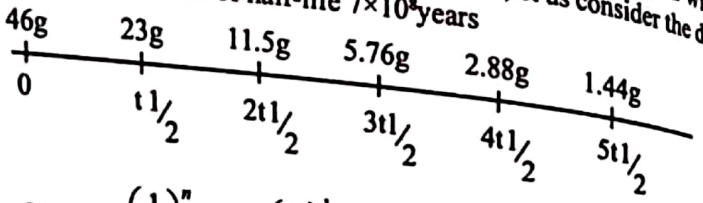
Do you agree?

Explanation:

No! After the half life undecayed nuclei is No/2, after 2 ($t_{\frac{1}{2}}$) ,the nuclei left will be No/4 and

after three half - lifes, the number of nuclei left undecayed will be No/8 and so on. To make the concept clear ,let us consider the decay of 46g of Tritium of half-life 7×10^{8}years



$$N = No\left(\frac{1}{2}\right)^{n} = 46\left(\frac{1}{2}\right)^{1} = 23g \longleftarrow t\frac{1}{2}$$

$$N = No\left(\frac{1}{2}\right)^{n} = 46\left(\frac{1}{2}\right)^{2} = 11.5g \longleftarrow 2t\frac{1}{2}$$

$$N = No\left(\frac{1}{2}\right)^{n} = 46\left(\frac{1}{2}\right)^{3} = 5.76g \longleftarrow 3t\,1/2$$

$$N = No\left(\frac{1}{2}\right)^{n} = 46\left(\frac{1}{2}\right)^{4} = 2.88g \longleftarrow 4t\,1/2$$

$$N = No\left(\frac{1}{2}\right)^{n} = 46\left(\frac{1}{2}\right)^{15} = 1.44g \longleftarrow 5t\,1/2, \textit{Okay!}$$

Illustrative Example

14. Compare the charges of alpha ,beta and gamma rays

Explanation

Alpha → Positive

Beta minus → Negative

Beta plus → positive

Gamma → Neutral

15. Compare the mass number and atomic number changes in alpha ,beta and gamma decays

Explanation :

Alpha → mass number decrease by 4 atomic number decreases by 2

Beta → atomic number changes by 1

Gamma → No change

16. What is the half life of radium of decay constant 1.07×10^{-4} per years.

Solution

$$t\frac{1}{2} = \frac{0.693}{\lambda} = \frac{0.693}{1.07 \times 10^{-4}}$$

$$t\frac{1}{2} = 6476 \text{ years}$$

17. What fraction of radioactive sample has decayed after two half - lifes

Solution

$$N = No\left(\frac{1}{2}\right)^{n},\ n = 2$$

$$\frac{N}{No} = \left(\frac{1}{2}\right)^{n} = \left(\frac{1}{2}\right)^{2}$$

$$\frac{N}{No} = \frac{1}{4}$$

18. A 16g of radon with half - life of 4 days is present at t = 0,what is the mass of radon left after 24 days?

Solution

$$N = N_{o}\left(\frac{1}{2}\right)^{n}$$

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$$n = \frac{t}{t\frac{1}{2}} = \frac{24}{4} = 6$$

$$N = N_o\left(\frac{1}{2}\right)^n = 16\left(\frac{1}{2}\right)^6$$

$$N = 0.25g$$

19. What is the activity of 2×10^8 atoms of C - 14 of decay constant 3.84×10^{-12}/s ?

Solution:

$$\text{Activity} = \frac{dN}{dt} = -\lambda N = (3.84 \times 10^{-12})(2 \times 10^8)$$

$$\frac{dN}{dt} = 7.68 \times 10^{-4}\ Bq$$

7.2 Nuclear Reaction

A nuclear reaction takes place when an incident particles causes a change to occur in a target nucleus .

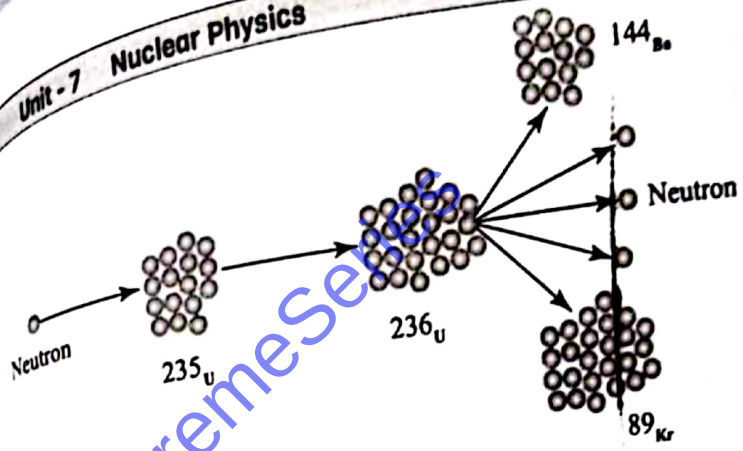
For example ,when an α particles strikes a nitrogen nucleus,an oxygen nucleus and a proton are produced.

$${}^4_2He + {}^{14}_7N \longrightarrow {}^{17}_8O + {}^1_1H$$

What is Nuclear Fission?

Nuclear fission is the process of splitting heavy nuclei into two or more nuclei.

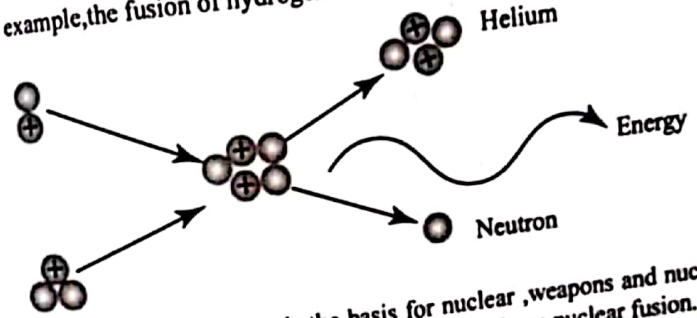
For example ,when a neutron bombards ^{235}U,^{236}U is produced and then splits into ^{144}Ba and ^{89}Kr



What is Nuclear Fusion?

Nuclear fusion is the process of combining two or more nuclei to form a heavy nucleus.

For example,the fusion of hydrogen isotopes to form a single nucleus.



Remember, nuclear fission is the basis for nuclear ,weapons and nuclear power plants while, the working of the sun depends on nuclear fusion.

7.3 Use of Nuclear Radiation

Nuclear reactors are used in agriculture,industries,medicine, water and sanitation, and power generation.

Agriculture and Food Processing

Radioisotopes are used to improve fertility of soil, increase yield, control insects, improve seed breeding, increase food safety and control animal

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production

Medical Application

Nuclear medicine is used to treat cancer ,heart disease, gastrointestinal endocrine or neurological disorder and other abnormalities. Moreover, nuclear diagnostic diagnoses cancer by using positron emitting nucleotides imaging.

Industeries

Radioisotopes are used to move machine parts, assess minerals and oil, measure pollution, disinfect machine components and optimize quality of production.

Nuclear Power

Nuclear generates electricity by the process of nuclear fission using ^{235}U as fuel for the reactor-during fusion heat is released and trapped to produce steam so as to turn turbine generator and produce electricity.

7.4 Safety Rules Against Nuclear Radiation

The international Atomic Agency has safety guideline for the proper use and storage of nuclear materials

Therefore,potential sources of hazard that trigger nuclear radiation should be identified and mitigated in advance. Since personal errors and external agents than anticipated can have a greater impact ,nuclear scientists take into account catastrophic events such as earth quake,terrorist attacks, cyberattack and so much more ,while using ,storing, transporting and disposing nuclear materials.

Illustrative Example

20. A nuclear reaction ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + e^{-}$

A) B - decay B) γ - decay C) α - decay D) Fusion

Answer :A

21. Which of the following is not a mode of radioactive decay?

A) Positron emission B) Fusion C) Electron Capture D) Alpha decay

Answer: C

22. The radioactive source is found to fall by 0.875 of its initial value in 210 seconds .What is the half life of the source ?

A) 30 seconds B) 70 seconds C) 60 seconds D) 120 seconds

Solution:

$N = 1 - 0.875 = 0.125$

$N = No\left(\frac{1}{2}\right)^{n} \Rightarrow 0.125 = 1\left(\frac{1}{2}\right)^{n}$

$\frac{125}{1000} = \left(\frac{1}{2}\right)^{3} = \frac{1n}{2} \Rightarrow n = 3$

$n = t/t_{\frac{1}{2}} \Rightarrow t_{\frac{1}{2}} = \frac{t}{n} = \frac{210}{3}$, $t_{\frac{1}{2}} = 70$ seconds

Answer :B

23. In the disintegration series ${}^{238}_{92}U \xrightarrow{\alpha} X \xrightarrow{B-} {}^{A}_{Z}U$ .The value of Z and A respectively will be?

A) 92,236 B) 90 ,234 C) 88 ,230 D) 91 ,234

Solution

α decrease mass number by 4 and atomic number by 2.And β - keeps mass number unchanged and increase atomic number by1

Answer :D

24. A sample contains 10mg of radioactive mate... half life 270days. After 540 days,the mass left will be?

A) 5mg B) 2.5mg C) 1.25m... D) Zero

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Solution

$$N = No\left(\frac{1}{2}\right)^n,\quad n = \frac{t}{t_{\frac{1}{2}}} = \frac{540}{270} = 2$$

$$N = 10\left(\frac{1}{2}\right)^2 = 2.5mg$$

25. Which of the following represents gamma decay ? **Answer :B**

A) ${}^{A}_{Z}X + \delta \longrightarrow {}^{A}_{Z-1}X + a + b$

B) ${}^{A}_{Z} + {}^{1}n_0 \longrightarrow {}^{A}X_{Z-2} + C$

C) ${}^{A}_{Z}X + \longrightarrow {}^{A}_{Z}X + \delta$

D) ${}^{A}_{Z}X + e^0_{-1} \longrightarrow {}^{A}X_{Z-2} + \alpha$

Hint: During gamma decay, neither A nor Z changes

26. In the ration : **Answer :C**

$${}^{2}_{1}H + {}^{2}_{1}H \longrightarrow {}^{4}_{2}He + x$$

x is ?

A) Electron B) Photon C) Proton D) Neutron

27. Determine the reaction equation of the decay of **Answer :B**

A) U - 238 by alpha emission

B) Th - 234 by beta minus emission?

Solution

A) ${}^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He$

B) ${}^{234}_{90}Th \longrightarrow {}^{234}_{91}Pa + {}^{0}_{-1}\beta$

28. Complete the following nuclear reactions

A) ${}^{9}_{4}Be + {}^{1}_{1}H \longrightarrow {}^{6}_{3}Li +$ _____

B) ${}^{10}_{5}B + {}^{4}_{2}He \longrightarrow {}^{13}_{7}N +$ _____

Solution

A) ${}^{9}_{4}Be + {}^{1}_{1}H \longrightarrow {}^{6}_{3}Li + {}^{4}_{2}He$

B) ${}^{10}_{5}B + {}^{4}_{2}He \longrightarrow {}^{13}_{7}N + {}^{1}_{0}n$

29. After how many years will the activity of cesium - 137 of half life 30 years change from 5.8×10^{15} Bq to 1.6×10^{6} Bq

Solution

$$N = \frac{N_o}{2^n} \Rightarrow 2^n = \frac{N_o}{N} = \frac{5.8 \times 10^{15}}{1.6 \times 10^{6}} = 3.625 \times 10^9$$

$$n = \frac{t}{t\,1/2} = 32,\ t = (30)\ (32) = 1016\ years$$

End of unit Questions and problems

I. Give short answer to the following Questions

1. Which one is massive? the electron or the proton?
2. Can we classify nuclei as massive or lighter? How
3. What is radioactivity?
4. What is nuclear fission?
5. What is nuclear fission?
6. Mension some application of nuclear reaction?
7. What is half - life of a radioactive atom
8. How do you relate nuclear radius with mass number ?
9. What is the relation between reduced mass and binding energy?
10. Which decay process emits photon?

II. Solve the Following Problem

11. What is the wavelength associated with the transition of electron from $n_1 = 4$ to $n_2 = 2$states
12. What is the frequency of spectral lines associated with transition of electron from $n = 4$ to $n_2 = 2$states

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13. Evaluate the energy levels at 532nm and 610nm wavelength of radiation

14. Determine x in the nuclear decays given below

A) $^{42}K \longrightarrow \underline{\quad X \quad} + {}_{-1}e^{0}$

B) $^{239}Pu \longrightarrow {}^{4}_{2}He \ \underline{\quad X \quad}$

15. The half - life of radium is 3.5days. What fraction of the sample remains undecayed after 14 days?

Short Answer and Explanations

1. Proton is 1836 times massive than the electron
2. Yes. Based on their atomic number, we can have, Lighter nuclei up to 20 atomic number Heavier nuclei greater than 20 atomic number
3. Radioactivity is the process of decay of unstable nuclei
4. Nuclear fusion is the process of splinting a nuclei into parts or fragments
5. Nuclear fusion is the process of combining lighter nuclei into heavier nuclei
6. Nuclear radiation is used in medcine, agriculture, defense, power generation. etc.
7. Half-life is the time it takes unstable nuclei to decay by half of its initial nuclei.
8. Nuclear radius $r = roA^{1/3}$,A is the mass number
9. Binding energy = (Reduced mass)$c^2 = Eb = mc^2$
10. Gamma decay.

Solution to the End of unit Problems

11. $\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\frac{1}{\lambda} = 1.1 \times 10^7\left(\frac{1}{2^2} - \frac{1}{4^3}\right) = 0.21 \times 10^7 m$

$\lambda = 4.84 \times 10^7 m$

12. $c = \lambda f$, $\lambda = 4.84 \times 10^{-7}$

$f = \frac{c}{\lambda}$.

$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{4.84 \times 10^{-7}}$

$f = 0.62 \times 10^{15}\ Hf$

13. $E = hf = h\frac{c}{\lambda}$

$E_1 = h\frac{c}{\lambda_1} = \frac{6.63 \times 10^{-13} \times 3 \times 10^8}{532 \times 10^{-9}}$

$E_1 = 0.037 \times 10^{-17}$ J ,and

$E_2 = h\frac{c}{\lambda_2} = \frac{6.63 \times 10^{-13} \times 3 \times 10^8}{610 \times 10^{-9}}$

$E_2 = 0.087 \times 10^{17}$ J

14. A. ^{42}x

B. ^{235}x

15. $N = N_o\left(\frac{1}{2}\right)^n$

$n = \frac{t}{t_{\frac{1}{2}}} = \frac{14}{3.5}$

$n = 4$

Therefore, $\frac{N}{N_o} = \left(\frac{1}{2}\right)^n$

$\frac{N}{N_o} = \left(\frac{1}{2}\right)^4$

$\frac{N}{N_o} = \frac{1}{16}$

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